(c) Connect the connector [7] of the new call switch assembly [6] to the in-line connection.

(d) Install the new call switch assembly [6] into the lavatory wall.

G. Test

SUBTASK 33-27-00-860-009

(1) Remove the safety tags and close the applicable circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	8	C01403	PASSENGER CABIN PASS CALL LEFT
B	8	C01404	PASSENGER CABIN PASS CALL RIGHT

(2) Do a test of the call switch assembly [6] and call light/reset switch assembly [4]:

(a) Set the call switch assembly [6] in the lavatory to the on mode.

1) Make sure the light in the call switch assembly [6] comes on.

2) Make sure the call light/reset switch assembly [4] outside the lavatory comes on.

(b) Push the call light/reset switch assembly [4] outside the lavatory to the off mode.

1) Make sure the call light/reset switch assembly [4]goes off.

2) Make sure the light in the call switch assembly [6] in the lavatory goes off.

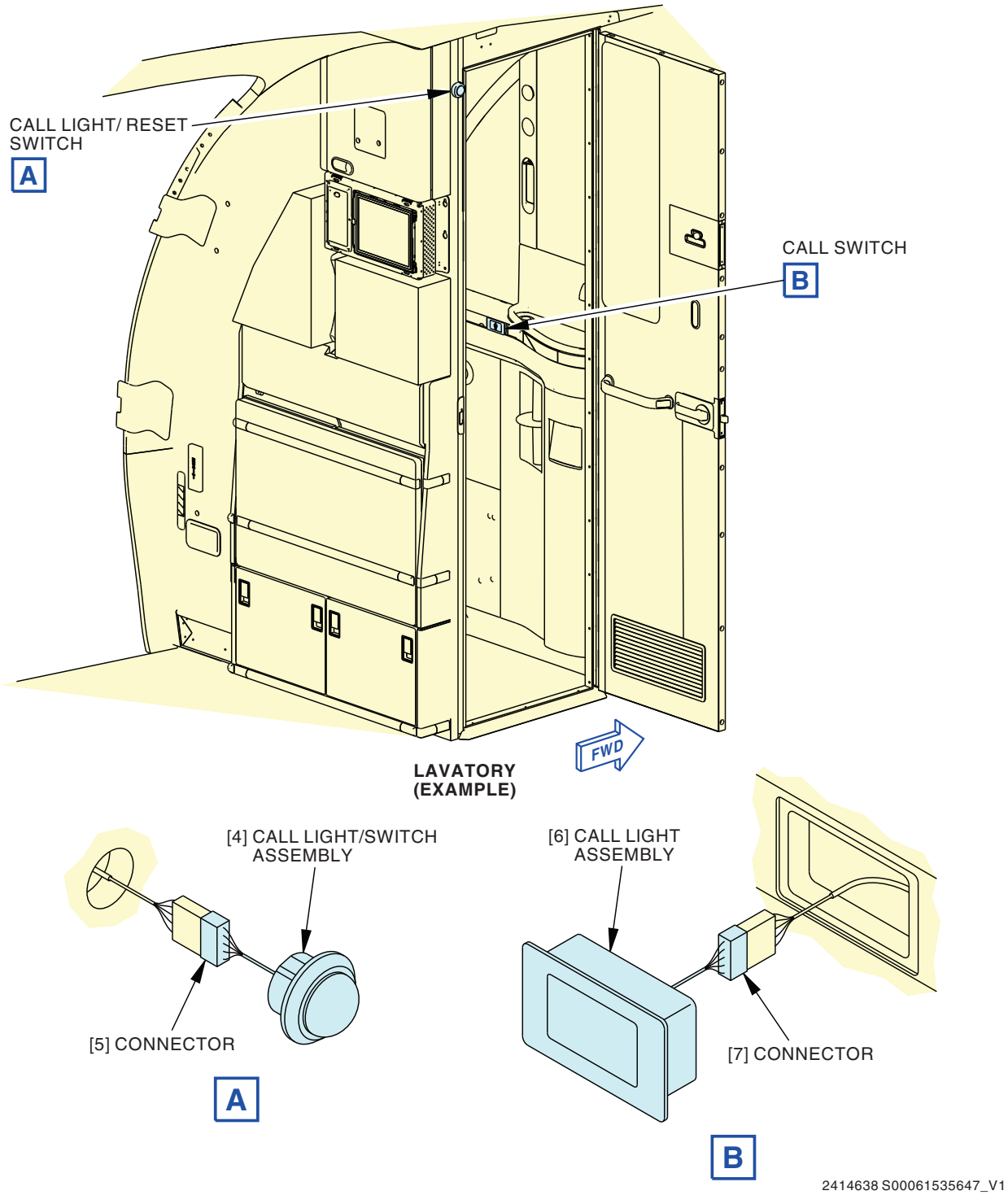
H. Put the Airplane Back to its Usual Condition

SUBTASK 33-27-00-860-010

(1) If it is not necessary, remove electrical power. To remove it, do this task: Remove Electrical Power, TASK 24-22-00-860-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Lavatory Call Light - Light/Switch Replacement
Figure 203/33-27-00-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENTRY LIGHTS - MAINTENANCE PRACTICES

1. General

A. This procedure has this task:

(1) Dim entry LED light assembly replacement.

TASK 33-29-00-960-801

2. Dim Entry Light Assembly Replacement

(Figure 201)

A. General

(1) This task is for the replacement of the dim entry light assembly.

(2) The Light Emitting Diode (LED) and its circuits are in the light assembly.

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Dim entry light assembly	33-29-00-01-020	PTF ALL
		33-29-00-15-010	PTF ALL

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Prepare for the Replacement

SUBTASK 33-29-00-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
E	9	C00155	ENTRY LIGHTS DIM

E. Dim Entry Light Assembly Replacement

SUBTASK 33-29-00-020-001

(1) Remove the dim entry light assembly [1] as follows:

(a) Lower the applicable panel to get access to the dim entry light assembly [1].

(b) Disconnect the electrical connector [4] from the dim entry light assembly [1].

(c) Hold the lens when you push the clips along the sides of the dim entry light assembly [1] to release the housing [5] from the retainer [3].

(d) Remove the housing [5] from the ceiling panel.

(e) Remove the retainer [3] and gasket [2].

SUBTASK 33-29-00-420-001

(2) Install the dim entry light assembly [1] as follows:

(a) Push the clips to remove the retainer [3] and gasket [2] from the housing [5].

(b) Move the housing [5] through the ceiling panel mounting hole.

(c) Put the gasket [2] and retainer [3] on the housing [5].

(d) Carefully push the lens and retainer [3] together until the clips latch the housing [5] to the retainer [3].

(e) Connect the electrical connector [4] to the dim entry light assembly [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Dim Entry Light Assembly Replacement Test

SUBTASK 33-29-00-710-001

(1) Do a test of the dim entry light assembly [1] as follows:

(a) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-4

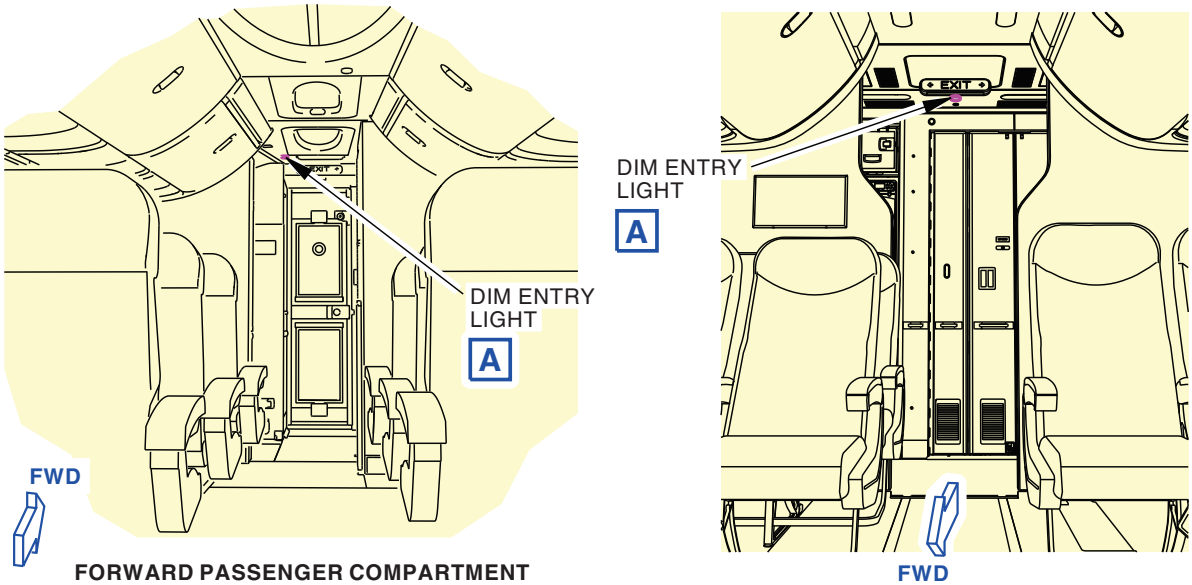
Row	Col	Number	Name
E	9	C00155	ENTRY LIGHTS DIM

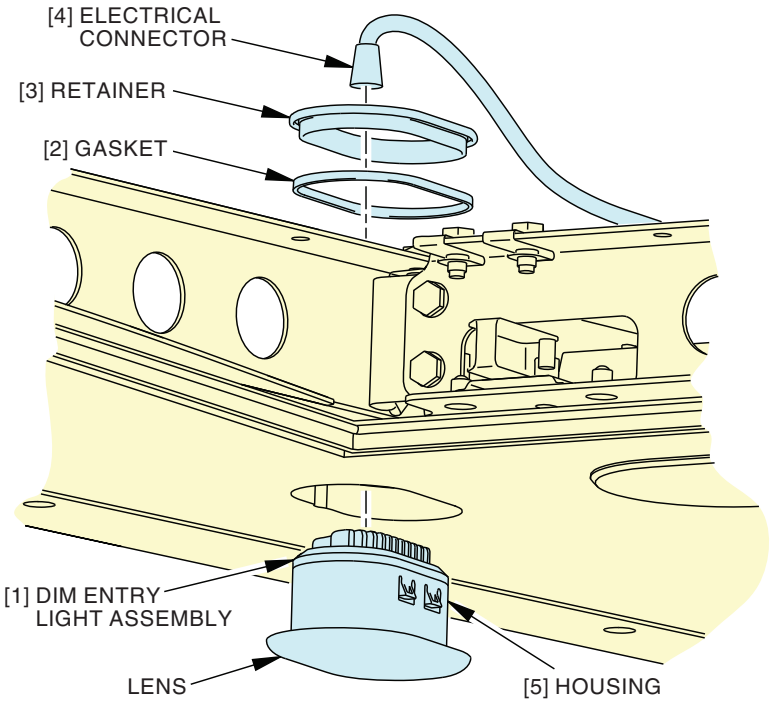
(b) Make sure that the light comes on.

NOTE: The light is energized by the 28 volt dc hot battery bus.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





DIM ENTRY LIGHT
(EXAMPLE)

A

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Dim Entry Light Assembly Installation
Figure 201/33-29-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WHEEL WELL LIGHTS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Wheel Well Lights - Deactivation

(2) Wheel Well Lights - Activation

(3) Wheel Well Lights - Light Assembly Replacement

(4) Wheel Well Lights - Operational Test.

TASK 33-32-00-040-801

2. Wheel Well Lights - Deactivation

(Figure 201)

A. General

(1) This procedure removes electrical power to the wheel well lights.

B. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
118	Electrical and Electronics Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Wheel Well Lights Deactivation

SUBTASK 33-32-00-010-001

(1) To get access to the P92 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-32-00-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	2	C00125	WHEEL WELL LT

E. Wheel Well Lights Tryout

NOTE: This tryout is to make sure that the Wheel Well Lights are in a zero energy stage.

SUBTASK 33-32-00-860-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Make sure that this circuit breaker is open and has safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	2	C00125	WHEEL WELL LT

SUBTASK 33-32-00-410-003

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-32-00-700-001

(3) Do the following steps for the Wheel Well Lights:

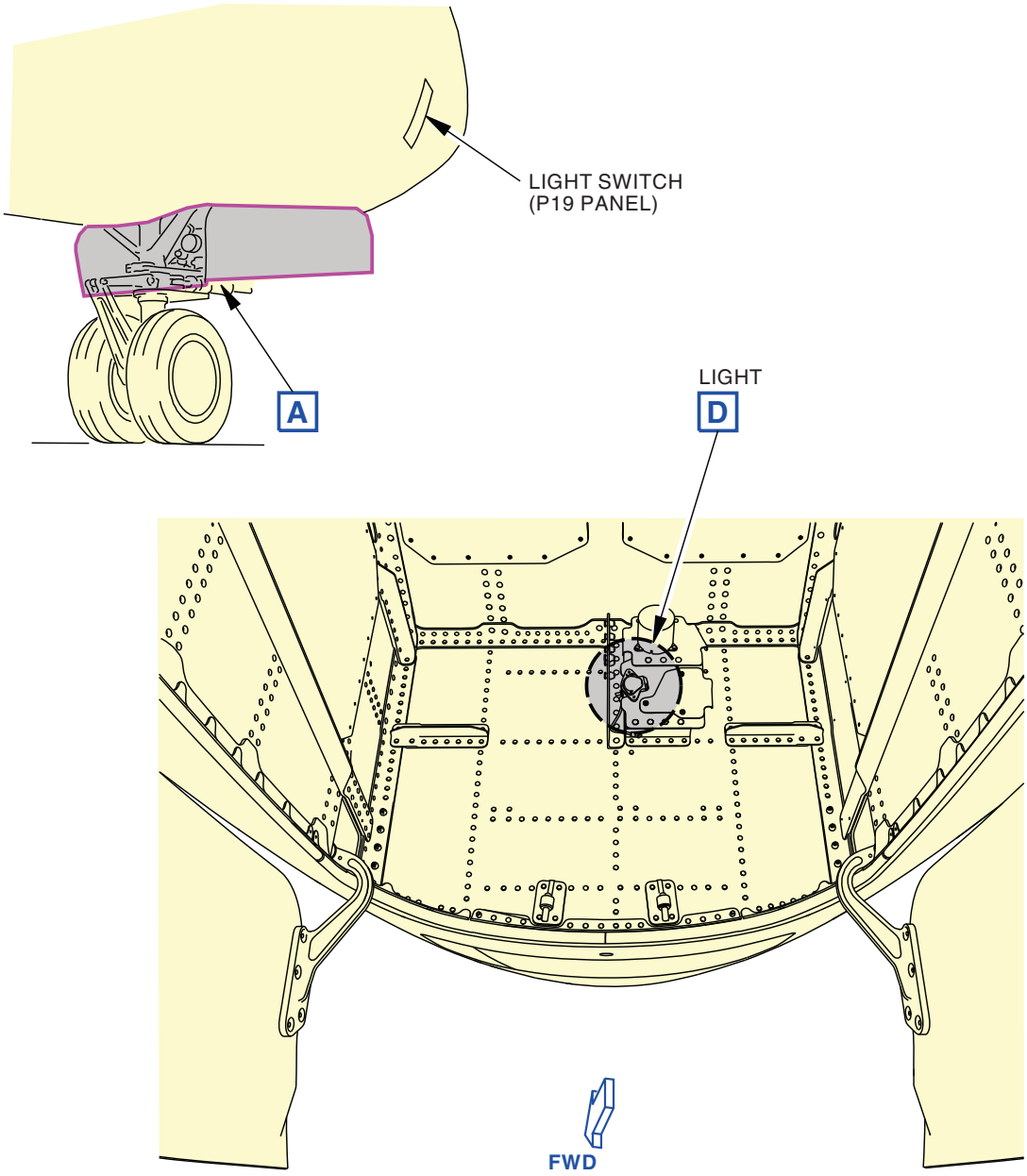
(a) At the overhead panel, P5, set the wheel well lights switch to the ON mode.

1) Make sure that the NLG wheel well light does not come on.

2) Make sure that none of the MLG wheel well lights come on.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



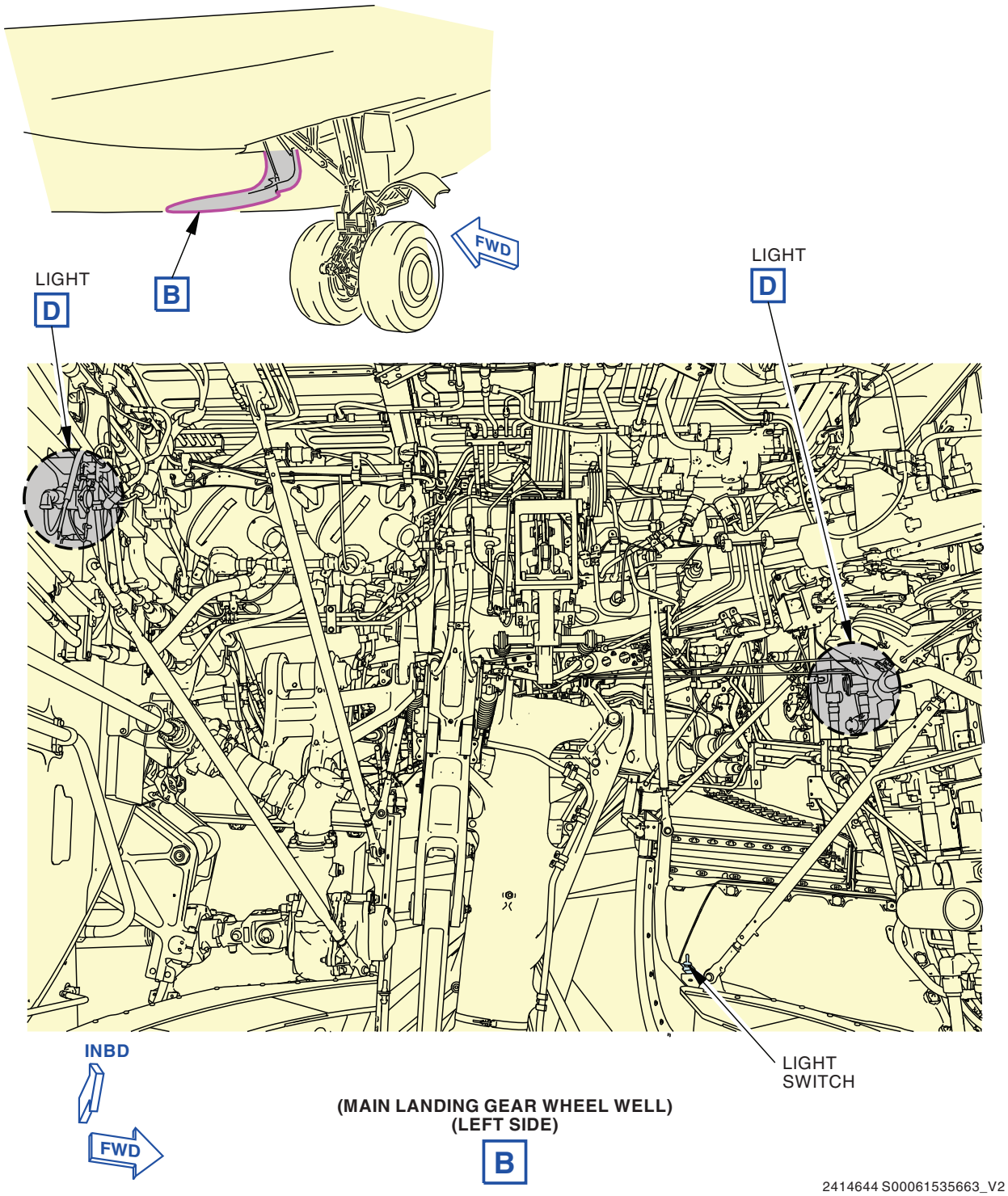
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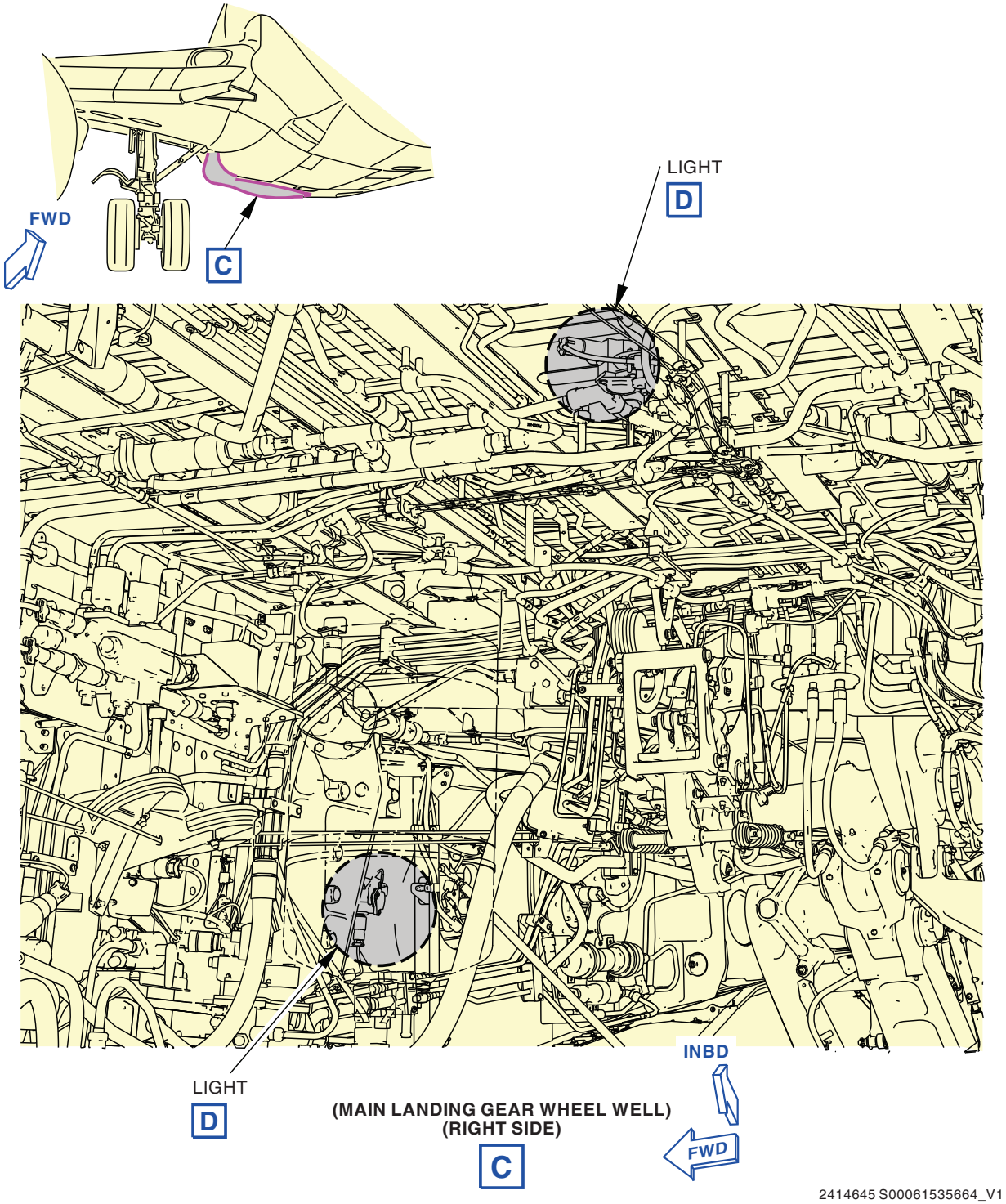
Wheel Well Lights
Figure 201/33-32-00-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



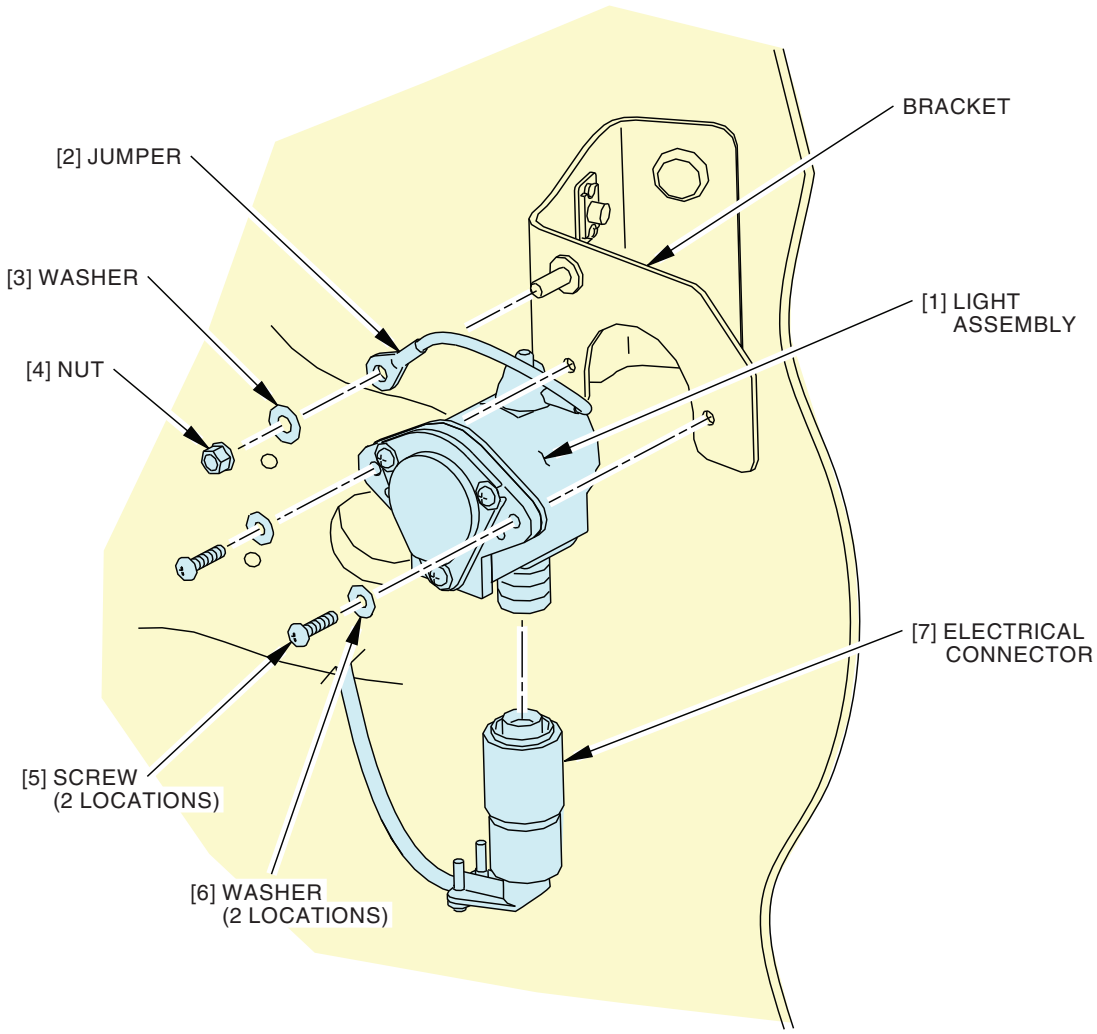
Wheel Well Lights
Figure 201/33-32-00-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Wheel Well Lights
Figure 201/33-32-00-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LIGHT (EXAMPLE)

D

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Wheel Well Lights
Figure 201/33-32-00-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-32-00-440-801

3. Wheel Well Lights - Activation

(Figure 201)

A. General

(1) This procedure adds electrical power to the wheel well lights.

B. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
118	Electrical and Electronics Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Wheel Well Lights Activation

SUBTASK 33-32-00-010-002

(1) To get access to the P92 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-32-00-860-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	2	C00125	WHEEL WELL LT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-32-00-410-001

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 33-32-00-960-801

4. Wheel Well Light - Light Assembly Replacement

(Figure 202)

A. References

Reference	Title
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-20-00, Paragraph 2	CLEANING OF FAY SURFACES
WDM 33-32-11	Wiring Diagram Manual

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426

C. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Light assembly	33-32-00-01-025	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
1 (cont.)		33-32-00-02B-040	PTF ALL

E. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

F. Access Panels

Number	Name/Location
114AR	External Power Receptacle Door

G. Wheel Well Light Assembly Replacement

SUBTASK 33-32-00-860-004

(1) Set the switch for the light to OFF or NORMAL (WDM 33-32-11):

(a) Use the switch on the left wall of the main wheel well.

(b) Use the switch on the P19 external power receptacle panel.

1) For access to the switch, do this step:

a) Open this access panel:

Number	Name/Location
114AR	External Power Receptacle Door

SUBTASK 33-32-00-020-001

(2) Do these steps to remove the light assembly [1]:

(a) Disconnect the electrical connector [7] from the light assembly [1].

(b) Remove the nut [4], washer [3], and jumper [2].

(c) Remove the screws [5] and washers [6].

(d) Remove the light assembly [1] from the bracket.

SUBTASK 33-32-00-100-001

(3) Clean the ground stud on the aluminum bracket (SWPM 20-20-00, Paragraph 2).

SUBTASK 33-32-00-420-001

(4) Do these steps to install the new light assembly [1] to the bracket:

(a) Install the screws [5] and washers [6].

(b) Install the nut [4], washer [3] to connect the jumper [2] to the bracket.

1) Use cotton wiper, G00034, moist in solvent, B00184, to clean the mating surfaces of the nut [4], washer [3], and jumper [2].

2) Apply sealant, A50155, sealant, A50110, sealant, A50296, or sealant, A50231, to the mating surfaces of the ground stud, nut [4], washer [3], and jumper [2].

3) Install the nut [4], washer [3], and jumper [2].

4) Tighten the nut [4] to 28 in-lb (3 N·m) - 35 in-lb (4 N·m).

a) Make sure that there is squeeze out around all the interfaces of each component.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

5) Measure the electrical resistance between jumper [2] and airplane structure (SWPM 20-20-00).
   a) Use a non-intrinsically safe bonding meter, COM-614.
   b) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohm) or less.

(c) Connect the electrical connector [7] to the light assembly [1].

SUBTASK 33-32-00-710-001

(5) Do a test of the light assembly [1]:
   (a) Set the switch for the light to ON.
      1) Make sure that the light assembly [1] comes on.
   (b) Set the switch for the light to OFF or NORMAL.
      1) Make sure that the light assembly [1] goes off.

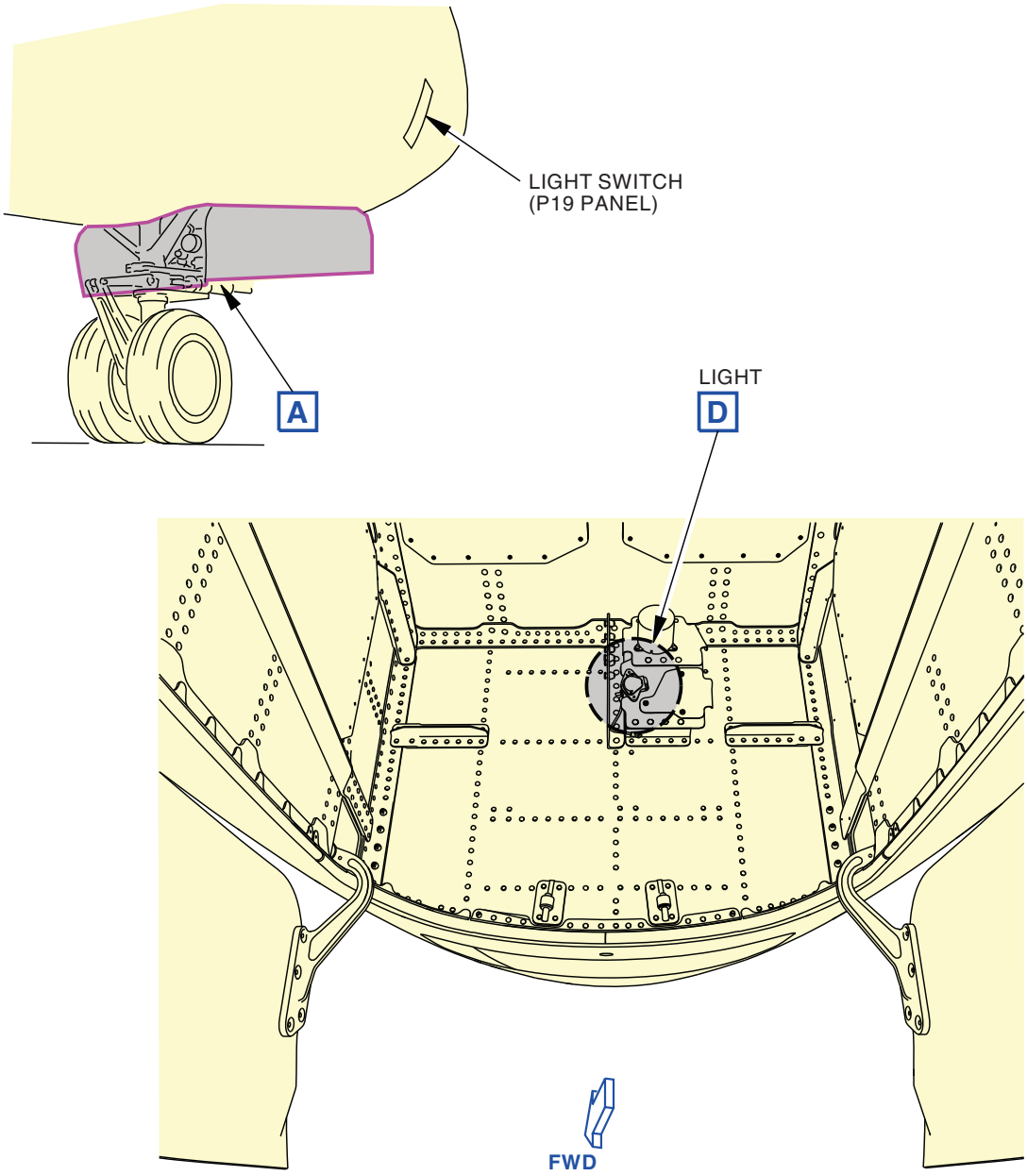
SUBTASK 33-32-00-410-002

(6) Close this access panel:

Number	Name/Location
114AR	External Power Receptacle Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



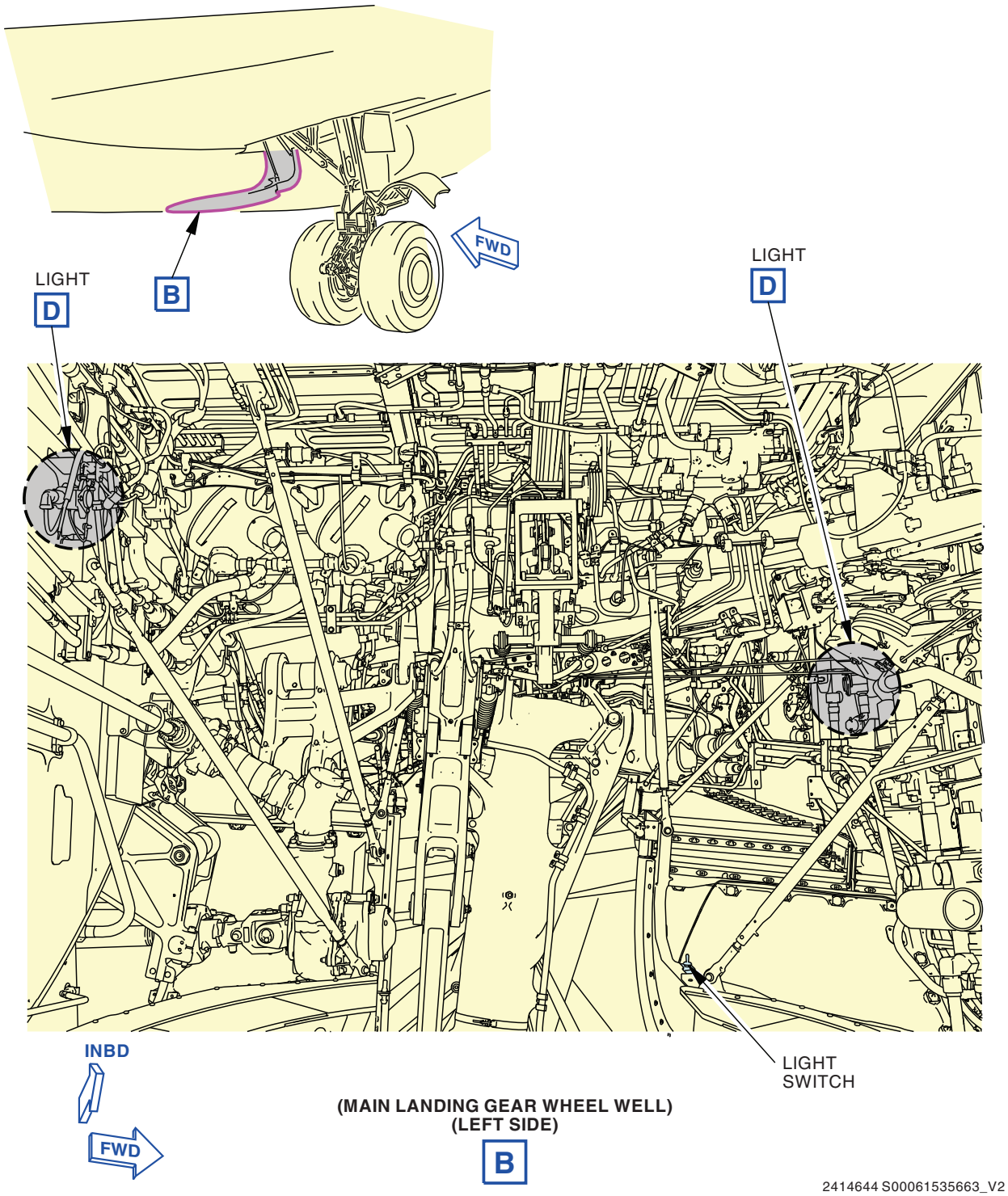
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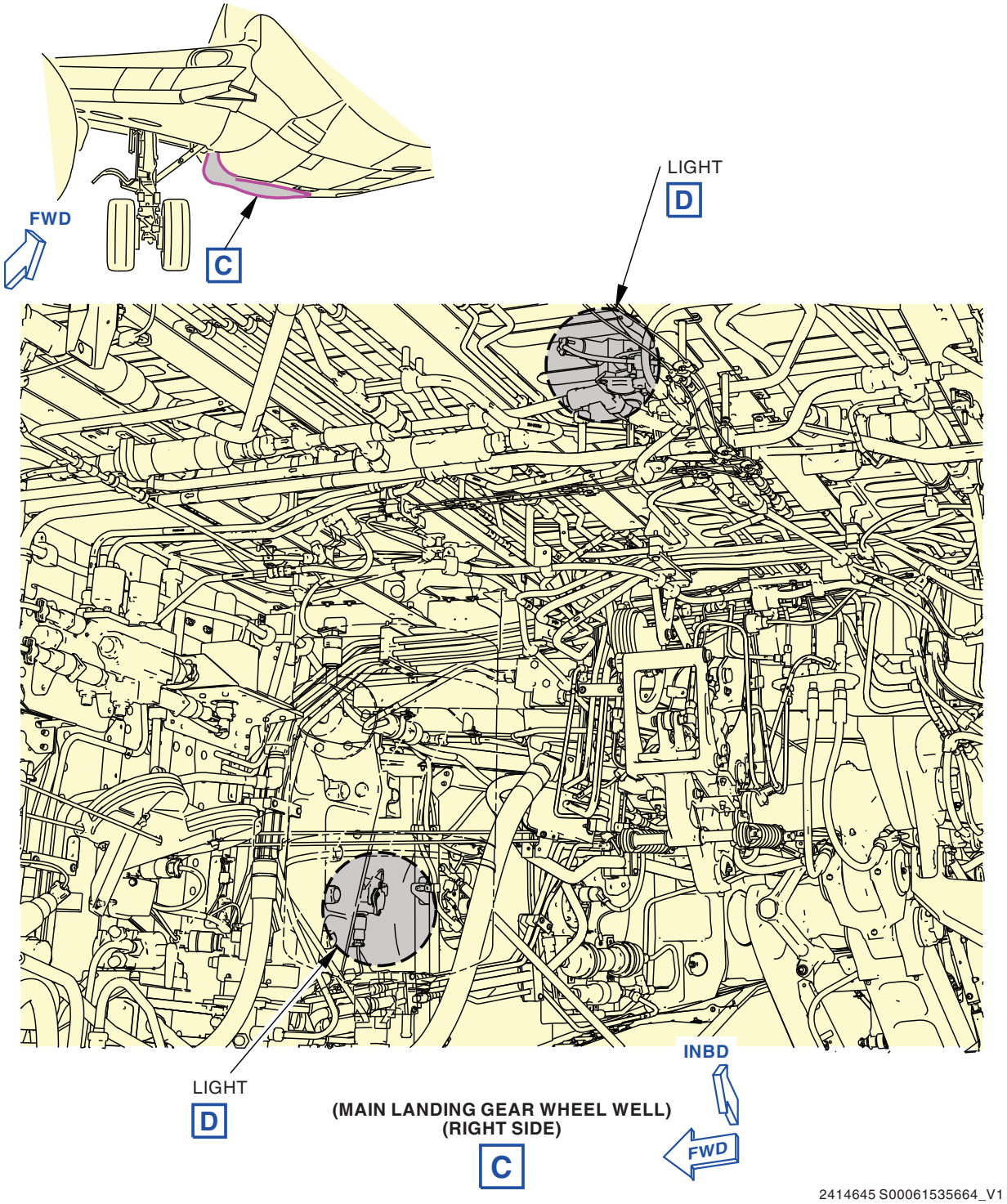
Wheel Well Light - Light Assembly Replacement
Figure 202/33-32-00-990-802 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



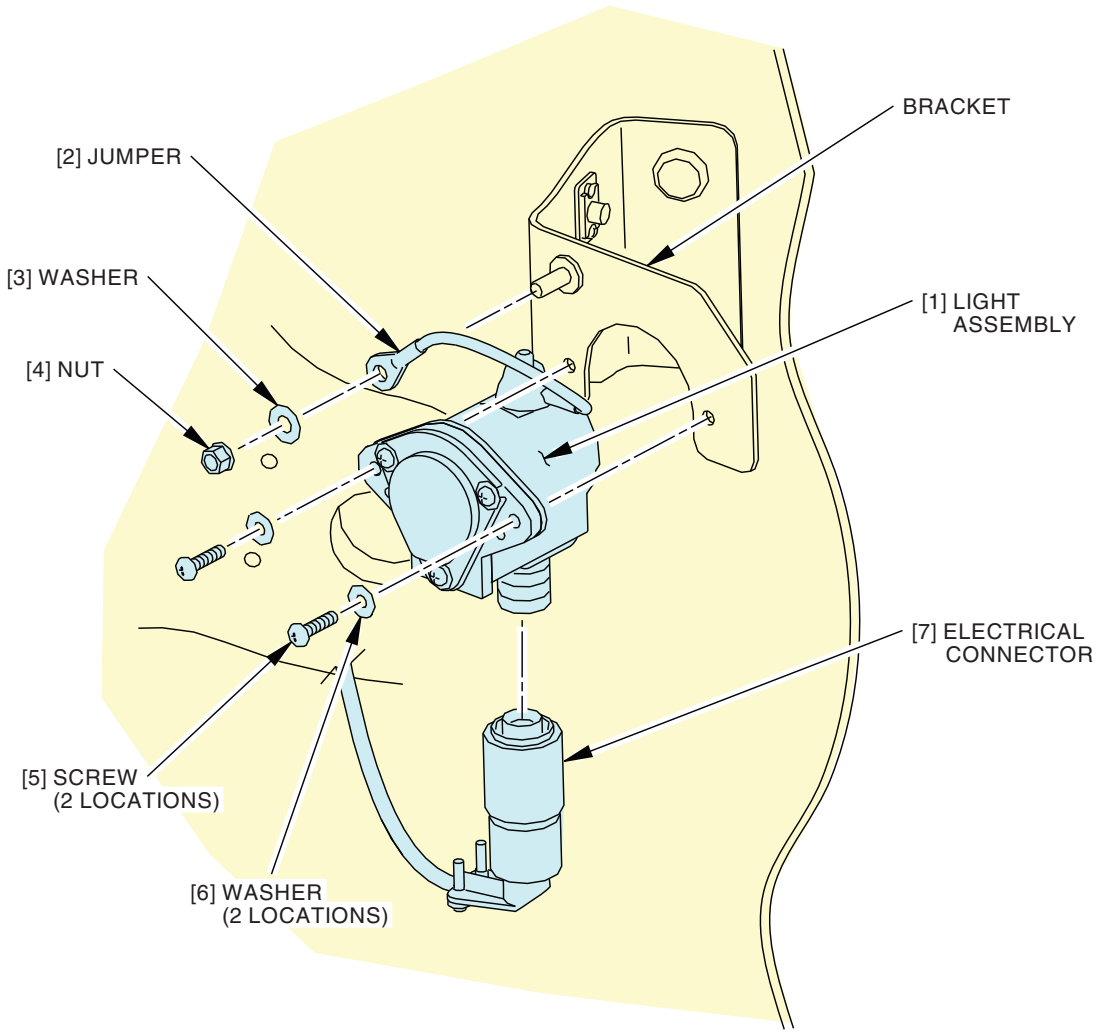
Wheel Well Light - Light Assembly Replacement
Figure 202/33-32-00-990-802 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Wheel Well Light - Light Assembly Replacement
Figure 202/33-32-00-990-802 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LIGHT (EXAMPLE)

D

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Wheel Well Light - Light Assembly Replacement
Figure 202/33-32-00-990-802 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-32-00-710-801

5. Wheel Well Lights - Operational Test

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Wheel Well Lights Operational Test

SUBTASK 33-32-00-710-002

(1) Do a test of the Nose Landing Gear (NLG) and Main Landing Gear (MLG) wheel well lights:

(a) At the overhead panel, P5, set the WHEEL WELL switch to ON.

1) Make sure that the light in the NLG wheel well comes on.

2) Make sure that the lights in the MLG wheel well comes on.

(b) Set the WHEEL WELL switch to OFF.

1) Make sure that the light in the NLG wheel well goes off.

2) Make sure that the lights in the MLG wheel well goes off.

(c) Set the MAIN WHEEL WELL LIGHT switch to ON.

1) Make sure that the lights in the left and right MLG come on.

(d) Set the MAIN WHEEL WELL LIGHT switch to NORMAL or OFF.

1) Make sure that the lights in the left and right MLG goes off.

(e) At the P19 external power receptacle panel, set the NWW LIGHT switch to ON.

1) Make sure that the light in the NLG wheel well come on.

(f) Set the NWW LIGHT switch to NORMAL.

1) Make sure that the light in the NLG wheel well goes off.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AIR CONDITIONING COMPARTMENT LIGHTS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Air conditioning compartment lights deactivation

(2) Air conditioning compartment lights activation

(3) Air conditioning compartment LED light assembly replacement.

TASK 33-33-00-040-801

2. Air Conditioning Compartment Lights Deactivation

(Figure 201)

A. General

(1) This task gives the instructions on how to deactivate the lights for the air conditioning compartment.

(2) This task makes sure that the air conditioning compartment lights are in a zero energy stage when deactivated.

(3) The air conditioning compartment is also referred to as the Equipment Cooling System (ECS) bay.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
192	Lower Wing-To-Body Fairing - Under Wing Box

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door
192JL	Air Conditioning Panel - Aft
192JR	Air Conditioning Panel - Aft

E. Prepare for the Deactivation

SUBTASK 33-33-00-010-008

(1) To get access to the P92 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-33-00-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	3	C00126	AIR COND COMPT LT

SUBTASK 33-33-00-410-006

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-33-00-010-001

(4) Open the access door to the left ECS bay:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 33-33-00-010-011

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(5) Open these access doors to the right ECS bay:

(TASK 06-41-00-800-801)

(a) Open this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access door:

Number	Name/Location
192CR	ECS Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-33-00-010-002

(6) Open these access panels:

Number	Name/Location
192JL	Air Conditioning Panel - Aft
192JR	Air Conditioning Panel - Aft

F. Air Conditioning Compartment Lights Deactivation

SUBTASK 33-33-00-700-001

(1) Make sure that the air conditioning compartment lights are deactivated as follow:

(a) At the forward end of the left ECS bay, set the light switch to the ON position.

(b) Make sure that the left conditioning compartment lights do not come on.

(c) Set the light switch to the OFF position.

(d) At the forward end of the right ECS bay, set the light switch to the ON position.

(e) Make sure that the right conditioning compartment lights do not come on.

(f) Set the light switch to the OFF position.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 33-33-00-410-001

(1) Close these access doors to the ECS bay:

(TASK 06-41-00-800-801)

(a) Close this access door:

Number	Name/Location
192CL	ECS Access Door

(b) Close this access door:

Number	Name/Location
192CR	ECS Access Door

(c) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

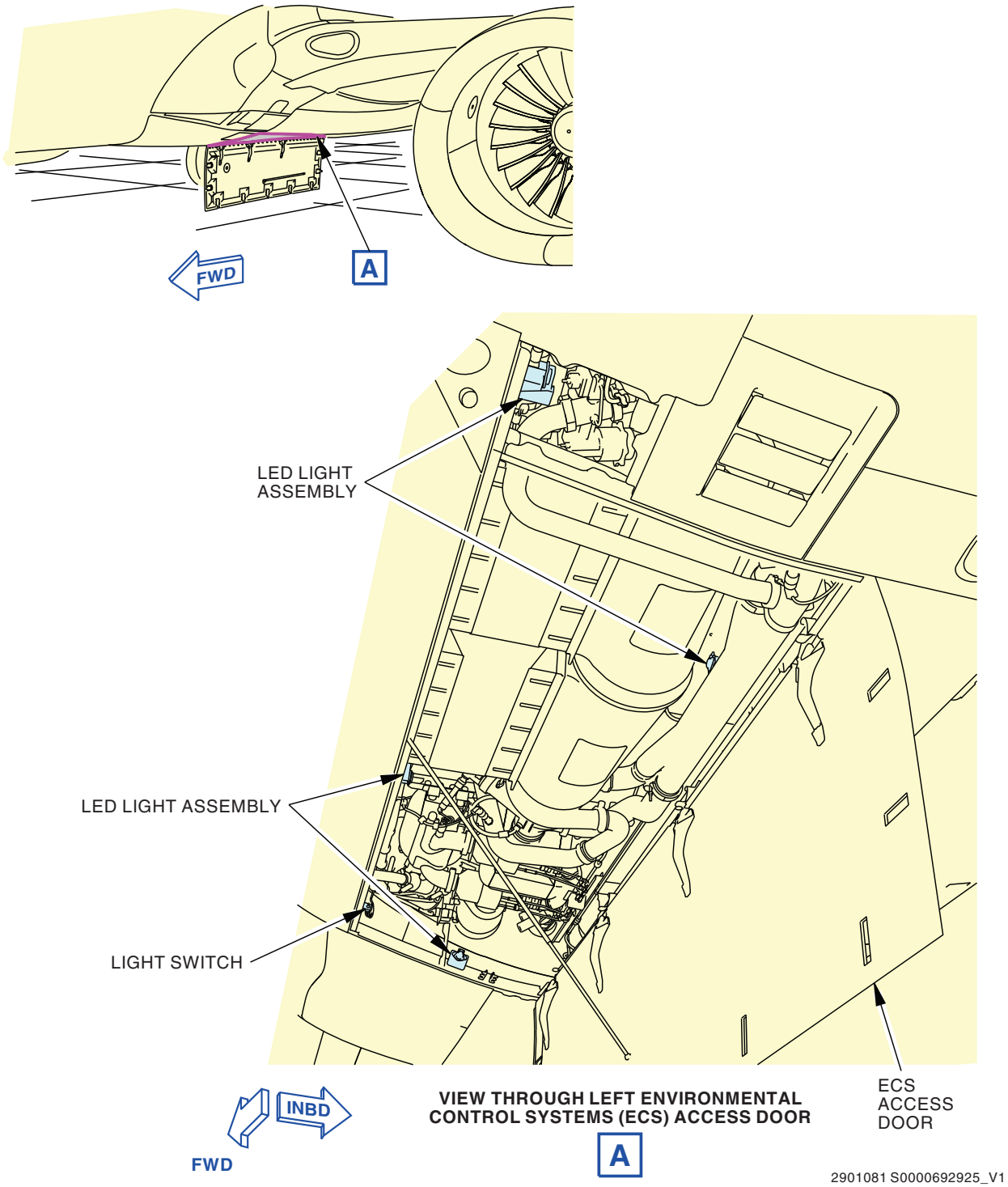
SUBTASK 33-33-00-410-002

(2) Close these access panels:

Number	Name/Location
192JL	Air Conditioning Panel - Aft
192JR	Air Conditioning Panel - Aft

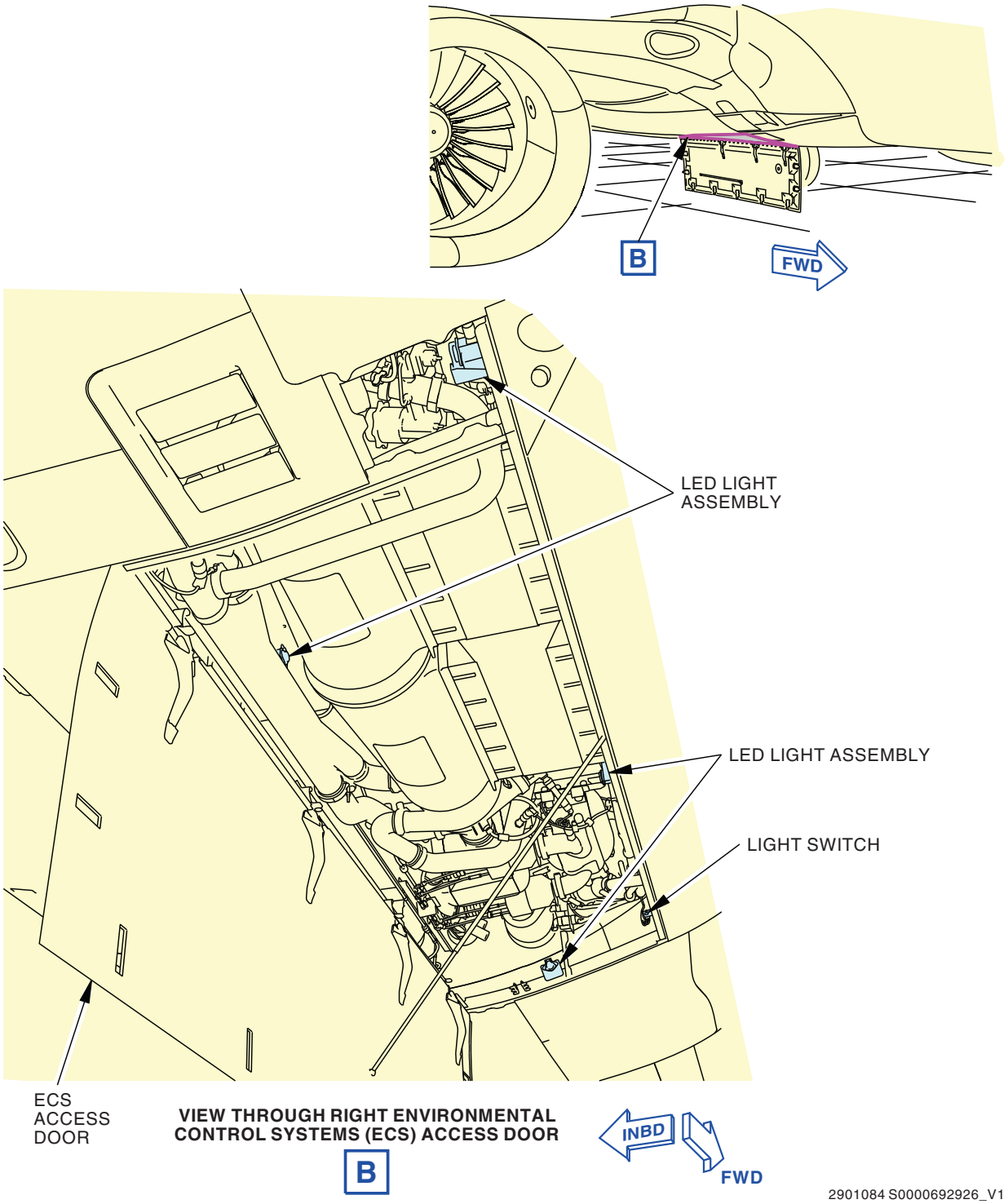
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Air Conditioning Compartment Lights
Figure 201/33-33-00-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Air Conditioning Compartment Lights
Figure 201/33-33-00-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-33-00-440-801

3. Air Conditioning Compartment Lights Activation

(Figure 201)

A. General

(1) This task gives the instructions on how to activate the lights for the air conditioning compartment.

(2) This task makes sure that electrical power is restored to the air conditioning compartment lights.

(3) The air conditioning compartment is also referred to as the Equipment Cooling System (ECS) bay.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
192	Lower Wing-To-Body Fairing - Under Wing Box

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door
192JL	Air Conditioning Panel - Aft
192JR	Air Conditioning Panel - Aft

E. Prepare for the Activation

SUBTASK 33-33-00-010-003

(1) Open the access door to the left ECS bay:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 33-33-00-010-012

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(2) Open these access doors to the right ECS bay:

(TASK 06-41-00-800-801)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

(b) Open this access door:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 33-33-00-010-004

(3) Open these access panels:

Number	Name/Location
192JL	Air Conditioning Panel - Aft
192JR	Air Conditioning Panel - Aft

SUBTASK 33-33-00-010-009

(4) To get access to the P92 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-33-00-860-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(5) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	3	C00126	AIR COND COMPT LT

SUBTASK 33-33-00-410-007

(6) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

F. Air Conditioning Compartment Lights Activation

SUBTASK 33-33-00-720-001

(1) Make sure that the air conditioning compartment lights work as follows:

(a) At the forward end of the left ECS bay, set the light switch to the ON position.

(b) Make sure that the left air conditioning compartment lights come on.

(c) Set the light switch to the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Make sure that the left air conditioning compartment lights go off.

(e) At the forward end of the right ECS bay, set the light switch to the ON position.

(f) Make sure that the right air conditioning compartment lights come on.

(g) Set the light switch to the OFF position.

(h) Make sure that the right air conditioning compartment lights go off.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 33-33-00-410-003

(1) Close these access doors to the ECS bay:

(TASK 06-41-00-800-801)

(a) Close this access door:

Number	Name/Location
192CL	ECS Access Door

(b) Close this access door:

Number	Name/Location
192CR	ECS Access Door

(c) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

SUBTASK 33-33-00-410-004

(2) Close these access panels:

Number	Name/Location
192JL	Air Conditioning Panel - Aft
192JR	Air Conditioning Panel - Aft

——— END OF TASK ———

TASK 33-33-00-960-801

4. Air Conditioning Compartment LED Light Assembly Replacement

(Figure 202)

A. General

(1) This task gives the instructions on how to replace the LED light assembly for the air conditioning compartment.

(2) The air conditioning compartment is also referred to as the Equipment Cooling System (ECS) bay.

B. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

Reference	Description
STD-6972	Milliohmmeter - 0.0 to 0.1 Ohm Range

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C

E. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
131	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Left
132	Center Section Wing Box, Body Station 540.00 to Body Station 663.75 - Right
192	Lower Wing-To-Body Fairing - Under Wing Box

F. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
192CL	ECS Access Door
192CR	ECS Access Door
192DR	ECS High Pressure Access Door
192JL	Air Conditioning Panel - Aft
192JR	Air Conditioning Panel - Aft

G. Prepare for the Replacement

SUBTASK 33-33-00-010-005

(1) Open the access door to the left ECS bay:

Number	Name/Location
192CL	ECS Access Door

SUBTASK 33-33-00-010-013

CAUTION

DO NOT OPEN THE ECS ACCESS DOOR 192CR, UNTIL YOU OPEN THE ECS HIGH PRESSURE ACCESS DOOR 192DR. IF YOU DO NOT OBEY, DAMAGE TO THE ECS HIGH PRESSURE ACCESS DOOR 192DR, ITS LATCHES AND HINGES CAN OCCUR.

(2) Open these access doors to the right ECS bay:

(TASK 06-41-00-800-801)

(a) Open this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Open this access door:

Number	Name/Location
192CR	ECS Access Door

SUBTASK 33-33-00-010-006

(3) If the light assembly is installed aft of the ECS access doors, also open the applicable access panel:

(a) For the left side, open this access panel:

Number	Name/Location
192JL	Air Conditioning Panel - Aft

(b) For the right side, open this access panel:

Number	Name/Location
192JR	Air Conditioning Panel - Aft

SUBTASK 33-33-00-010-010

(4) To get access to the P92 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-33-00-860-006

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(5) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	3	C00126	AIR COND COMPT LT

H. Air Conditioning Compartment LED Light Assembly Replacement

SUBTASK 33-33-00-980-001

(1) Set the light switch at the forward end of the ECS bay to the OFF position.

SUBTASK 33-33-00-960-001

(2) Do these steps to replace the LED light assembly:

(a) Remove the nut [4], washer [5] and the jumper cable [7] from the stud on the mounting bracket.

(b) Remove the screws [2] and washers [3] from the LED light assembly [1].

(c) Remove the LED light assembly [1] from the mounting bracket.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Disconnect the electrical connector [6] from the LED light assembly [1].

(e) Connect the electrical connector [6] to the new LED light assembly [1].

(f) Install the LED light assembly [1] to the mounting bracket with the washers [3] and screws [2].

(g) Install the jumper cable [7] to the mounting bracket as follows:

1) Apply a bead of sealant, A50110, to the contact surfaces of the stud, terminal of jumper cable [7], washer [5] and nut [4].

NOTE: The sealant, A50155, sealant, A50231, and sealant, A50296, are equivalent alternatives for the sealant, A50110.

2) Install the jumper cable [7], washer [5] and nut [4] to the stud.

a) Tighten the nut [4].

3) Make sure that there is a continuous squeeze-out of sealant around the stud, terminal of jumper cable [7], washer [5] and nut [4].

a) If necessary, add more sealant around the contact surfaces.

(h) Measure the resistance between the terminal of jumper cable [7] and mounting bracket with a milliohmmeter, STD-6972.

1) Make sure that the resistance is not more than 1.0 milliohms.

SUBTASK 33-33-00-860-007

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	3	C00126	AIR COND COMPT LT

SUBTASK 33-33-00-410-008

(4) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-33-00-710-001

(5) Test the new LED light assembly as follows:

(a) On the forward end of the ECS bay, set the switch for the light to the ON position.

1) Make sure that the light comes on.

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(b) Set the switch to the OFF position.

1) Make sure that the light goes off.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 33-33-00-410-005

(1) Close these access doors to the ECS bay:

(TASK 06-41-00-800-801)

(a) Close this access door:

Number	Name/Location
192CL	ECS Access Door

(b) Close this access door:

Number	Name/Location
192CR	ECS Access Door

(c) Close this access door:

Number	Name/Location
192DR	ECS High Pressure Access Door

NOTE: The ECS high pressure access door, 192DR, will not latch in its position if the ECS access door, 192CR, is not closed.

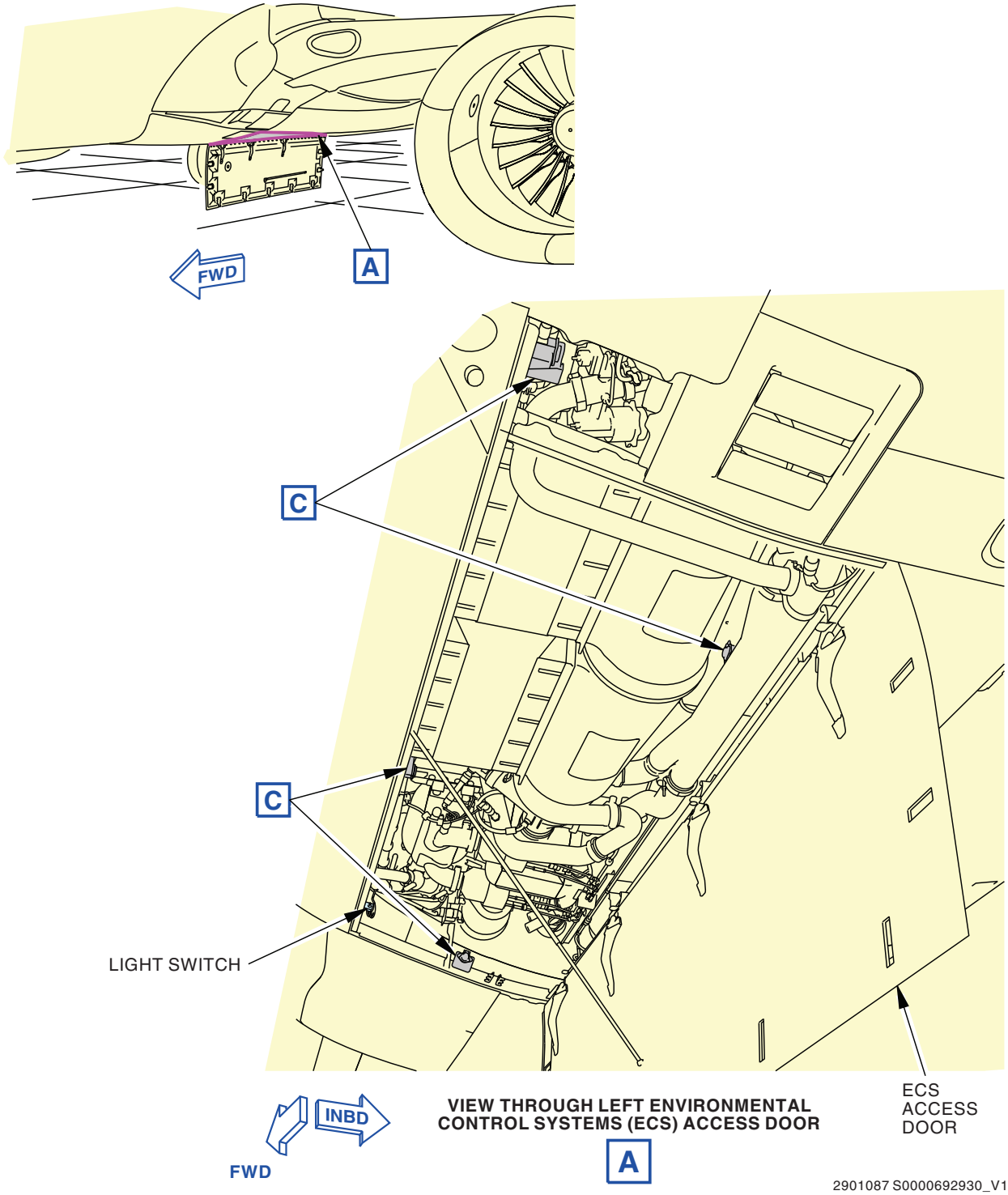
SUBTASK 33-33-00-010-007

(2) Close the access panel, if opened:

Number	Name/Location
192JL	Air Conditioning Panel - Aft
192JR	Air Conditioning Panel - Aft

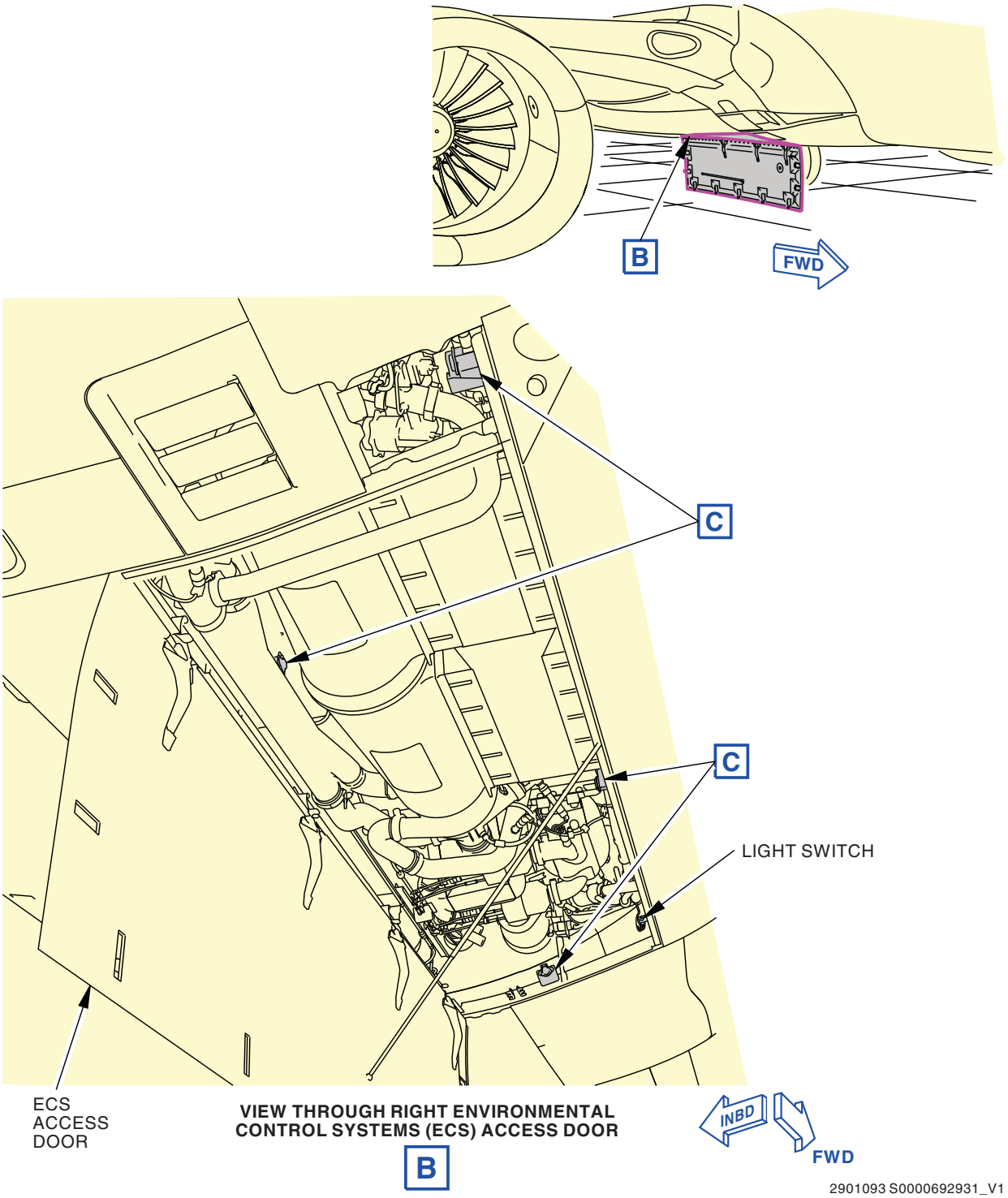
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



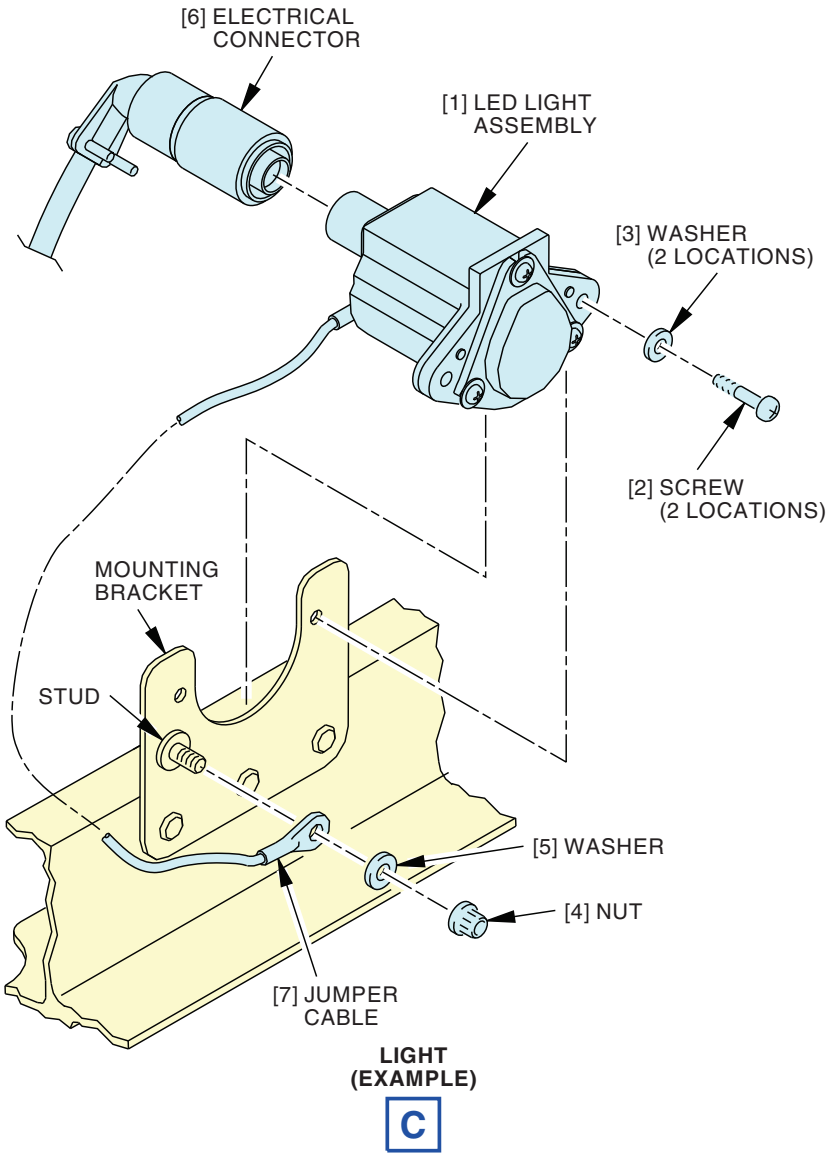
Air Conditioning Compartment Light Lamp Replacement
Figure 202/33-33-00-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Air Conditioning Compartment Light Lamp Replacement
Figure 202/33-33-00-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Air Conditioning Compartment Light Lamp Replacement
Figure 202/33-33-00-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELECTRONIC EQUIPMENT COMPARTMENT LIGHTS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A Deactivation of the Electronic Equipment Compartment Lights

(2) An Activation of the Electronic Equipment Compartment Lights

(3) A Replacement of the Electronic Equipment Compartment Light Lamp.

TASK 33-34-00-040-801

2. Electronic Equipment Compartment Lights Deactivation

(Figure 201)

A. General

(1) This task gives the instructions on how to deactivate the lights for the electronic equipment compartment.

(2) This task makes sure that the electronic equipment compartment lights are in a zero energy stage when deactivated.

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
117	Electrical and Electronics Compartment - Left

C. Access Panels

Number	Name/Location
112A	Forward Access Door
117A	Electronic Equipment Access Door

D. Prepare for the Deactivation

SUBTASK 33-34-00-010-004

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-34-00-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	2	C00139	ELEC RACK LT-FWD

SUBTASK 33-34-00-010-002

(3) Open this access panel:

Number	Name/Location
112A	Forward Access Door

E. Electronic Equipment Compartment Lights Deactivation

SUBTASK 33-34-00-700-001

(1) Make sure that the electronic equipment compartment lights are deactivated as follows:

(a) At the forward access door, set the light switch to the ON position.

(b) Make sure that the forward equipment compartment lights do not come on.

(c) Set the light switch to the OFF position

(d) At the electronic equipment compartment service panel, set the light switch to the ON position.

(e) Make sure that the electronic equipment compartment lights do not come on.

(f) Set the light switch to the OFF position.

F. Put the Airplane Back to Its Usual Condition

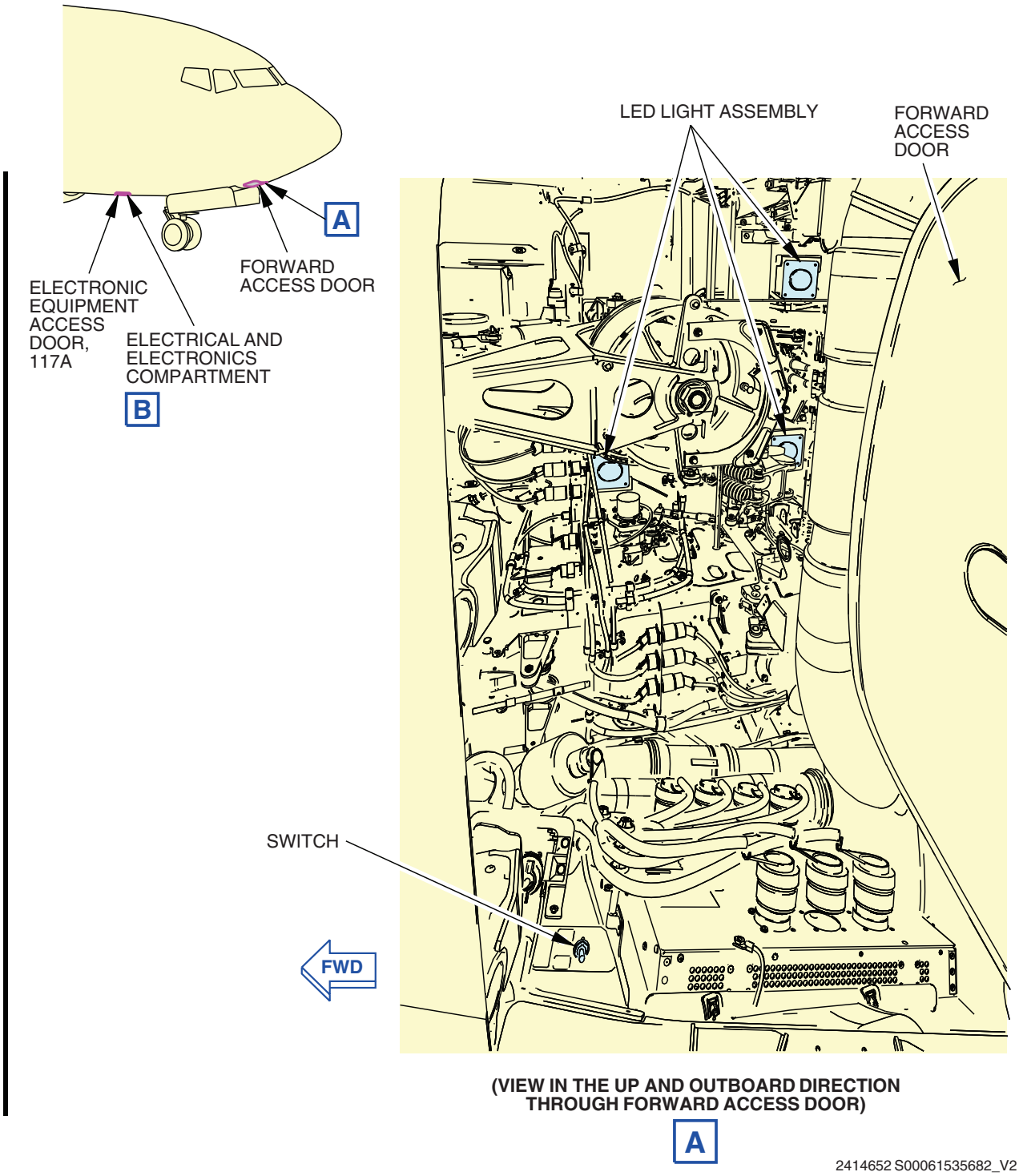
SUBTASK 33-34-00-410-001

(1) Close these access panels:

Number	Name/Location
112A	Forward Access Door
117A	Electronic Equipment Access Door

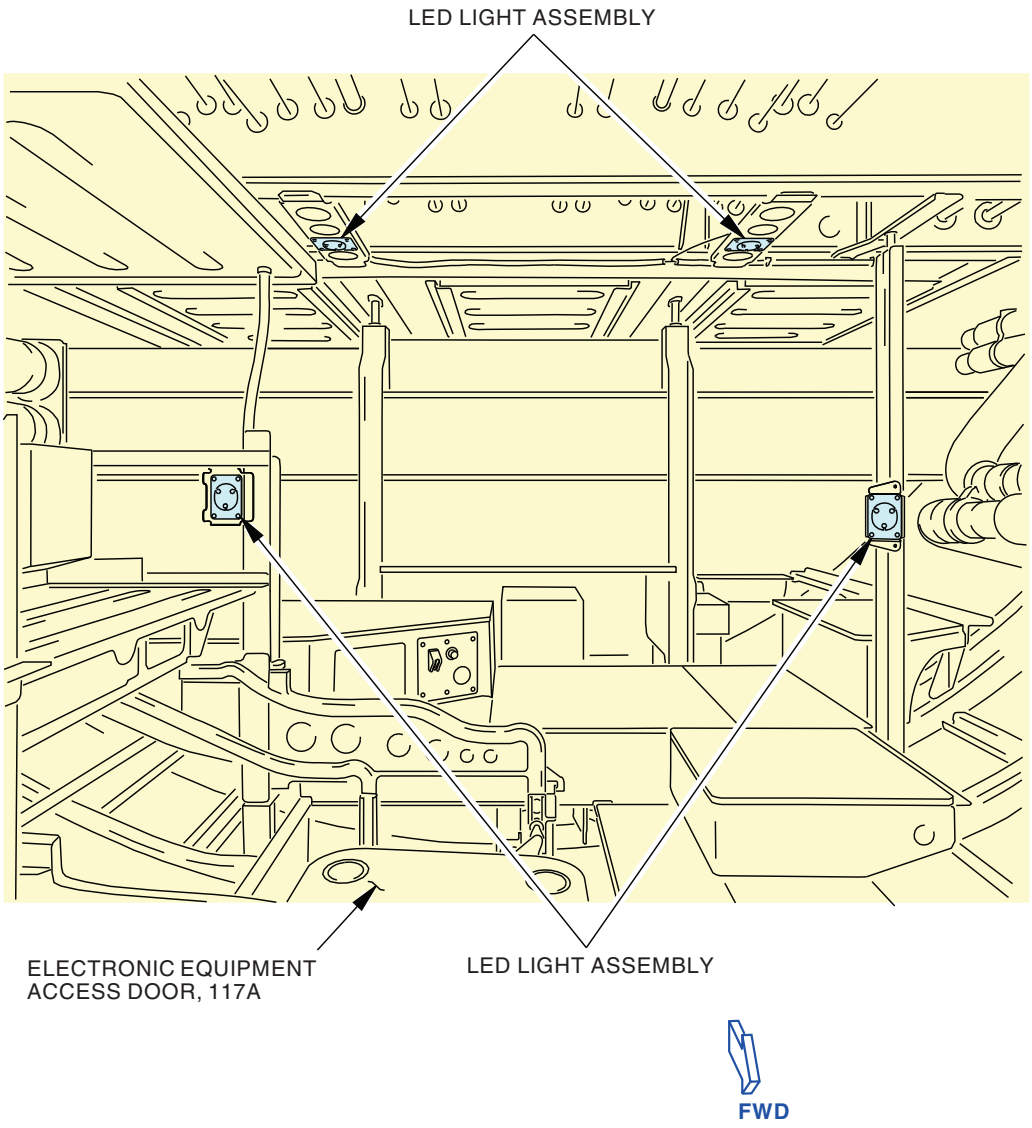
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Electronic Equipment Compartment Lights
Figure 201/33-34-00-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ELECTRICAL AND ELECTRONICS COMPARTMENT

B

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Electronic Equipment Compartment Lights
Figure 201/33-34-00-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-34-00-440-801

3. Electronic Equipment Compartment Lights Activation

(Figure 201)

A. General

(1) This task gives the instructions on how to activate the lights for the electronic equipment compartment.

(2) This task makes sure that electrical power is restored for the electronic equipment compartment lights.

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
117	Electrical and Electronics Compartment - Left

C. Access Panels

Number	Name/Location
112A	Forward Access Door
117A	Electronic Equipment Access Door

D. Prepare for the Activation

SUBTASK 33-34-00-010-003

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

E. Electronic Equipment Compartment Lights Activation

SUBTASK 33-34-00-860-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	2	C00139	ELEC RACK LT-FWD

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-34-00-010-005

(2) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 33-34-00-440-001

(3) Make sure that the electronic compartment lights work as follow:

(a) At the forward access door, set the light switch to the ON position.

(b) Make sure that the forward equipment compartment lights come on.

(c) Set the light switch to the OFF position

(d) Make sure that the forward equipment compartment lights go off.

(e) At the electronic equipment compartment service panel, set the light switch to the ON position.

(f) Make sure that the electronic equipment compartment lights come on.

(g) Set the light switch to the OFF position.

(h) Make sure that the electronic equipment compartment lights go off.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 33-34-00-410-002

(1) Close these access panels:

Number	Name/Location
112A	Forward Access Door
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 33-34-00-960-801

4. Electronic Equipment Compartment LED Light Assembly Replacement

(Figure 202)

A. General

(1) This task gives the instructions on how to replace the LED light assembly for the electronic equipment compartment light.

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
117	Electrical and Electronics Compartment - Left

C. Access Panels

Number	Name/Location
112A	Forward Access Door
117A	Electronic Equipment Access Door

D. Prepare for the Replacement

SUBTASK 33-34-00-960-001

(1) If you will replace a LED light assembly in the area forward of the nose landing gear wheel well, do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 33-34-00-960-002

(2) If you will replace a LED light assembly in the electrical and electronics compartment, do this step:

(a) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-34-00-040-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	2	C00139	ELEC RACK LT-FWD

E. Electronic Equipment Compartment LED Light Assembly Replacement

SUBTASK 33-34-00-860-003

(1) If you will replace a LED light assembly in the area forward of the nose landing gear wheel well, do this step:

(a) Set the light switch at the forward access door to the OFF position.

SUBTASK 33-34-00-860-004

(2) If you will replace a LED light assembly in the electrical and electronics compartment, do this step:

(a) Set the light switch at the service panel to the OFF position.

SUBTASK 33-34-00-960-003

(3) Do these steps to replace LED light assembly:

(a) For post mounted LED light assembly, remove nut [5], washer [6], and ground terminal [7].

(b) Remove screws [3] and washers [4].

(c) Remove LED light assembly [1].

(d) Disconnect electrical connector [2] from the LED light assembly [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Connect electrical connector [2] to the new LED light assembly [1].

(f) Install the new LED light assembly [1].

(g) Install the washers [4] and screws [3].

(h) Install the ground terminal [7], washer [6], and nut [5].

SUBTASK 33-34-00-440-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	2	C00139	ELEC RACK LT-FWD

SUBTASK 33-34-00-710-001

(5) Do this test if you replaced an LED light assembly in the area forward of the nose landing gear wheel well:

(a) Set the light switch at the forward access door to the ON position.

1) Make sure that the lamp comes on.

(b) Set the switch to the OFF position.

1) Make sure that the lamp goes off.

(6) Do this test if you replaced an LED light assembly in the electrical and electronics compartment:

(a) Set the light switch at the service panel to the ON position.

1) Make sure that the lamp comes on.

(b) Set the switch to the OFF position.

1) Make sure that the lamp goes off.

F. Put the Airplane Back to Its Usual Condition

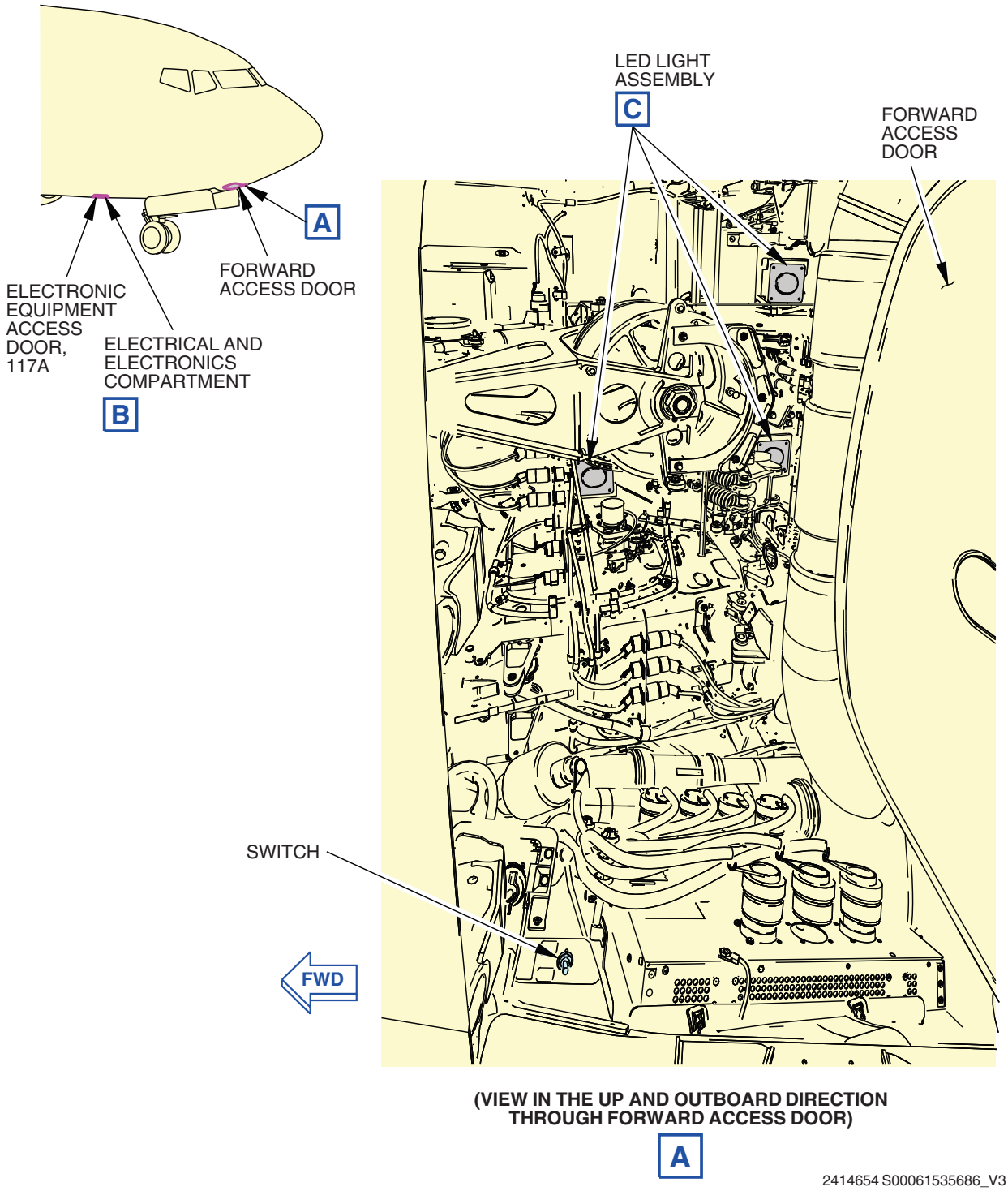
SUBTASK 33-34-00-410-003

(1) Close these access panels:

Number	Name/Location
112A	Forward Access Door
117A	Electronic Equipment Access Door

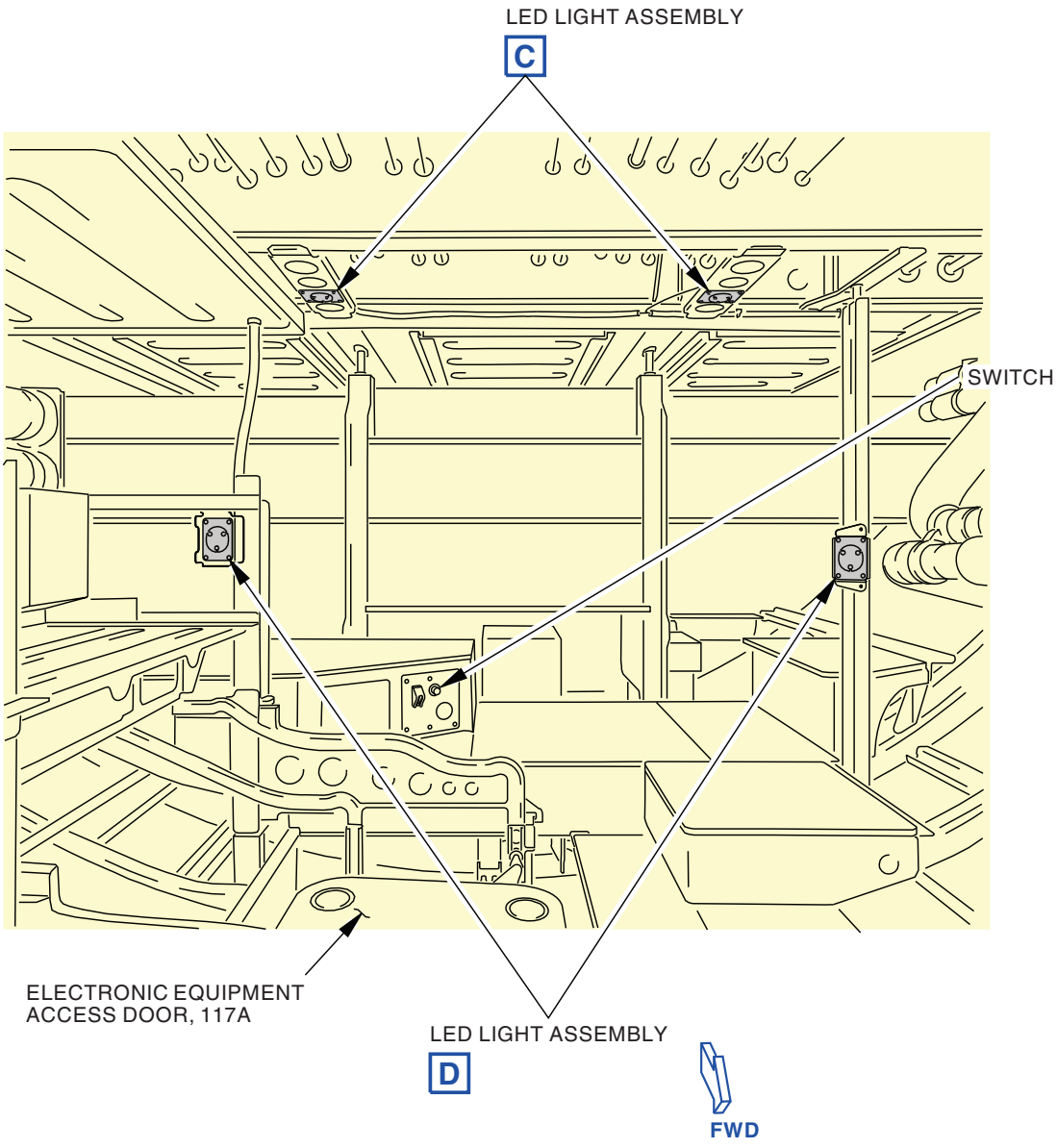
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Electronic Equipment Compartment LED Light Assembly Replacement
Figure 202/33-34-00-990-802 (Sheet 1 of 3)

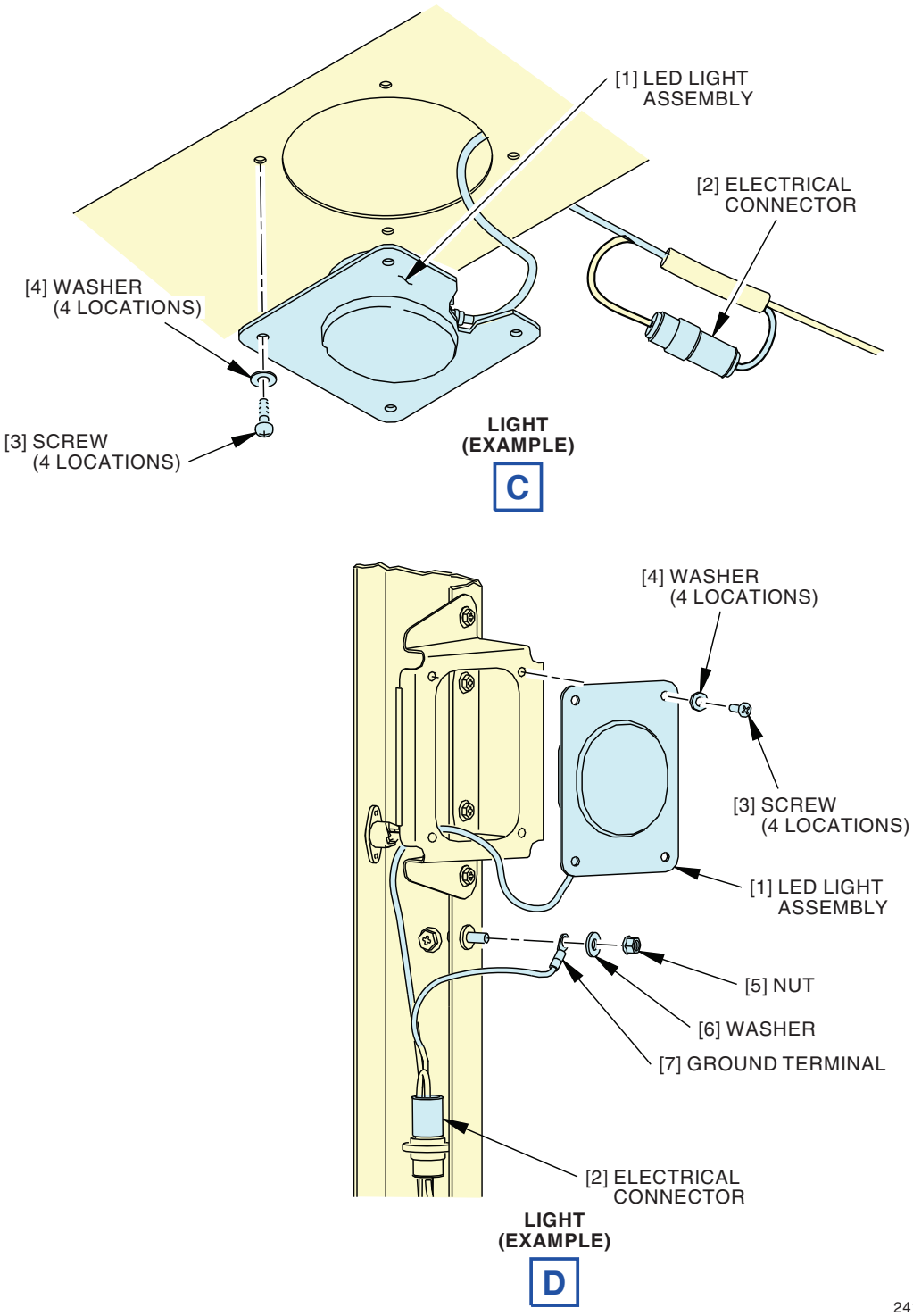
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414655 S00061535687_V2

Electronic Equipment Compartment LED Light Assembly Replacement
Figure 202/33-34-00-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414656 S00061535688_V3

Electronic Equipment Compartment LED Light Assembly Replacement
Figure 202/33-34-00-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ACCESSORY COMPARTMENT LIGHTS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A Deactivation of the Accessory Compartment Lights

(2) An Activation of the Accessory Compartment Lights

(3) A Replacement of the Accessory Compartment Light Lamp.

TASK 33-35-00-040-801

2. Accessory Compartment Lights - Deactivation

(Figure 201)

A. General

(1) This procedure removes electrical power to the Accessory Compartment Lights.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
310	Fuselage - Body Station 1016.00 to Body Station 1260.00
311	Area Aft of Pressure Bulkhead - Left
315	APU Compartment - Left
318	Tail Cone Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
311BL	Stabilizer Trim Access Door
315A	APU Cowl Door
318BR	Tailcone Access Door

D. Accessory Compartment Lights Deactivation

SUBTASK 33-35-00-010-001

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-35-00-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	5	C00389	ACCESS COMPT LT

E. Accessory Compartment Lights - Tryout

NOTE: This tryout is to make sure that the Accessory Compartment Lights are in a zero energy state.

SUBTASK 33-35-00-010-002

(1) Open these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
315A	APU Cowl Door
318BR	Tailcone Access Door

SUBTASK 33-35-00-860-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Make sure that this circuit breaker is open and has safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	5	C00389	ACCESS COMPT LT

SUBTASK 33-35-00-410-003

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-35-00-700-001

(4) At the tail cone access door, toggle the light switch to the ON mode.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the tail cone compartment light does not come on.

(5) At the FWD end of the APU cowl door, toggle the light switch to the ON mode.

(a) Make sure that none of the APU compartment lights come on.

(6) At the AFT end of the accessory compartment door, press the light switch button to the ON mode.

(a) Make sure that none of the accessory compartment lights come on.

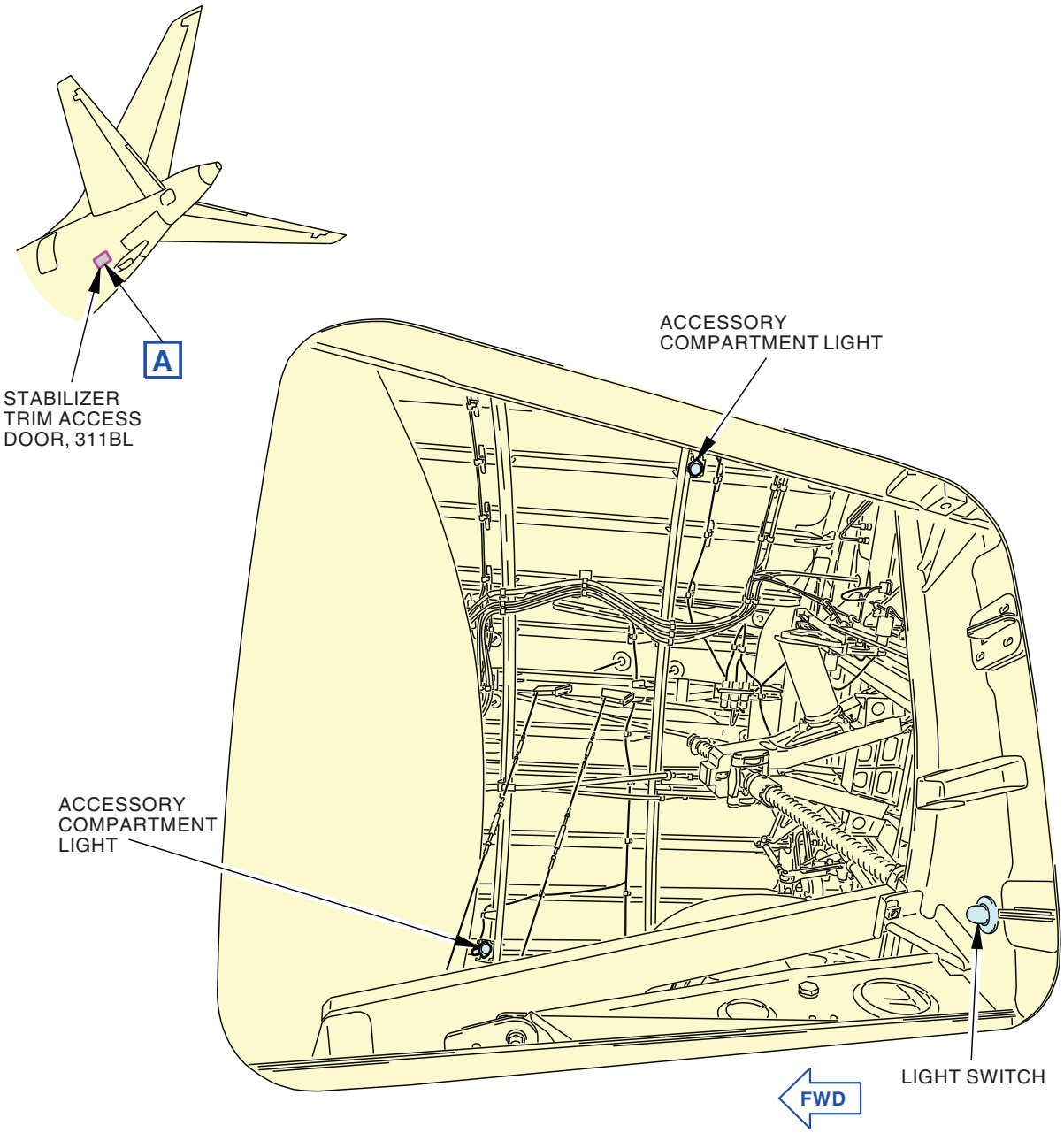
SUBTASK 33-35-00-410-004

(7) Close these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
315A	APU Cowl Door
318BR	Tailcone Access Door

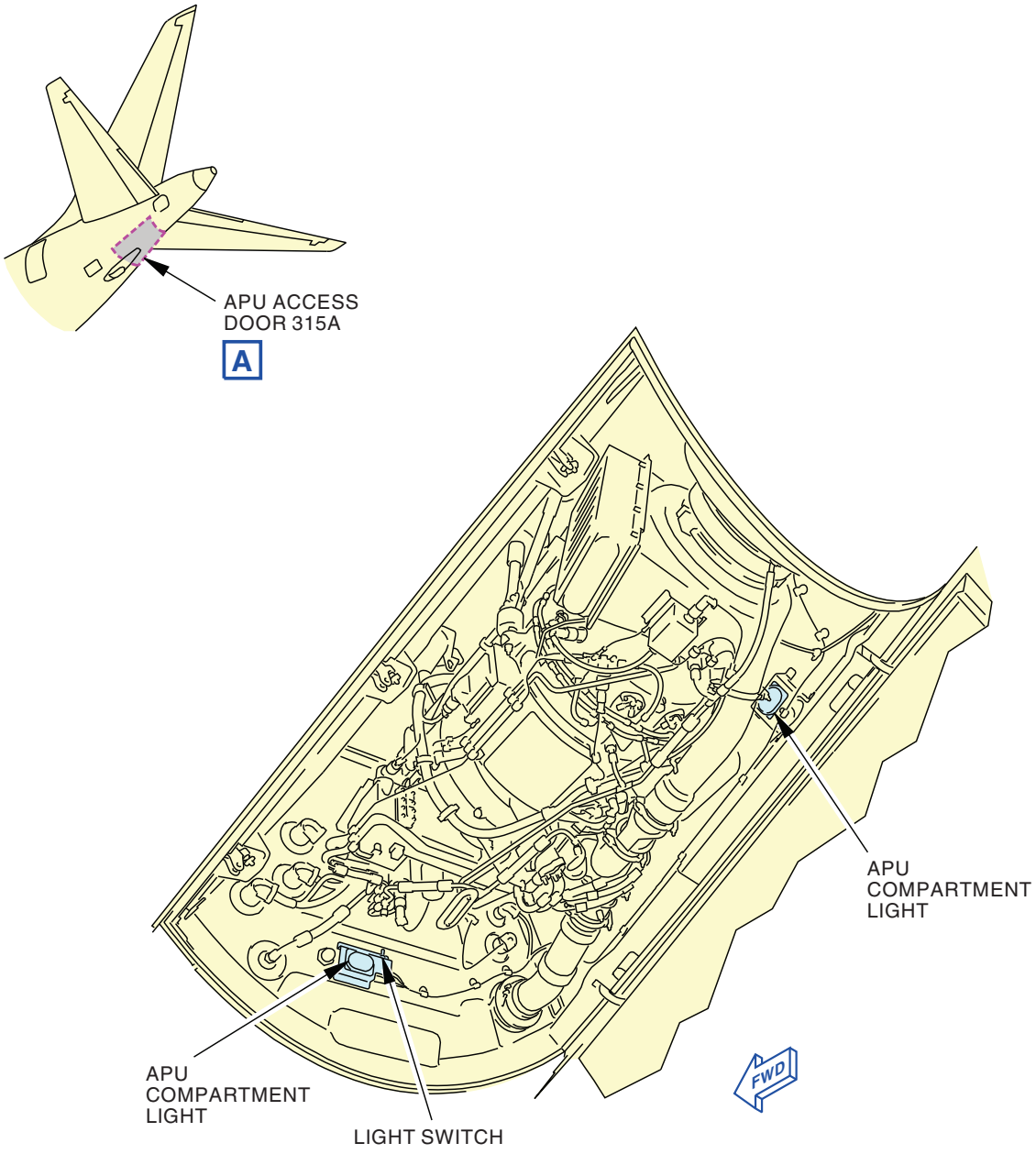
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Accessory Compartment Lights
Figure 201/33-35-00-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

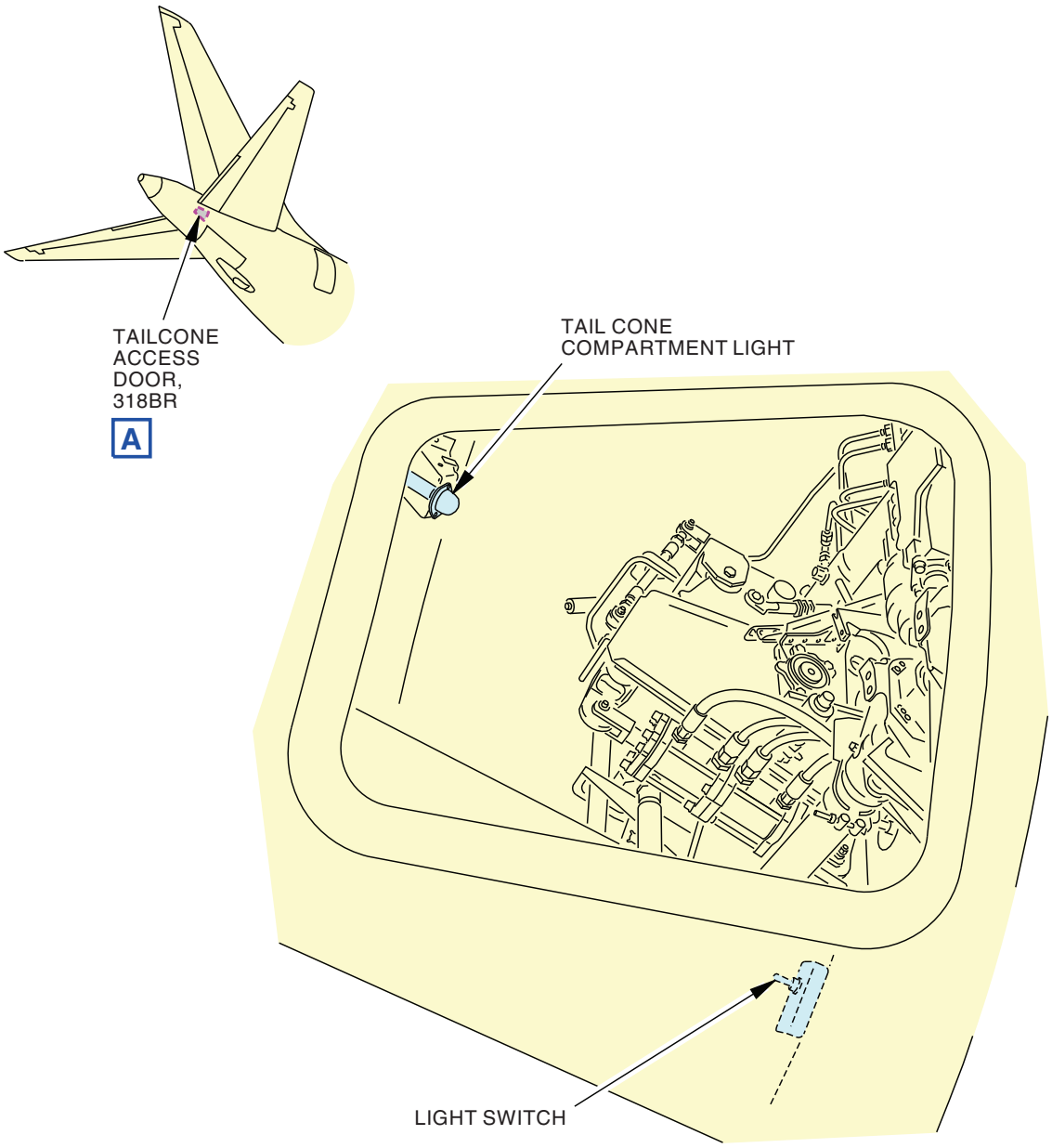


APU ACCESS DOOR 315A
(VIEW LOOKING INBOARD)
A

2414658 S00061535694_V2

Accessory Compartment Lights
Figure 201/33-35-00-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**TAIL CONE ACCESS DOOR, 318BR
(VIEW LOOKING INBOARD)**

A

2414659 S00061535695_V2

**Accessory Compartment Lights
Figure 201/33-35-00-990-801 (Sheet 3 of 3)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-35-00-440-801

3. Accessory Compartment Lights - Activation

(Figure 201)

A. General

(1) This procedure adds electrical power to the Accessory Compartment Lights.

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Accessory Compartment Lights Activation

SUBTASK 33-35-00-010-008

(1) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-35-00-860-003

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	5	C00389	ACCESS COMPT LT

SUBTASK 33-35-00-410-001

(3) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-35-00-960-801

4. Accessory Compartment Light - Lamp Replacement

(Figure 202)

A. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
310	Fuselage - Body Station 1016.00 to Body Station 1260.00

B. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
311BL	Stabilizer Trim Access Door
315A	APU Cowl Door
318BR	Tailcone Access Door

C. Accessory Compartment Light Lamp Replacement

SUBTASK 33-35-00-010-003

(1) For access to the area aft of the bulkhead, do this step:

(a) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

SUBTASK 33-35-00-010-004

(2) For access to the Auxiliary Power Unit (APU) compartment, do this step:

(a) Open this access panel:

Number	Name/Location
315A	APU Cowl Door

SUBTASK 33-35-00-010-005

(3) For access to the tailcone, do this step:

(a) Open this access panel:

Number	Name/Location
318BR	Tailcone Access Door

SUBTASK 33-35-00-010-009

(4) To get access to the P91 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-35-00-010-006

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(5) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	5	C00389	ACCESS COMPT LT

SUBTASK 33-35-00-860-004

(6) Set the light switch at the access door to the OFF mode.

SUBTASK 33-35-00-960-001

(7) Do these steps to replace service LED light assembly:

(a) Disconnect electrical connector [6] from light assembly [2].

(b) Remove nut [3], washer [1] and ground terminal [7].

(c) Remove fasteners [5], washers [4] and light assembly [2].

(d) Install the new light assembly [2], washers [4], and fasteners [5].

(e) Install ground terminal [7], washer [1], and nut [3]

(f) Connect electrical connector [6] to the new light assembly [2].

SUBTASK 33-35-00-960-002

(8) Do these steps to replace the lamp:

(a) Loosen the applicable fasteners [8].

(b) Remove the applicable lens assembly [9].

(c) Replace the applicable lamp [10].

(d) Install the applicable lens assembly [9].

1) Check the gasket, if one is installed.

2) If the gasket is damaged, replace it.

(e) Tighten the applicable fasteners [8].

SUBTASK 33-35-00-010-007

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(9) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 1, P91

Row	Col	Number	Name
C	5	C00389	ACCESS COMPT LT

SUBTASK 33-35-00-410-005

(10) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-35-00-710-001

(11) Do this test of the new lamp:

(a) At the access door, set the switch for the light to the ON mode.

1) Make sure that the lamp comes on.

(b) Set the switch to the OFF mode.

1) Make sure that the lamp goes off.

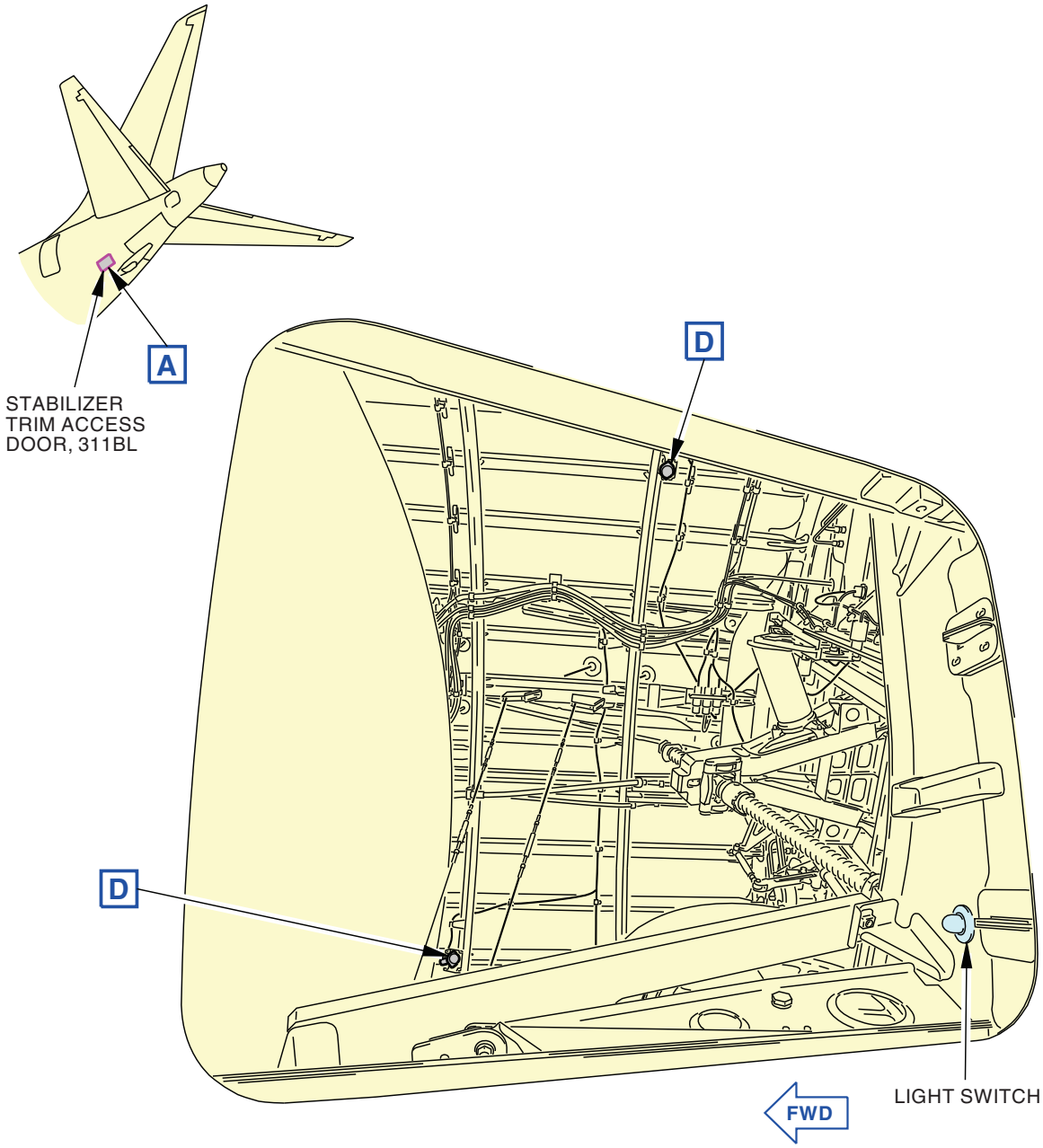
SUBTASK 33-35-00-410-002

(12) Close these access panels:

Number	Name/Location
311BL	Stabilizer Trim Access Door
315A	APU Cowl Door
318BR	Tailcone Access Door

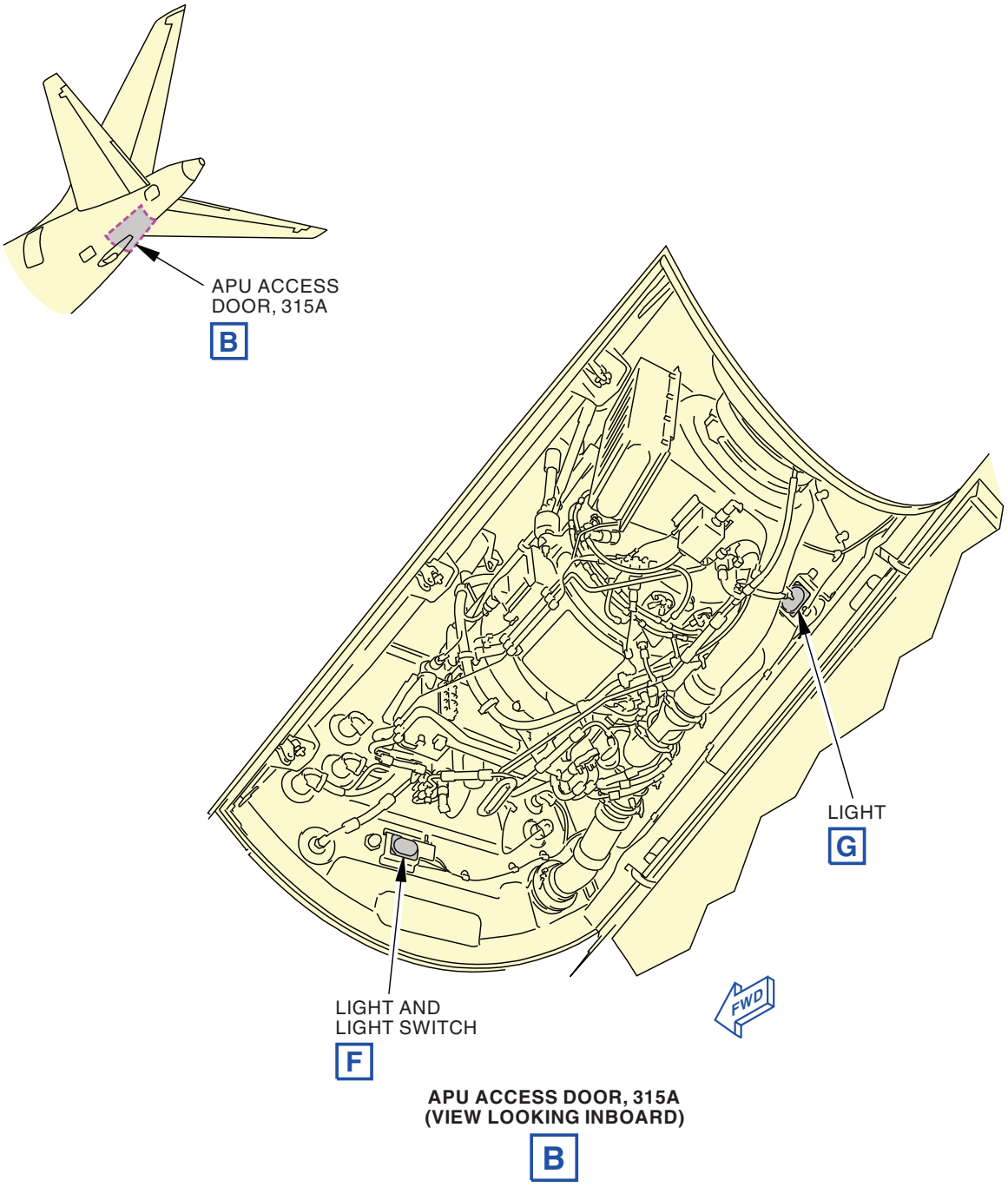
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ECCN 9E991 BOEING PROPRIETARY - See title page for details



Accessory Compartment Light - Lamp Replacement
Figure 202/33-35-00-990-802 (Sheet 1 of 5)

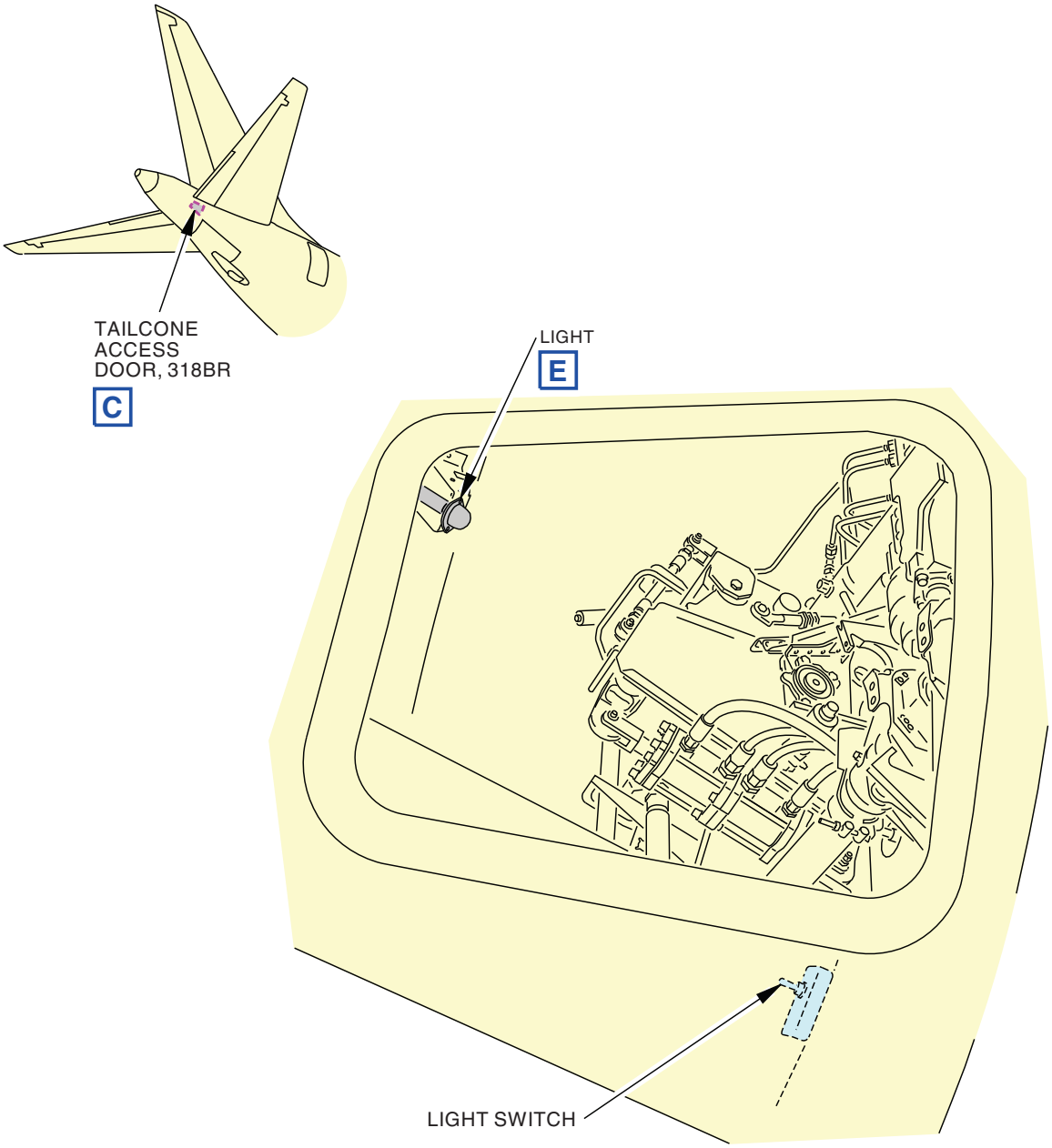
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414661 S00061535699_V2

Accessory Compartment Light - Lamp Replacement
Figure 202/33-35-00-990-802 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

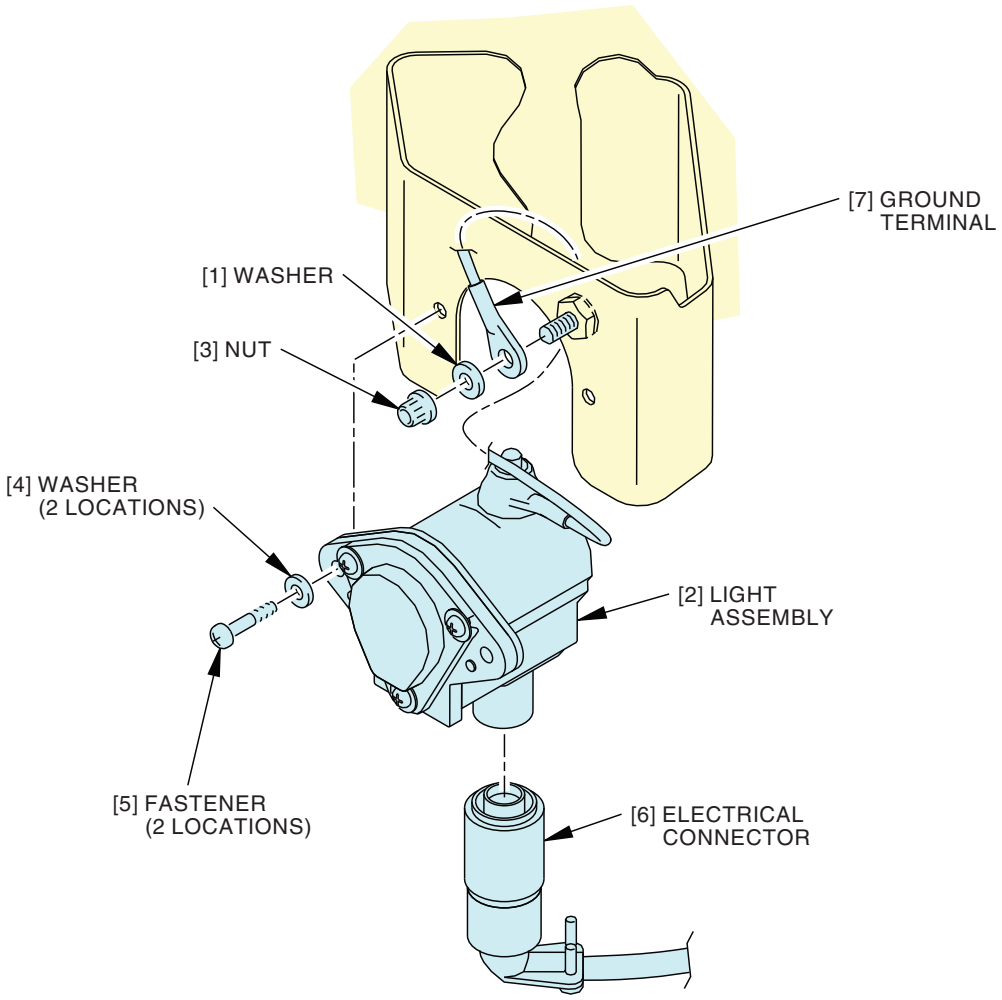


TAIL CONE ACCESS DOOR, 318BR
(VIEW LOOKING INBOARD)
C

2414662 S00061535700_V2

Accessory Compartment Light - Lamp Replacement
Figure 202/33-35-00-990-802 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



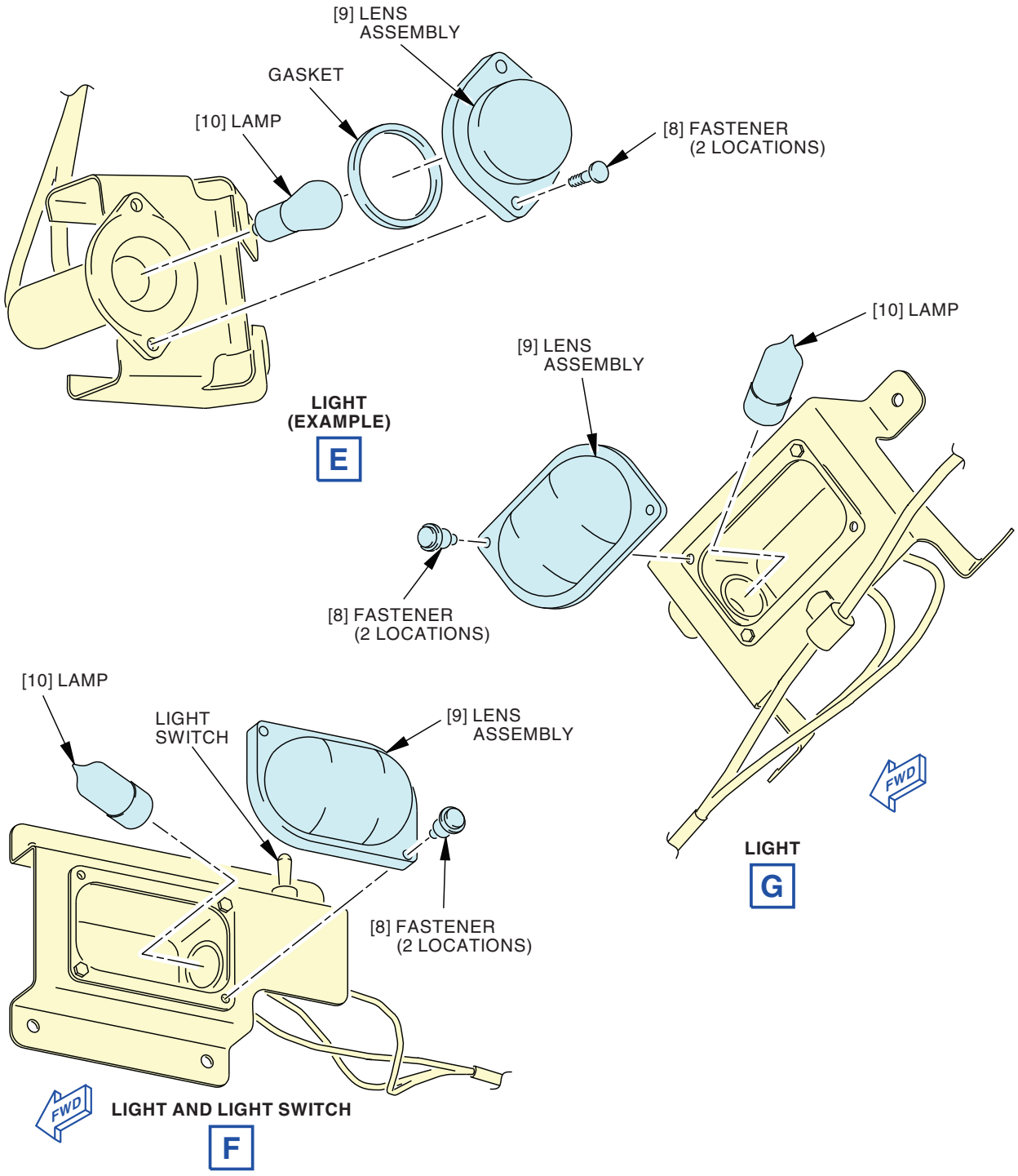
LIGHT (EXAMPLE)

D

2475695 S0000580259_V2

Accessory Compartment Light - Lamp Replacement
Figure 202/33-35-00-990-802 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2480573 S0000582630_V1

Accessory Compartment Light - Lamp Replacement
Figure 202/33-35-00-990-802 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CARGO COMPARTMENT LIGHTS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Cargo compartment lights deactivation

(2) Cargo compartment lights activation

(3) Cargo compartment ceiling light assembly replacement

(4) Cargo compartment door sill light assembly replacement.

TASK 33-36-00-040-801

2. Cargo Compartment Lights Deactivation

A. General

(1) This task provides the instructions on how to deactivate the cargo compartment lights.

(2) This make sure that the cargo compartment lights are in a zero energy state when it is necessary.

B. References

Reference	Title
52-31-00-580-801	Open the Cargo Door (P/B 201)

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Deactivation

SUBTASK 33-36-00-010-001

(1) To get access to the P92 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-36-00-010-002

(2) Open the forward and aft cargo doors (TASK 52-31-00-580-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Cargo Compartment Lights Deactivation

SUBTASK 33-36-00-860-001

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Open these circuit breakers and install safety tags:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	4	C00127	CARGO COMPT LT-FWD
C	5	C00128	CARGO COMPT LT-AFT
C	10	C01419	CARGO LT CONT

G. Put the Airplane Back to Its Usual Condition

SUBTASK 33-36-00-410-001

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

END OF TASK

TASK 33-36-00-440-801

3. Cargo Compartment Lights Activation

A. General

(1) This task provides the instructions on how to activate the cargo compartment lights.

(2) This restores the cargo compartment lights to the operational condition.

B. References

Reference	Title
52-31-00-580-801	Open the Cargo Door (P/B 201)

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Activation

SUBTASK 33-36-00-010-003

(1) To get access to the P92 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-36-00-010-004

(2) Open the forward and aft cargo doors (TASK 52-31-00-580-801).

F. Cargo Compartment Lights Activation

SUBTASK 33-36-00-860-002

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tags and close these circuit breakers:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	4	C00127	CARGO COMPT LT-FWD
C	5	C00128	CARGO COMPT LT-AFT
C	10	C01419	CARGO LT CONT

SUBTASK 33-36-00-210-001

(2) Make sure that the forward cargo compartment lights are activated:

(a) Move the forward cargo compartment lights switch (S67) to ON.

NOTE: The location of the forward cargo compartment lights switch is near the bottom on the forward side of the cargo door frame.

1) Make sure that the forward cargo compartment lights come on.

2) If the forward cargo compartment lights are not necessary, set the forward cargo compartment lights switch (S67) to OFF.

(3) Make sure that the aft cargo compartment lights are activated:

(a) Move the aft cargo compartment lights switch (S68) to ON.

NOTE: The location of the aft cargo compartment lights switch is near the bottom on the forward side of the cargo door frame.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the aft cargo compartment lights come on.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 33-36-00-410-002

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

——— END OF TASK ———

TASK 33-36-00-960-801

4. Cargo Compartment Ceiling Light Assembly Replacement

(Figure 201)

A. General

(1) This task gives the instructions on how to replace a light assembly [1] on the cargo compartment ceiling.

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
141	Aft Cargo Compartment - Left
142	Aft Cargo Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
821	Forward Cargo Door
822	Aft Cargo Door

D. Prepare for the Replacement

SUBTASK 33-36-00-010-005

(1) To get access to the P92 panel, open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-36-00-860-003

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open the applicable circuit breaker and install the safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	4	C00127	CARGO COMPT LT-FWD
C	5	C00128	CARGO COMPT LT-AFT

SUBTASK 33-36-00-010-006

(3) Open the applicable cargo door:

Number	Name/Location
821	Forward Cargo Door
822	Aft Cargo Door

SUBTASK 33-36-00-860-004

(4) Set the forward or aft control switch for the cargo compartment lights to OFF.

NOTE: The forward or aft control switch is located near the bottom of the cargo door frame.

E. Cargo Compartment Ceiling Light Assembly Replacement

SUBTASK 33-36-00-010-007

(1) Remove the closeout ring assembly [4].

(a) Remove the screws [2] that attach the closeout ring assembly [4] and the cargo liner [7] to the bracket assembly [5].

(b) Remove the closeout ring assembly [4].

SUBTASK 33-36-00-960-001

(2) Replace the light assembly [1].

(a) Remove the screws [3] that attach the light assembly [1] to the bracket assembly [5].

(b) Disconnect the electrical connector [6] from the light assembly [1].

(c) Remove the light assembly [1] from the bracket assembly [5].

(d) Connect the electrical connector [6] to the new light assembly [1].

(e) Attach the new light assembly [1] to the bracket assembly [5] with the screws [3].

SUBTASK 33-36-00-410-003

CAUTION

DO NOT COVER THE CLOSEOUT-RING OR LENS WITH TAPE, PAPER, PLASTIC, AND OTHER ITEMS. IF YOU DO, IT CAN CAUSE THE LIGHT TO BECOME TOO HOT.

(3) Attach the closeout ring assembly [4] and the cargo liner [7] to the bracket assembly [5] with the screws [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Cargo Compartment Ceiling Light Assembly Installation Test

SUBTASK 33-36-00-860-005

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tag and close the applicable circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	4	C00127	CARGO COMPT LT-FWD
C	5	C00128	CARGO COMPT LT-AFT

SUBTASK 33-36-00-860-007

(2) Do a test of the new light assembly [1]:

(a) Set the forward or aft control switch for the cargo compartment lights to ON.

NOTE: The forward or aft control switch is located near the bottom of the cargo door frame.

(b) Make sure that the light comes on.

(c) Set the forward or aft control switch to OFF.

(d) Make sure that the light goes off.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 33-36-00-410-004

(1) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

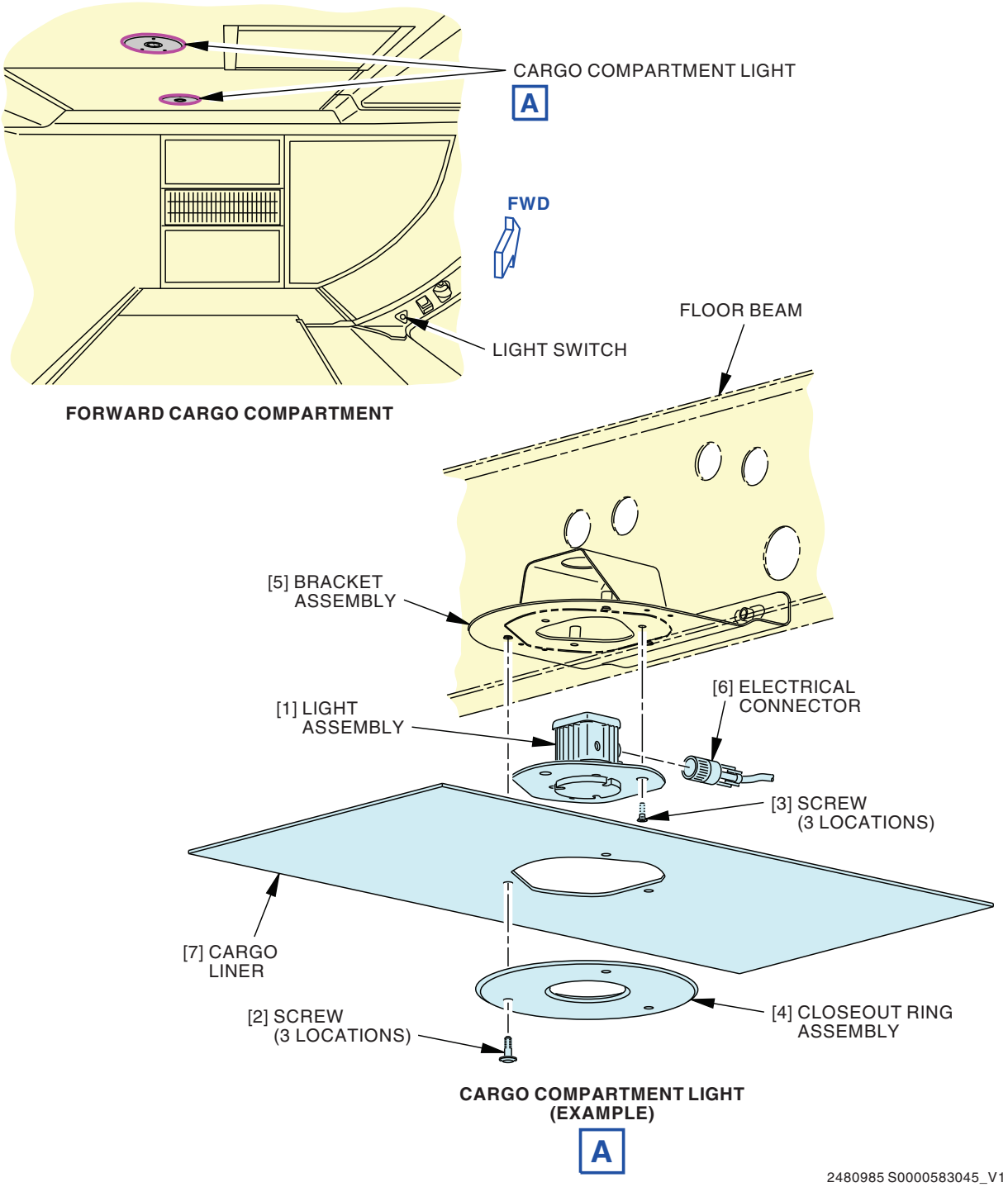
SUBTASK 33-36-00-410-005

(2) Close the applicable cargo door:

Number	Name/Location
821	Forward Cargo Door
822	Aft Cargo Door

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Cargo Compartment Ceiling Light Assembly Installation
Figure 201/33-36-00-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-36-00-960-802

5. Cargo Compartment Door Sill Light Assembly Replacement

(Figure 202)

A. General

(1) This task gives the instructions on how to replace the light assembly on the cargo compartment forward and aft door sills.

B. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
122	Forward Cargo Compartment - Right
142	Aft Cargo Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door
821	Forward Cargo Door
822	Aft Cargo Door

D. Prepare for the Replacement

SUBTASK 33-36-00-010-008

(1) To get access to the P92 panel, open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-36-00-860-009

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(2) Open the applicable circuit breakers and install the safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	4	C00127	CARGO COMPT LT-FWD
C	5	C00128	CARGO COMPT LT-AFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-36-00-010-009

(3) Open the applicable cargo door:

Number	Name/Location
821	Forward Cargo Door
822	Aft Cargo Door

E. Cargo Compartment Door Sill Light Assembly Replacement

SUBTASK 33-36-00-960-002

(1) Replace the forward cargo compartment door sill light assembly [8].

(a) Remove the screws [10] and washers [11].

(b) Remove the light assembly [8], gasket [13] and spacers [12] from the forward cargo door upper main sill.

(c) Put identification tags on each electrical connector for the correct connections during installation.

(d) Disconnect the electrical connectors [14] from the light assembly [8].

(e) Remove the gasket [13] and spacers [12] from the light assembly [8].

(f) Put the gasket [13] on the new light assembly [8].

(g) Insert the spacers [12] into the gasket mount holes.

(h) Connect the electrical connectors [14] to the new light assembly [8].

(i) Remove identification tags from the electrical connectors.

(j) Align the four holes on the new light assembly [8] with the holes on the gasket [13].

(k) Install the new light assembly [8] on the forward cargo door upper main sill with the screws [10] and washers [11].

SUBTASK 33-36-00-960-003

(2) Replace the aft cargo compartment door sill light assembly [9].

(a) Remove the screws [17] and washers [16].

(b) Remove the light assembly [9] from the aft cargo door upper main sill.

(c) Put identification tags on each electrical connector for the correct connections during installation.

(d) Disconnect the electrical connectors [15] from the light assembly [9].

(e) Connect the electrical connectors [15] to the new light assembly [9].

(f) Remove identification tags from the electrical connectors.

(g) Install the new light assembly [9] on the aft cargo door upper main sill with the screws [17] and washers [16].

F. Cargo Compartment Door Sill Light Assembly Replacement Test

SUBTASK 33-36-00-860-010

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tag and close the applicable circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
C	4	C00127	CARGO COMPT LT-FWD
C	5	C00128	CARGO COMPT LT-AFT

SUBTASK 33-36-00-710-001

(2) Do a test of the new light assembly as follows:

(a) Set the control switch for the light to ON.

NOTE: The forward and aft cargo light switches are located near the bottom of the cargo door frames.

(b) Make sure that the light comes on.

(c) Set the switch to OFF.

(d) Make sure that the light goes off.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 33-36-00-410-006

(1) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

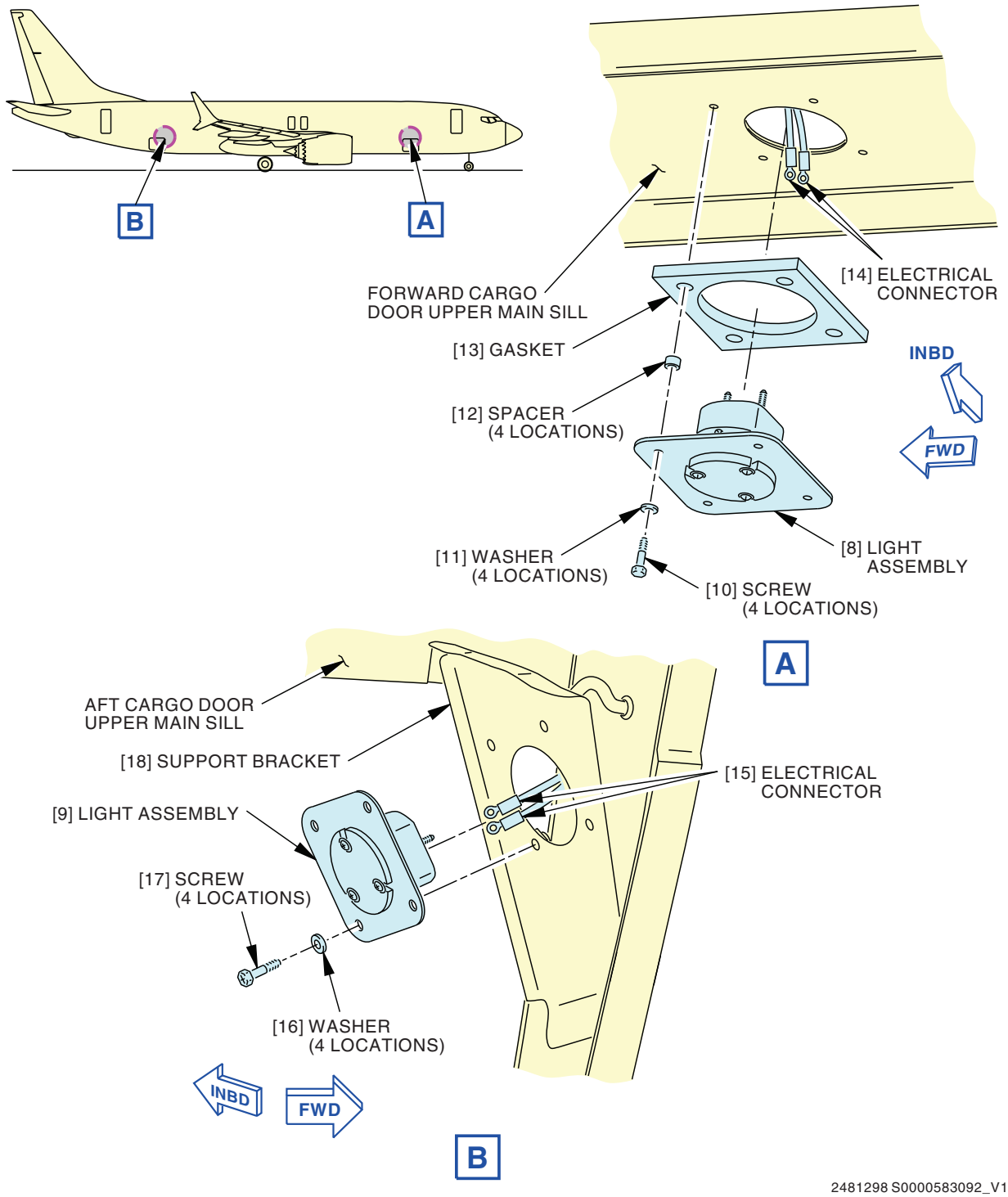
SUBTASK 33-36-00-410-007

(2) Close the applicable cargo door:

Number	Name/Location
821	Forward Cargo Door
822	Aft Cargo Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Cargo Compartment Door Sill Light Assembly Installation
Figure 202/33-36-00-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WING ILLUMINATION LIGHTS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Wing Illumination Light - Deactivation

(2) Wing Illumination Light - Activation

PTF ALL; AIRPLANES WITH LAMP LIGHTS

(3) Wing Illumination Light - Lamp Replacement

PTF ALL; AIRPLANES WITH LED MODULE ASSEMBLY LIGHTS

(4) Wing Illumination Light - LED Module Assembly Replacement

PTF ALL; AIRPLANES WITH LAMP LIGHTS

(5) Wing Illumination Light - Transformer Replacement

PTF ALL

(6) Wing Illumination Light - Light Assembly Replacement

(7) Wing Illumination Lights - Operational Test.

TASK 33-41-00-040-801

2. Wing Illumination Light - Deactivation

A. General

(1) This procedure removes electrical power from the wing illumination lights.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
511	Left Wing - Leading Edge To Front Spar
611	Right Wing - Leading Edge to Front Spar

C. Procedure

SUBTASK 33-41-00-860-001

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	14	C00110	EXTERIOR LIGHTING WING SCAN

D. Wing Illumination Light Deactivation Test

NOTE: This test is to make sure that the wing illumination lights are in a zero energy condition.

SUBTASK 33-41-00-200-001

(1) Do the following steps to make sure that the power is off:

(a) At the overhead panel, P5, set the wing light switch to ON position.

1) Make sure that the wing illumination lights (left/right) stay off.

(b) At the overhead panel, P5, set the wing light switch to OFF position.

1) Make sure that the wing illumination lights (left/right) stay off.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-41-00-440-801

3. Wing Illumination Light - Activation

A. General

(1) This procedure supplies electrical power to the wing illumination lights.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
511	Left Wing - Leading Edge To Front Spar
611	Right Wing - Leading Edge to Front Spar

C. Procedure

SUBTASK 33-41-00-860-002

(1) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	14	C00110	EXTERIOR LIGHTING WING SCAN

D. Wing Illumination Light - Tryout

NOTE: This tryout is to make sure that the wing illumination lights operate.

SUBTASK 33-41-00-710-006

(1) Do the following steps to make sure that the power is on:

(a) At the overhead panel, P5, set the wing light switch to ON position

1) Make sure that the wing illumination lights (left/right) come on.

(b) At the overhead panel, P5, set the wing light switch to OFF position

1) Make sure that the wing illumination lights (left/right) go off.

——— END OF TASK ———

PTF ALL; AIRPLANES WITH LAMP LIGHTS

TASK 33-41-00-960-801

4. Wing Illumination Light - Lamp Replacement

(Figure 201)

A. References

Reference	Title
WDM 33-41-11	Wiring Diagram Manual

B. Consumable Materials

Reference	Description	Specification
G01306	Gloves - Lint-free	
G02309	Material - Moldable Plastic Shim	BMS8-338

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Lamp	33-41-00-02-045	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH LAMP LIGHTS (Continued)

D. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

E. Wing Illumination Light Lamp Replacement

SUBTASK 33-41-00-860-003

(1) Open this circuit breaker and install safety tag:

(WDM 33-41-11)

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	14	C00110	EXTERIOR LIGHTING WING SCAN

SUBTASK 33-41-00-960-001

(2) Do these steps to replace the lamp [1]:

(a) Do not touch the new lamp with your hands.

1) Use a lint-free gloves, G01306.

(b) Remove the lens assembly [2].

1) Loosen the screws [6].

2) Do not remove the lanyard.

(c) Loosen the retainer [7] to release the socket.

(d) Disconnect the electrical connector from the lamp [1].

(e) Remove the lamp [1].

(f) If the shims on the lens gasket are damaged, replace them.

1) Remove the unwanted shim material.

2) Apply new shim material, G02309, in the applicable areas of the rubber gasket.

NOTE: Use only the quantity necessary. Adjacent compartments in the rubber gasket will help contain the shim material. Keep shim material away from the lens fasteners.

3) Install the lens assembly [2].

a) Tighten the screws [9] until the lens assembly [2] is aligned with the fuselage.

4) Let the shim material dry for six hours.

5) Remove the lens assembly [2].

a) Loosen the screws [6].

6) Clean and examine the shims.

a) Make sure that the shim thickness is no more than 0.125 in. (3.18 mm).

(g) Install the new lamp [1].

(h) Connect the electrical connector to the lamp [1].

(i) Put retainer [7] back to hold the socket.

(j) Install the lens assembly [2].

1) Tighten the screws [6] to 18 in-lb (2 N·m) - 21 in-lb (2 N·m).

2) Make sure that the UP label is at the top of the lens assembly [2].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH LAMP LIGHTS (Continued)

SUBTASK 33-41-00-860-004

(3) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

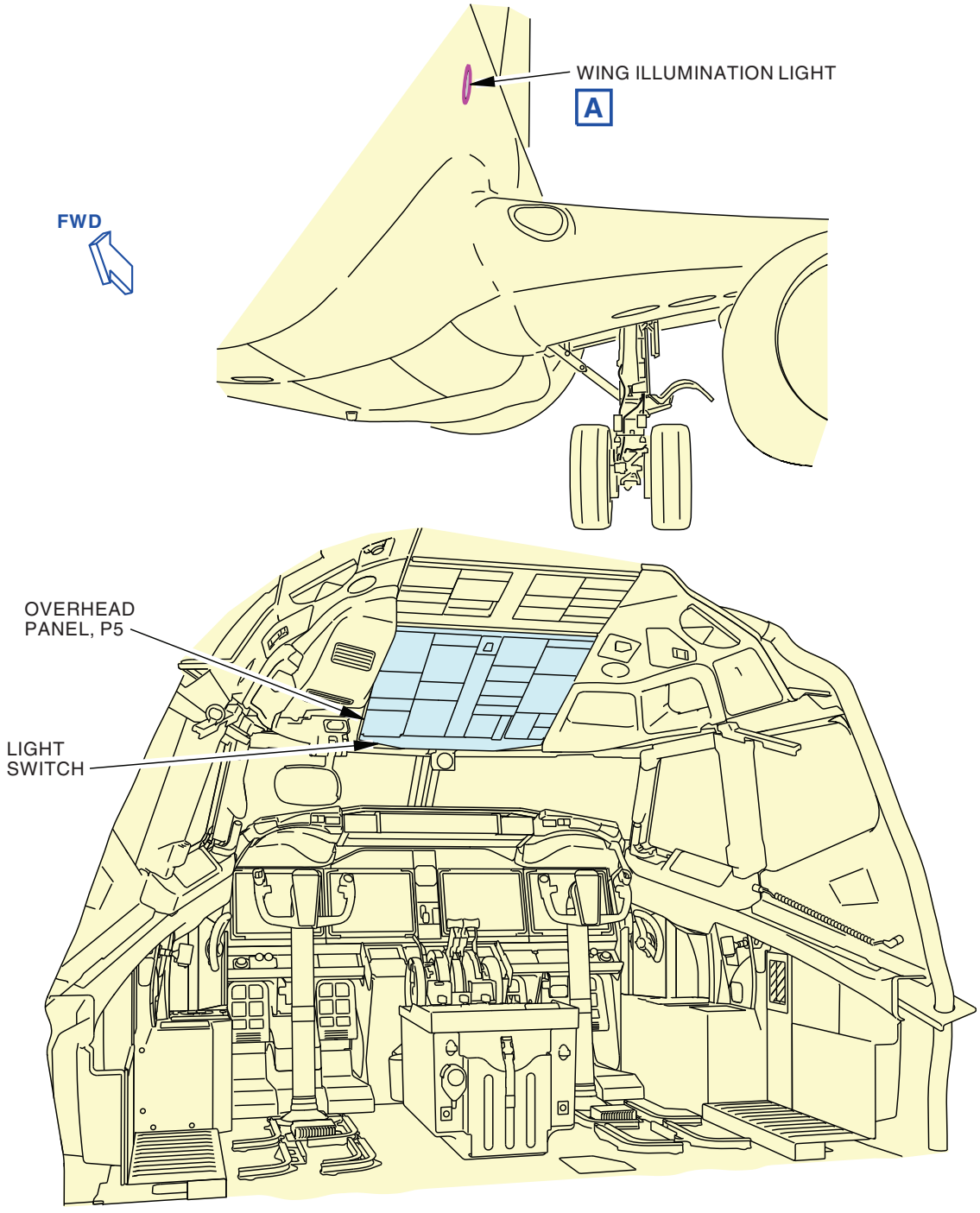
Row	Col	Number	Name
A	14	C00110	EXTERIOR LIGHTING WING SCAN

SUBTASK 33-41-00-710-001

(4) Do this task: Wing Illumination Light - Operational Test, TASK 33-41-00-710-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

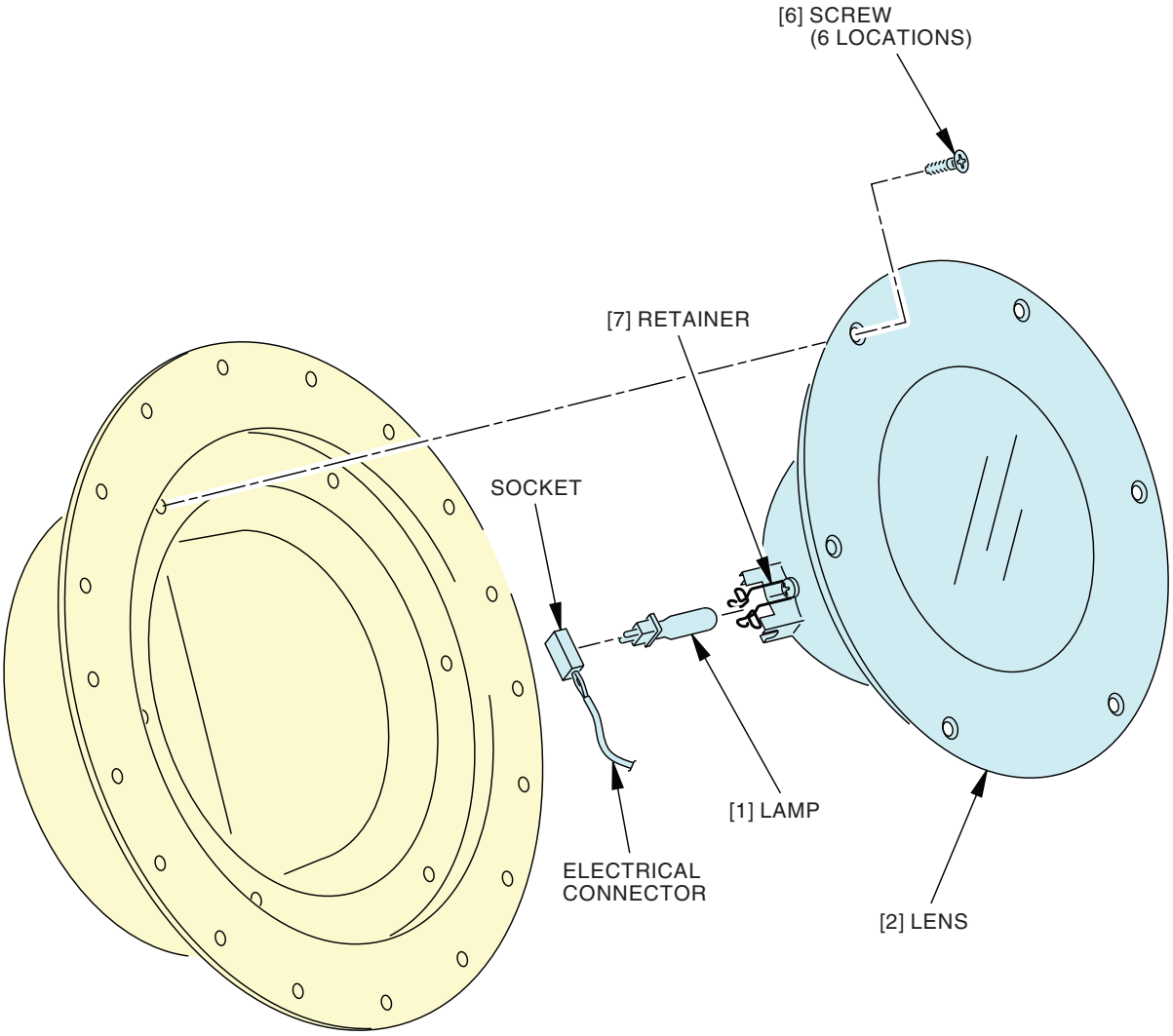


FLIGHT COMPARTMENT

2414668 S00061535722_V1

Wing Illumination Light - Lamp Replacement
Figure 201/33-41-00-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



WING ILLUMINATION LIGHT
(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

A

2414669 S00061535723_V2

Wing Illumination Light - Lamp Replacement
Figure 201/33-41-00-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH LED MODULE ASSEMBLY LIGHTS

TASK 33-41-00-960-804

5. Wing Illumination Light - LED Module Assembly Replacement

(Figure 202)

A. General

(1) This procedure gives the steps necessary for the wing illumination light LED module assembly replacement.

B. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

C. Procedure

SUBTASK 33-41-00-860-015

(1)

Make sure that this circuit breaker is open and has safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	14	C00110	EXTERIOR LIGHTING WING SCAN

SUBTASK 33-41-00-960-004

(2) Do these steps to replace the LED module assembly [30]:

(a) Remove the captive screws [31].

(b) Remove the lens assembly [32].

(c) Remove the captive screws [33].

(d) Disconnect the connector [34].

(e) Pull up the LED module assembly [30].

(f) Install a new LED module assembly [30].

(g) Secure the LED module assembly [30] with the captive screws [33].

(h) Connect the connector [34].

(i) Align the lens assembly [32] over the LED module assembly [30] and secure using the captive screws [31].

1) Tighten the captive screws [31] to 18 in-lb (2 N·m) - 21 in-lb (2 N·m).

2) Make sure the UP label is at the top of the lens assembly [32].

SUBTASK 33-41-00-860-016

(3) Make sure that this circuit breaker is closed:

CAPT Electrical System Panel, P18-3

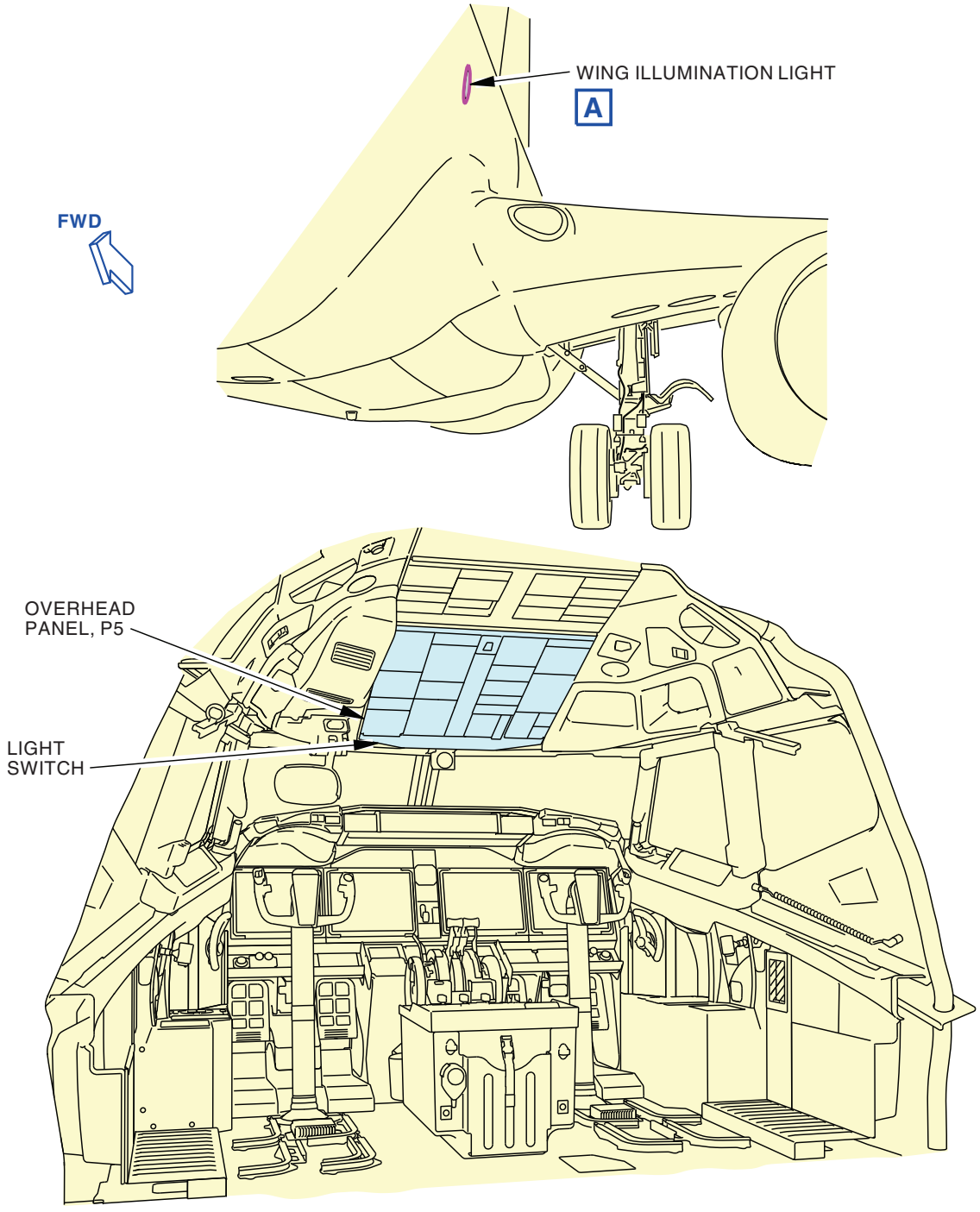
Row	Col	Number	Name
A	14	C00110	EXTERIOR LIGHTING WING SCAN

SUBTASK 33-41-00-710-007

(4) Do this task: Wing Illumination Light - Operational Test, TASK 33-41-00-710-801.

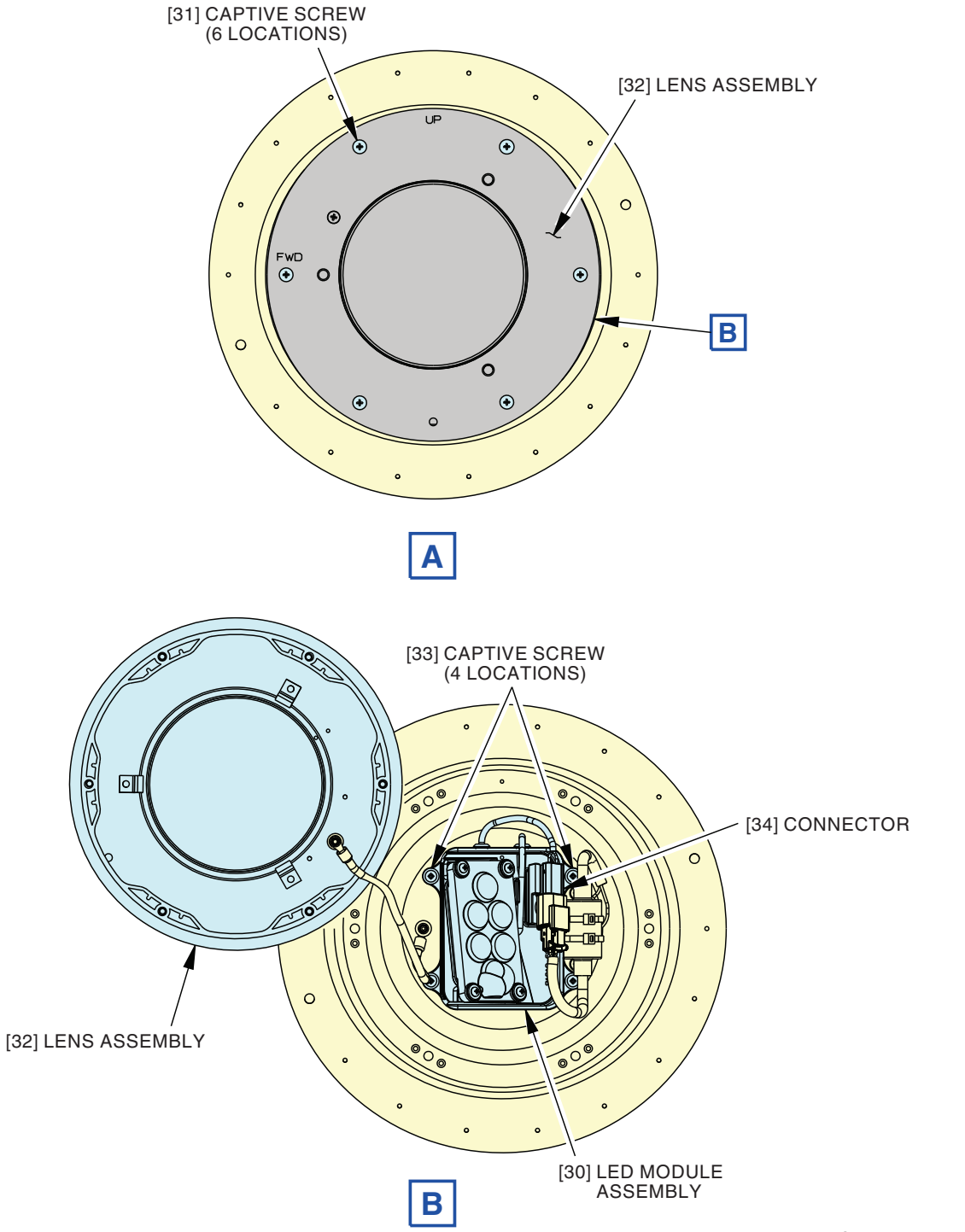
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Wing Illumination Light - LED Module Assembly Replacement
Figure 202/33-41-00-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Wing Illumination Light - LED Module Assembly Replacement
Figure 202/33-41-00-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH LAMP LIGHTS

TASK 33-41-00-960-802

6. Wing Illumination Light - Transformer Replacement

(Figure 201, Figure 203)

A. References

Reference	Title
05-51-91-790-801	Cabin Pressure Leak Test (P/B 201)
25-21-46-000-801	Sidewall Panel - Removal (P/B 401)
25-21-46-400-801	Sidewall Panel - Installation (P/B 401)

B. Tools/Equipment

Reference	Description
STD-810	Spatula - Fillet Smoothing, Hardwood or Plastic

C. Consumable Materials

Reference	Description	Specification
A00281	Adhesive - Dow Corning 3145 RTV	MIL-A-46146 (BAC5010 Type 79)
B00062	Solvent - Acetone (99.5% Grade)	ASTM D 329 (Supersedes O-A-51)
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
21	Transformer	33-41-00-02-105	PTF ALL

E. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

F. Wing Illumination Light Transformer Replacement

SUBTASK 33-41-00-860-017

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	14	C00110	EXTERIOR LIGHTING WING SCAN

SUBTASK 33-41-00-020-005

(2) Do these steps out of the airplane at the wing illumination light:

(a) Remove the lens assembly [2].

1) Loosen the screws [6].

(b) Disconnect the electrical connector from the lamp [1].

(c) Remove the screws [23], washers [24], washers [25], and lanyard from the transformer [21].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL

SUBTASK 33-41-00-010-007

(3) Remove the sidewall panel inboard of the wing illumination light transformer (TASK 25-21-46-000-801).

PTF ALL; AIRPLANES WITH LAMP LIGHTS

SUBTASK 33-41-00-010-008

(4) Pull away the liner for access to the back of the wing illumination light.

SUBTASK 33-41-00-960-005

(5) Do these steps to replace the transformer [21]:

(a) Disconnect the electrical wires [22].

(b) Use a hardwood or plastic fillet smoothing spatula, STD-810, to break the seal around the base of the transformer [21].

(c) Remove the transformer [21].

(d) Clean the mating surfaces with solvent, B00062, or solvent, B00184.

(e) Put the transformer [21] in its position on the back of the light assembly.

SUBTASK 33-41-00-420-004

(6) Do these steps out of the airplane at the wing illumination light:

(a) Install the screws [23], washers [24], washers [25], and lanyard on the transformer [21].

(b) Connect the electrical connector to the lamp [1].

(c) Install the lens assembly [2].

1) Tighten the screws [6] from 18 in-lb (2.0 N·m) to 21 in-lb (2.4 N·m).

2) Make sure that the UP label is at the top of the lens assembly [2].

SUBTASK 33-41-00-860-018

(7) Do these steps in the airplane:

(a) Apply Dow Corning 3145 RTV adhesive, A00281, (or equivalent) around the base of the transformer [21].

(b) Connect the electrical wires [22].

SUBTASK 33-41-00-860-019

(8) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	14	C00110	EXTERIOR LIGHTING WING SCAN

SUBTASK 33-41-00-710-008

(9) Do this task: Wing Illumination Light - Operational Test, TASK 33-41-00-710-801.

SUBTASK 33-41-00-790-001

(10) Do this task: Cabin Pressure Leak Test, TASK 05-51-91-790-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL

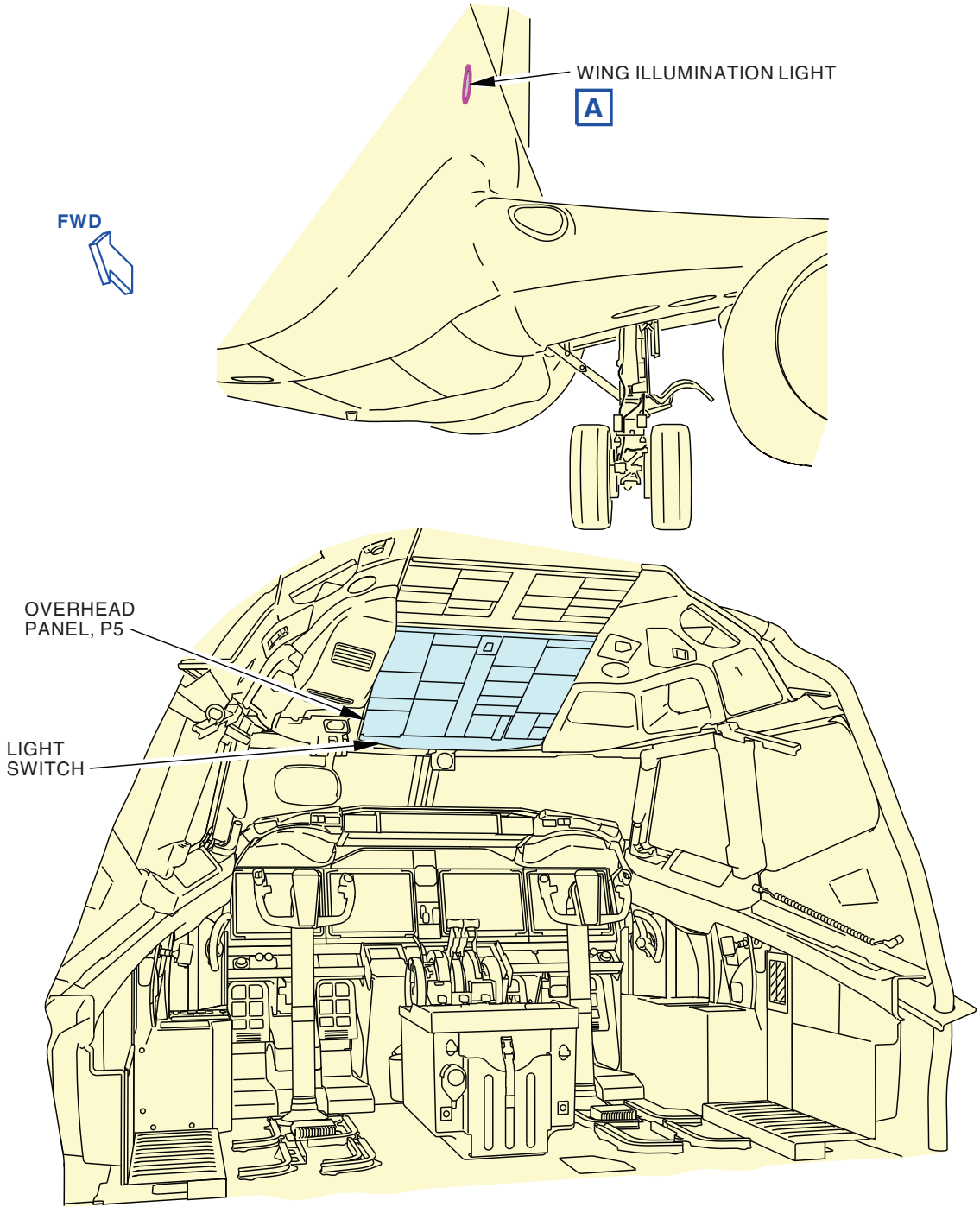
SUBTASK 33-41-00-410-005

(11) To install the sidewall panel, do this task: Sidewall Panel - Installation, TASK 25-21-46-400-801.

PTF ALL; AIRPLANES WITH LAMP LIGHTS

——— END OF TASK ———

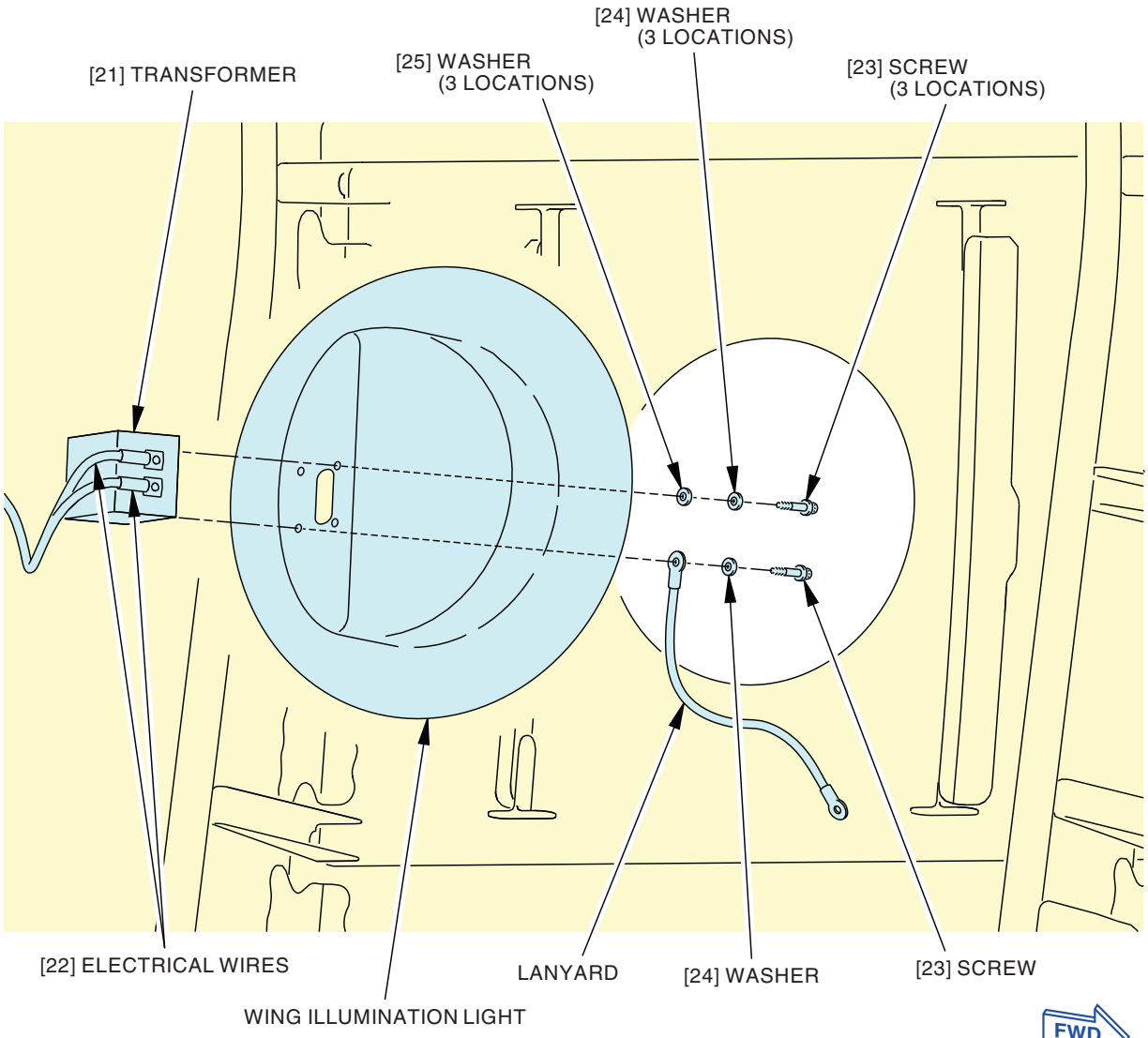
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414668 S00061535722_V1

Wing Illumination Light - Transformer Replacement
Figure 203/33-41-00-990-807 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



WING ILLUMINATION LIGHT TRANSFORMER
(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

A

2414671 S00061535726_V1

Wing Illumination Light - Transformer Replacement
Figure 203/33-41-00-990-807 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL

TASK 33-41-00-960-803

7. Wing Illumination Light - Light Assembly Replacement

(Figure 203, Figure 204)

A. References

Reference	Title
05-51-91-790-801	Cabin Pressure Leak Test (P/B 201)
25-21-46-000-801	Sidewall Panel - Removal (P/B 401)
25-21-46-400-801	Sidewall Panel - Installation (P/B 401)
51-22-00-160-801	Prepare For Sealing (P/B 201)

B. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Light assembly	33-41-00-02-010	PTF ALL
		33-41-00-02-150	PTF ALL

D. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

E. Wing Illumination Light - Light Assembly Replacement

SUBTASK 33-41-00-860-011

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	14	C00110	EXTERIOR LIGHTING WING SCAN

SUBTASK 33-41-00-010-003

(2) Remove the sidewall panel inboard of the wing illumination light (TASK 25-21-46-000-801).

SUBTASK 33-41-00-010-004

(3) Pull away the liner for access to the back of the wing illumination light.

SUBTASK 33-41-00-020-002

(4) Remove the electrical wires [22] from the light assembly [1].

PTF ALL; AIRPLANES WITH LAMP LIGHTS

NOTE: The electrical wires are installed on the transformer of the light assembly.

PTF ALL

SUBTASK 33-41-00-020-003

(5) Remove the bolts [2] from the light assembly [1].

SUBTASK 33-41-00-020-004

(6) Remove the light assembly [1] from the airplane skin.

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SUBTASK 33-41-00-160-001

(7) Clean the sealant from the airplane skin (TASK 51-22-00-160-801).

SUBTASK 33-41-00-390-001

(8) Apply sealant, A00247, between the light assembly [1] and airplane skin for a faying surface seal.

SUBTASK 33-41-00-390-002

(9) Apply a layer of sealant, A00247, under the heads of the bolts [2].

SUBTASK 33-41-00-420-002

(10) Install the new light assembly [1] with the bolts [2].

(a) If the bolt holes in the light assembly [1] are too small, drill the bolt holes.

SUBTASK 33-41-00-420-003

(11) Install the electrical wires [22] to the light assembly [1].

F. Wing Illumination Light Installation Test

SUBTASK 33-41-00-860-012

(1) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	14	C00110	EXTERIOR LIGHTING WING SCAN

SUBTASK 33-41-00-710-003

(2) Do this task: Wing Illumination Light - Operational Test, TASK 33-41-00-710-801.

SUBTASK 33-41-00-790-002

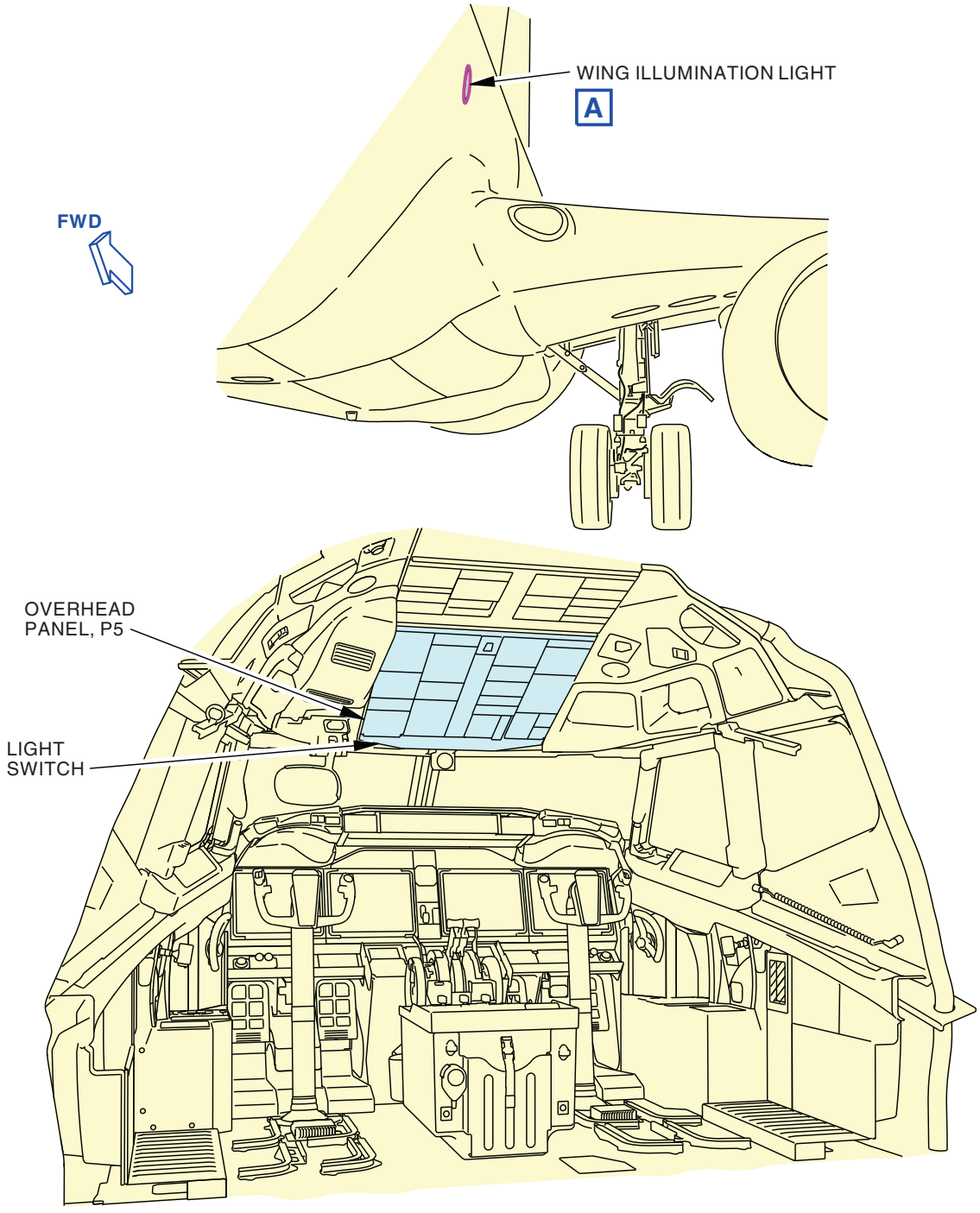
(3) Do this task: Cabin Pressure Leak Test, TASK 05-51-91-790-801.

SUBTASK 33-41-00-410-002

(4) To install the sidewall panel, do this task: Sidewall Panel - Installation, TASK 25-21-46-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

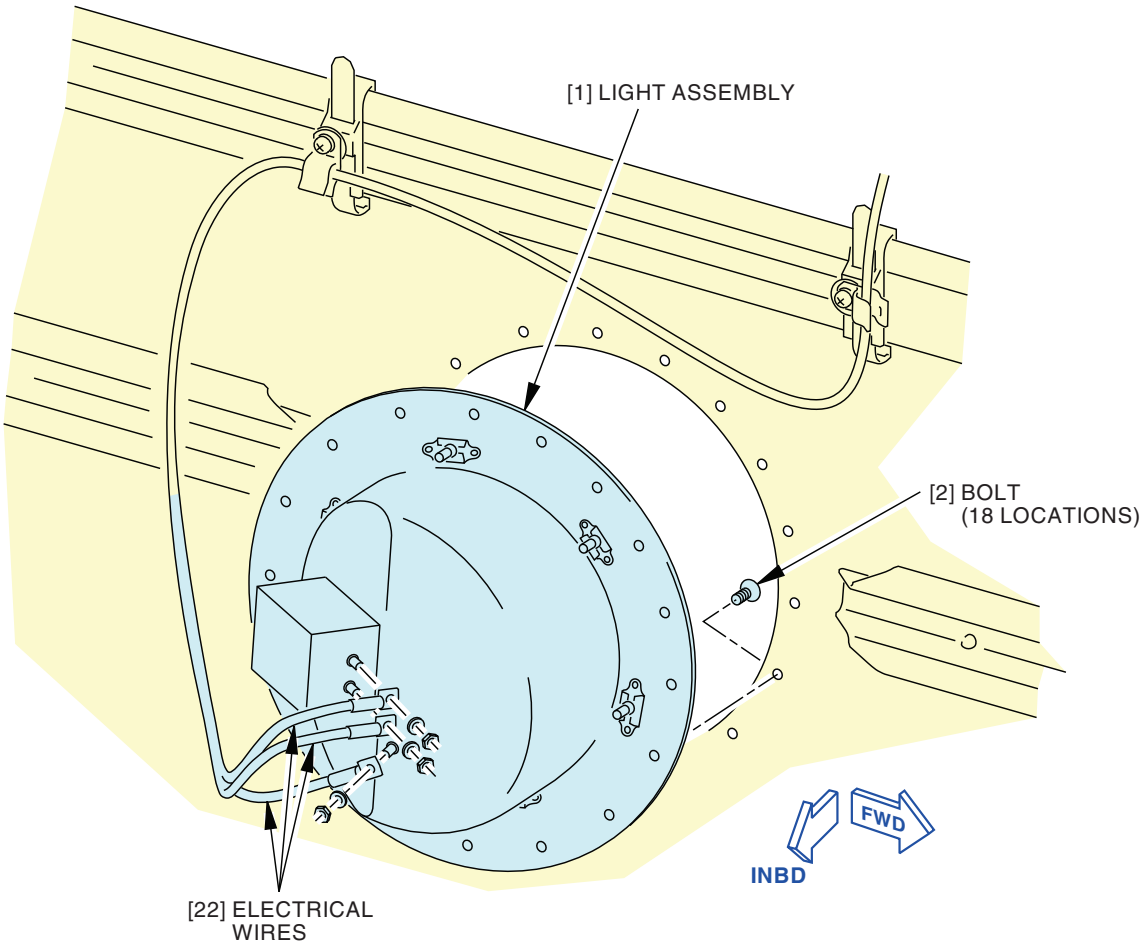


FLIGHT COMPARTMENT

2414668 S00061535722_V1

Wing Illumination Light - Light Assembly Replacement
Figure 204/33-41-00-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**WING ILLUMINATION LIGHT ASSEMBLY
(EXAMPLE)**

A

2414672 S00061535728_V2

**Wing Illumination Light - Light Assembly Replacement
Figure 204/33-41-00-990-803 (Sheet 2 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-41-00-710-801

8. Wing Illumination Light - Operational Test

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Procedure

SUBTASK 33-41-00-710-005

(1) Do a test of the wing illumination lights:

(a) At the overhead panel, P5, set the switch for the wing illumination lights to the on mode.

1) Make sure that the lights come on.

(b) Set the switch to the off mode.

1) Make sure that the lights go off.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LED LANDING, TAXI, AND RUNWAY TURNOFF LIGHTS (LTRTL) - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A deactivation of the fixed landing light

(2) An activation of the fixed landing light

(3) A replacement of the fixed landing light main array assembly

(4) A replacement of the fixed landing light power supply

(5) A temporary repair of the fixed landing light lens

(6) A deactivation of the runway turnoff lights

(7) An activation of the runway turnoff lights

(8) A replacement of the runway turnoff light assembly

(9) A replacement of the taxi light relay.

B. The main array assemblies are used for both landing and taxi light functions.

TASK 33-42-21-040-801

2. Fixed Landing Light - Deactivation

(Figure 201)

A. General

(1) This procedure removes the electrical power to the Fixed Landing Lights.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
511	Left Wing - Leading Edge To Front Spar
611	Right Wing - Leading Edge to Front Spar

C. Procedure

SUBTASK 33-42-21-860-029

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	14	C01968	EXTERIOR LIGHTING LTRTL - RIGHT
B	15	C01969	EXTERIOR LIGHTING LTRTL - LEFT
D	12	C01518	EXTERIOR LIGHTING GENERAL LIGHT CONT

D. Fixed Landing Light - Tryout

SUBTASK 33-42-21-700-002

NOTE: This tryout is to make sure that the Fixed Landing Light is in a zero energy state.

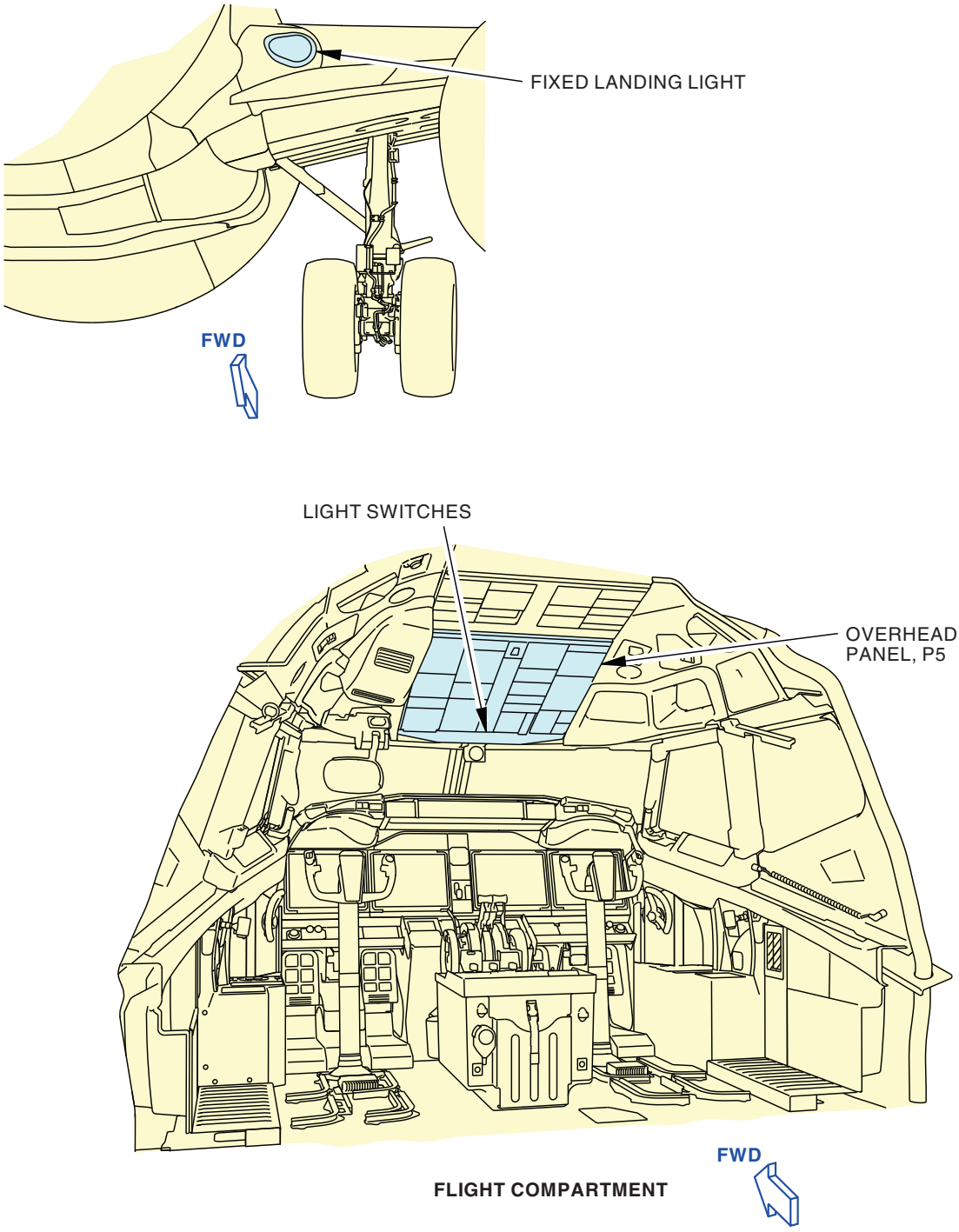
(1) Do the following steps to make sure that the power has been removed:

(a) At the overhead panel, P5, set the fixed landing lights switch to the ON Position.

(b) Make sure that the fixed landing lights in both wings are off.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fixed Landing Lights
Figure 201/33-42-21-990-810

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-42-21-440-801

3. Fixed Landing Light - Activation

(Figure 201)

A. General

(1) This procedure adds the electrical power to the Fixed Landing Lights.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
511	Left Wing - Leading Edge To Front Spar
611	Right Wing - Leading Edge to Front Spar

C. Procedure

SUBTASK 33-42-21-860-002

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	14	C01968	EXTERIOR LIGHTING LTRTL - RIGHT
B	15	C01969	EXTERIOR LIGHTING LTRTL - LEFT
D	12	C01518	EXTERIOR LIGHTING GENERAL LIGHT CONT

——— END OF TASK ———

TASK 33-42-21-960-801

4. Fixed Landing Light - Main Array Assembly Replacement

(Figure 202)

A. General

(1) This task gives the instructions to replace the main array assembly.

(2) The main array also does the taxi light function.

(3) The inboard and outboard array assemblies are the parts of the main array and can be replaced apart.

B. References

Reference	Title
20-40-12-000-801	ESDS Handling for Printed Circuit Board Removal (P/B 201)
24-22-00-860-801	Supply Electrical Power (P/B 201)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
SWPM 20-20-00, Paragraph 2.E.	Standard Wiring Practices Manual
SWPM 20-20-00, Section 21-A	Resistance of Faying Surface Bonds

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50066	Tape - Thick Ribbed Tape with Adhesive (Skyflex GSC-21-95241-011)	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
50	Packing	33-42-21-01-060	PTF ALL
51	Packing	33-42-21-01-060	PTF ALL
52	Packing	33-42-21-01-055	PTF ALL
53	Packing	33-42-21-01-055	PTF ALL

F. Location Zones

Zone	Area
511	Left Wing - Leading Edge To Front Spar
611	Right Wing - Leading Edge to Front Spar

G. Prepare for the Replacement

SUBTASK 33-42-21-860-030

WARNING DO THE DEACTIVATION PROCEDURE FOR THE LEADING EDGE FLAPS AND SLATS. THE FLAPS AND SLATS MOVE QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do one of these steps to deactivate the leading edge flaps and slats in the retracted (UP) position:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) If activation is not necessary to the trailing edge flaps, do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

NOTE: This task deactivates both the leading edge flaps/slats and the trailing edge flaps.

(b) If activation is necessary to the trailing edge flaps, do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

NOTE: This task only deactivates the leading edge flaps and slats. The trailing edge flaps can still operate.

SUBTASK 33-42-21-860-031

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	14	C01968	EXTERIOR LIGHTING LTRTL - RIGHT
B	15	C01969	EXTERIOR LIGHTING LTRTL - LEFT
D	12	C01518	EXTERIOR LIGHTING GENERAL LIGHT CONT

SUBTASK 33-42-21-910-007

(3) For devices that are sensitive to electrostatic discharge, do this task: ESDS Handling for Printed Circuit Board Removal, TASK 20-40-12-000-801.

H. Fixed Landing Light Main Array Assembly Replacement

SUBTASK 33-42-21-010-004

(1) Loosen the studs [4] and remove the lens assembly [3].

(a) Make sure that the lanyard [5] stays connected.

(b) Keep the lens assembly [3] and lanyard [5] away from the wing leading edge.

SUBTASK 33-42-21-020-006

(2) Do these steps to replace the main array:

(a) Remove the nut [7], washer [6], jumper [8], nut [43], washer [44], and jumper [45].

(b) Loosen the screws [9] and screws [37].

(c) Remove the screws [12] and washers [11] to disconnect the inboard array assembly [1] and outboard array assembly [2].

(d) Remove the inboard array assembly [1] with the captive fasteners from the brackets.

(e) Remove the screws [9], washers [10], packings [50] (if installed), screws [37], washers [38], and packings [51] (if installed) from the inboard array assembly [1].

1) If packings [50] and packings [51] have signs of damage or wear, discard them.

(f) Disconnect the electrical connector [13] from the inboard array assembly [1].

(g) Disconnect the clamp [54] from the inboard array assembly [1].

(h) Loosen the screws [39] and screws [41].

(i) Remove the outboard array assembly [2] with the captive fasteners from the brackets.

(j) Remove the screws [39], washers [40], packings [52] (if installed), screws [41], washers [42], and packings [53] (if installed) from the outboard array assembly [2].

1) If packings [52] and packings [53] have signs of damage or wear, discard them.

(k) Disconnect the electrical connector [46] from the outboard array assembly [2].

(l) Disconnect the clamp [54] from the outboard array assembly [2].

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(m) Install the screws [39], washers [40], packings [52], screws [41], washers [42], and packings [53] into holes for the fasteners of the new outboard array assembly [2].

1) Install new packings [52] and new packings [53], if necessary.

NOTE: As an optional procedure, it is permitted to install the array assembly without packings.

(n) Connect the electrical connector [46] to the new outboard array assembly [2].

(o) Connect the clamp [54] to the new outboard array assembly [2] to secure the wire.

(p) Install the new outboard array assembly [2] with the captive fasteners to the brackets.

(q) Install the screws [9], washers [10], packings [50], screws [37], washers [38], and packings [51] into holes for the fasteners of the new inboard array assembly [1].

1) Install new packings [50] and new packings [51], if necessary.

NOTE: As an optional procedure, it is permitted to install the array assembly without packings.

(r) Connect the electrical connector [13] to the new inboard array assembly [1].

(s) Connect the clamp [54] to the new inboard array assembly [1] to secure the wire.

(t) Install the new inboard array assembly [1] with the captive fasteners into the brackets.

(u) Install the screws [12] and washers [11] to connect the inboard array assembly [1] and outboard array assembly [2].

1) Tighten the screws [12], screws [9], screws [37], screws [39], and screws [41] to 35 in-lb (3.95 N·m) - 36 in-lb (4.07 N·m).

SUBTASK 33-42-21-120-003

(3) Do these steps to prepare the ground stud on the aluminum bracket, fasteners, jumper [8], and jumper [45] for the electrical bond installation:

(a) Clean the washer [6], nut [7], jumper [8], washer [44], nut [43], jumper [45], ground stud, and fay surfaces of the aluminum bracket.

1) Clean the washer [6], nut [7], jumper [8], washer [44], nut [43], and jumper [45] (SWPM 20-20-00, Paragraph 2.E.).

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

a) Use a new cotton wiper, G00034, made moist with solvent, B00184, or solvent, B00083.

b) Use a new cotton wiper, G00034, to dry the surfaces.

c) Make sure that you remove the remaining corrosion inhibiting compound on the bracket.

2) Clean the ground stud threads and head of the ground stud assembly (SWPM 20-20-00, Paragraph 2.E.).

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WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

a) Use a new cotton wiper, G00034, made moist with solvent, B00184, or solvent, B00083.

b) Use a new cotton wiper, G00034, to dry the surfaces.

c) Continue to clean and dry the surface until the cotton wiper shows clean.

SUBTASK 33-42-21-420-007

(4) Do these steps to install the washer [6], nut [7], and jumper [8] on the bracket:

(a) Apply sealant, A50231, to the threads of the ground stud, washer [6], and jumper [8].

(b) Install the jumper [8], nut [7], and washer [6] on the ground stud.

1) Tighten the nut [7] to 28 in-lb (3.16 N·m) - 35 in-lb (3.95 N·m).

SUBTASK 33-42-21-420-008

(5) Do these steps to install the washer [44], nut [43], and jumper [45] on the bracket:

(a) Apply sealant, A50231, to the threads of the ground stud, washer [44], and jumper [45].

(b) Install the jumper [45], nut [43], and washer [44] on the ground stud.

1) Tighten the nut [43] to 28 in-lb (3.16 N·m) - 35 in-lb (3.95 N·m).

SUBTASK 33-42-21-760-003

(6) Measure the electrical bonding resistance between the jumper [8] and fuselage (SWPM 20-20-00, Section 21-A).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the resistance is 0.0010 ohm (1.0 milliohm) or less.

SUBTASK 33-42-21-760-004

(7) Measure the electrical bonding resistance between the jumper [45] and fuselage (SWPM 20-20-00, Section 21-A).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the resistance is 0.0010 ohm (1.0 milliohm) or less.

SUBTASK 33-42-21-860-032

(8) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	14	C01968	EXTERIOR LIGHTING LTRTL - RIGHT
B	15	C01969	EXTERIOR LIGHTING LTRTL - LEFT
D	12	C01518	EXTERIOR LIGHTING GENERAL LIGHT CONT

SUBTASK 33-42-21-860-033

(9) Make sure that electrical power is supplied to the airplane.

(a) If it is necessary, do this task: Supply Electrical Power, TASK 24-22-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-42-21-710-007

WARNING DO NOT LOOK DIRECTLY AT LIGHTS. HIGH INTENSITY OF LIGHTS CAN CAUSE EYE DAMAGE.

(10) Do a test of the main arrays.

(a) Set left LANDING switch, at the P5 overhead panel, to the ON position.

1) Make sure that the left main array comes on.

(b) Set left LANDING switch, at the P5 overhead panel, to the FLASH position.

1) Make sure that the left main array alternately flash at approximately 22 (±1) flashes per minute.

(c) Set left LANDING switch, at the P5 overhead panel, to the OFF position.

1) Make sure that the left main array goes off.

(d) Set right LANDING switch, at the P5 overhead panel, to the ON position.

1) Make sure that the right main array comes on.

(e) Set right LANDING switch, at the P5 overhead panel, to the FLASH position.

1) Make sure that the right main array alternately flash at approximately 22 (±1) flashes per minute.

(f) Set right LANDING switch, at the P5 overhead panel, to the OFF position.

1) Make sure that the right main array goes off.

(g) Set the TAXI switch, at the P5 overhead panel, to the ON or AUTO position.

1) Make sure that the left and right main arrays come on but at a lower power level than when the LANDING switches are in the ON position.

(h) Set the TAXI switch, at the P5 overhead panel, to the OFF position.

1) Make sure that the left and right main arrays go off.

SUBTASK 33-42-21-410-004

(11) Do these steps to install the lens assembly [3]:

(a) If the tape around the perimeter of the lens opening is damaged, do these steps:

1) Remove and discard the damaged tape.

a) Apply Skyflex tape, G50066, around the perimeter of the lens opening.

NOTE: If the tape is too long, fold it to fit the opening, but not more than four times.

<1> Do not stretch the tape.

2) Use a sharp tool to make a 0.25 in. (6.35 mm) cut in the tape at the 15 holes.

a) Make sure that no other cuts to the tape.

(b) Install the lens assembly [3] with the stud [4].

1) Tighten each stud [4] to 20 in-lb (2.3 N·m) - 25 in-lb (2.8 N·m).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 33-42-21-860-034

(1) Do these steps to reactivate the leading edge flaps and slats:

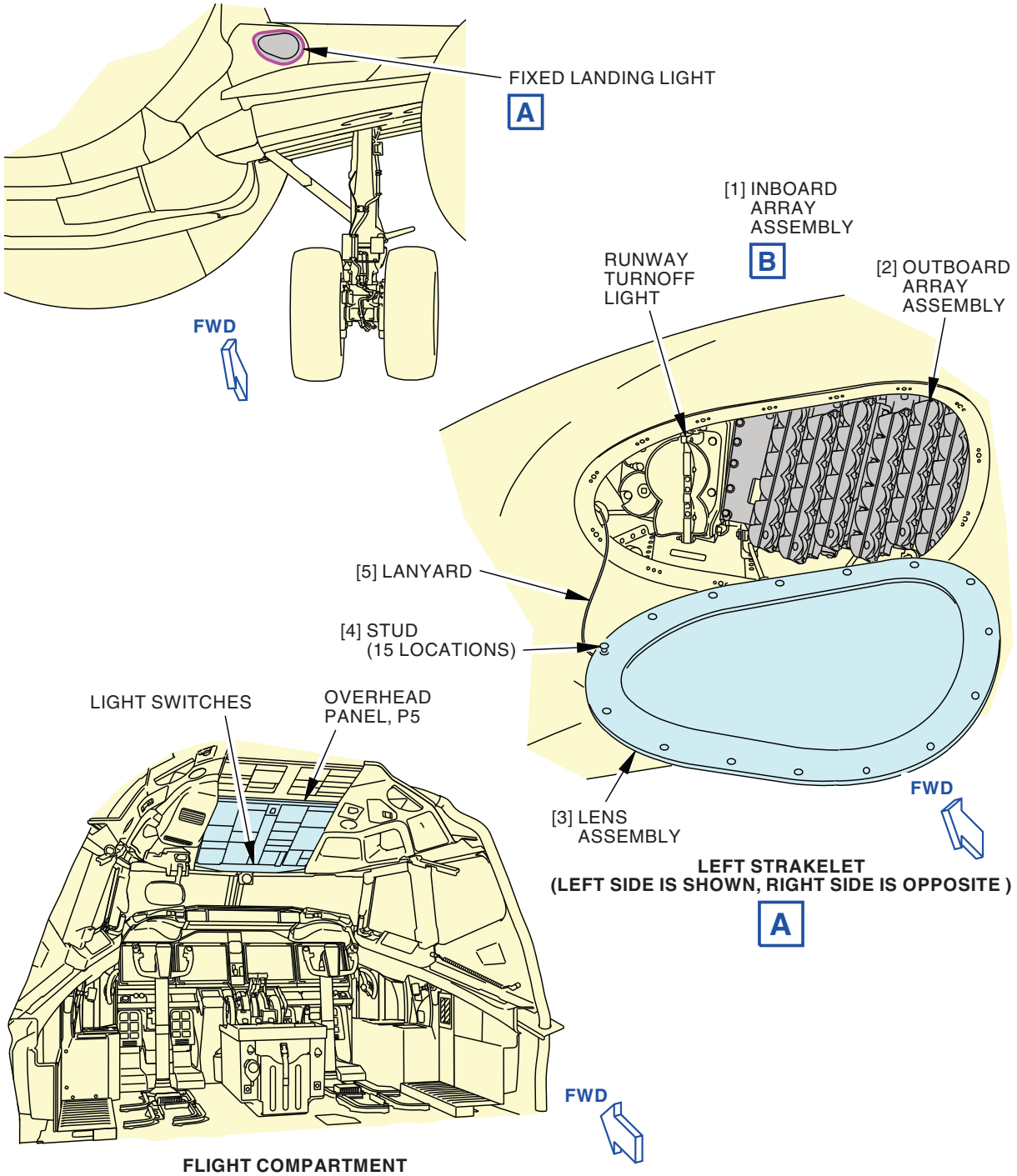
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) If both the leading edge flaps/slats and the trailing edge flaps were deactivated, do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(b) If only leading edge flaps and slats were deactivated, do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

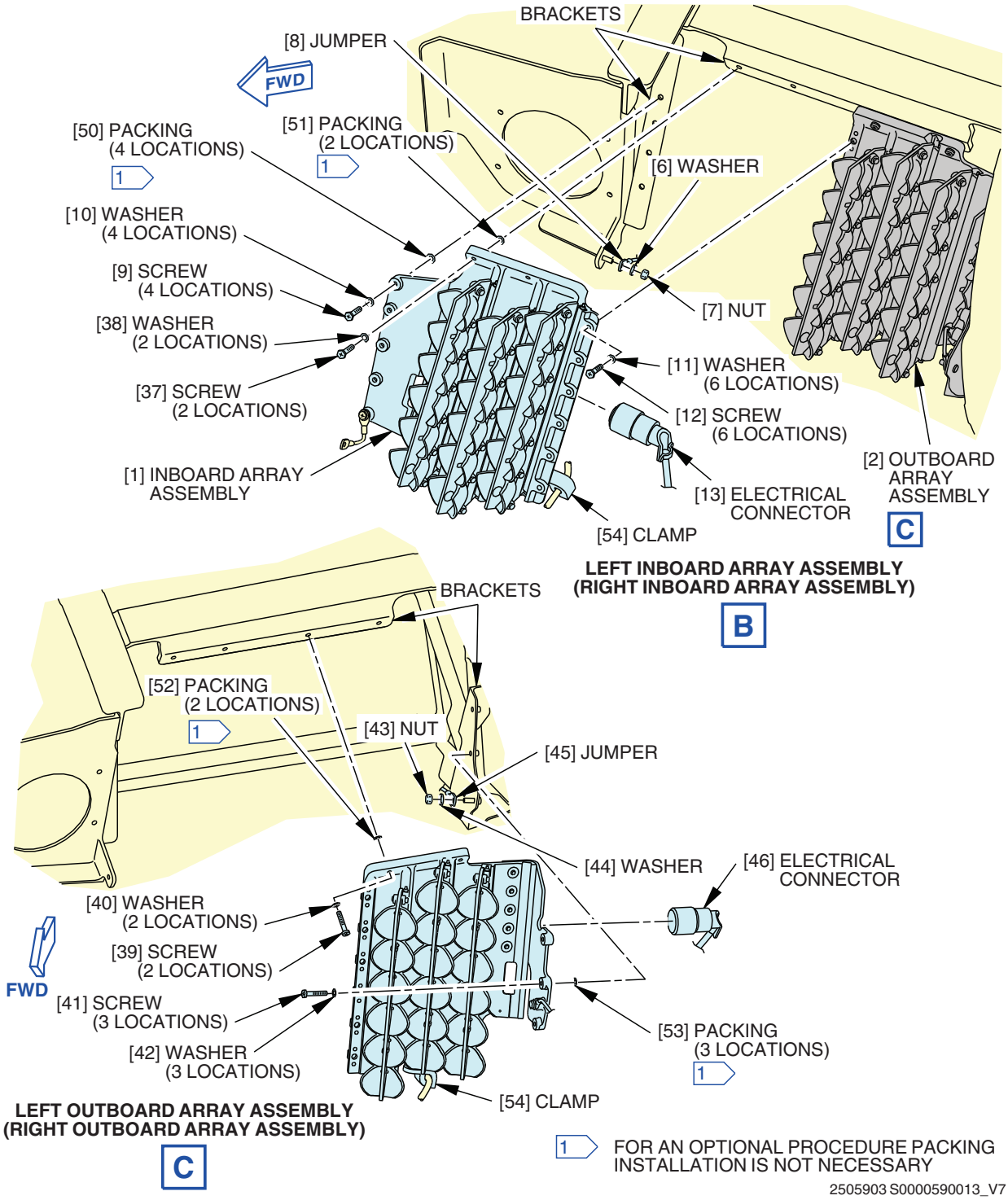
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fixed Landing Light - Main Array Assembly Replacement
Figure 202/33-42-21-990-811 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fixed Landing Light - Main Array Assembly Replacement
Figure 202/33-42-21-990-811 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-42-21-960-802

5. Fixed Landing Light - Power Supply Replacement

(Figure 203)

A. General

(1) This task has the steps to replace the power supply.

(2) The power supply provides power and control to the main array and runway turnoff light assemblies.

(3) The power supply is located behind the main array.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
24-22-00-860-801	Supply Electrical Power (P/B 201)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
51-21-41-370-802	Apply Bonderite M-CR 600, Bonderite M-CR 1200, Bonderite M-CR 600RTU, or Bonderite M-CR 1200S Solution (P/B 701)
51-21-72-370-804	BMS10-11 Primer - Application (P/B 701)
51-22-00-390-806	Fillet Seal Application (P/B 201)
SWPM 20-20-00, Paragraph 2.A.	Cleaning Procedure 1
SWPM 20-20-00, Paragraph 2.E.	Standard Wiring Practices Manual
SWPM 20-20-00, Section 21-A	Resistance of Faying Surface Bonds

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description	Specification
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50066	Tape - Thick Ribbed Tape with Adhesive (Skyflex GSC-21-95241-011)	
G50398	Pad - Adbrasive, Scotch-Brite Type S, Abrasive Pad	
G50492	Pad - 3M Scotch Brite 7448 Ultra Fine Pad	MIL-A-9962A Type III Grade AAA
G50961	Coating - Corrosion Inhibiting Epoxy Primer - (BAC 377 Yellow Color)	BMS10-11 Type I

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
14	Power supply	33-42-21-10-035	PTF ALL

F. Location Zones

Zone	Area
511	Left Wing - Leading Edge To Front Spar
611	Right Wing - Leading Edge to Front Spar

G. Prepare for the Replacement

SUBTASK 33-42-21-860-008

WARNING DO THE DEACTIVATION PROCEDURE FOR THE LEADING EDGE FLAPS AND SLATS. THE FLAPS AND SLATS MOVE QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do one of these steps to deactivate the leading edge flaps and slats in the retracted (UP) position:

(a) If activation is not necessary to the trailing edge flaps, do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

NOTE: This task deactivates both the leading edge flaps/slats and the trailing edge flaps.

(b) If activation is necessary to the trailing edge flaps, do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

NOTE: This task only deactivates the leading edge flaps and slats. The trailing edge flaps can still operate.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-42-21-860-009

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	14	C01968	EXTERIOR LIGHTING LTRTL - RIGHT
B	15	C01969	EXTERIOR LIGHTING LTRTL - LEFT
D	12	C01518	EXTERIOR LIGHTING GENERAL LIGHT CONT

SUBTASK 33-42-21-010-002

(3) Remove the main array assembly (Fixed Landing Light - Main Array Assembly Replacement, TASK 33-42-21-960-801).

SUBTASK 33-42-21-910-002

CAUTION

DO NOT TOUCH THE UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE UNIT.

(4) Before the power supply [14] is touched, do this task for devices that are sensitive to electrostatic discharge (ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802).

H. Power Supply Replacement

SUBTASK 33-42-21-020-002

(1) Do these steps to replace the power supply [14]:

(a) Remove the nut [17], washer [18], and jumper [19].

(b) Remove the bolts [15] and washers [16].

(c) Remove the power supply [14] from the bracket.

(d) Disconnect the electrical connector [20] from the power supply [14].

(e) Connect the electrical connector [20] to the new power supply [14].

(f) Install the power supply [14] to the bracket.

(g) Install the bolts [15] and washers [16].

1) Tighten the bolts [15] to 35 in-lb (3.95 N·m) - 36 in-lb (4.07 N·m).

SUBTASK 33-42-21-120-002

(2) Do these steps to prepare the ground stud on the aluminum bracket, fasteners, and jumper [19] for the electrical bond installation:

(a) Clean the nut [17], washer [18], jumper [19], ground stud, and fay surfaces of the aluminum bracket:

1) Use Cleaning Procedure 1 to abrasively clean the fay surfaces of the aluminum bracket (SWPM 20-20-00, Paragraph 2.A.).

a) Use a very fine Scotch-Brite Type S pad, G50398, or an ultra-fine Scotch-Brite 7448 pad, G50492.

b) Move the pad in a circular movement while you press with your thumb.

2) Use Cleaning Procedure 5 to clean the fay surface of the aluminum bracket, nut [17], washer [18], and jumper [19] (SWPM 20-20-00, Paragraph 2.E.).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

a) Use a new cotton wiper, G00034, made moist with solvent, B00184, or solvent, B00148.

b) Use a new cotton wiper, G00034, to dry the surfaces.

c) Continue to clean and dry the surface until the cotton wiper shows clean.

SUBTASK 33-42-21-420-002

(3) Install the nut [17], washer [18], and jumper [19].

(a) Tighten the nut [17] to 65 in-lb (7 N·m) - 70 in-lb (8 N·m).

SUBTASK 33-42-21-760-002

(4) Measure the electrical bonding resistance between the power supply and fuselage (SWPM 20-20-00, Section 21-A).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the bonding resistance is 0.0010 ohms (1 milliohm) or less.

SUBTASK 33-42-21-390-001

(5) Apply sealant, A50231, to the nut [17], washer [18], and jumper [19] (TASK 51-22-00-390-806).

SUBTASK 33-42-21-910-003

(6) Apply protective coating, C00064, to the bare metal areas of the aluminum bracket (TASK 51-21-41-370-802).

(a) Apply protective coating to the area where the finish was removed from the electrical bonding probe touching the surface.

SUBTASK 33-42-21-910-004

(7) Apply primer coating, G50961, to the bare metal areas of the aluminum bracket (TASK 51-21-72-370-804).

(a) Apply primer to the area where the finish was removed from the electrical bonding probe touching the surface.

SUBTASK 33-42-21-420-003

(8) Install the main array (TASK 33-42-21-960-801).

SUBTASK 33-42-21-860-010

(9) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	14	C01968	EXTERIOR LIGHTING LTRTL - RIGHT
B	15	C01969	EXTERIOR LIGHTING LTRTL - LEFT
D	12	C01518	EXTERIOR LIGHTING GENERAL LIGHT CONT

SUBTASK 33-42-21-860-011

(10) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-42-21-710-002

WARNING DO NOT LOOK DIRECTLY AT LIGHTS. HIGH INTENSITY OF LIGHTS CAN CAUSE EYE DAMAGE.

(11) Do a test of the main array and the runway turnoff light assemblies.

(a) At the P5 overhead panel, set left LANDING switch to the ON position.

1) Make sure that the left main array comes on.

(b) At the P5 overhead panel, set left LANDING switch to the FLASH position.

1) Make sure that the left main array alternately flash at approximately 22 (±1) flashes per minute.

(c) At the P5 overhead panel, set left LANDING switch to the OFF position.

1) Make sure that the left main array goes off.

(d) At the P5 overhead panel, set right LANDING switch to the ON position.

1) Make sure that the right main array comes on.

(e) At the P5 overhead panel, set right LANDING switch to the FLASH position.

1) Make sure that the right main array alternately flash at approximately 22 (±1) flashes per minute.

(f) At the P5 overhead panel, set right LANDING switch to the OFF position.

1) Make sure that the right main array goes off.

(g) At the P5 overhead panel, set the TAXI switch to the ON or AUTO position.

1) Make sure that the left and right main arrays come on but at a lower power level than when the LANDING switches are in the ON position.

(h) At the P5 overhead panel, set the TAXI switch to the OFF position.

1) Make sure that the left and right main arrays go off.

(i) At the P5 overhead panel, set the left RUNWAY TURNOFF switch to the ON position.

1) Make sure that the left runway turnoff light comes on.

(j) At the P5 overhead panel, set the left RUNWAY TURNOFF switch to the OFF position.

1) Make sure that the left runway turnoff light goes off.

(k) At the P5 overhead panel, set the right RUNWAY TURNOFF switch to the ON position.

1) Make sure that the right runway turnoff light comes on.

(l) At the P5 overhead panel, set the right RUNWAY TURNOFF switch to the OFF position.

1) Make sure that the right runway turnoff light goes off.

SUBTASK 33-42-21-410-002

(12) Do these steps to install the lens assembly:

(a) If the tape around the perimeter of the lens opening is damaged, do these steps:

1) Remove and discard the damaged tape.

a) Apply Skyflex tape, G50066, around the perimeter of the lens opening.

b) If the tape is too long, fold it to fit the opening, but not more than four times.

<1> Do not stretch the tape.

2) Use a sharp tool to make a 0.25 in. (6.35 mm) cut in the tape at the 15 holes.

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a) Make no other cuts to the tape.

(b) Install the lens assembly [3] with the studs [4].

1) Tighten each studs [4] to 20 in-lb (2.3 N·m) - 25 in-lb (2.8 N·m).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 33-42-21-860-012

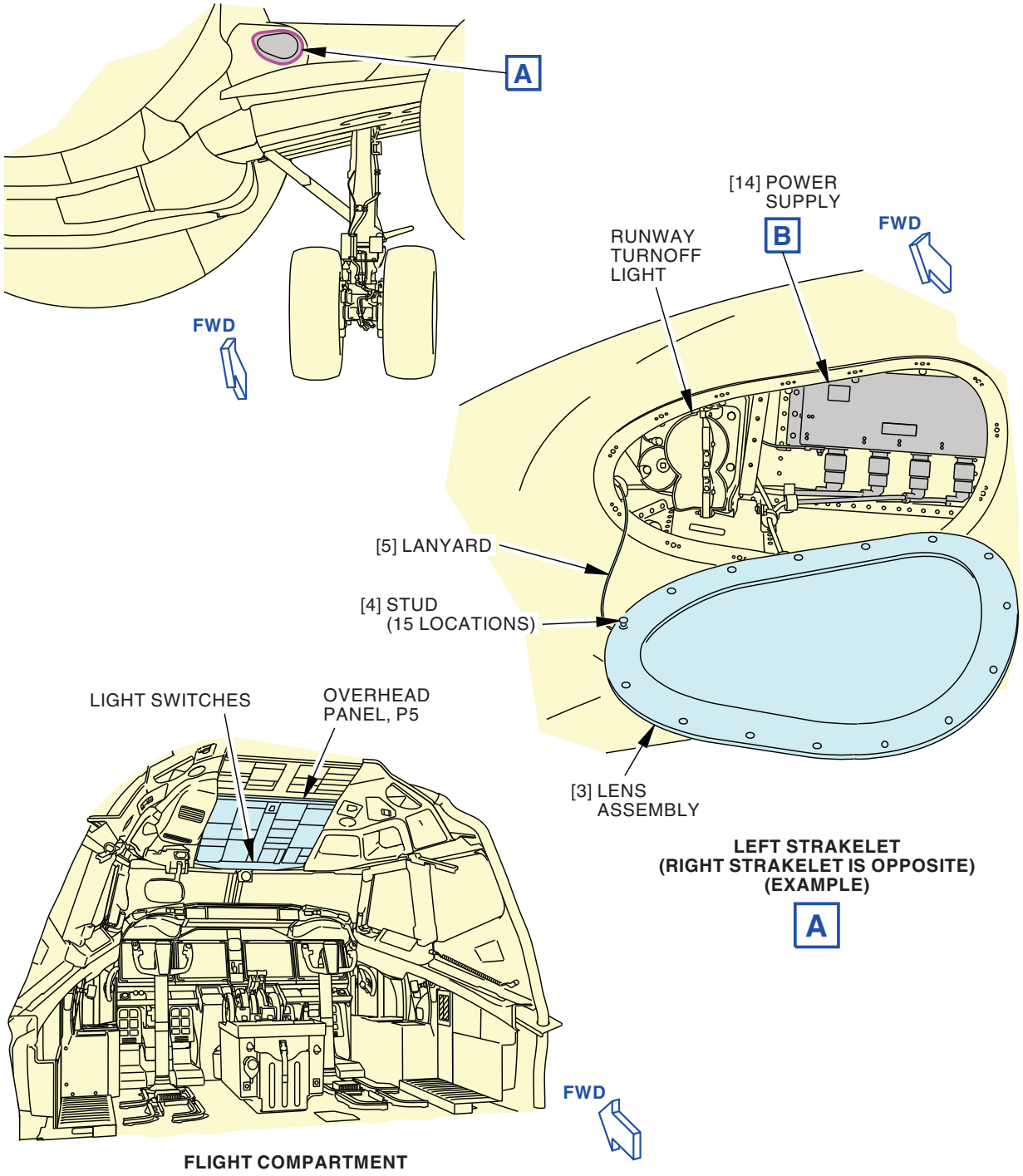
(1) Do these steps to reactivate the leading edge flaps and slats:

(a) If both the leading edge flaps/slats and the trailing edge flaps were deactivated, do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(b) If only leading edge flaps and slats were deactivated, do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

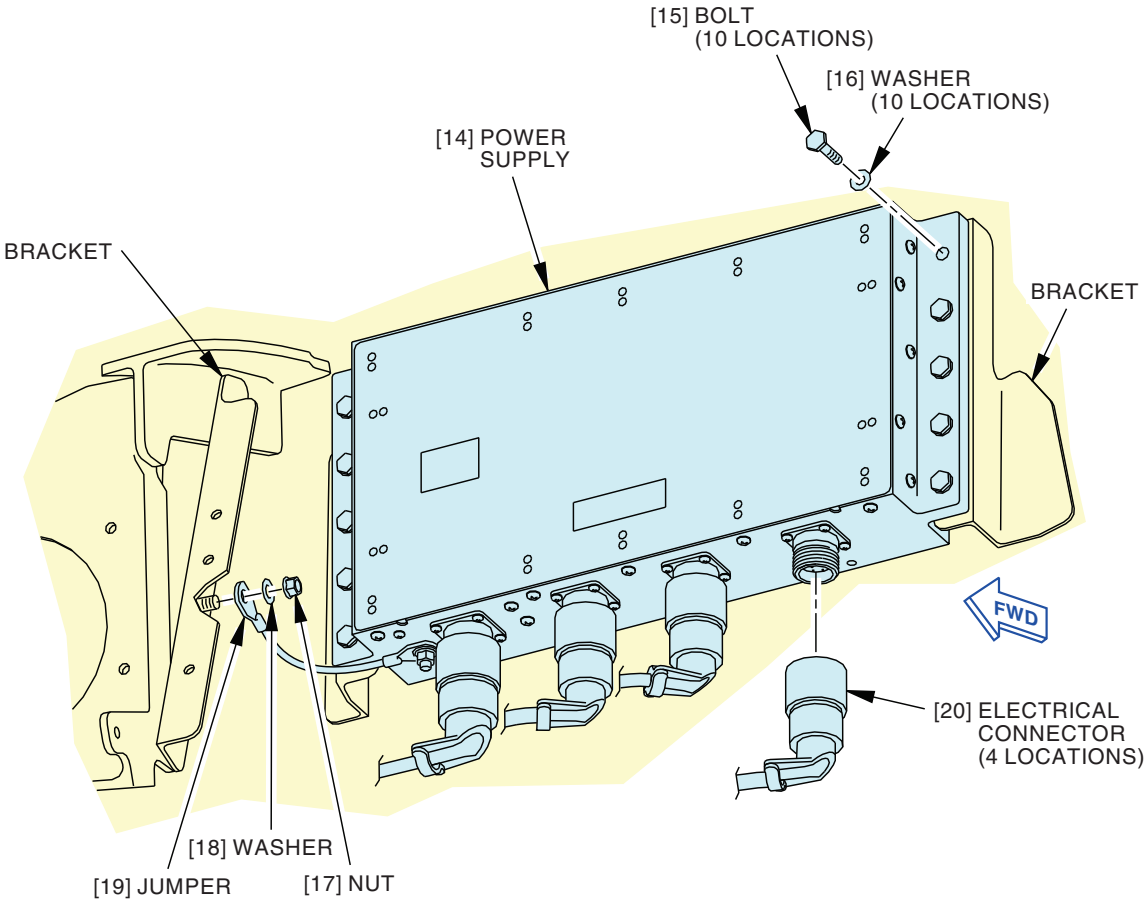
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fixed Landing Light - Power Supply Assembly Replacement
Figure 203/33-42-21-990-803 (Sheet 1 of 2)

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LEFT POWER SUPPLY
(RIGHT POWER SUPPLY IS OPPOSITE)

B

2508642 S0000591212_V1

Fixed Landing Light - Power Supply Assembly Replacement
Figure 203/33-42-21-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-42-21-350-801

6. Fixed Landing Light - Lens Temporary Repair

(Figure 204)

A. General

(1) Use this task to make a temporary cover for the fixed landing light when the lens assembly is broken and no spare is available.

B. References

Reference	Title
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
G00111	Sheet - Mylar	
G00270	Tape - Scotch Flatback Masking 250	ASTM D6123 (Supersedes A-A-883)
G02042	Material - Aluminum Alloy 2024 Sheet	MS14361
G02329	Tape - Aluminum Foil, Pressure Sensitive - Vibration Damping Tape 434	

D. Location Zones

Zone	Area
511	Left Wing - Leading Edge To Front Spar
611	Right Wing - Leading Edge to Front Spar

E. Prepare for the Fixed Landing Light Lens Temporary Repair

SUBTASK 33-42-21-860-013

WARNING DO THE DEACTIVATION PROCEDURE FOR THE LEADING EDGE FLAPS AND SLATS. THE FLAPS AND SLATS MOVE QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do one of these steps to deactivate the leading edge flaps and slats in the retracted (UP) position:

(a) If activation is not necessary to the trailing edge flaps, do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

NOTE: This task deactivates both the leading edge flaps/slats and the trailing edge flaps.

(b) If activation is necessary to the trailing edge flaps, do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

NOTE: This task only deactivates the leading edge flaps and slats. The trailing edge flaps can still operate.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-42-21-860-014

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	14	C01968	EXTERIOR LIGHTING LTRTL - RIGHT
B	15	C01969	EXTERIOR LIGHTING LTRTL - LEFT

F. Fixed Landing Light Lens Temporary Repair

SUBTASK 33-42-21-020-003

(1) Do these steps to remove the broken lens assembly [3]:

(a) Loosen the studs [4].

(b) Disconnect the lanyard [5].

SUBTASK 33-42-21-020-004

(2) Remove the broken lens assembly from the airplane:

(a) Disassemble the studs [4] and rings [21] from the lens assembly [3].

1) Keep the studs [4] for the installation of the temporary cover [22].

SUBTASK 33-42-21-350-001

(3) Do these steps to make a temporary cover [22]:

(a) Make a four-piece mylar template.

1) Cut the mylar sheet, G00111, into four strips.

2) Make each mylar strip align with the contour of the broken lens assembly [3] then attach the strips with the Scotch Flatback Masking Tape 250, G00270.

NOTE: The edges between adjacent mylar strips align on the flange, midway between the fastener holes and perpendicular to the rim.

3) Cut the mylar strips the same as the rim around the lens assembly [3].

4) Carefully make a mark for each fastener holes.

5) Remove the mylar strips from the lens assembly [3].

(b) Make the four-piece lanyard [5] the same as the mylar template.

1) Make the temporary cover from aluminum sheet material, G02042 0.094 ±0.031 in. (2.388 ±0.788 mm) thick.

2) Drill 0.274 ±0.006 in. (6.960 ±0.153 mm) diameter holes and countersink to 0.4015 ±0.0025 in. (10.1981 ±0.0635 mm) diameter x 100 degree head.

3) If the thickness of the aluminum material is less than 0.09 in. (2.29 mm), make a shim out of similar material in the shape of the rim.

SUBTASK 33-42-21-350-002

(4) Do these steps to install the temporary cover [22]:

(a) Make each aluminum strip align with the contour of the broken lens assembly.

(b) Attach the shim (if one is necessary) and each aluminum strip with the studs [4].

1) Do not use the rings [21].

2) Bring the aluminum strips up to contour, 0.126 ±0.040 in. (3.200 ±1.016 mm).

(c) Cover all spaces with speed Vibration Damping Tape 434 tape, G02329.

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1) The spaces between the aluminum strips must be 0.10 ±0.10 in. (2.54 ±2.54 mm) wide. The spaces between the aluminum strips and the opening should be 0.060 ±0.060 in. (1.524 ±1.524 mm) wide.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 33-42-21-860-015

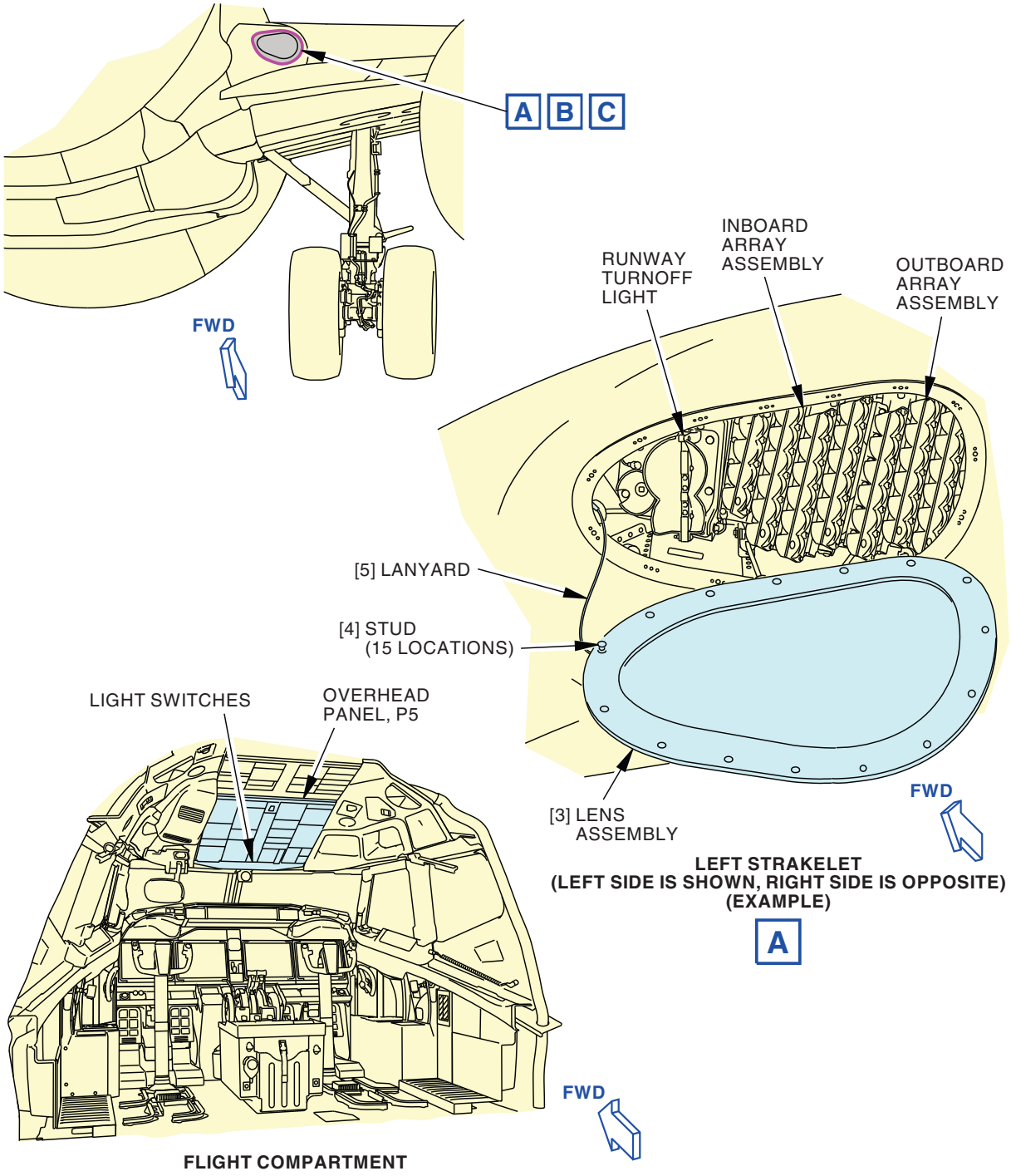
(1) Do these steps to reactivate the leading edge flaps and slats:

(a) If both the leading edge flaps/slats and the trailing edge flaps were deactivated, do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(b) If only leading edge flaps and slats were deactivated, do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

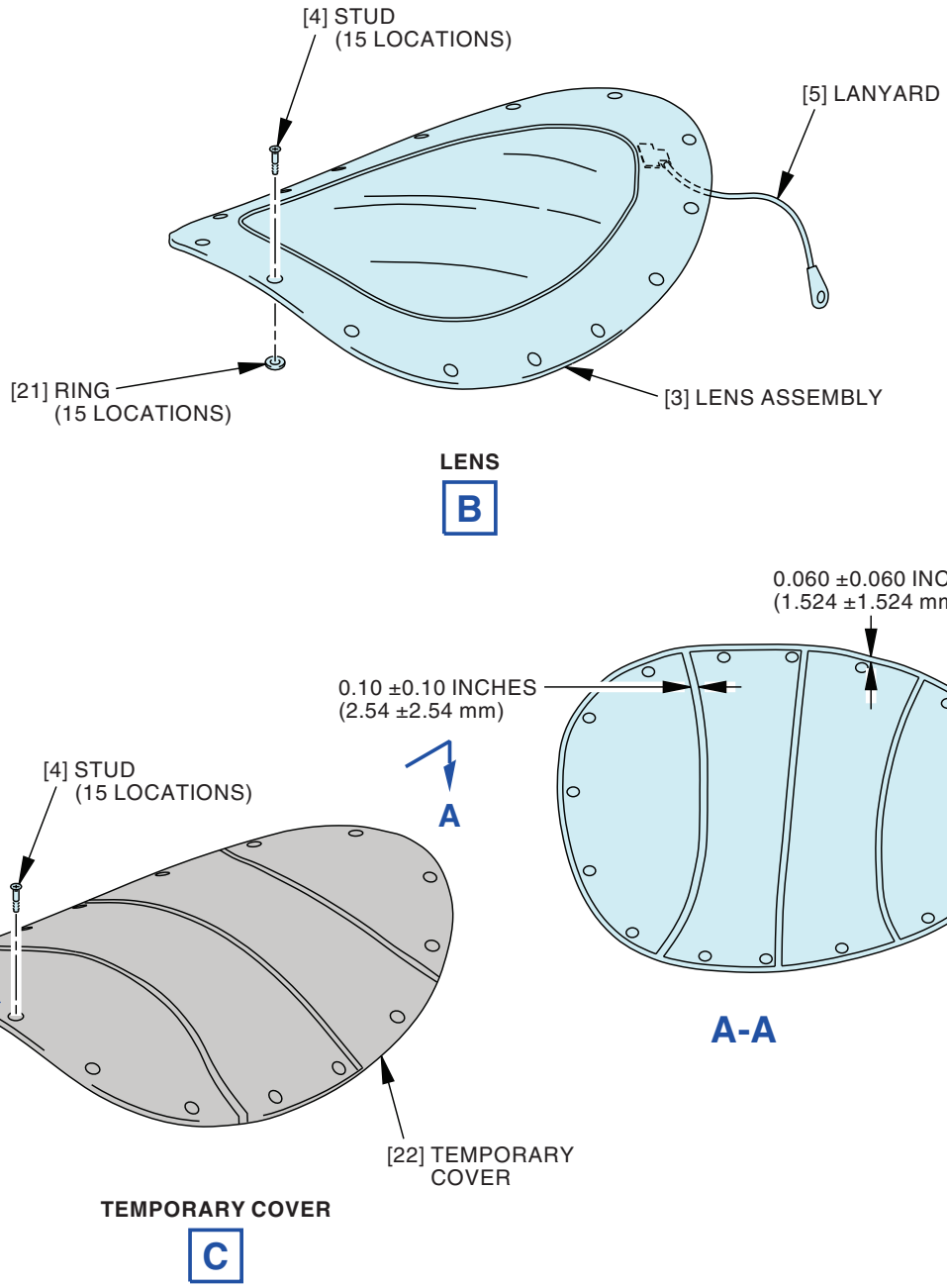
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2517429 S0000591328_V2

Fixed Landing Light - Lens Temporary Repair
Figure 204/33-42-21-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2517453 S0000591329_V1

Fixed Landing Light - Lens Temporary Repair
Figure 204/33-42-21-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-42-21-040-802

7. Runway Turnoff Lights - Deactivation

(Figure 205)

A. General

(1) This procedure removes electrical power to the runway turnoff lights.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
511	Left Wing - Leading Edge To Front Spar
611	Right Wing - Leading Edge to Front Spar

C. Procedure

SUBTASK 33-42-21-860-016

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	14	C01968	EXTERIOR LIGHTING LTRTL - RIGHT
B	15	C01969	EXTERIOR LIGHTING LTRTL - LEFT
D	12	C01518	EXTERIOR LIGHTING GENERAL LIGHT CONT

D. Runway Turnoff Lights - Tryout

NOTE: This tryout is to make sure that the Runway Turnoff Lights are in a zero energy state.

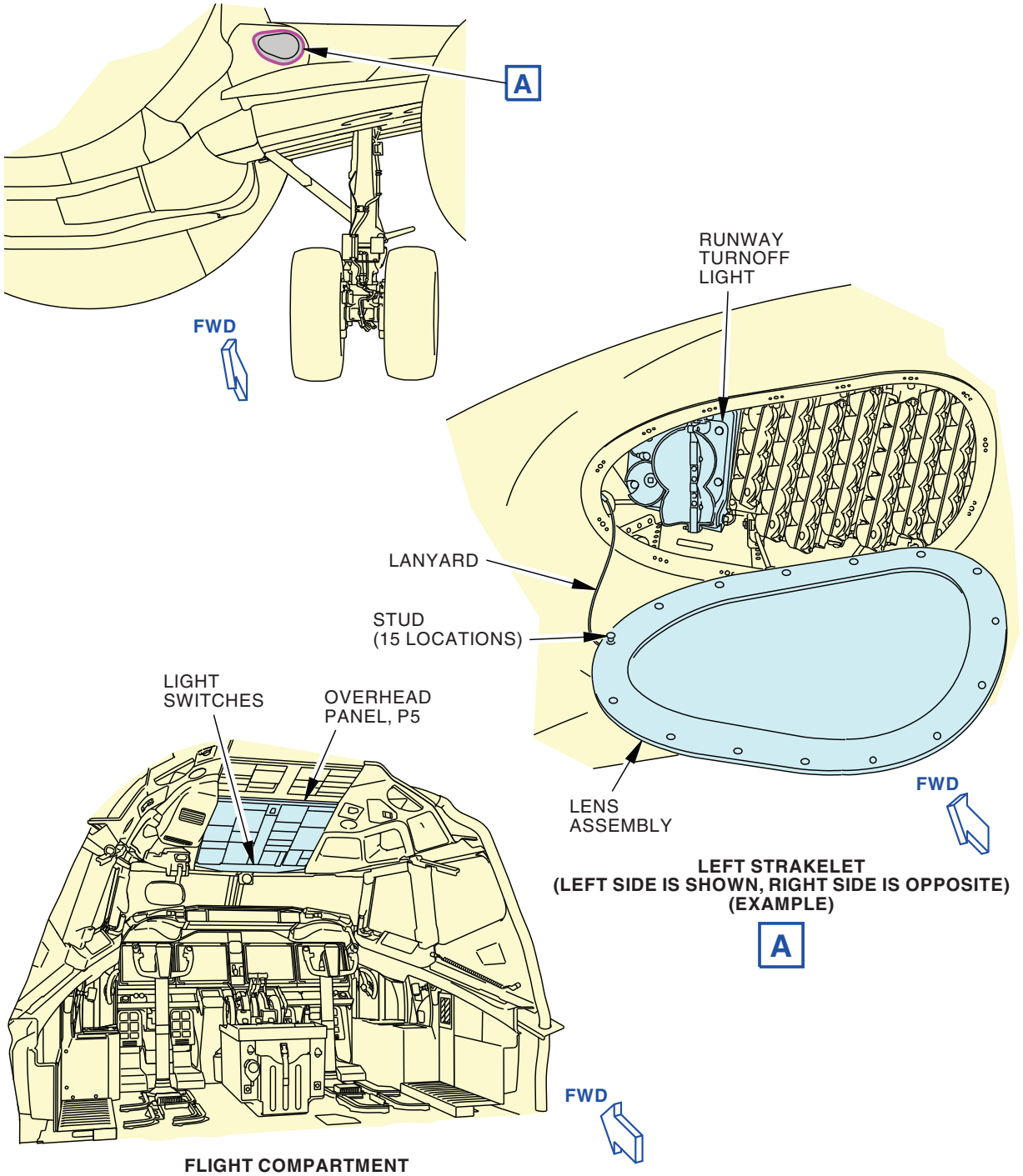
SUBTASK 33-42-21-700-001

(1) At the overhead panel, P5, set the runway turnoff light switches to the ON mode.

(a) Make sure that none of the runway turnoff lights come on.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Runway Turnoff Lights
Figure 205/33-42-21-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-42-21-440-802

8. Runway Turnoff Lights - Activation

(Figure 205)

A. General

(1) This procedure adds electrical power to the runway turnoff lights.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
511	Left Wing - Leading Edge To Front Spar
611	Right Wing - Leading Edge to Front Spar

C. Procedure

SUBTASK 33-42-21-860-017

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	14	C01968	EXTERIOR LIGHTING LTRTL - RIGHT
B	15	C01969	EXTERIOR LIGHTING LTRTL - LEFT
D	12	C01518	EXTERIOR LIGHTING GENERAL LIGHT CONT

——— END OF TASK ———

TASK 33-42-21-960-803

9. Runway Turnoff Light - Light Assembly Replacement

(Figure 206)

A. General

(1) This procedure contains the steps to replace the runway turnoff light assembly.

B. References

Reference	Title
20-40-12-000-801	ESDS Handling for Printed Circuit Board Removal (P/B 201)
24-22-00-860-801	Supply Electrical Power (P/B 201)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
SWPM 20-20-00, Paragraph 2.E.	Standard Wiring Practices Manual
SWPM 20-20-00, Section 21-A	Resistance of Faying Surface Bonds

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50066	Tape - Thick Ribbed Tape with Adhesive (Skyflex GSC-21-95241-011)	

E. Location Zones

Zone	Area
511	Left Wing - Leading Edge To Front Spar
611	Right Wing - Leading Edge to Front Spar

F. Prepare for the Replacement

SUBTASK 33-42-21-860-018

WARNING DO THE DEACTIVATION PROCEDURE FOR THE LEADING EDGE FLAPS AND SLATS. THE FLAPS AND SLATS MOVE QUICKLY. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do one of these steps to deactivate the leading edge flaps and slats in the retracted (UP) position:

(a) If activation is not necessary to the trailing edge flaps, do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

NOTE: This task deactivates both the leading edge flaps/slats and the trailing edge flaps.

(b) If activation is necessary to the trailing edge flaps, do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

NOTE: This task only deactivates the leading edge flaps and slats. The trailing edge flaps can still operate.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-42-21-860-019

(2) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	14	C01968	EXTERIOR LIGHTING LTRTL - RIGHT
B	15	C01969	EXTERIOR LIGHTING LTRTL - LEFT
D	12	C01518	EXTERIOR LIGHTING GENERAL LIGHT CONT

SUBTASK 33-42-21-910-005

(3) Before you touch the light assembly, do the procedure for devices that are sensitive to electrostatic discharge (TASK 20-40-12-000-801).

G. Runway Turnoff Light Light Assembly Replacement

SUBTASK 33-42-21-020-005

(1) Loosen the studs [4] and remove the lens assembly [3].

(a) The lanyard [5] stays connected.

(b) Keep the lens assembly [3] and lanyard [5] away from the wing leading edge.

SUBTASK 33-42-21-960-001

(2) Do these steps to replace the runway turnoff light:

(a) Remove the nut [27], washer [28], jumper [29].

(b) Remove the screws [26] and washers [24].

(c) Remove the runway turnoff light [23] from the bracket.

(d) Disconnect the electrical connector [25] from the runway turnoff light [23].

(e) Connect the electrical connector [25] to the new runway turnoff light [23].

(f) Install the new runway turnoff light [23] to the bracket.

(g) Install the screws [26] and washers [24].

1) Tighten the screws [26] to 30 in-lb (3.4 N·m) - 31 in-lb (3.5 N·m).

SUBTASK 33-42-21-110-001

(3) Do these steps to prepare the ground stud on the aluminum bracket, fasteners, and jumper [29] for the electrical bond installation:

(a) Clean the nut [27], washer [28], jumper [29], ground stud, and the fay surfaces of the aluminum bracket.

1) Use Cleaning Procedure 5 to clean the nut [27], washer [28], and jumper [29] (SWPM 20-20-00, Paragraph 2.E.).

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

a) Use a new cotton wiper, G00034, made moist with solvent, B00184, or solvent, B00148.

b) Use a new cotton wiper, G00034, to dry the surfaces.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

c) Make sure that you remove the remaining corrosion inhibiting compound on the bracket.

2) Use Cleaning Procedure 5 to clean the ground stud threads and head of the ground stud assembly (SWPM 20-20-00, Paragraph 2.E.).

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE OR CAUSE DAMAGE TO THE ENVIRONMENT. REFER TO THE MATERIAL SAFETY DATA SHEETS (MSDS) AND THE LOCAL SAFETY PRECAUTIONS.

a) Use a new cotton wiper, G00034, made moist with solvent, B00184, or solvent, B00148.

b) Use a new cotton wiper, G00034, to dry the surfaces.

c) Continue to clean and dry the surface until the cotton wiper shows clean.

SUBTASK 33-42-21-420-004

(4) Do these steps to install the nut [27], washer [28], and jumper [29] on the bracket:

(a) Apply sealant, A50231, to the contacts of the jumper [29], washer [28], and nut [27].

(b) Install the jumper [29], washer [28], and nut [27].

1) Tighten the nut [27] to 28 in-lb (3.2 N·m) - 35 in-lb (4.0 N·m).

SUBTASK 33-42-21-910-006

(5) Measure the electrical bonding resistance between the jumper [29] and fuselage (SWPM 20-20-00, Section 21-A).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the bonding resistance is less than 0.0010 ohms.

SUBTASK 33-42-21-860-020

(6) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	14	C01968	EXTERIOR LIGHTING LTRTL - RIGHT
B	15	C01969	EXTERIOR LIGHTING LTRTL - LEFT
D	12	C01518	EXTERIOR LIGHTING GENERAL LIGHT CONT

SUBTASK 33-42-21-860-021

(7) Make sure that electrical power is supplied to the airplane, do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 33-42-21-710-004

(8) Do a test of the runway turnoff light:

(a) Set the left RUNWAY TURNOFF switch, on the P5 overhead panel, to the ON position.

1) Make sure that the left runway turnoff light comes on.

(b) Set the left RUNWAY TURNOFF switch, on the P5 overhead panel, to the OFF position.

1) Make sure that the left runway turnoff light goes off.

(c) Set the right RUNWAY TURNOFF switch, on the P5 overhead panel, to the ON position.

1) Make sure that the right runway turnoff light comes on.

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(d) Set the right RUNWAY TURNOFF switch, on the P5 overhead panel, to the OFF position.

1) Make sure that the right runway turnoff light goes off.

SUBTASK 33-42-21-420-005

(9) Do these steps to install the lens assembly:

(a) If the tape around the perimeter of the lens opening is damaged, do these steps:

1) Remove and discard the damaged tape.

2) Apply Skyflex tape, G50066, around the perimeter of the lens opening.

a) If the tape is too long, fold it to fit the opening, but not more than four times.

<1> Do not stretch the tape.

3) Use a sharp tool to make a 0.25 in. (6.35 mm) cut in the tape at the 15 holes.

a) Make no other cuts to the tape.

(b) Install the lens assembly [3] with the studs [4].

1) Tighten each studs [4] to 20 in-lb (2.3 N·m) - 25 in-lb (2.8 N·m).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 33-42-21-860-022

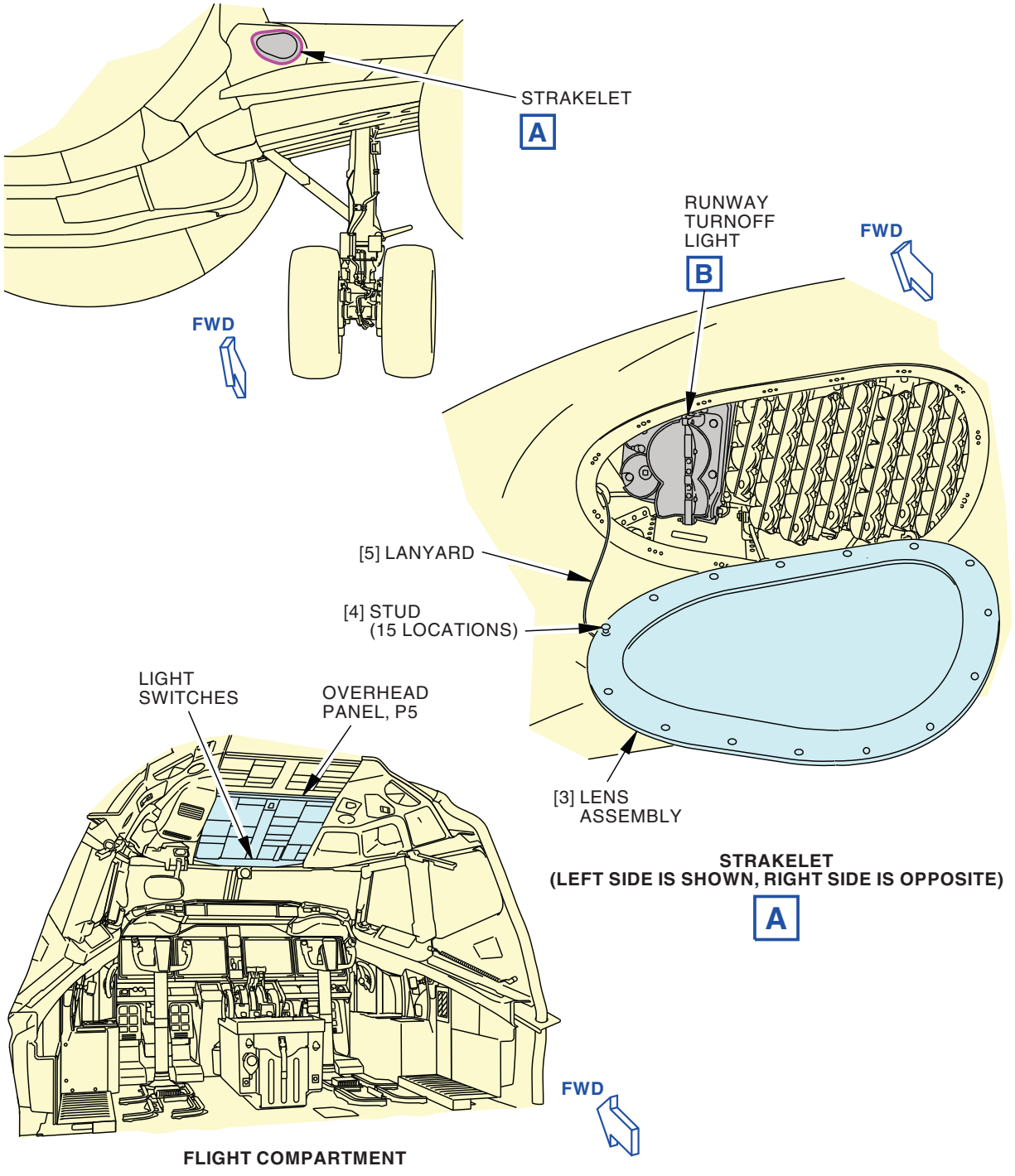
(1) Do these steps to reactivate the leading edge flaps and slats:

(a) If both the leading edge flaps/slats and the trailing edge flaps were deactivated, do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(b) If only leading edge flaps and slats were deactivated, do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STRAKELET
(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

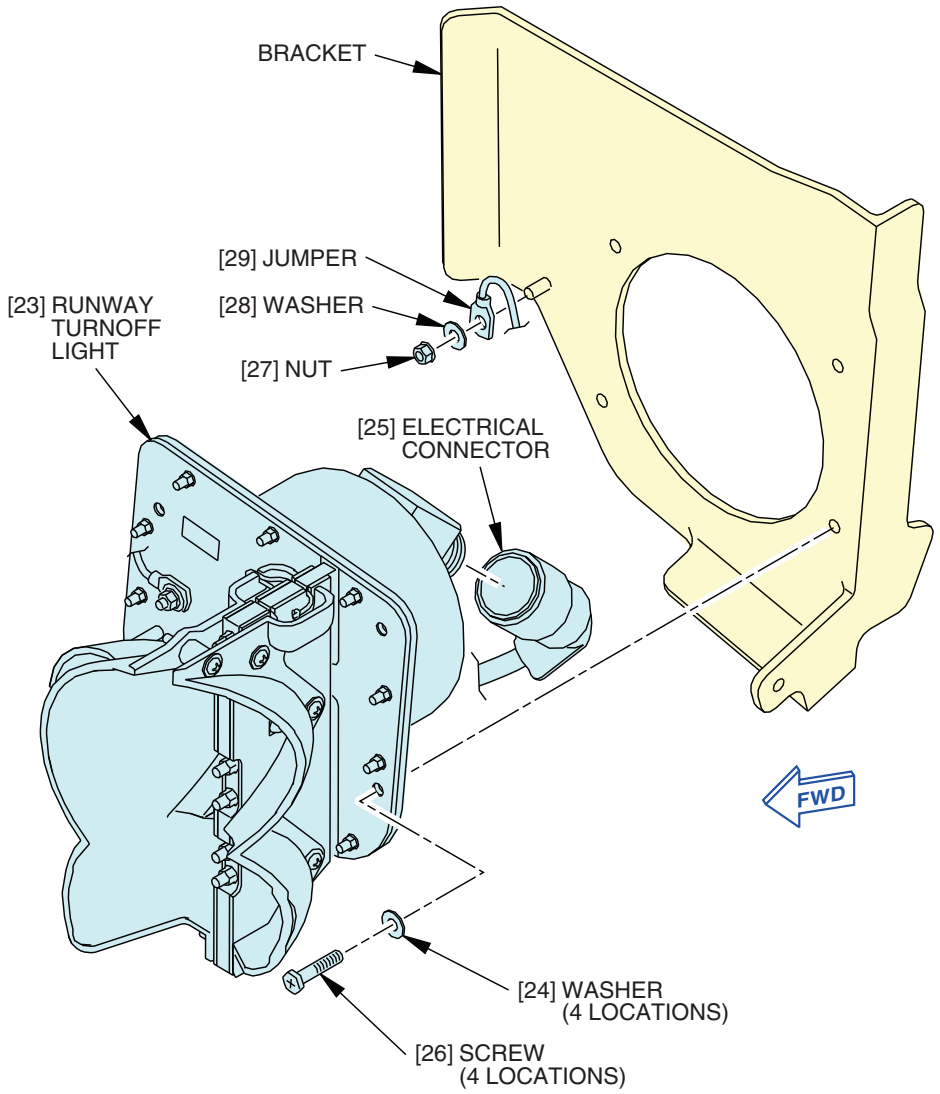
A

FLIGHT COMPARTMENT

2518335 S0000591904_V3

Runway Turnoff Light - Light Assembly Replacement
Figure 206/33-42-21-990-806 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RUNWAY TURNOFF LIGHT
(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

B

3060136 S0000823504_V3

Runway Turnoff Light - Light Assembly Replacement
Figure 206/33-42-21-990-806 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-42-21-960-804

10. Taxi Light Relay - Replacement

(Figure 207)

A. References

Reference	Title
33-42-21-710-802	Taxi Light Relay - Operational Test (P/B 501)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
32	Relay	33-42-51-01-005	PTF ALL

C. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
114BW	Forward Nose Wheel Well Panel

E. Taxi Light Relay Replacement

SUBTASK 33-42-21-860-025

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
D	12	C01518	EXTERIOR LIGHTING GENERAL LIGHT CONT

SUBTASK 33-42-21-010-003

(2) Open this access panel:

(TASK 53-14-01-020-801)

Number	Name/Location
114BW	Forward Nose Wheel Well Panel

SUBTASK 33-42-21-960-002

(3) Do these steps to replace the relay [32]:

(a) Loosen the fasteners [30] and remove the cover [31] from the J4 junction box.

(b) Remove the screws [33] on the relay [32].

(c) Replace the relay [32].

(d) Install the screws [33] on the relay [32].

(e) Put the cover [31] on the J4 junction box and tighten the fasteners [30].

SUBTASK 33-42-21-410-003

(4) Close this access panel:

(TASK 53-14-01-420-801)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Number	Name/Location
114BW	Forward Nose Wheel Well Panel

SUBTASK 33-42-21-860-026

(5) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

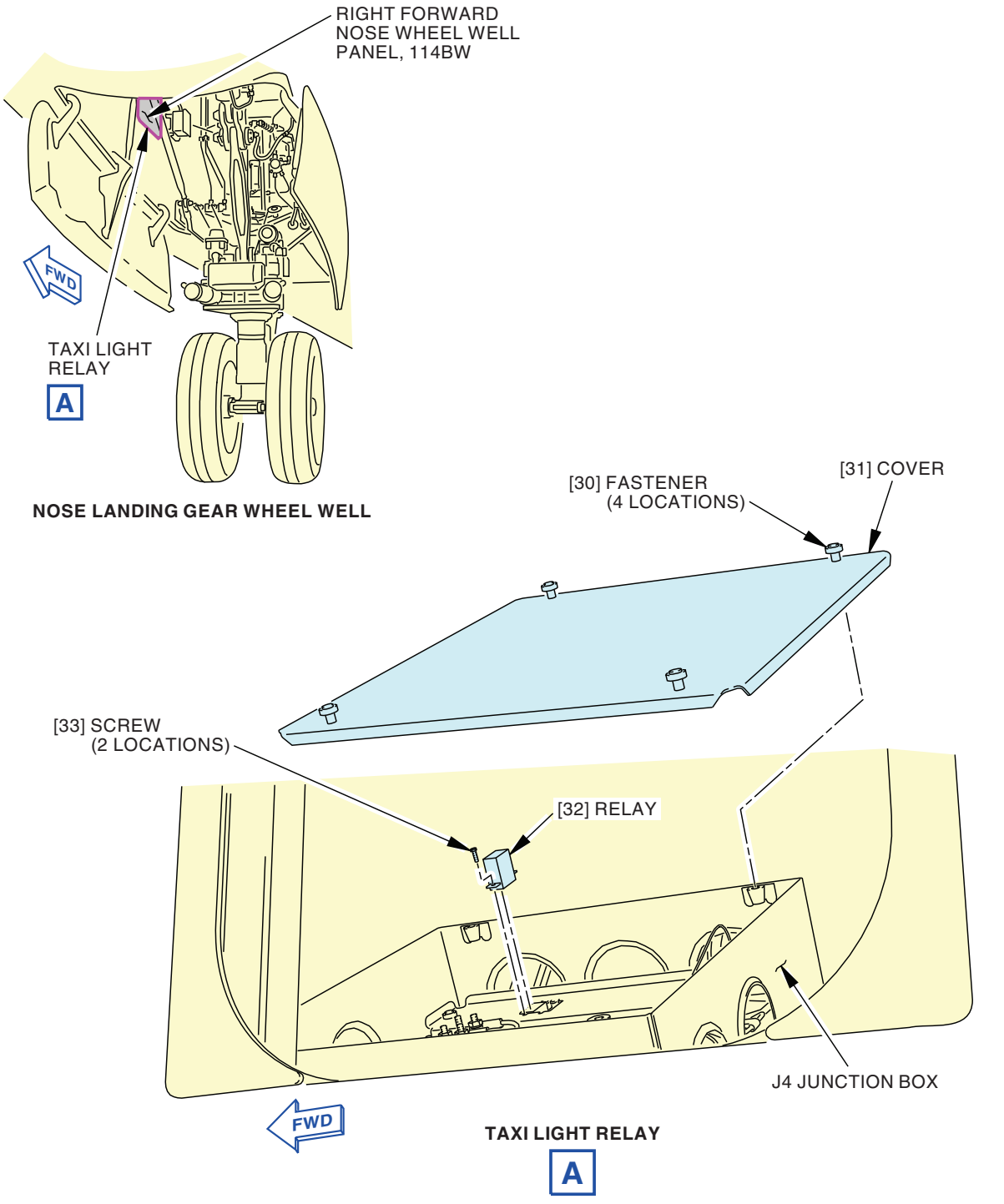
Row	Col	Number	Name
D	12	C01518	EXTERIOR LIGHTING GENERAL LIGHT CONT

SUBTASK 33-42-21-710-006

(6) Do this task: Taxi Light Relay - Operational Test, TASK 33-42-21-710-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Taxi Light Relay - Replacement
Figure 207/33-42-21-990-808

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LED LANDING, TAXI, AND RUNWAY TURNOFF LIGHTS (LTRTL) - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) LTRTL Main Array - Operational Test

(2) Taxi Light Relay - Operational Test.

TASK 33-42-21-710-801

2. LTRTL Main Array - Operational Test

A. General

(1) This task contains the steps for the operational test for the main array.

(2) The main array has the inboard and outboard arrays.

(3) The operational test is automated by the power supply.

(4) The inboard or outboard array will come on when 50% or more of the nominal array voltage is maintained. The power supply measures this automatically. The power supply stops the array when the array does not meet this condition.

(5) The inboard or outboard array is serviceable when:

(a) The array goes on and off correctly.

NOTE: It is possible that the inboard or outboard array could have multiple Light Emitting Diode (LED) failures and the array will continue to operate normally. The array is considered serviceable in this condition.

NOTE: It is recommended to replace LED array if more than 18 LEDs are non-operational.

(6) The inboard or outboard array is unserviceable when the array does not turn on.

B. References

Reference	Title
33-42-21-440-801	Fixed Landing Light - Activation (P/B 201)

C. Prepare for the Operational Test

SUBTASK 33-42-21-440-001

(1) Operate the main arrays, do this task: Fixed Landing Light - Activation, TASK 33-42-21-440-801.

D. LTRTL Main Array Operational Test

SUBTASK 33-42-21-710-003



WARNING

DO NOT LOOK DIRECTLY AT LIGHTS. HIGH INTENSITY OF LIGHTS CAN CAUSE EYE DAMAGE.

(1) Do the following steps to test the main arrays:

(a) On the P5 overhead panel, set the left LANDING switch to the ON position.

1) Make sure that the left main array comes on.

(b) Set the left LANDING switch to the OFF position.

1) Make sure that the left main array goes off.

(c) On the P5 overhead panel, set the right LANDING switch to the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the right main array comes on.

(d) Set the right LANDING switch to the OFF position.

1) Make sure that the right main array goes off.

(e) On the P5 overhead panel, set the TAXI switch to the AUTO or ON position.

1) Make sure that both right and left main arrays come on.

NOTE: When the TAXI switch is in the AUTO or ON position, the intensity of the main array is less than when the LANDING switches are in the ON position.

(f) Set the TAXI switch to the OFF position.

1) Make sure that both right and left main arrays go off.

(g) On the P5 overhead panel, set the RUNWAY TURNOFF L switch to the ON position.

1) Make sure that the left runway turnoff light comes on.

(h) Set the RUNWAY TURNOFF L switch to the OFF position.

1) Make sure that the left runway turnoff light goes off.

(i) On the P5 overhead panel, set the RUNWAY TURNOFF R switch to the ON position.

1) Make sure that the right runway turnoff light comes on.

(j) Set the RUNWAY TURNOFF R switch to the OFF position.

1) Make sure that the right runway turnoff light goes off.

——— END OF TASK ———

TASK 33-42-21-710-802

3. Taxi Light Relay - Operational Test

(Figure 501)

A. General

(1) This task includes the steps to make sure that the taxi light relay operates correctly.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1690	Actuators/Deactuators Set - Proximity Sensor Part #: 8-758-01 Supplier: 08748 Part #: A27092-139 Supplier: 81205 Part #: A27092-146 Supplier: 81205 Opt Part #: A27092-106 Supplier: 81205

C. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Taxi Light Relay Operational Test

SUBTASK 33-42-21-710-005

(1) Do a test of the taxi light relay as follows:

(a) On the P5 overhead panel, set the TAXI switch to the AUTO or ON position.

(b) Make sure that the taxi light comes on.

(c) Attach the deactuators to the face of these proximity sensors on the nose landing gear:

NOTE: The deactuators are part of the proximity sensor test set, SPL-1690.

1) S845 #1 NOSE GEAR DOWN

2) S846 #1 NOSE GEAR LOCKED.

(d) Make sure that the taxi light goes off.

(e) Remove the deactuators.

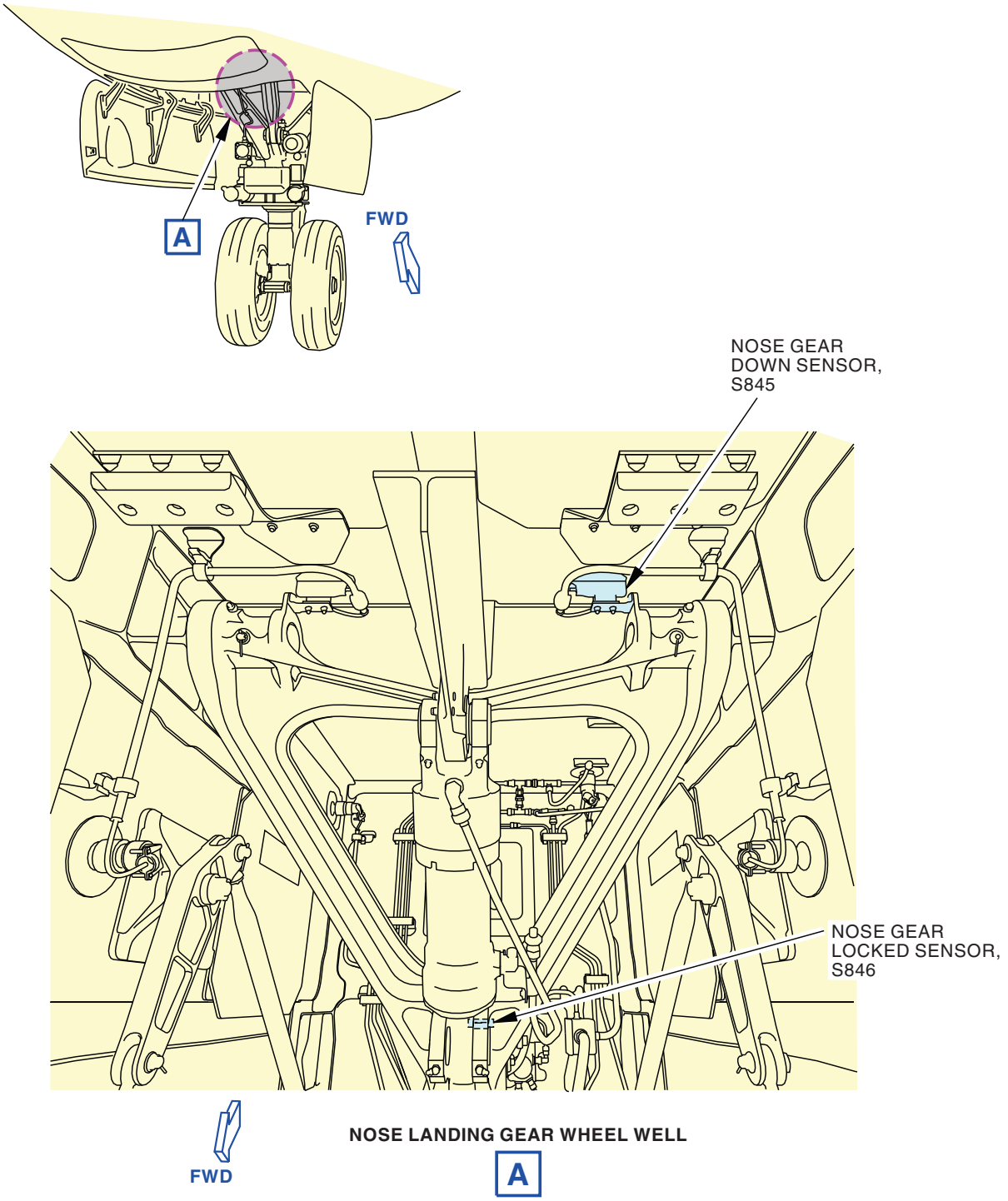
(f) Make sure that the taxi light comes on.

(g) On the P5 overhead panel, set the TAXI switch to the OFF position.

(h) Make sure that the taxi light goes off.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Taxi Light Relay - Operational Test
Figure 501/33-42-21-990-807

ECCN 9E991 BOEING PROPRIETARY - See title page for details

POSITION LIGHTS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Position Lights Deactivation

(2) Position Lights Activation

(3) Position Lights Transformer Replacement

(4) Position Lights Power Supply Unit Replacement

(5) Position Lights Operational Test

(6) Winglet (AFT) Marker and Winglet Anti-Collision Lights - Disconnect the Electrical Connectors

(7) Winglet (AFT) Marker and Winglet Anti-Collision Lights - Connect the Electrical Connectors.

TASK 33-43-10-040-801

2. Position Lights Deactivation

A. General

(1) This procedure removes electrical power from the position lights.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
310	Fuselage - Body Station 1016.00 to Body Station 1260.00
527	Left Winglet
627	Right Winglet

C. Procedure

SUBTASK 33-43-10-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	12	C04052	EXTERIOR LIGHTING POSITION RIGHT
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT

D. Position Lights Deactivation Test

NOTE: This test is to make sure that the Position lights are in a zero energy condition.

SUBTASK 33-43-10-200-001

(1) Do these steps to make sure that the power is off:

(a) Set the switch for the position lights to STEADY on the P5 overhead panel.

1) Make sure that the forward position, winglet marker (aft), and tailcone position lights (left/right) stay off.

(b) Set the switch for the position lights to STROBE & STEADY on the P5 overhead panel.

1) Make sure that the forward position, winglet marker (aft), and tailcone position lights (left/right) stay off.

(c) Set the switch for the position lights to OFF.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the forward position, winglet marker (aft), and tailcone position lights (left/right) stay off.

——— END OF TASK ———

TASK 33-43-10-440-801

3. Position Lights Activation

A. General

(1) This procedure supplies electrical power to the position lights.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
310	Fuselage - Body Station 1016.00 to Body Station 1260.00
527	Left Winglet
627	Right Winglet

C. Procedure

SUBTASK 33-43-10-860-002

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	12	C04052	EXTERIOR LIGHTING POSITION RIGHT
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT

D. Position Lights Activation Test

NOTE: This test is to make sure that the Position lights operate.

SUBTASK 33-43-10-710-001

(1) Do these steps to make sure that the power is on:

(a) Set the switch for the position lights to STEADY on the P5 overhead panel.

1) Make sure that the forward position, winglet marker (aft), and tailcone position lights (left/right) come on.

(b) Set the switch for the position lights to STROBE & STEADY on the P5 overhead panel.

1) Make sure that the forward position, winglet marker (aft), and tailcone position lights (left/right) come on.

(c) Set the switch for the position lights to OFF.

1) Make sure that the forward position, winglet marker (aft), and tailcone position lights (left/right) go off.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-43-10-960-801

4. Position Lights Transformer Replacement

(Figure 201)

A. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
33-43-13-960-802	Tailcone Position Lights Transformer Replacement (P/B 201)
51-21-41-370-802	Apply Bonderite M-CR 600, Bonderite M-CR 1200, Bonderite M-CR 600RTU, or Bonderite M-CR 1200S Solution (P/B 701)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
WDM 33-43-11	Wiring Diagram Manual

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Consumable Materials

Reference	Description	Specification
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50337	Sealant - Fuel Tank	BMS5-45 Class B
A50359	Sealant - Low Density, Non-Chromate Type	BMS5-142 Type II Class B
B00068	Alcohol - Denatured, Ethyl (Ethanol)	AMS 3002, MIL-E-51454 Type II
B00130	Alcohol - Isopropyl	TT-I-735
C00862	Coating - Chemical Conversion - Bonderite M-CR 600 Aero (Formerly Alodine 600)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3
G50262	Wiper - Cleaning	BMS15-5
G50961	Coating - Corrosion Inhibiting Epoxy Primer - (BAC 377 Yellow Color)	BMS10-11 Type I
G51464	Abrasive - Aluminum Oxide paper, 60 grit or finer	

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D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Forward position light transformer	33-43-10-01-020	PTF ALL

E. Location Zones

Zone	Area
527	Left Winglet
627	Right Winglet

F. Access Panels

Number	Name/Location
521AAB	Lower Leading Edge Access Panel
621AAB	Lower Leading Edge Access Panel

G. Position Lights Transformer Replacement

SUBTASK 33-43-10-860-015

(1) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 33-43-10-860-003

(2) Open the applicable circuit breaker and install the safety tag:

(WDM 33-43-11)

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	12	C04052	EXTERIOR LIGHTING POSITION RIGHT
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT

SUBTASK 33-43-10-960-001

(3) Do these steps to replace the forward position light transformer [1]:

(a) Open the applicable access panel:

Number	Name/Location
521AAB	Lower Leading Edge Access Panel
621AAB	Lower Leading Edge Access Panel

(b) Disconnect the electrical connector [4].

(c) Remove the bolts [3] and washers [2].

(d) Remove the forward position light transformer [1].

(e) Clean the mating surfaces of the forward position light transformer [1] and mounting bracket with 60 grit or finer wet/dry aluminum oxide sandpaper, G51464.

(f) Clean the mating surfaces of the forward position light transformer [1] and mounting bracket with wiper, G50262, moist in alcohol, B00130, or alcohol, B00068.

1) If it is necessary, use clean wiper, G50262, to remove remaining solvent.

(g) Install the bolts [3] and washers [2] that attach the forward position light transformer [1] to the mounting bracket.

(h) Measure the electrical bonding resistance between the forward position light transformer [1] and airplane structure (SWPM 20-20-00):

1) Use the intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the resistance is 0.00050 ohm (0.50 milliohm) or less.

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(i) Apply a layer of Bonderite M-CR 600 Aero coating, C00862, to bare surfaces of the forward position light transformer [1] and mounting bracket (TASK 51-21-41-370-802).

(j) Apply fillet seal with sealant, A50337, or sealant, A50231, or sealant, A50359, to the forward position light transformer [1].

(k) Apply coating, G50961, to the bare metal surfaces of the forward position light transformer [1] and mounting bracket.

(l) Connect the electrical connector [4].

SUBTASK 33-43-10-860-017

(4) Close these access panels:

Number	Name/Location
521AAB	Lower Leading Edge Access Panel
621AAB	Lower Leading Edge Access Panel

SUBTASK 33-43-10-960-005

(5) For the tailcone position light transformer replacement:

(a) Do this task: Tailcone Position Lights Transformer Replacement, TASK 33-43-13-960-802.

SUBTASK 33-43-10-860-004

(6) Remove the safety tag and close the applicable circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	12	C04052	EXTERIOR LIGHTING POSITION RIGHT
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT

SUBTASK 33-43-10-710-002

(7) Do a test of the new forward position light transformer [1]:

(a) Set the switch for the position lights, on the P5 overhead panel, to the STEADY position.

(b) Make sure that the forward position, winglet marker (aft), and tailcone position lights come on.

(c) Set the switch for the position lights, on the P5 overhead panel, to the STROBE & STEADY position.

(d) Make sure that the forward position, winglet marker (aft), and tailcone position lights come on.

(e) Set the switch for the position lights to the OFF position.

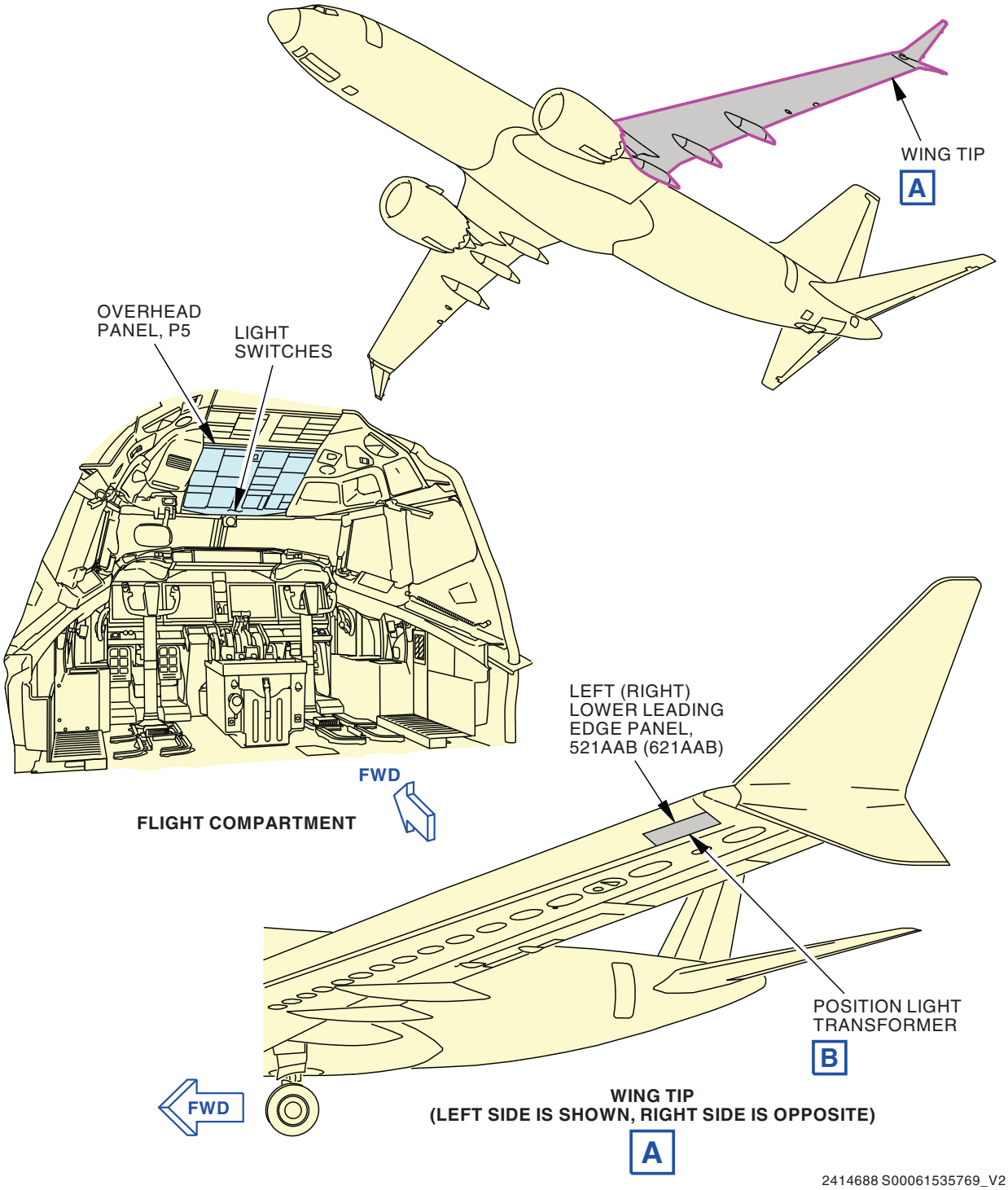
(f) Make sure that the forward position, winglet marker (aft), and tailcone position lights go off.

SUBTASK 33-43-10-860-016

(8) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

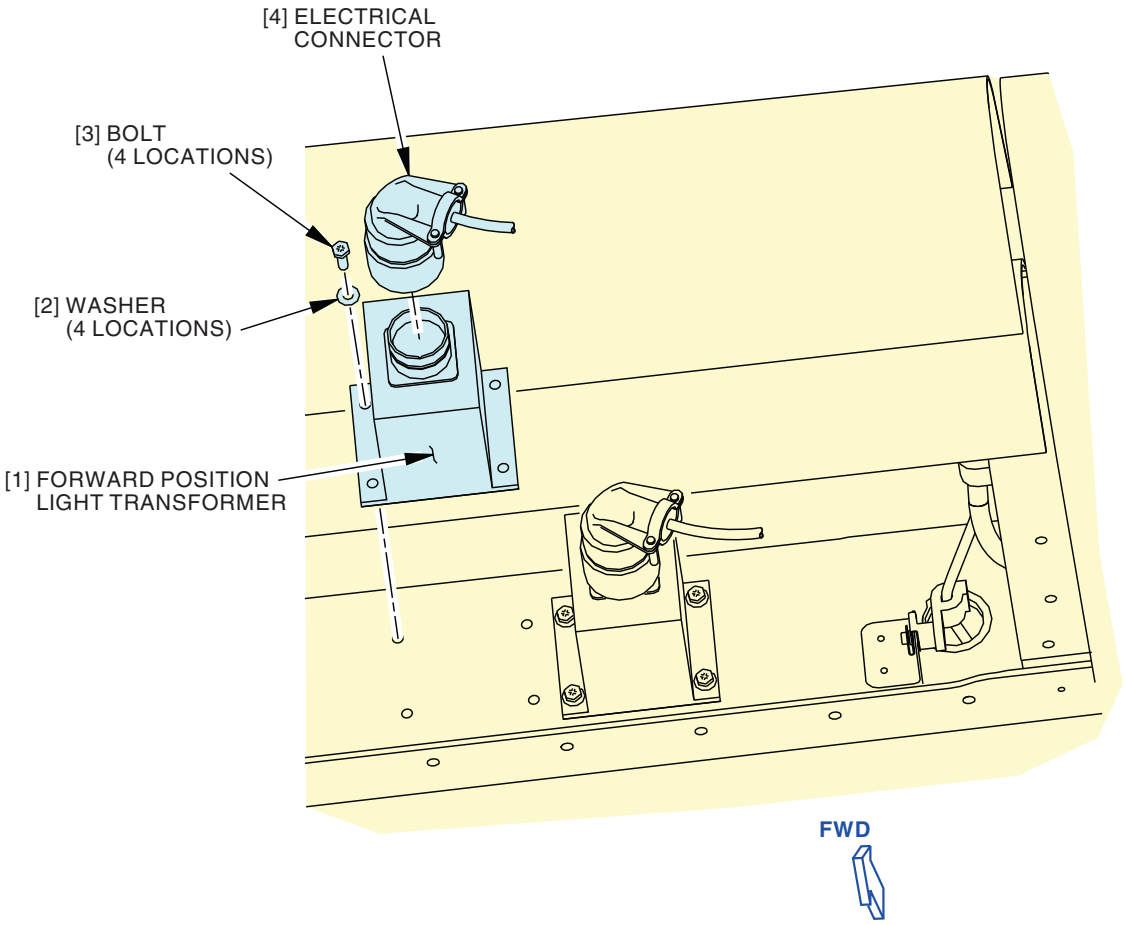
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Position Light Transformer Replacement
Figure 201/33-43-10-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



POSITION LIGHT TRANSFORMER

B

2414689 S00061535770_V3

Position Light Transformer Replacement
Figure 201/33-43-10-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-43-10-960-803

5. Forward Position Lights Power Supply Unit Replacement

(Figure 202)

A. General

(1) This task contains instructions for the replacement of the power supply unit installed on the winglet forward position light.

B. References

Reference	Title
33-43-11-960-801	Forward Position Lights LED Module Replacement (P/B 201)

C. Tools/Equipment

Reference	Description
STD-420	Gloves - White Cotton

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
50	Power supply assembly	33-43-11-02-180	PTF ALL
		33-43-11-02-185	PTF ALL

E. Location Zones

Zone	Area
527	Left Winglet
627	Right Winglet

F. Access Panels

Number	Name/Location
521ABB	Lower Leading Edge Access Panel
621ABB	Lower Leading Edge Access Panel

G. Forward Position Lights Power Supply Unit Replacement

SUBTASK 33-43-10-860-011

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	12	C04052	EXTERIOR LIGHTING POSITION RIGHT
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT

SUBTASK 33-43-10-860-012

(2) Open these access panels:

Number	Name/Location
521ABB	Lower Leading Edge Access Panel
621ABB	Lower Leading Edge Access Panel

SUBTASK 33-43-10-020-004

(3) Remove the LED module assembly, do this task: Forward Position Lights LED Module Replacement, TASK 33-43-11-960-801.

SUBTASK 33-43-10-960-004

(4) Do these steps to replace the power supply assembly [50]:

(a) Use white cotton gloves, STD-420, to handle lens assembly and LED modules.

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(b) Remove the screws [51] and washers [52] that attach the power supply assembly [50] to the forward position housing.

(c) Pull the power supply assembly [50] forward and disconnect the electrical connector [53].

(d) Replace the power supply assembly [50].

(e) Connect the electrical connector [53].

(f) Install the washers [52] and screws [51] to the forward position housing.

(g) Tighten the screws [51] until fully seated.

SUBTASK 33-43-10-420-004

(5) Install the LED module assembly, do this task: Forward Position Lights LED Module Replacement, TASK 33-43-11-960-801.

SUBTASK 33-43-10-860-014

(6) Close these access panels:

Number	Name/Location
521ABB	Lower Leading Edge Access Panel
621ABB	Lower Leading Edge Access Panel

SUBTASK 33-43-10-860-013

(7) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	12	C04052	EXTERIOR LIGHTING POSITION RIGHT
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT

SUBTASK 33-43-10-710-006

(8) Do a test of the new power supply assembly [50]:

(a) Set the switch for the position lights to the STEADY position, on the P5 overhead panel.

1) Make sure that the forward position, winglet marker (aft) and tailcone position lights come on.

(b) Set the switch for the position lights to the STROBE & STEADY position, on the P5 overhead panel.

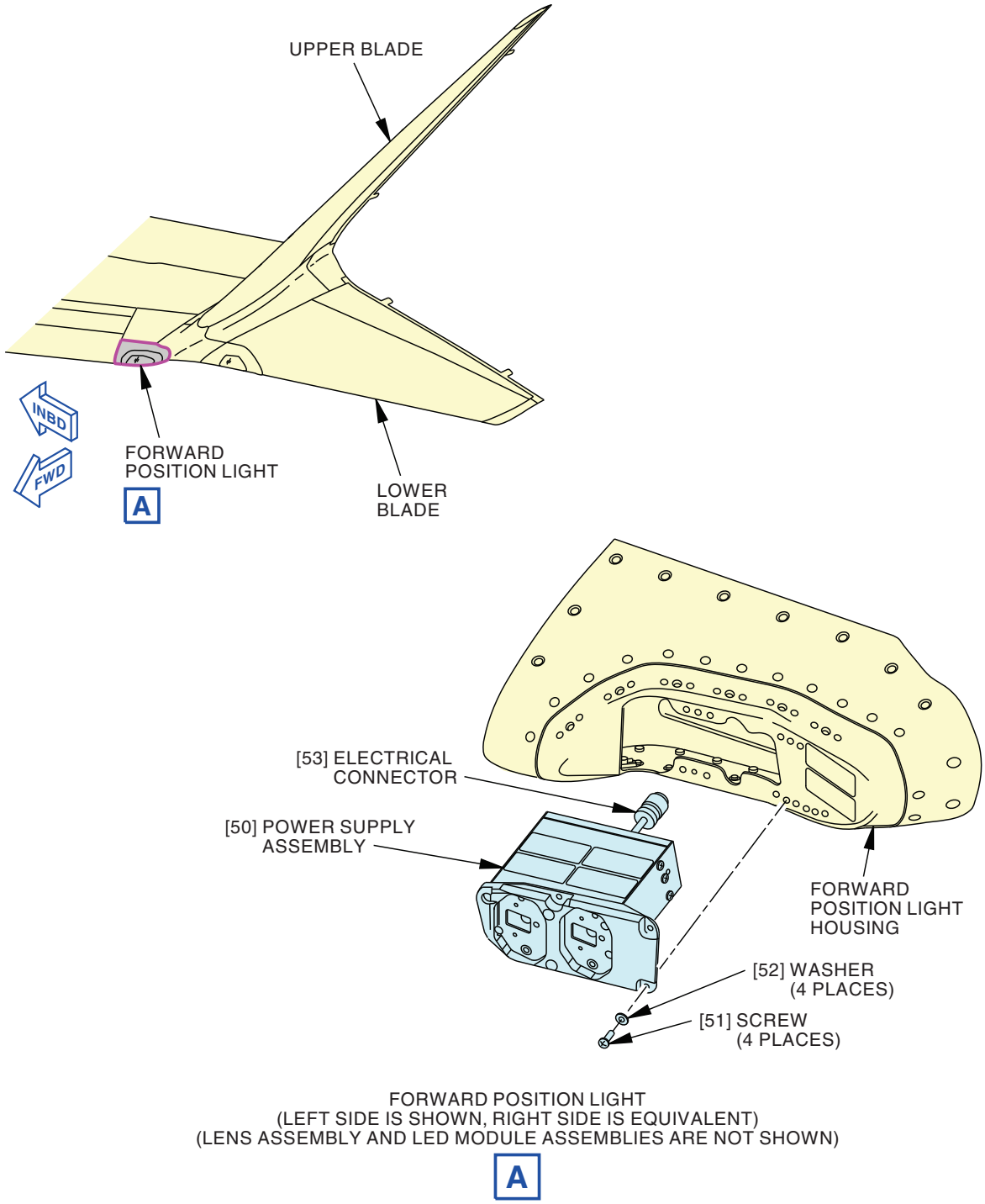
1) Make sure that the forward position, winglet marker (aft) and tailcone position lights come on.

(c) Set the switch for the position lights to the OFF position, on the P5 overhead panel.

1) Make sure that the forward position, winglet marker (aft) and tailcone position lights go off.

——— END OF TASK ———

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FORWARD POSITION LIGHT
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)
(LENS ASSEMBLY AND LED MODULE ASSEMBLIES ARE NOT SHOWN)

A

2965275 S0000744049_V1

Forward Position Lights Power Supply Unit Replacement
Figure 202/33-43-10-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-43-10-710-801

6. Position Lights Operational Test

A. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

B. Position Lights Operational Test

SUBTASK 33-43-10-710-003

(1) Test the forward position, winglet (aft) marker, and tailcone position lights (left/right):

(a) Set the switch for the position lights, on the P5 overhead panel, to the STEADY position.

(b) Make sure that the forward position, winglet marker (aft), and tailcone position lights (left/right) come on.

NOTE: There are two LED module assemblies in each winglet. All position lights in the modules must be operational per certification requirement. A single LED intensity is not sufficient to meet the requirement.

(c) Set the switch for the position lights, on the P5 overhead panel, to the STROBE & STEADY position.

(d) Make sure that the forward position, winglet marker (aft), and tailcone position lights (left/right) come on.

WARNING DO NOT LET THE LIGHTS FLASH DIRECTLY IN YOUR EYES. THE INTENSITY OF THE LIGHT CAN TEMPORARILY CAUSE INJURIES TO YOUR EYES.

(e) Make sure that the winglet anti-collision and tailcone anti-collision lights (left/right) flash.

(f) Set the switch for the position lights, on the P5 overhead panel, to the OFF position.

(g) Make sure that the forward position, winglet (aft) marker, and tailcone position lights (left/right) go off.

(h) Make sure that the winglet anti-collision and tailcone anti-collision lights (left/right) go off.

——— END OF TASK ———

TASK 33-43-10-020-801

7. Winglet (AFT) Marker and Winglet Anti-Collision Lights - Disconnect the Electrical Connectors

(Figure 203)

A. General

(1) Use this procedure when it is necessary to remove the winglet.

B. References

Reference	Title
57-31-25-010-801	Transition Fairing Removal (P/B 401)
57-50-02-000-803	Outboard Fixed Trailing Edge Lower Panel Removal (P/B 401)
WDM 33-43-11	Wiring Diagram Manual

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C. Access Panels

Number	Name/Location
571FB	Lower Outboard Fixed Trailing Edge Wedge Access Panel
671FB	Lower Outboard Fixed Trailing Edge, Wedge Access Panel

D. Winglet (AFT) Marker and Winglet Anti-Collision Lights - Disconnect the Electrical Connectors

SUBTASK 33-43-10-860-005

(1) Open the applicable circuit breaker and install the safety tag:

(WDM 33-43-11)

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	12	C04052	EXTERIOR LIGHTING POSITION RIGHT
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE
C	13	C04054	ANTI COLLISION LIGHTS-WINGLET-WHITE

SUBTASK 33-43-10-010-003

(2) Do this task: Transition Fairing Removal, TASK 57-31-25-010-801.

SUBTASK 33-43-10-010-001

(3) Remove these access panels:

(TASK 57-50-02-000-803)

Number	Name/Location
571FB	Lower Outboard Fixed Trailing Edge Wedge Access Panel
671FB	Lower Outboard Fixed Trailing Edge, Wedge Access Panel

SUBTASK 33-43-10-020-001

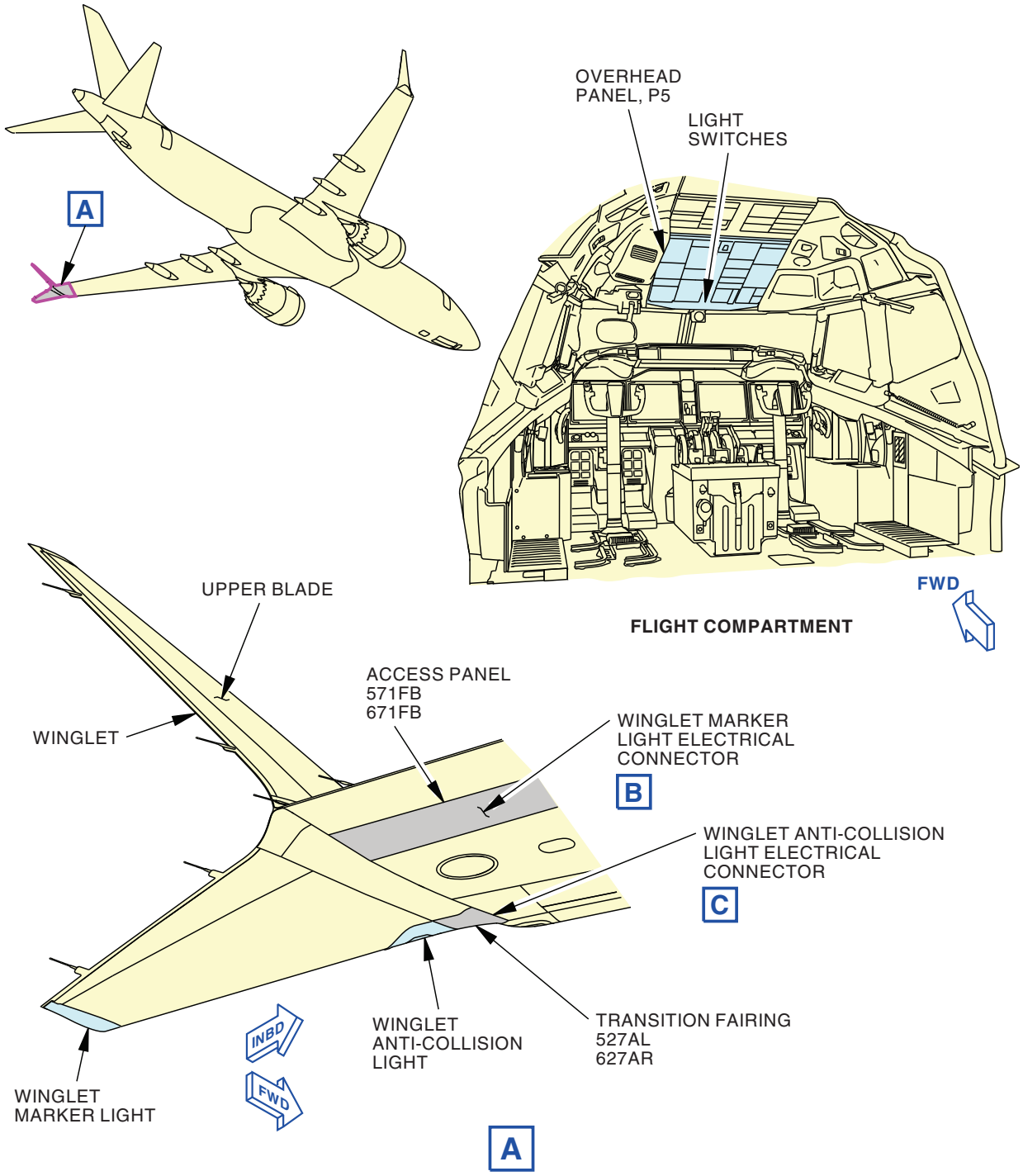
(4) Disconnect the electrical connector [21].

SUBTASK 33-43-10-020-002

(5) Disconnect the electrical connector [22].

——— END OF TASK ———

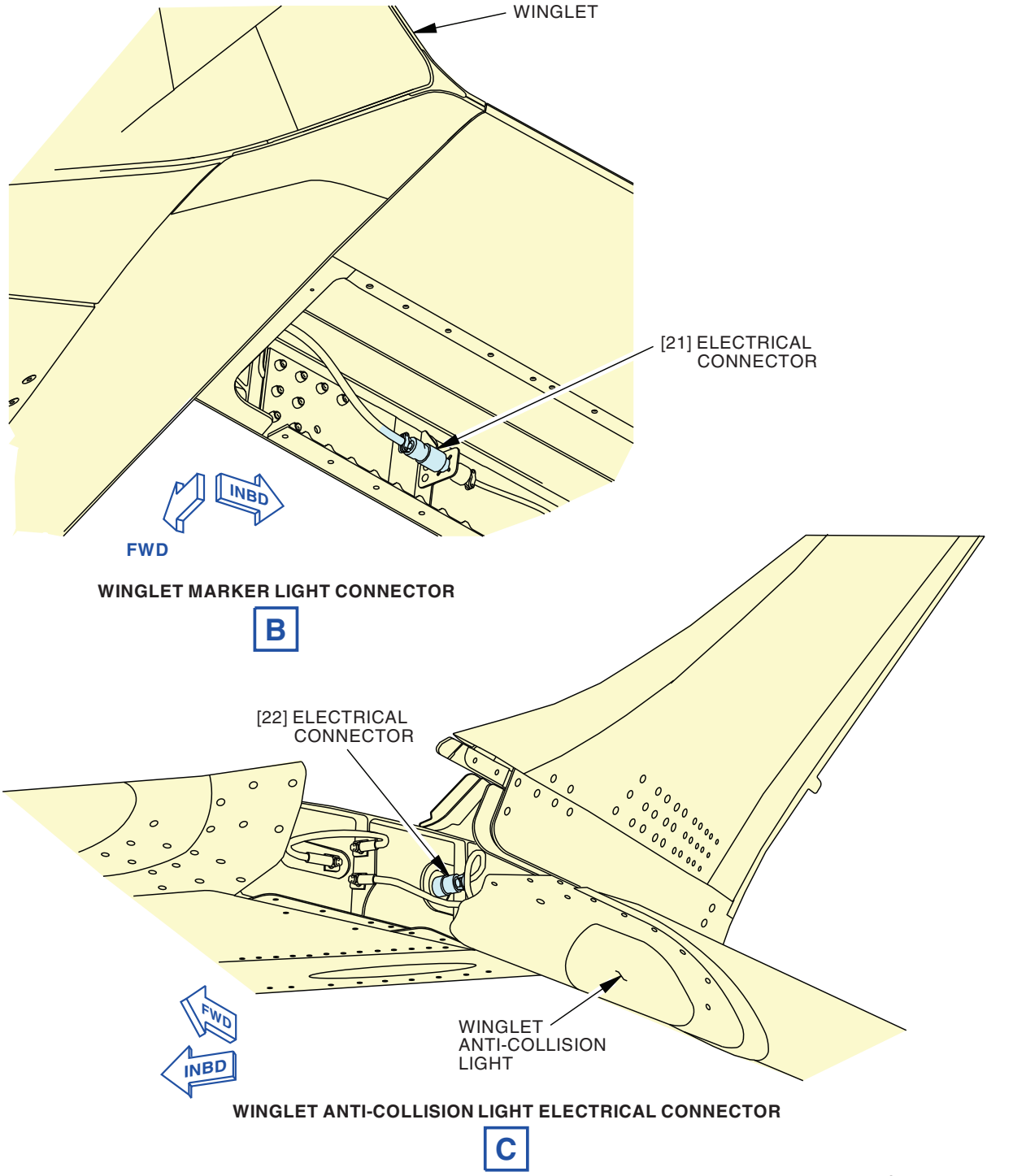
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Winglet (AFT) Marker and Winglet Anti-Collision Lights - Disconnect/Connect the Electrical Connectors
Figure 203/33-43-10-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Winglet (AFT) Marker and Winglet Anti-Collision Lights - Disconnect/Connect the Electrical Connectors
Figure 203/33-43-10-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-43-10-420-801

8. Winglet (AFT) Marker and Winglet Anti-Collision Lights - Connect the Electrical Connectors

(Figure 203)

A. General

(1) Use this procedure when it is necessary to install the winglet.

B. References

Reference	Title
57-31-25-410-801	Transition Fairing Installation (P/B 401)
57-50-02-400-803	Outboard Fixed Trailing Edge Lower Panel Installation (P/B 401)

C. Location Zones

Zone	Area
527	Left Winglet
627	Right Winglet

D. Access Panels

Number	Name/Location
571FB	Lower Outboard Fixed Trailing Edge Wedge Access Panel
671FB	Lower Outboard Fixed Trailing Edge, Wedge Access Panel

E. Winglet (AFT) Marker and Winglet Anti-Collision Lights - Connect the Electrical Connector

SUBTASK 33-43-10-420-001

(1) Connect the electrical connector [21].

SUBTASK 33-43-10-420-002

(2) Connect the electrical connector [22].

F. Winglet (AFT) Marker and Winglet Anti-Collision Lights - Connect the Electrical Connector Test

SUBTASK 33-43-10-010-002

(1) Install these access panels:

(TASK 57-50-02-400-803)

Number	Name/Location
571FB	Lower Outboard Fixed Trailing Edge Wedge Access Panel
671FB	Lower Outboard Fixed Trailing Edge, Wedge Access Panel

SUBTASK 33-43-10-410-001

(2) Do this task: Transition Fairing Installation, TASK 57-31-25-410-801.

SUBTASK 33-43-10-860-006

(3) Remove the safety tag and close the applicable circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	12	C04052	EXTERIOR LIGHTING POSITION RIGHT
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE
C	13	C04054	ANTI COLLISION LIGHTS-WINGLET-WHITE

SUBTASK 33-43-10-710-004

(4) Do a test of the electrical connection:

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(a) Set the switch for the position lights, on the P5 overhead panel, to the STEADY position.

(b) Make sure that the forward position, winglet marker (aft), and tailcone position lights (left/right) come on.

(c) Set the switch for the position lights, on the P5 overhead panel, to the STROBE & STEADY position.

(d) Make sure that the forward position, winglet marker (aft), and tailcone position lights (left/right) come on.

WARNING DO NOT LET THE LIGHTS FLASH DIRECTLY IN YOUR EYES. THE INTENSITY OF THE LIGHT CAN TEMPORARILY CAUSE INJURIES TO YOUR EYES.

(e) Make sure that the winglet anti-collision and tailcone anti-collision lights (left/right) flash.

(f) Set the switch for the position lights, on the P5 overhead panel, to the OFF position.

(g) Make sure that these lights go off.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FORWARD POSITION LIGHTS - MAINTENANCE PRACTICE

1. General

A. This procedure has this task:

(1) Forward Position Lights LED Module Replacement.

TASK 33-43-11-960-801

2. Forward Position Lights LED Module Replacement

(Figure 201)

A. General

(1) This task contains instructions for the replacement of the LED modules installed on the winglet forward position light.

B. References

Reference	Title
20-30-11-910-801	Adhesives, Cements, and Sealants (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
SOPM 20-50-11	Standard Overhaul Practices Manual
SOPM 20-50-19	General Sealing

C. Tools/Equipment

Reference	Description
STD-420	Gloves - White Cotton

D. Consumable Materials

Reference	Description	Specification
A00160	Sealant - Firewall - Hydraulic Fluid Resistant	BMS5-63
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50343	Sealant - PPG Aerospace PR-1772 Class B Low Weight Fuselage Sealant	BMS5-142 Type II Class B
G00291	Tape - Aluminum Foil, 3M 425	AMS-T-23397 / L-T-80
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

E. Location Zones

Zone	Area
212	Flight Compartment - Right
527	Left Winglet
627	Right Winglet

F. Forward Position Lights LED Module Replacement

SUBTASK 33-43-11-860-009

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	12	C04052	EXTERIOR LIGHTING POSITION RIGHT
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-43-11-860-011

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 33-43-11-020-003

(3) Remove the lens assembly [1].

(a) Use white cotton gloves, STD-420, to handle lens assembly [1] and LED module assemblies [5].

(b) Remove the bolts [2].

(c) Remove the lens assembly [1] from the forward position housing.

SUBTASK 33-43-11-960-002

(4) Do these steps to replace the LED module assembly [5]:

(a) Use white cotton gloves, STD-420, to handle lens assembly [1] and LED module assemblies [5].

(b) Remove the screws [3] and washers [4] that attach the LED module assembly [5] to the power supply assembly [6].

(c) Pull the LED module assembly [5] to disconnect it from the power supply assembly [6].

(d) Replace with the new LED module assembly [5].

(e) Install the washers [4] and screws [3] that hold the LED module assembly [5] to the power supply assembly [6].

(f) Tighten the screws [3] until fully seated.

SUBTASK 33-43-11-420-003

(5) Install the lens assembly [1].

(a) Use white cotton gloves, STD-420, to handle lens assembly [1] and LED module assemblies [5].

(b) Put the lens assembly [1] back to its position.

(c) Apply a layer of corrosion inhibiting material, G50136, to the bolts [2].

(d) Install the bolts [2] that hold the lens assembly [1] to the forward position housing.

(e) Apply an aerodynamic fillet seal of PR-1772 Class B Sealant, A50343, or sealant, A00160, or sealant, A50231, to the mating surface between the lens and the forward position light housing or to the external gap around the lens (SOPM 20-50-19, SOPM 20-50-11).

1) If it is necessary to send the airplane before the sealant is fully cured, install 3M 425 Aluminum Foil Tape, G00291, on the top of the sealant.

a) Remove the 3M 425 Aluminum Foil Tape, G00291, after the sealant fully dries (TASK 20-30-11-910-801).

b) Carefully examine the sealant to make sure that it bonded to the edges and dried correctly.

SUBTASK 33-43-11-860-010

(6) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	12	C04052	EXTERIOR LIGHTING POSITION RIGHT
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-43-11-710-002

(7) Do a test of the new LED module assembly [5].

(a) Set the switch for the position lights to the STEADY position, on the P5 overhead panel.

1) Make sure that the forward position, winglet marker (aft) and tailcone position lights come on.

(b) Set the switch for the position lights to the STROBE & STEADY position, on the P5 overhead panel.

1) Make sure that the forward position, winglet marker (aft) and tailcone position lights come on.

(c) Set the switch for the position lights to the OFF position, on the P5 overhead panel.

1) Make sure that the forward position, winglet marker (aft) and tailcone position lights go off.

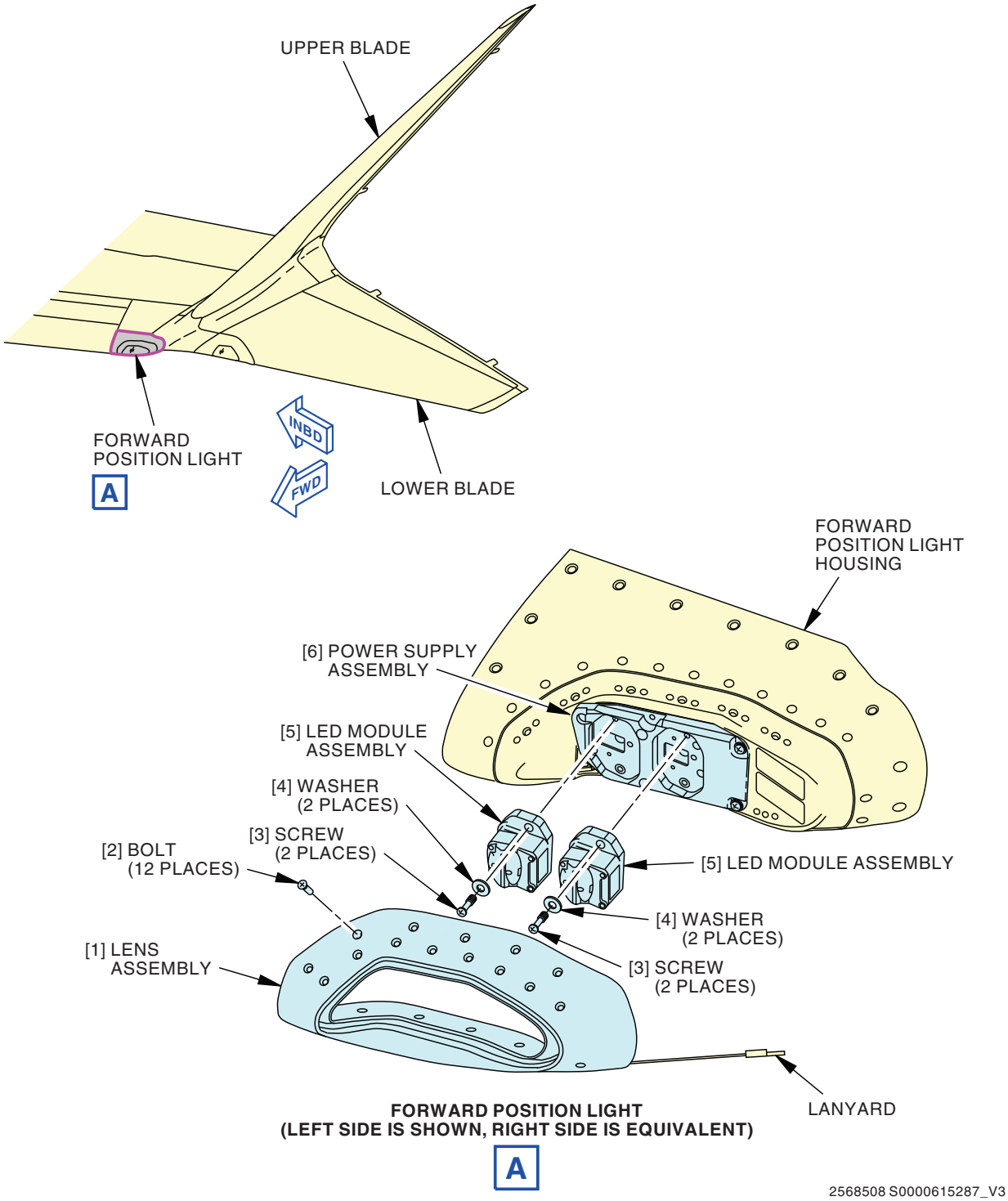
G. Put the Airplane Back to Its Usual Condition

SUBTASK 33-43-11-860-012

(1) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Forward Position Light LED Module Replacement
Figure 201/33-43-11-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FORWARD POSITION LIGHT - REMOVAL/INSTALLATION

1. General

A. This procedure has the these tasks:

(1) Forward Position Light Removal

(2) Forward Position Light Installation.

TASK 33-43-11-000-801

2. Forward Position Light Removal

(Figure 401)

A. General

(1) This task contains instructions for the removal of the forward position light from the forward edge of the winglet.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-659	Platform - Maintenance Lift, 3 ft Minimum Height, 12.5 ft Maximum Height Part #: 1135-8000 Supplier: 00994 Opt Part #: 8662-010 Supplier: 00994
COM-2480	Platform - Mobile Elevating Work Platform SJ III Series Part #: SJIII 4726 Supplier: 3AF08 Part #: SJIII 4732 Supplier: 3AF08 Opt Part #: SJIII 4620 Supplier: 3AF08 Opt Part #: SJIII 4626 Supplier: 3AF08 Opt Part #: SJIII 4632 Supplier: 3AF08
SPL-768	Sealant Removal Tool, Non-Metallic Part #: ST982L-9 Supplier: 81205 Part #: ST982LF-9 Supplier: 81205 Part #: ST982LH-A-1 Supplier: 81205

D. Location Zones

Zone	Area
500	Left Wing
527	Left Winglet
600	Right Wing
627	Right Winglet

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Removal

SUBTASK 33-43-11-860-004

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	12	C04052	EXTERIOR LIGHTING POSITION RIGHT
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT

SUBTASK 33-43-11-860-006

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 33-43-11-010-001

(3) Get a ladder, work platform, COM-2480, or maintenance platform, COM-659.

SUBTASK 33-43-11-010-002

(4) Do these steps to prepare metal support equipment such as work platforms, work/maintenance stands, ladders.

NOTE: These steps apply to all metal support equipment within a 50 ft (15 m) radius of an open fuel tank.

(a) All support equipment must be in place before you begin the procedure.

(b) Bond the support equipment at an approved airplane bonding location.

(c) Ground the support equipment to the same earth ground as the airplane.

F. Forward Position Light Removal

SUBTASK 33-43-11-000-001

(1) Remove the forward position light [1].

(a) Remove the screws [2] from the forward position light [1].

(b) Remove the forward position light [1] from the wing.

SUBTASK 33-43-11-020-001

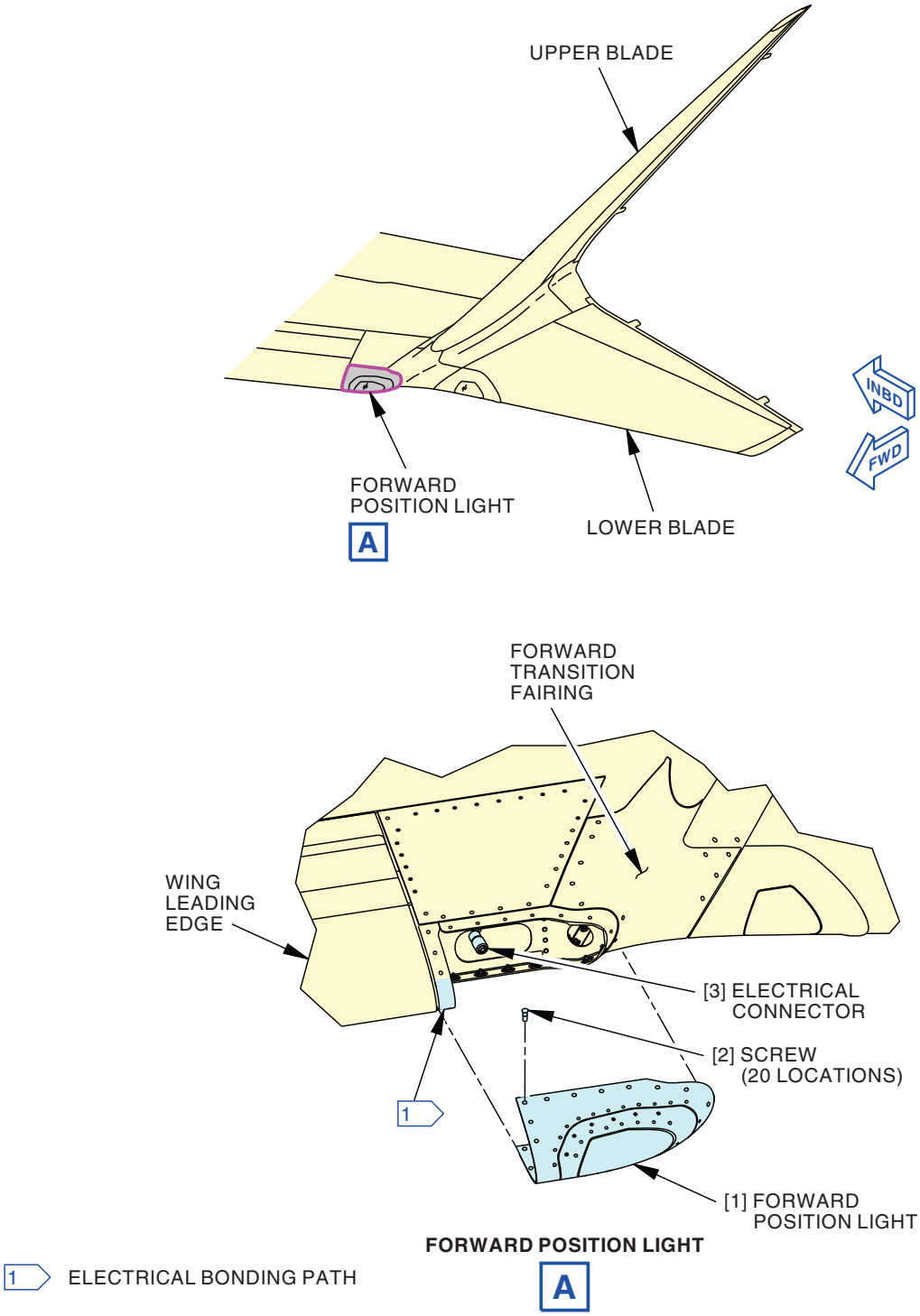
(2) Disconnect the electrical connector [3] from the forward position light [1].

SUBTASK 33-43-11-100-002

(3) Remove all sealant from leading edge panel and the forward position light [1] with a sealant removal tool, SPL-768.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Forward Position Light Installation
Figure 401/33-43-11-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-43-11-400-801

3. Forward Position Light Installation

(Figure 401)

A. General

(1) This task contains instructions for the installation of the forward position light on the forward edge of the winglet.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
33-43-10-710-801	Position Lights Operational Test (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-659	Platform - Maintenance Lift, 3 ft Minimum Height, 12.5 ft Maximum Height Part #: 1135-8000 Supplier: 00994 Opt Part #: 8662-010 Supplier: 00994
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
COM-2480	Platform - Mobile Elevating Work Platform SJ III Series Part #: SJIII 4726 Supplier: 3AF08 Part #: SJIII 4732 Supplier: 3AF08 Opt Part #: SJIII 4620 Supplier: 3AF08 Opt Part #: SJIII 4626 Supplier: 3AF08 Opt Part #: SJIII 4632 Supplier: 3AF08

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
G00291	Tape - Aluminum Foil, 3M 425	AMS-T-23397 / L-T-80
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Forward position light	33-43-11-02-125	PTF ALL
		33-43-11-02-130	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Location Zones

Zone	Area
500	Left Wing
527	Left Winglet
600	Right Wing
627	Right Winglet

G. Forward Position Light Installation

SUBTASK 33-43-11-100-001

(1) Clean the designated bond area of the forward position light [1] and the leading edge to make an electrical bond.

SUBTASK 33-43-11-390-001

(2) Apply a layer of sealant, A50110, to the mounting surface to make a fay surface seal between the mounting surface and the forward position light.

(a) If you must send the airplane before the sealant is fully cured, install 3M 425 Aluminum Foil Tape, G00291, on the top of the sealant.

1) Remove 3M 425 Aluminum Foil Tape, G00291, when the sealant is fully cured.

2) Carefully inspect the sealant and make sure that the sealant is adhered to the edges and that it is cured properly.

SUBTASK 33-43-11-420-001

(3) Connect the electrical connector [3] to the forward position light [1].

SUBTASK 33-43-11-400-001

(4) Install the forward position light [1]:

(a) Put the forward position light [1] on the mounting surface of the wing leading edge.

(b) Apply a layer of corrosion inhibiting material, G50136, to the screws [2].

(c) Install the screws [2].

(d) Tighten the screws [2] to 30 ±1 in-lb (3.4 ±0.1 N·m).

SUBTASK 33-43-11-760-001

(5) Measure the electrical resistance between the forward position light [1] and the leading edge seal rib (SWPM 20-20-00).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the electrical resistance is 0.0005 ohm (0.5 milliohm) or less.

SUBTASK 33-43-11-860-001

(6) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	12	C04052	EXTERIOR LIGHTING POSITION RIGHT
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT

SUBTASK 33-43-11-700-001

(7) Do this task: Position Lights Operational Test, TASK 33-43-10-710-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 33-43-11-410-001

(1) Remove the ladders, work platform, COM-2480, or maintenance platform, COM-659.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-43-11-860-003

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WINGLET AFT MARKER LIGHTS - MAINTENANCE PRACTICES

1. General

A. This procedure has this task:

(1) An aft marker light LED optic assembly replacement.

TASK 33-43-12-960-801

2. Aft Marker Light LED Optic Assembly Replacement

(Figure 201)

A. General

(1) This task contains instructions for the replacement of the LED optic assembly installed on the aft marker light.

(2) Before LED optic assembly replacement, carefully look at the unit to see its position. If it is necessary, make records. This will help you do the LED optic assembly replacement easier.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
33-43-10-710-801	Position Lights Operational Test (P/B 201)
33-43-12-000-801	Aft Marker Light Removal (P/B 401)
33-43-12-400-801	Aft Marker Light Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-659	Platform - Maintenance Lift, 3 ft Minimum Height, 12.5 ft Maximum Height Part #: 1135-8000 Supplier: 00994 Opt Part #: 8662-010 Supplier: 00994
COM-2480	Platform - Mobile Elevating Work Platform SJ III Series Part #: SJIII 4726 Supplier: 3AF08 Part #: SJIII 4732 Supplier: 3AF08 Opt Part #: SJIII 4620 Supplier: 3AF08 Opt Part #: SJIII 4626 Supplier: 3AF08 Opt Part #: SJIII 4632 Supplier: 3AF08
STD-420	Gloves - White Cotton

D. Consumable Materials

Reference	Description	Specification
A50119	Adhesive - Silicone Rubber - RTV 103 (Black)	MIL-A-46106
A50250	Sealant - P/S 870 Class B-1/2 Corrosion Inhibitive Sealant	BMS5-95 Class B-1/2
A50416	Sealant - RTV 732, Type 1, Clear Adhesive-Sealant	MIL-A-46106
A50479	Sealant - P/S 870 Class B-2 Corrosion Inhibitive Sealant	BMS5-95 Class B-2

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E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
6	LED optic assembly	33-43-12-10-020	PTF ALL
		33-43-12-10-025	PTF ALL

F. Prepare for the Replacement

SUBTASK 33-43-12-860-007

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	12	C04052	EXTERIOR LIGHTING POSITION RIGHT
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT

SUBTASK 33-43-12-860-008

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 33-43-12-940-003

(3) Get a ladder, work platform, COM-2480, or maintenance platform, COM-659.

SUBTASK 33-43-12-940-004

(4) Do these steps to prepare metal support equipment such as work platforms, work/maintenance stands, ladders.

NOTE: These steps apply to all metal support equipment within a 50 ft (15 m) radius of an open fuel tank.

(a) All support equipment must be in position before you begin the procedure.

(b) Bond the support equipment at an approved airplane bonding location.

(c) Ground the support equipment to the same earth ground as the airplane.

SUBTASK 33-43-12-020-004

(5) Remove the aft marker light assembly, do this task: Aft Marker Light Removal, TASK 33-43-12-000-801.

G. LED Optic Assembly Replacement

SUBTASK 33-43-12-960-001

(1) Do these steps to replace the LED optic assembly [6]:

(a) Use white cotton gloves, STD-420, to handle LED module.

(b) Remove the LED optic assembly [6] from the mounting bracket [7] and tipcap assembly.

1) Remove the screws [4], washer [3], and screws [5].

2) Disconnect the terminal end [2].

(c) If it is necessary, remove the mounting bracket [7] from tipcap assembly.

1) Remove the screws [8].

(d) Pull the LED optic assembly [6] from the aft marker light assembly.

(e) Replace with a new LED optic assembly [6].

1) Clean all the parts before installation.

a) Remove all contamination, oils signs, and other unwanted material from the light assembly.

b) Clean tools from all oils and contamination.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Put the mounting bracket [7] into tipcap assembly.

WARNING USE PROTECTIVE GLOVES AROUND CORROSION-INHIBITING COMPOUND. IF YOU DO NOT OBEY, SKIN IRRITATION CAN OCCUR.

1) Apply a thin layer of P/S 870 Class B-1/2 sealant, A50250, or P/S 870 Class B-2 sealant, A50479, to shank and underside of heads the screws [8].
2) If removed, install the screws [8] to secure the mounting bracket [7] to tipcap assembly.
   a) Fully tighten the screws [8].

WARNING USE PROTECTIVE GLOVES AROUND CORROSION-INHIBITING COMPOUND. IF YOU DO NOT OBEY, SKIN IRRITATION CAN OCCUR.

3) Apply a thin layer of P/S 870 Class B-1/2 sealant, A50250, or P/S 870 Class B-2 sealant, A50479, to heads of screws [8].

(g) Install the LED optic assembly [6] in position on tipcap assembly and mounting bracket [7].

1) Put the LED optic assembly [6] in its position.

WARNING USE PROTECTIVE GLOVES AROUND CORROSION-INHIBITING COMPOUND. IF YOU DO NOT OBEY, SKIN IRRITATION CAN OCCUR.

2) Apply a thin layer of P/S 870 Class B-1/2 sealant, A50250, or P/S 870 Class B-2 sealant, A50479, to underside of heads of screws [5].
3) Install screws [5] to the mounting holes.
   a) Fully tighten the screws [5].

WARNING USE PROTECTIVE GLOVES AROUND CORROSION-INHIBITING COMPOUND. IF YOU DO NOT OBEY, SKIN IRRITATION CAN OCCUR.

4) Apply a thin layer of P/S 870 Class B-1/2 sealant, A50250, or P/S 870 Class B-2 sealant, A50479, to heads of screws [5].

(h) Install terminal end [2] with washer [3] and screw [4] on the forward mounting hole of the chassis assembly [1].

WARNING USE PROTECTIVE GLOVES AROUND CORROSION-INHIBITING COMPOUND. IF YOU DO NOT OBEY, SKIN IRRITATION CAN OCCUR.

1) Apply a thin layer of P/S 870 Class B-1/2 sealant, A50250, or P/S 870 Class B-2 sealant, A50479, to underside of head of screw [4].
2) Fully tighten the screw [4].

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WARNING USE PROTECTIVE GLOVES AROUND CORROSION-INHIBITING COMPOUND. IF YOU DO NOT OBEY, SKIN IRRITATION CAN OCCUR.

3) Apply a thin layer of P/S 870 Class B-1/2 sealant, A50250, or P/S 870 Class B-2 sealant, A50479, to head of the screw [4].

(i) Install remaining screws [4] to the mounting holes at the wide end of the chassis assembly [1] and inner mounting hole at the forward end of the chassis assembly [1].

WARNING USE PROTECTIVE GLOVES AROUND CORROSION-INHIBITING COMPOUND. IF YOU DO NOT OBEY, SKIN IRRITATION CAN OCCUR.

1) Apply a thin layer of P/S 870 Class B-1/2 sealant, A50250, or P/S 870 Class B-2 sealant, A50479, to underside of heads of screws [4].

2) Fully tighten the screws [4].

WARNING USE PROTECTIVE GLOVES AROUND CORROSION-INHIBITING COMPOUND. IF YOU DO NOT OBEY, SKIN IRRITATION CAN OCCUR.

3) Apply a thin layer of P/S 870 Class B-1/2 sealant, A50250, or P/S 870 Class B-2 sealant, A50479, to heads of the screws [4].

WARNING USE PROTECTIVE GLOVES AROUND RTV ADHESIVE SEALANT. IF YOU DO NOT OBEY, SKIN IRRITATION CAN OCCUR.

(j) Apply RTV 103 adhesive, A50119, or RTV 732 adhesive, A50416, to the gap between the LED optic assembly [6] and tipcap.

(k) Install the aft marker light assembly, do this task: Aft Marker Light Installation, TASK 33-43-12-400-801.

H. LED Optic Assembly Installation Test

SUBTASK 33-43-12-860-009

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	12	C04052	EXTERIOR LIGHTING POSITION RIGHT
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT

SUBTASK 33-43-12-710-002

(2) Do a test of the LED optic assembly [6], do this task: Position Lights Operational Test, TASK 33-43-10-710-801.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 33-43-12-840-002

(1) Remove the ladder, work platform, COM-2480, or maintenance platform, COM-659.

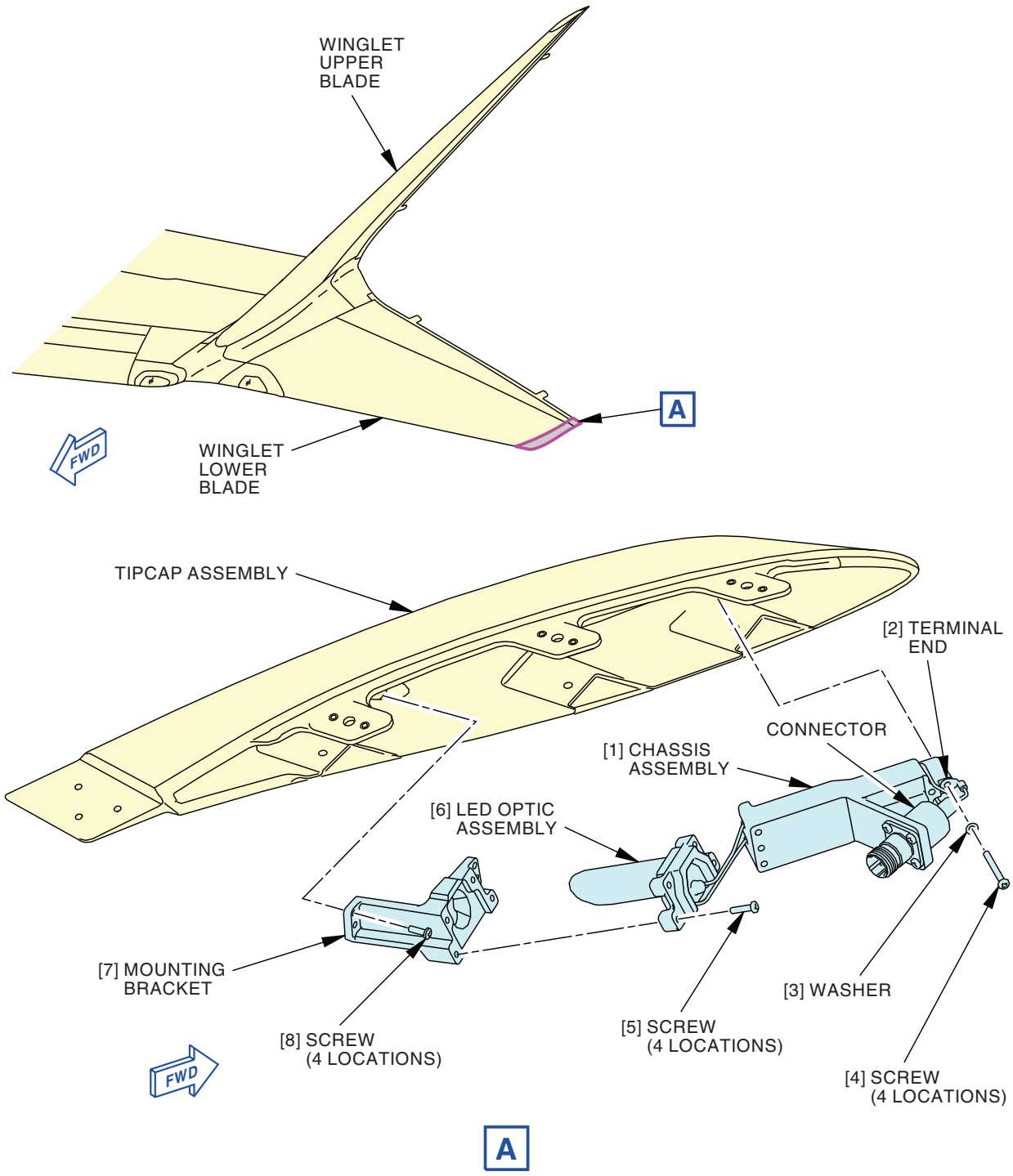
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-43-12-860-010

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Aft Marker Light LED Optic Assembly Replacement
Figure 201/33-43-12-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WINGLET AFT MARKER LIGHTS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Winglet Aft Marker Light Removal

(2) Winglet Aft Marker Light Installation.

TASK 33-43-12-000-801

2. Aft Marker Light Removal

(Figure 401)

A. General

(1) This procedure gives the task to remove the Aft Marker Light from the lower blade of the winglet.

B. References

Reference	Title
57-31-24-000-801	Winglet Erosion Protection Removal (P/B 401)
57-31-24-000-802	Winglet Erosion Protection Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-659	Platform - Maintenance Lift, 3 ft Minimum Height, 12.5 ft Maximum Height Part #: 1135-8000 Supplier: 00994 Opt Part #: 8662-010 Supplier: 00994
COM-2480	Platform - Mobile Elevating Work Platform SJ III Series Part #: SJIII 4726 Supplier: 3AF08 Part #: SJIII 4732 Supplier: 3AF08 Opt Part #: SJIII 4620 Supplier: 3AF08 Opt Part #: SJIII 4626 Supplier: 3AF08 Opt Part #: SJIII 4632 Supplier: 3AF08
SPL-768	Sealant Removal Tool, Non-Metallic Part #: ST982L-9 Supplier: 81205 Part #: ST982LF-9 Supplier: 81205 Part #: ST982LH-A-1 Supplier: 81205

D. Location Zones

Zone	Area
500	Left Wing
527	Left Winglet
600	Right Wing
627	Right Winglet

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Removal

SUBTASK 33-43-12-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	12	C04052	EXTERIOR LIGHTING POSITION RIGHT
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT

SUBTASK 33-43-12-940-001

(2) Get a ladder, work platform, COM-2480, or maintenance platform, COM-659.

SUBTASK 33-43-12-860-011

(3) Do this task: Winglet Erosion Protection Removal, TASK 57-31-24-000-801 or Winglet Erosion Protection Removal, TASK 57-31-24-000-802

SUBTASK 33-43-12-940-002

(4) Do these steps to prepare metal support equipment such as work platforms, work/maintenance stands, ladders.

NOTE: These steps apply to all metal support equipment within a 50 ft (15 m) radius of an open fuel tank.

(a) All support equipment must be in position before you begin the procedure.

(b) Bond the support equipment at an approved airplane bonding location.

(c) Ground the support equipment to the same earth ground as the airplane.

F. Aft Marker Light Removal

SUBTASK 33-43-12-980-001

(1) Remove and discard the rivets [43] that attach the aft end of the aft marker light [44] to the arrowhead [41].

SUBTASK 33-43-12-020-001

(2) Remove the aft marker light [44].

(a) Remove the upper bolts [42] and the lower bolts [46] that attach the aft marker light [44] to the lower blade of the winglet.

(b) Remove the aft marker light [44] from the lower blade.

SUBTASK 33-43-12-020-002

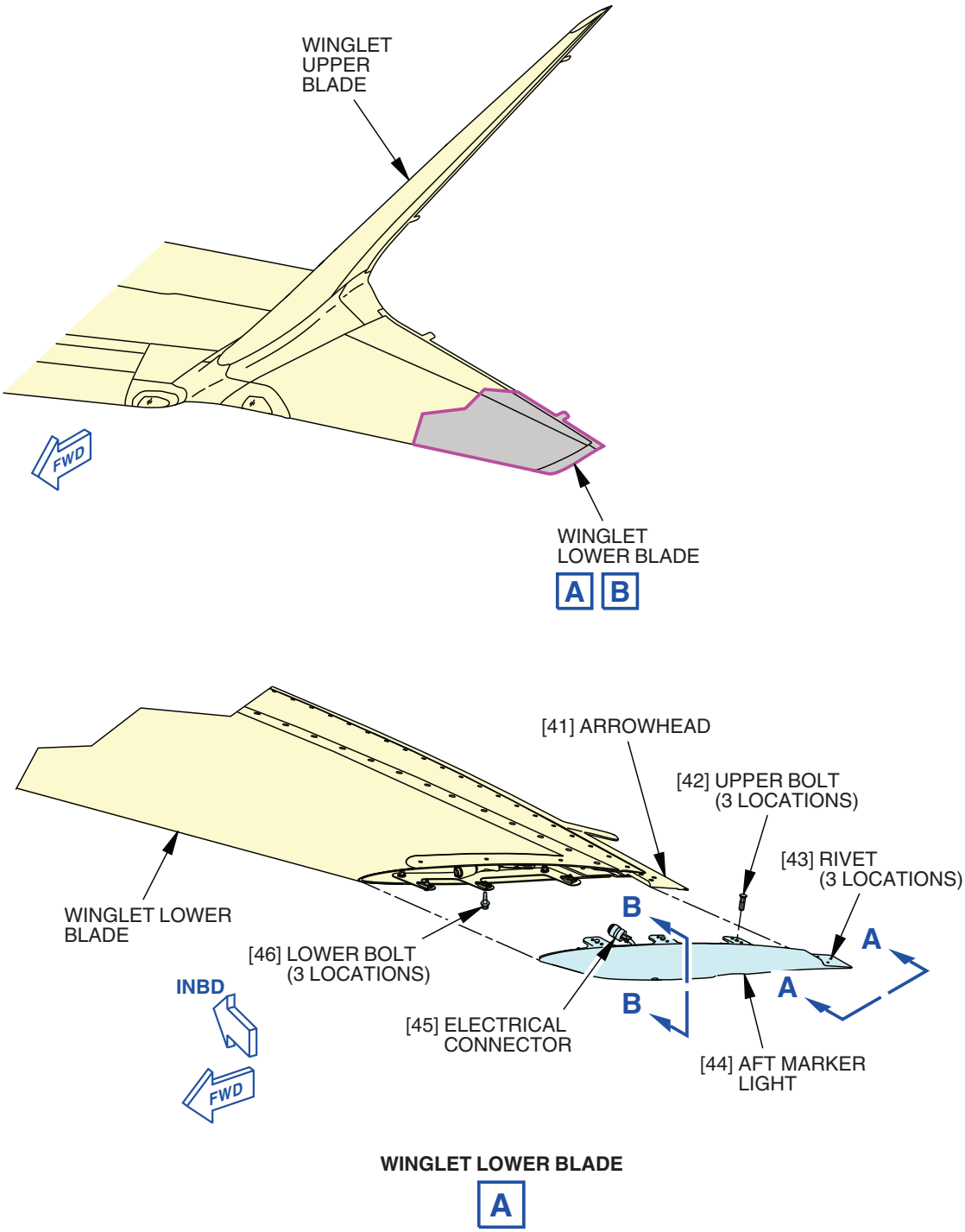
(3) Disconnect the electrical connector [45] from the aft marker light [44] .

SUBTASK 33-43-12-020-003

(4) Remove all sealant from lower blade of the winglet and the aft marker light [44] with a sealant removal tool, SPL-768.

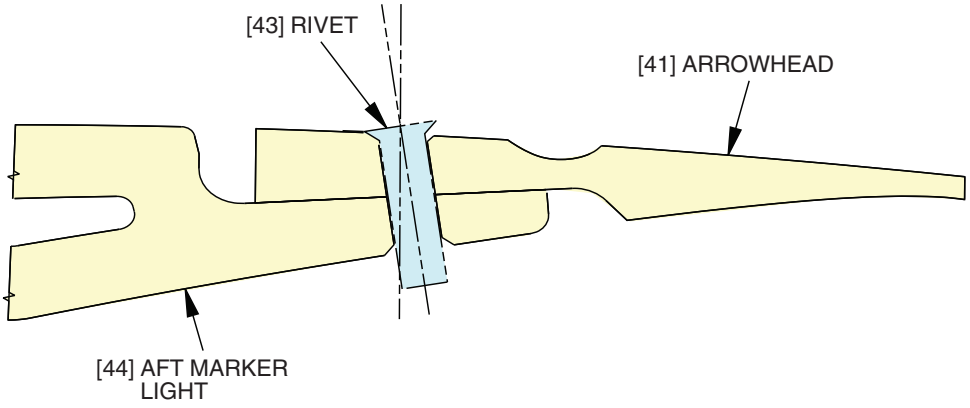
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

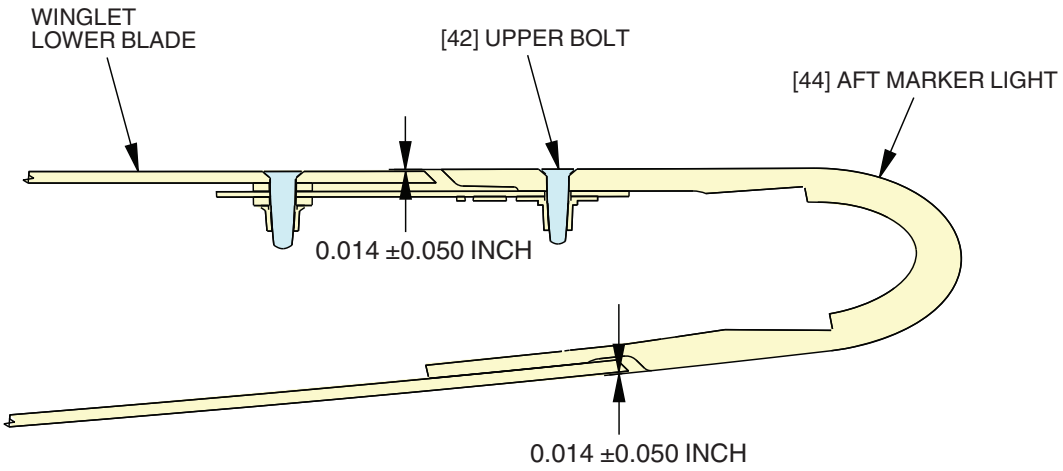


Aft Marker Light Installation
Figure 401/33-43-12-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A-A



B-B

2453490 S0000570282_V2

Aft Marker Light Installation
Figure 401/33-43-12-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-43-12-400-801

3. Aft Marker Light Installation

(Figure 401)

A. General

(1) This procedure gives the task to install the Aft Marker Light on the lower blade of the winglet.

B. References

Reference	Title
33-43-10-710-801	Position Lights Operational Test (P/B 201)
57-31-21-390-801	Winglet Aerodynamic Sealant Repair (P/B 801)
57-31-24-400-801	Winglet Erosion Protection Installation (P/B 401)
57-31-24-400-802	Winglet Erosion Protection Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-659	Platform - Maintenance Lift, 3 ft Minimum Height, 12.5 ft Maximum Height Part #: 1135-8000 Supplier: 00994 Opt Part #: 8662-010 Supplier: 00994
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
COM-2480	Platform - Mobile Elevating Work Platform SJ III Series Part #: SJIII 4726 Supplier: 3AF08 Part #: SJIII 4732 Supplier: 3AF08 Opt Part #: SJIII 4620 Supplier: 3AF08 Opt Part #: SJIII 4626 Supplier: 3AF08 Opt Part #: SJIII 4632 Supplier: 3AF08

D. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
A50528	Sealant - Fast Cure	BMS5-184
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
43	Rivet	33-43-12-01-050	PTF ALL
		33-43-12-05-050	PTF ALL
44	Aft marker light	33-43-12-01-030	PTF ALL
		33-43-12-01-035	PTF ALL

F. Location Zones

Zone	Area
500	Left Wing
527	Left Winglet
600	Right Wing
627	Right Winglet

G. Aft Marker Light Installation

SUBTASK 33-43-12-370-001

(1) Do a check of the fastener holes on the upper surface of the lower blade:

(a) If it is necessary, apply primer, C00259, to the fastener holes.

SUBTASK 33-43-12-400-001

(2) Install the aft marker light [44]:

(a) Apply a layer of corrosion inhibiting material, G50136, to the mating surfaces of the aft marker light [44], lower blade of the winglet and arrowhead [41].

(b) Apply a layer of corrosion inhibiting material, G50136, to the shank and threads of the upper bolts [42] and the holes for the lower bolts [46].

(c) Connect the electrical connector [45] to the aft marker light [44].

(d) Put the aft marker light [44] in its position on the lower blade.

(e) Install the upper bolts [42] and the lower bolts [46] to attach the aft marker light [44].

(f) Tighten the upper bolts [42] to 30 in-lb (3.4 N·m).

(g) Tighten the lower bolts [46] to 15 in-lb (1.7 N·m).

(h) Install the new rivets [43] to attach the aft marker light [44] to the arrowhead [41].

SUBTASK 33-43-12-980-002

(3) Make sure that the aft marker light [44] and the lower blade align in the limit of 0.0140 ±0.0500 in. (0.3556 ±1.2700 mm).

SUBTASK 33-43-12-760-001

(4) Do a check of the electrical resistance between the aft marker light [44] and the arrowhead [41].

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the electrical resistance is 0.0005 ohm (0.5 milliohm) or less.

SUBTASK 33-43-12-390-001

(5) Apply sealant, A02315, or sealant, A50528, between the lower blade of the winglet and aft marker light [44] (TASK 57-31-21-390-801).

SUBTASK 33-43-12-980-003

(6) Shave the heads of the rivets [43] so they align with the arrowhead [41] +0.0020 in. (0.0508 mm)/ -0.0000 in. (0.0000 mm).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Aft Marker Light Installation Test

SUBTASK 33-43-12-860-004

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	12	C04052	EXTERIOR LIGHTING POSITION RIGHT
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT

SUBTASK 33-43-12-710-001

(2) Do this task: Position Lights Operational Test, TASK 33-43-10-710-801.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 33-43-12-860-012

(1) Do this task: Winglet Erosion Protection Installation, TASK 57-31-24-400-801 or Winglet Erosion Protection Installation, TASK 57-31-24-400-802.

SUBTASK 33-43-12-840-001

(2) Remove the ladder, work platform, COM-2480, or maintenance platform, COM-659.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TAILCONE POSITION LIGHTS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Tailcone Position Lights Deactivation

(2) Tailcone Position Lights Activation

(3) Tailcone Position Lights Replacement

(4) Tailcone Position Lights Transformer Replacement

(5) Tailcone Position Lights LED Module Replacement

(6) Tailcone Position Lights Housing Assembly Replacement.

B. The tailcone position light and tailcone anti-collision light are in the same light assembly.

TASK 33-43-13-040-801

2. Tailcone Position Lights Deactivation

A. General

(1) This procedure removes electrical power from the tailcone position lights.

B. Location Zones

Zone	Area
310	Fuselage - Body Station 1016.00 to Body Station 1260.00
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

C. Procedure

SUBTASK 33-43-13-860-009

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

D. Tailcone Position Lights Deactivation Test

NOTE: This test is to make sure that the tailcone position lights are in a zero energy condition.

SUBTASK 33-43-13-200-001

(1) Do these steps to make sure that the power is off:

(a) Set the POSITION switch for the tailcone position/anti-collision lights to STEADY on the P5 overhead panel.

1) Make sure that the tailcone position/anti-collision, winglet position and winglet (AFT) marker lights stay off.

(b) Set the POSITION switch for the tailcone position/anti-collision lights to STROBE & STEADY on the P5 overhead panel.

1) Make sure that the tailcone position/anti-collision, winglet position and winglet (AFT) marker lights stay off.

(c) Set the POSITION switch for the tailcone position/anti-collision lights to OFF on the P5 overhead panel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the tailcone position/anti-collision, winglet position and winglet (AFT) marker lights stay off.

——— END OF TASK ———

TASK 33-43-13-440-801

3. Tailcone Position Lights Activation

A. General

(1) This procedure supplies electrical power to the tailcone position lights.

B. Location Zones

Zone	Area
310	Fuselage - Body Station 1016.00 to Body Station 1260.00
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

C. Procedure

SUBTASK 33-43-13-860-010

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

D. Tailcone Position Lights Activation Test

NOTE: This test is to make sure that the tailcone position lights operate.

SUBTASK 33-43-13-710-005

(1) Do these steps to make sure that the power is on:

(a) Set the POSITION switch for the tailcone position/anti-collision lights to STEADY on the P5 overhead panel.

1) Make sure that the tailcone position/anti-collision, winglet position and winglet (AFT) marker lights come on.

(b) Set the POSITION switch for the tailcone position/anti-collision lights to STROBE & STEADY on the P5 overhead panel.

1) Make sure that the tailcone position/anti-collision, winglet position and winglet (AFT) marker lights come on.

(c) Set the POSITION switch for the tailcone position/anti-collision lights to OFF.

1) Make sure that the tailcone position/anti-collision, winglet position and winglet (AFT) marker lights go off.

——— END OF TASK ———

TASK 33-43-13-960-801

4. Tailcone Position Lights Replacement

(Figure 201)

A. References

Reference	Title
51-22-00-160-801	Prepare For Sealing (P/B 201)

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(Continued)

Reference	Title
51-22-00-390-809	Aerodynamic Smoother Application (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
WDM 33-43-11	Wiring Diagram Manual

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
STD-810	Spatula - Fillet Smoothing, Hardwood or Plastic

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
6	Light assembly	33-44-04-02-045	PTF ALL

E. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

F. Prepare for the Replacement

SUBTASK 33-43-13-860-024

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

SUBTASK 33-43-13-860-025

(2) Make sure that the Auxiliary Power Unit (APU) master switch, on the P5 forward overhead panel, is in the OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Install the DO NOT OPERATE tag, STD-858, on the APU master switch.

SUBTASK 33-43-13-860-026

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

G. Tailcone Position Lights Replacement

SUBTASK 33-43-13-000-001

(1) Do these steps to replace the light assembly (WDM 33-43-11):

NOTE: Do not discard the light assembly. Refer to the vendor manual for possible repairs. Do the repairs in a shop.

(a) Remove the screws [7].

(b) Disconnect the electrical connector [10].

(c) Replace the light assembly [6].

(d) Connect the electrical connector [10]:

1) Clean the electrical connector [10] (SWPM 20-60-01).

2) Connect the electrical connector [10] (SWPM 20-60-06).

3) Measure the electrical bonding resistance between the electrical connector braid and housing (SWPM 20-20-00).

a) Use an intrinsically safe approved bonding meter, COM-1550.

b) Make sure that the electrical bonding resistance is 0.003 ohm (3 milliohms) or less.

(e) Abrasive clean the sealant from the light assembly [6] and fuselage surface (TASK 51-22-00-160-801).

(f) Apply sealant, A00247, between the light assembly [6] and the fuselage for an electrical faying surface seal.

(g) Install the light assembly [6] to the fuselage with the screws [7].

(h) Measure the electrical bonding resistance between the metal components of the light assembly [6] and fuselage surface (SWPM 20-20-00).

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the electrical bonding resistance is 0.0005 ohm (0.5 milliohms) or less.

(i) Apply an aerodynamic fillet seal around the baseplate [9] with sealant, A00247 (TASK 51-22-00-390-809).

(j) Use the hardwood or plastic fillet smoothing spatula, STD-810, to make a smooth 45-degree fillet.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

DO NOT SEAL LOWER DRAIN HOLE OF THE TAILCONE ANTI-COLLISION LIGHT. LOWER DRAIN HOLE TO BE CLEAR OF ALL SEALANT.

(k) Apply sealant, A00247, to the heads of the screws [7] and the upper drain hole (TASK 51-22-00-390-809).

(l) If the lens is dirty, clean it with a clean, soft, dry cotton wiper, G00034.

H. Tailcone Position Lights Replacement Test

SUBTASK 33-43-13-860-001

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

SUBTASK 33-43-13-710-001

(2) Do a test of the new light assembly [6].

(a) Set the POSITION switch for the tailcone position/anti-collision light, on the P5 overhead panel, to the STROBE & STEADY mode.

1) Make sure that the tailcone position/anti-collision light (left/right) flashes continuously.

(b) Set the POSITION switch for the tailcone position/anti-collision light, on the P5 overhead panel, to the STEADY mode.

1) Make sure that the tailcone position/anti-collision light (left/right) comes on and stays on.

(c) Set the POSITION switch to the OFF mode.

1) Make sure that the tailcone position/anti-collision lights (left/right) goes off.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 33-43-13-860-022

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

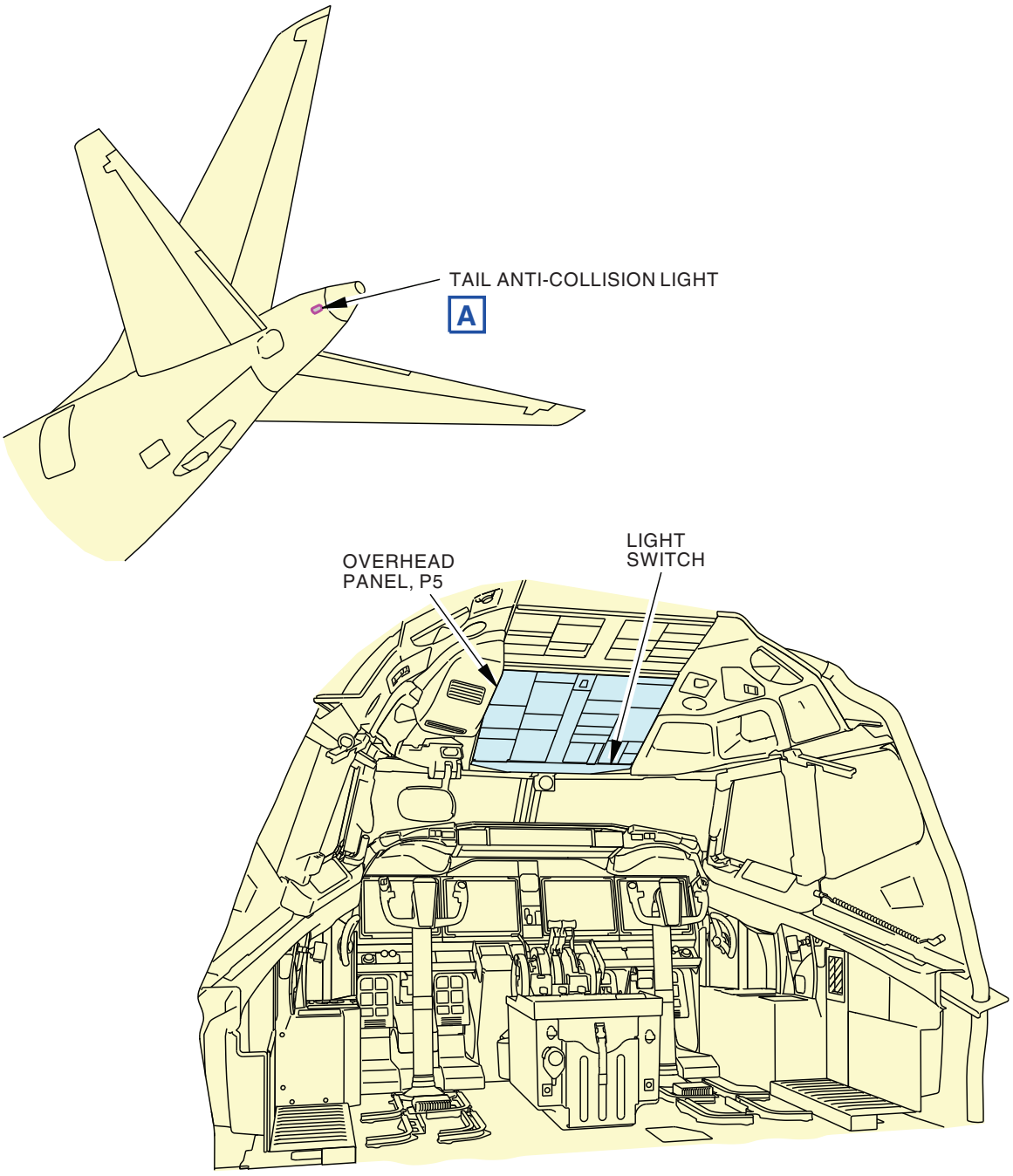
Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 33-43-13-860-023

(2) Remove the DO NOT OPERATE tag from the APU master switch, on the P5 forward overhead panel.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

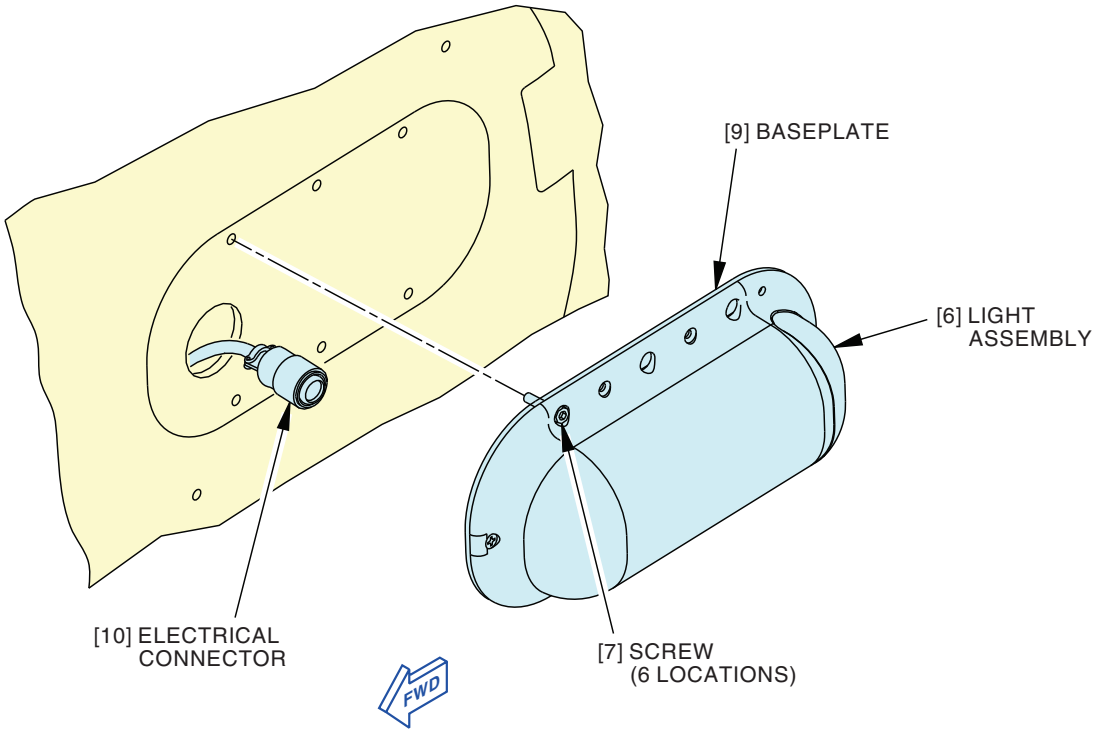


FLIGHT COMPARTMENT

2414735 S00061535859_V1

Tailcone Position - Lights Assembly Replacement
Figure 201/33-43-13-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TAIL ANTI-COLLISION LIGHT
(LEFT SIDE SHOWN, RIGHT SIDE OPPOSITE)

A

2414736 S00061535860_V1

Tailcone Position - Lights Assembly Replacement
Figure 201/33-43-13-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-43-13-960-802

5. Tailcone Position Lights Transformer Replacement

(Figure 202)

A. References

Reference	Title
WDM 33-43-11	Wiring Diagram Manual

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Consumable Materials

Reference	Description	Specification
A50473	Sealant - Brush (BMS5-95 Type I Class A-2 (See BAC5000 8.5.3 a. for preparation instructions))	BAC5000 Section 8.5.3 a.
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Transformer	33-43-13-01-010	PTF ALL

E. Location Zones

Zone	Area
310	Fuselage - Body Station 1016.00 to Body Station 1260.00
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

F. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Prepare for the Replacement

SUBTASK 33-43-13-860-003

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

SUBTASK 33-43-13-010-001

(2) Open this access door:

Number	Name/Location
311BL	Stabilizer Trim Access Door

H. Tailcone Position Lights Transformer Replacement

SUBTASK 33-43-13-960-001

(1) Do these steps to remove the transformer [1] (WDM 33-43-11):

(a) Disconnect the electrical connector [2].

(b) Remove the sealant between the transformer [1] and the mounting bracket.

(c) Remove the bolts [3] and washers [4].

(d) Remove the transformer [1].

(2) Do these steps to install a new transformer [1]:

(a) Clean the fay surface bonding area with solvent, B00083, and a clean cotton wiper, G00034.

(b) Put the new transformer [1] on the mounting bracket.

(c) Install the washers [4] and bolts [3].

(d) With the intrinsically safe approved bonding meter, COM-1550, make sure that the resistance between the transformer [1] and the fuselage measures 0.0025 ohm (2.5 milliohms) or less.

(e) Apply a polysulfide sealant, A50473, fillet seal between transformer [1] and the mounting bracket.

(f) Connect the electrical connector [2].

1) Clean the electrical connector [2] (SWPM 20-60-01).

2) Connect the electrical connector [2] (SWPM 20-60-06).

3) Use an intrinsically safe approved bonding meter, COM-1550, to measure the resistance between the electrical connector [2] braid and the housing (SWPM 20-20-00).

a) Make sure that the resistance is 0.003 ohm (3 milliohms) or less.

I. Tailcone Position Lights Transformer Replacement Test

SUBTASK 33-43-13-860-004

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

SUBTASK 33-43-13-710-002

(2) Do a test of the new transformer [1]:

(a) At the P5 overhead panel, set the POSITION switch for the tailcone position/anti-collision light to the STEADY mode.

1) Make sure that the forward position, winglet marker (aft), and tailcone position/anti-collision lights come on.

(b) At the P5 overhead panel, set the POSITION switch for the tailcone position/anti-collision light to the STROBE & STEADY mode.

1) Make sure that the forward position, winglet marker (aft), and tailcone position/anti-collision lights come on.

(c) Set the POSITION switch to the OFF mode.

1) Make sure that the forward position, winglet marker (aft), and tailcone position/anti-collision lights go off.

J. Put the Airplane Back to Its Usual Condition

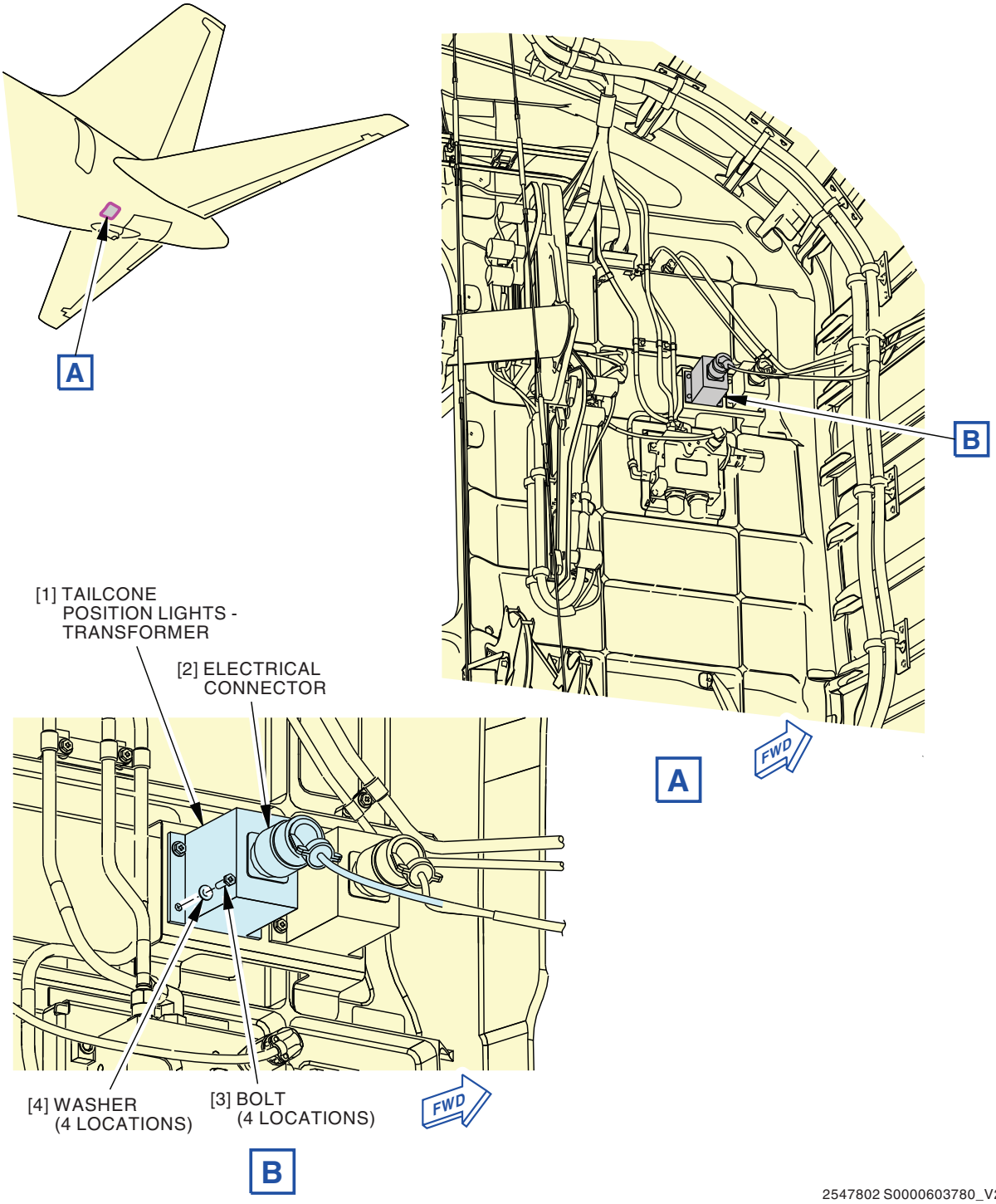
SUBTASK 33-43-13-410-001

(1) Close this access door:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2547802 S0000603780_V2

Tailcone Position Light Transformer Replacement
Figure 202/33-43-13-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-43-13-960-803

6. Tailcone Position Lights LED Module Replacement

(Figure 203)

A. General

(1) This procedure contains instructions for the replacement of the LED modules installed on the tailcone position light.

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
20	LED module assembly	33-44-04-02-050	PTF ALL
30	Housing assembly	33-44-04-02-055	PTF ALL

D. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

E. Prepare for the Replacement

SUBTASK 33-43-13-860-005

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

SUBTASK 33-43-13-860-016

(2) Make sure that the Auxiliary Power Unit (APU) master switch, on the P5 forward overhead panel, is in the OFF position.

(a) Install the DO NOT OPERATE tag, STD-858, on the APU master switch.

SUBTASK 33-43-13-860-017

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

F. Tailcone Position Lights LED Module Replacement

SUBTASK 33-43-13-020-001

(1) Do these steps to remove the housing assembly [30]:

(a) Use white cotton gloves to handle housing assembly and LED modules.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Do this task: Tailcone Position Lights Housing Assembly Replacement, TASK 33-43-13-960-804.

SUBTASK 33-43-13-960-002

(2) Do these steps to replace the LED module assembly [20]:

(a) Use white cotton gloves to handle housing assembly and LED modules.

(b) Remove the screws [21] that attach the LED module assembly [20] to the housing assembly [30].

(c) Pull the LED module assembly [20] to disconnect it from the housing assembly [30].

(d) Replace the LED module assembly [20].

(e) Install the screws [21] that hold the LED module assembly [20] to the housing assembly [30].

(f) Tighten the screws [21] untill fully seated.

SUBTASK 33-43-13-420-001

(3) Do these steps to install the housing assembly [30]:

(a) Use white cotton gloves to handle the housing assembly and LED modules.

(b) Do this task: Tailcone Position Lights Housing Assembly Replacement, TASK 33-43-13-960-804.

SUBTASK 33-43-13-860-018

(4) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

SUBTASK 33-43-13-710-003

(5) Do a test of the new LED module assembly [20]:

(a) At the P5 overhead panel, set the POSITION switch for the tailcone position/anti-collision light to the STEADY mode.

1) Make sure that the forward position, winglet marker (aft), and tailcone position/anti-collision lights come on.

(b) At the P5 overhead panel, set the POSITION switch for the tailcone position/anti-collision light to the STROBE & STEADY mode.

1) Make sure that the forward position, winglet marker (aft), and tailcone position/anti-collision lights come on.

(c) Set the POSITION switch to the OFF mode.

1) Make sure that the forward position, winglet marker (aft), and tailcone position/anti-collision lights go off.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 33-43-13-860-019

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-4

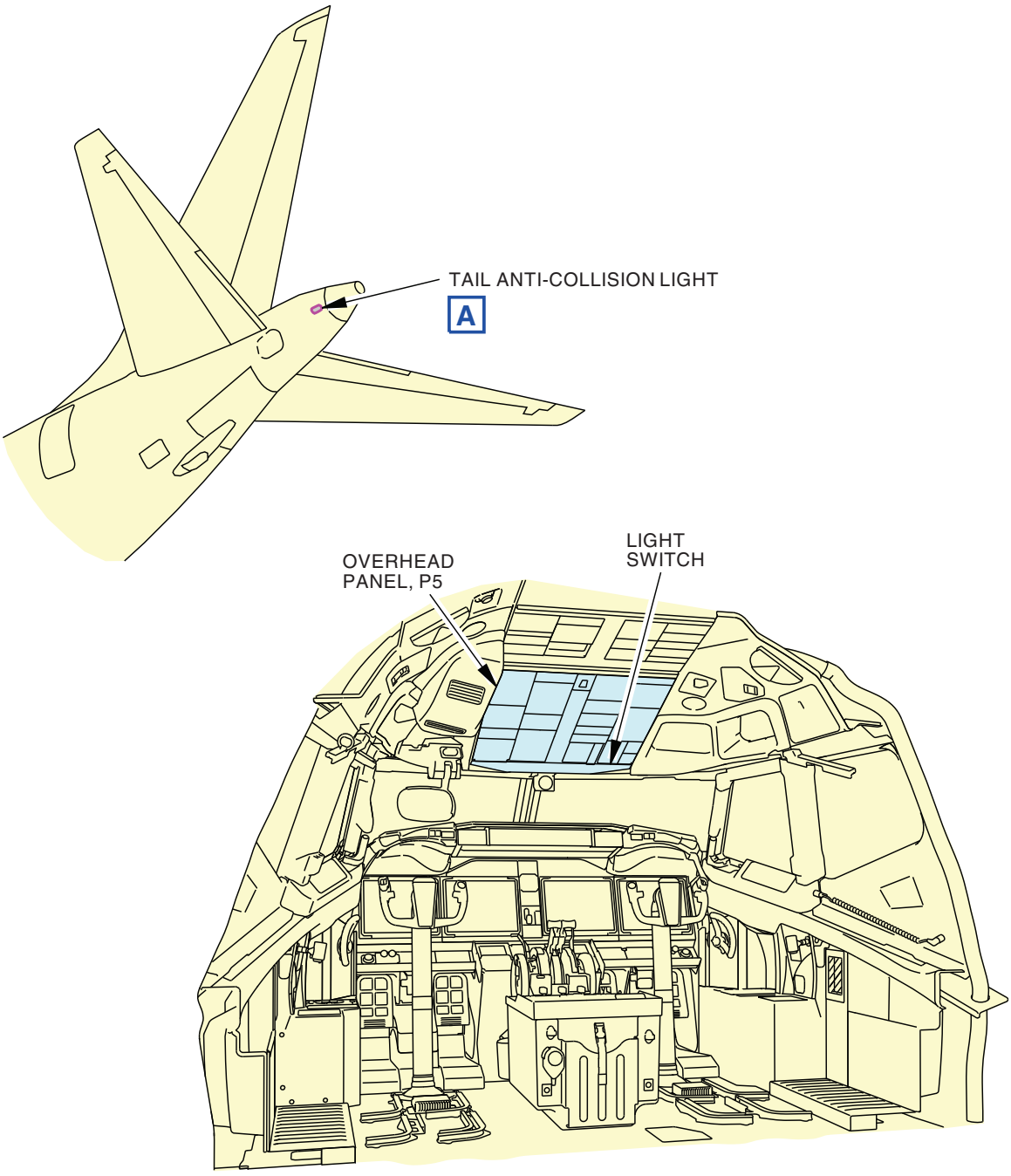
Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 33-43-13-860-020

(2) Remove the DO NOT OPERATE tag from the APU master switch, on the P5 forward overhead panel.

——— END OF TASK ———

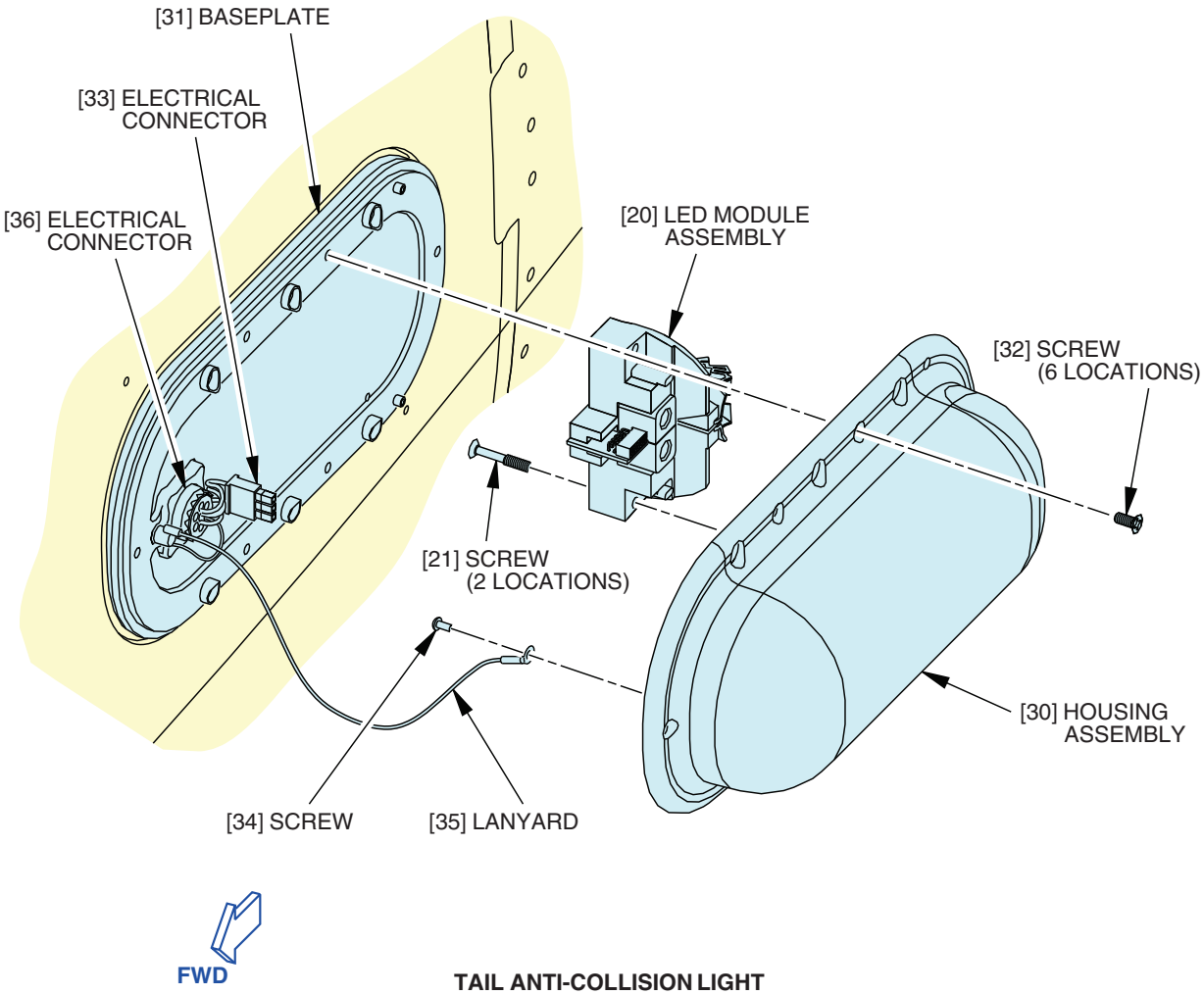
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414735 S00061535859_V1

Tailcone Position Light LED Module Replacement
Figure 203/33-43-13-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TAIL ANTI-COLLISION LIGHT
(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

A

2570017 S0000615941_V2

Tailcone Position Light LED Module Replacement
Figure 203/33-43-13-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-43-13-960-804

7. Tailcone Position Lights Housing Assembly Replacement

(Figure 203)

A. General

(1) This procedure contains instructions for the replacement of the housing assembly installed on the tailcone position light.

B. References

Reference	Title
51-22-00-390-809	Aerodynamic Smoother Application (P/B 201)

C. Tools/Equipment

Reference	Description
STD-810	Spatula - Fillet Smoothing, Hardwood or Plastic
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
30	Housing assembly	33-44-04-02-055	PTF ALL

F. Location Zones

Zone	Area
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

G. Prepare for the Replacement

SUBTASK 33-43-13-860-007

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

SUBTASK 33-43-13-860-011

(2) Make sure that the Auxiliary Power Unit (APU) master switch, on the P5 forward overhead panel, is in the OFF position.

(a) Install the DO NOT OPERATE tag, STD-858, on the APU master switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-43-13-860-012

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

H. Tailcone Position Lights Housing Assembly Replacement

SUBTASK 33-43-13-960-003

(1) Do these steps to replace the housing assembly [30]:

(a) Use white cotton gloves to handle the housing assembly and LED modules.

(b) Remove the screws [32] that attach the housing assembly [30] to the baseplate [31].

(c) Disconnect the electrical connector [33].

(d) Pull the housing assembly [30] to disconnect it from the baseplate [31].

(e) Replace the new housing assembly [30].

(f) Connect the electrical connector [33].

(g) Install the screws [32] that hold the housing assembly [30] to the baseplate [31].

(h) Clean the housing assembly [30] with a clean, soft, dry cotton wiper, G00034.

(i) Apply an aerodynamic fillet seal around the baseplate [31] with sealant, A00247 (TASK 51-22-00-390-809).

(j) Use the hardwood or plastic fillet smoothing spatula, STD-810, to make a smooth 45-degree fillet.

CAUTION DO NOT SEAL LOWER DRAIN HOLE OF THE TAILCONE ANTI-COLLISION LIGHT. LOWER DRAIN HOLE TO BE CLEAR OF ALL SEALANT.

(k) Apply sealant, A00247, to the heads of the screws [32] and upper drain hole(TASK 51-22-00-390-809).

(l) If the lens is dirty, clean it with a clean, soft, dry cotton wiper, G00034.

SUBTASK 33-43-13-860-013

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

SUBTASK 33-43-13-710-004

(3) Do a test of the new housing assembly [30]:

(a) At the P5 overhead panel, set the POSITION switch for the tailcone position/anti-collision light to the STEADY mode.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the forward position, winglet marker (aft), and tailcone position/anti-collision lights come on.

(b) At the P5 overhead panel, set the POSITION switch for the tailcone position/anti-collision light to the STROBE & STEADY mode.

1) Make sure that the forward position, winglet marker (aft), and tailcone position/anti-collision lights come on.

(c) Set the POSITION switch to the OFF mode.

1) Make sure that the forward position, winglet marker (aft), and tailcone position/anti-collision lights go off.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 33-43-13-860-014

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 33-43-13-860-015

(2) Remove the DO NOT OPERATE tag from the APU master switch, on the P5 forward overhead panel.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ANTI-COLLISION LIGHTS - TEST

1. General

A. This procedure has one task:

(1) Anti-Collision Lights - Operational Test.

TASK 33-44-00-710-801

2. Anti-Collision Lights - Operational Test

(Figure 501)

A. General

(1) This test makes sure that the anti-collision lights operate.

B. References

Reference	Title
WDM 33-44-11	Wiring Diagram Manual
WDM 33-44-12	Wiring Diagram Manual

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Procedure

SUBTASK 33-44-00-710-001

WARNING DO NOT LET THE LIGHTS FLASH DIRECTLY IN YOUR EYES. THE INTENSITY OF THE LIGHT CAN TEMPORARILY CAUSE INJURIES TO YOUR EYES.

(1) Do a test of the upper and lower anti-collision lights (WDM 33-44-11):

(a) At the overhead panel, P5, set the ANTI COLLISION switch for the upper and lower anti-collision lights to the ON position.

NOTE: FOR THE UPPER, LOWER, OR WING ANTI-COLLISION LIGHT:

Let the light flash for 20 seconds to stabilize the system. If no misfire occurs, and the light flashed 40 to 45 times a minute, the system operates correctly. If a misfire occurs before the system becomes stable, let the light flash for 60 more seconds. There must be no other misfires after the system becomes stable for the system to operate correctly.

1) Make sure that the red anti-collision lights on the top and bottom of the fuselage flash 40 to 45 times a minute.

(b) Set the ANTI COLLISION switch to the OFF position.

1) Make sure that lights go off.

SUBTASK 33-44-00-710-002

WARNING DO NOT LET THE LIGHTS FLASH DIRECTLY IN YOUR EYES. THE INTENSITY OF THE LIGHT CAN TEMPORARILY CAUSE INJURIES TO YOUR EYES.

(2) Do a test of the wing and tail anti-collision lights (WDM 33-44-12):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) At the overhead panel, P5, set the POSITION switch for wing and tail anti-collision lights to the STROBE & STEADY position.

NOTE: FOR THE UPPER, LOWER, OR WING ANTI-COLLISION LIGHT:

Let the light flash for 20 seconds to stabilize the system. If no misfire occurs, and the light flashed 40 to 45 times a minute, the system operates correctly. If a misfire occurs before the system becomes stable, let the light flash for 60 more seconds. There must be no other misfires after the system becomes stable for the system to operate correctly.

1) Make sure that the white anti-collision lights on each wing flash 40 to 45 times a minute.

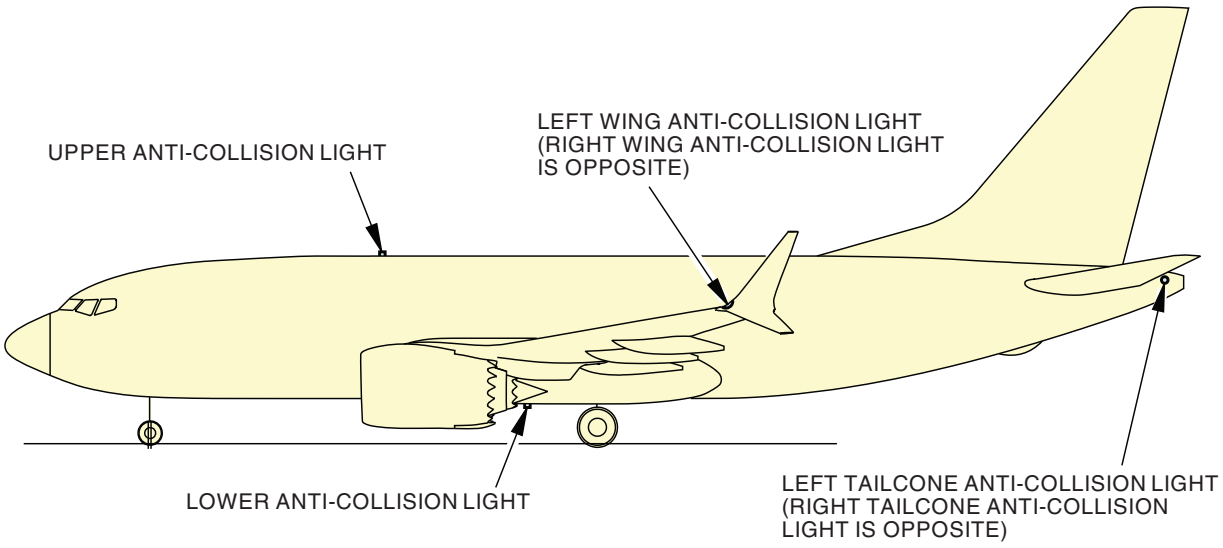
2) Make sure that the white anti-collision lights on each side of the tail flash 40 to 45 times a minute.

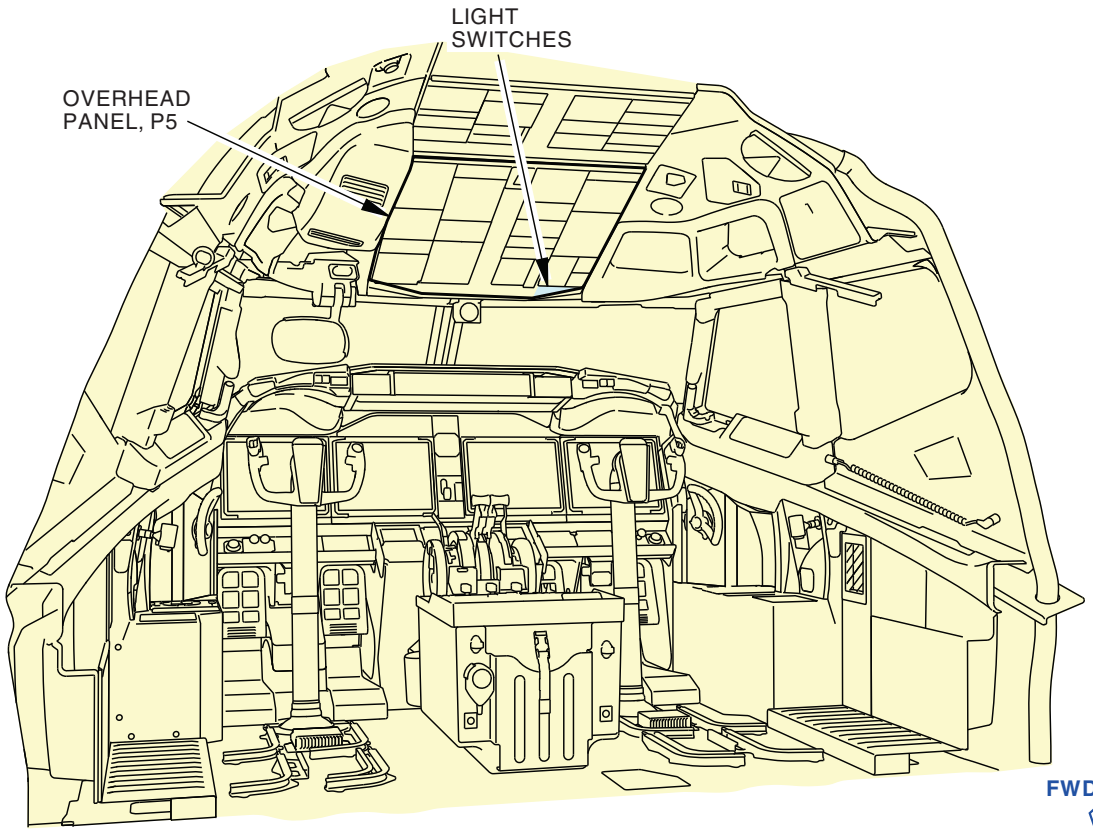
(b) Set the POSITION switch to the OFF position.

1) Make sure that the lights go off.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





FLIGHT COMPARTMENT

2414708 S00061535811_V2

Anti-Collision Lights - Test
Figure 501/33-44-00-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

UPPER ANTI-COLLISION LIGHT - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A deactivation of the upper anti-collision light

(2) An activation of the upper anti-collision light

(3) A replacement of the upper anti-collision light lamp assembly

(4) A replacement of the upper anti-collision light light assembly

(5) A replacement of the upper anti-collision light lens

(6) A replacement of the upper anti-collision light power supply

(7) A replacement of the upper anti-collision light adapter plate.

TASK 33-44-01-040-801

2. Upper Anti-Collision Light Deactivation

A. General

(1) This procedure removes electrical power from the upper anti-collision light.

B. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

C. Procedure

SUBTASK 33-44-01-860-015

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

D. Upper Anti-Collision Light Deactivation Test

NOTE: This test is to make sure that the upper anti-collision light is in a zero energy condition.

SUBTASK 33-44-01-200-001

(1) Do these steps to make sure that the power is off:

(a) On the P5 overhead panel, set the ANTI COLLISION switch for the upper anti-collision light to ON position.

1) Make sure that the upper anti-collision light stays off.

(b) Set the ANTI COLLISION switch for the upper anti-collision light to OFF position.

1) Make sure that the upper anti-collision light stays off.

——— END OF TASK ———

TASK 33-44-01-440-801

3. Upper Anti-Collision Light Activation

A. General

(1) This procedure supplies electrical power to the upper anti-collision light.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

C. Procedure

SUBTASK 33-44-01-860-016

(1) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

D. Upper Anti-Collision Light Activation Test

NOTE: This test is to make sure that the upper anti-collision light operates.

SUBTASK 33-44-01-710-005

(1) Do these steps to make sure that the power is on:

(a) On the P5 overhead panel, set the ANTI COLLISION switch for the upper anti-collision light to ON position.

1) Make sure that the upper anti-collision light comes on.

(b) Set the ANTI COLLISION switch for the upper anti-collision light to OFF position.

1) Make sure that the upper anti-collision light goes off.

——— END OF TASK ———

TASK 33-44-01-960-806

4. Upper Anti-Collision Light Lamp Assembly Replacement

(Figure 201)

A. General

(1) The lamp assembly in this procedure is also referred to as a reflector assembly by the vendor.

(2) The lamp assembly can be replaced from inside the airplane or from outside the airplane.

B. References

Reference	Title
21-23-01-000-801	Overhead Distribution Duct Removal (P/B 201)
21-23-01-400-801	Overhead Distribution Duct Installation (P/B 201)
24-22-00-860-801	Supply Electrical Power (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)
25-21-45-000-801	Main Ceiling Panel - Removal (P/B 401)
25-21-45-400-801	Main Ceiling Panel - Installation (P/B 401)
33-44-00-710-801	Anti-Collision Lights - Operational Test (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
WDM 33-44-11	Wiring Diagram Manual

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

F. Upper Anti-Collision Light Lamp Assembly Replacement

SUBTASK 33-44-01-860-013

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

SUBTASK 33-44-01-860-021

(2) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

SUBTASK 33-44-01-000-001

WARNING WAIT 10 MINUTES AFTER YOU REMOVE ELECTRICAL POWER AND BEFORE YOU OPEN THE ANTI-COLLISION LIGHT UPPER AND LOWER FUSELAGE STROBE. THE ELECTRICAL POWER CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) To replace the lamp assembly [22] from inside the airplane, do the following steps:

(a) Do one of the steps below to get access to the upper anti-collision light:

NOTE: The overhead distribution duct is installed below and near the upper anti-collision light, which makes light removal difficult.

1) To remove the ceiling panel only, do this task: Main Ceiling Panel - Removal, TASK 25-21-45-000-801.

2) To remove the ceiling panel and the section of the duct below the light, do this task: Overhead Distribution Duct Removal, TASK 21-23-01-000-801.

(b) Do these steps to replace the lamp assembly [22] (WDM 33-44-11):

NOTE: Do not discard the lamp assembly. Refer to the vendor manual for possible repairs. Do the repairs in a shop.

1) Loosen the bolts [25] on the bottom of the light assembly [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Turn the light assembly [3] clockwise and pull it down from the adapter plate [20].
3) Carefully loosen the fastener [21] on top of the lamp assembly [22].
4) Replace the lamp assembly [22].
5) Tighten the fastener [21].
6) If the lens [7] is dirty, clean it with a clean, soft, dry cotton wiper, G00034.
7) Put the light assembly [3] in the adapter plate [20] and turn it counterclockwise.
8) Tighten the bolts [25].
9) Measure the electrical resistance between the light assembly [3] and adapter plate [20] (SWPM 20-20-00).
   a) Use an intrinsically safe approved bonding meter, COM-1550.
   b) Make sure that resistance is less than 0.0015 ohm (1.5 miliohms) or less.

(c) Install the items that were removed for access to the light:
1) If the ceiling panel and the section of the duct below the light were removed, do this task: Overhead Distribution Duct Installation, TASK 21-23-01-400-801.
2) If the ceiling panel only was removed, do this task: Main Ceiling Panel - Installation, TASK 25-21-45-400-801.

SUBTASK 33-44-01-000-002

(4) To replace the lamp assembly [22] from outside the airplane do the following steps:
(a) For access to the lens [7], go out of the airplane, on top of the fuselage.
(b) Do these steps to remove the lens [7]:
1) Loosen the screws [26] and lift off the lens [7].
(c) Do these steps to replace the lamp assembly [22] (WDM 33-44-11):

NOTE: Do not discard the lamp assembly. Refer to the vendor manual for possible repairs. Do the repairs in a shop.

1) Loosen the fastener [21] on top of the lamp assembly [22].
2) Replace the lamp assembly [22].
3) Tighten the fastener [21].
4) If the lens [7] is dirty, clean it with a clean, soft, dry cotton wiper, G00034.
5) Put on the lens [7] and tighten the screws [26].

SUBTASK 33-44-01-860-014

(5) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

SUBTASK 33-44-01-860-022

(6) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 33-44-01-710-004

(7) Do a test of the new lamp assembly [22].
(a) At the P5 overhead panel, set the ANTI COLLISION switch for the upper anti-collision light to ON position.
1) Make sure that the upper anti-collision light flashes (TASK 33-44-00-710-801).

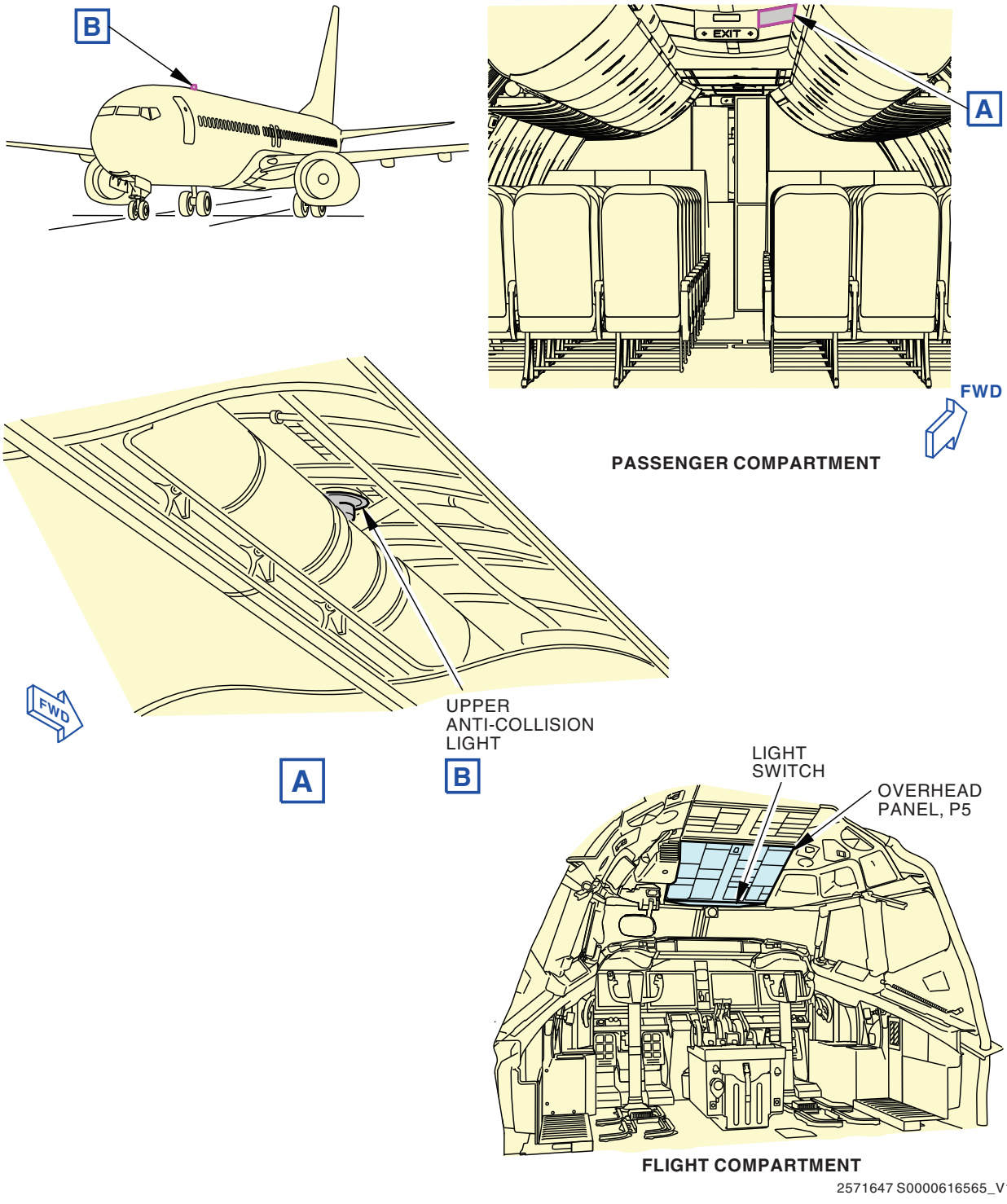
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Set the ANTI COLLISION switch to OFF position.

1) Make sure that the upper anti-collision light goes off.

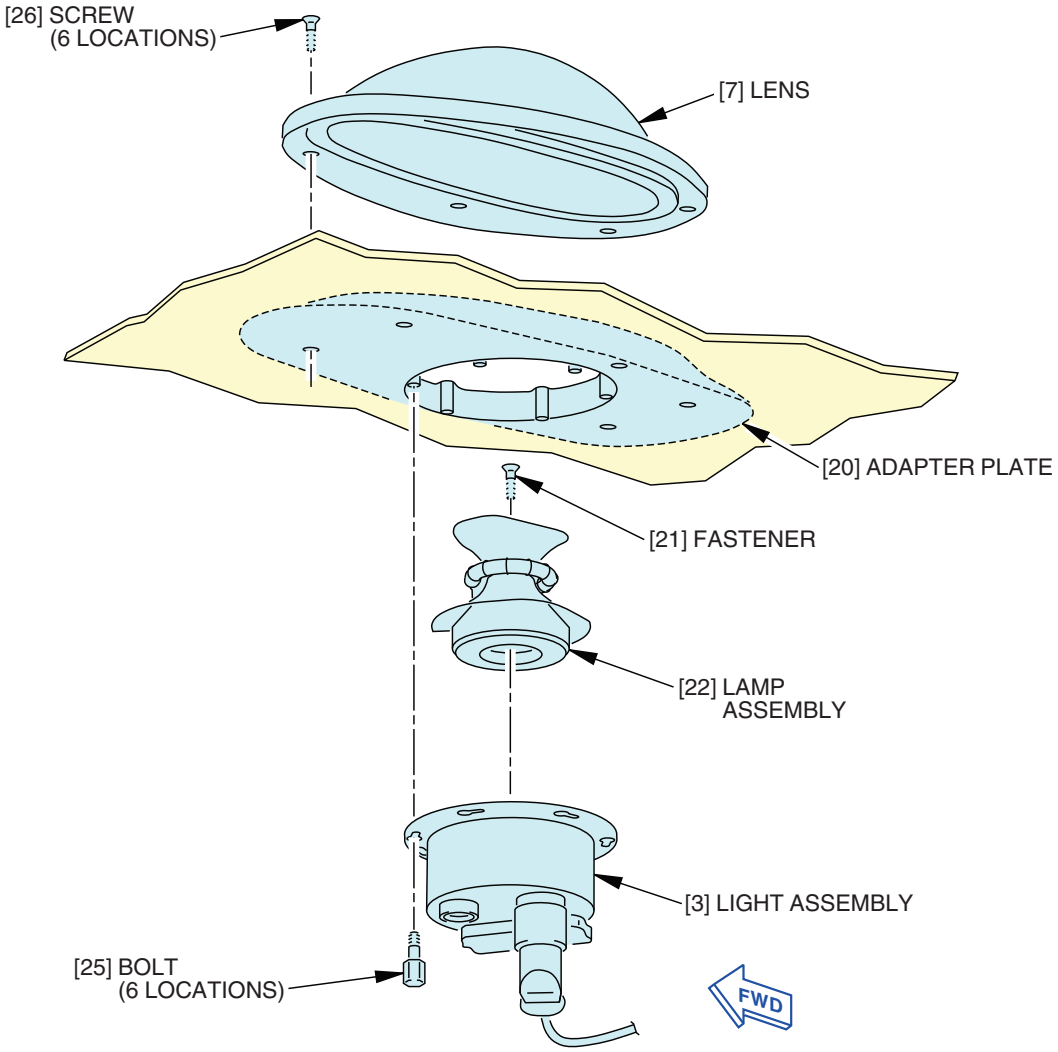
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Upper Anti-Collision Light Lamp Assembly Replacement
Figure 201/33-44-01-990-808 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



UPPER ANTI-COLLISION LIGHT

B

2571650 S0000616564_V2

Upper Anti-Collision Light Lamp Assembly Replacement
Figure 201/33-44-01-990-808 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-44-01-960-801

5. Upper Anti-Collision Light - Light Assembly Replacement

(Figure 202, Figure 201, Figure 203)

A. References

Reference	Title
21-23-01-000-801	Overhead Distribution Duct Removal (P/B 201)
21-23-01-400-801	Overhead Distribution Duct Installation (P/B 201)
24-22-00-860-801	Supply Electrical Power (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)
25-21-45-000-801	Main Ceiling Panel - Removal (P/B 401)
25-21-45-400-801	Main Ceiling Panel - Installation (P/B 401)
33-44-00-710-801	Anti-Collision Lights - Operational Test (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
WDM 33-44-11	Wiring Diagram Manual

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Consumable Materials

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Light assembly	33-44-01-01A-080	PTF ALL

E. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Upper Anti-Collision Light Light Assembly Replacement

SUBTASK 33-44-01-860-001

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

SUBTASK 33-44-01-860-023

(2) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

SUBTASK 33-44-01-010-001

WARNING WAIT 10 MINUTES AFTER YOU REMOVE ELECTRICAL POWER AND BEFORE YOU OPEN THE ANTI-COLLISION LIGHT UPPER AND LOWER FUSELAGE STROBE. THE ELECTRICAL POWER CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Do one of these steps to get access to the upper anti-collision light:

NOTE: The overhead distribution duct is installed below and near the upper anti-collision light, which makes light not easily removed.

(a) To remove the ceiling panel only, do this task: Main Ceiling Panel - Removal, TASK 25-21-45-000-801.

(b) To remove the ceiling panel and the section of the duct below the light, do this task: Overhead Distribution Duct Removal, TASK 21-23-01-000-801.

SUBTASK 33-44-01-960-001

(4) Do these steps to replace the light assembly [3] (WDM 33-44-11):

NOTE: Do not discard the light assembly. Refer to the vendor manual for possible repairs. Do the repairs in a shop.

(a) Disconnect the electrical connector [8].

(b) Loosen the bolts [2] on the bottom of the light assembly [3].

(c) Turn the light assembly [3] clockwise and pull it down from the adapter plate [4].

(d) Replace the new light assembly [3].

(e) If the lens [7] is dirty, clean it with a clean, soft, dry cotton wiper, G00034.

(f) Put the light assembly [3] in the adapter plate [4] and turn it counterclockwise.

(g) Tighten the bolts [2].

(h) Connect the electrical connector [8].

(i) Measure the electrical resistance between the light assembly [3] and adapter plate [4] (SWPM 20-20-00).

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that resistance is 0.0025 ohm (2.5 milliohms) or less.

SUBTASK 33-44-01-410-001

(5) Install the items that were removed for access to the light:

(a) If the ceiling panel and the section of the duct below the light were removed, do this task: Overhead Distribution Duct Installation, TASK 21-23-01-400-801.

(b) If the ceiling panel only was removed, do this task: Main Ceiling Panel - Installation, TASK 25-21-45-400-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-44-01-860-002

(6) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

SUBTASK 33-44-01-860-024

(7) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 33-44-01-710-001

(8) Do a test of the new light assembly [3]:

(a) At the overhead panel, P5, set the ANTI COLLISION switch for the upper anti-collision light to ON position.

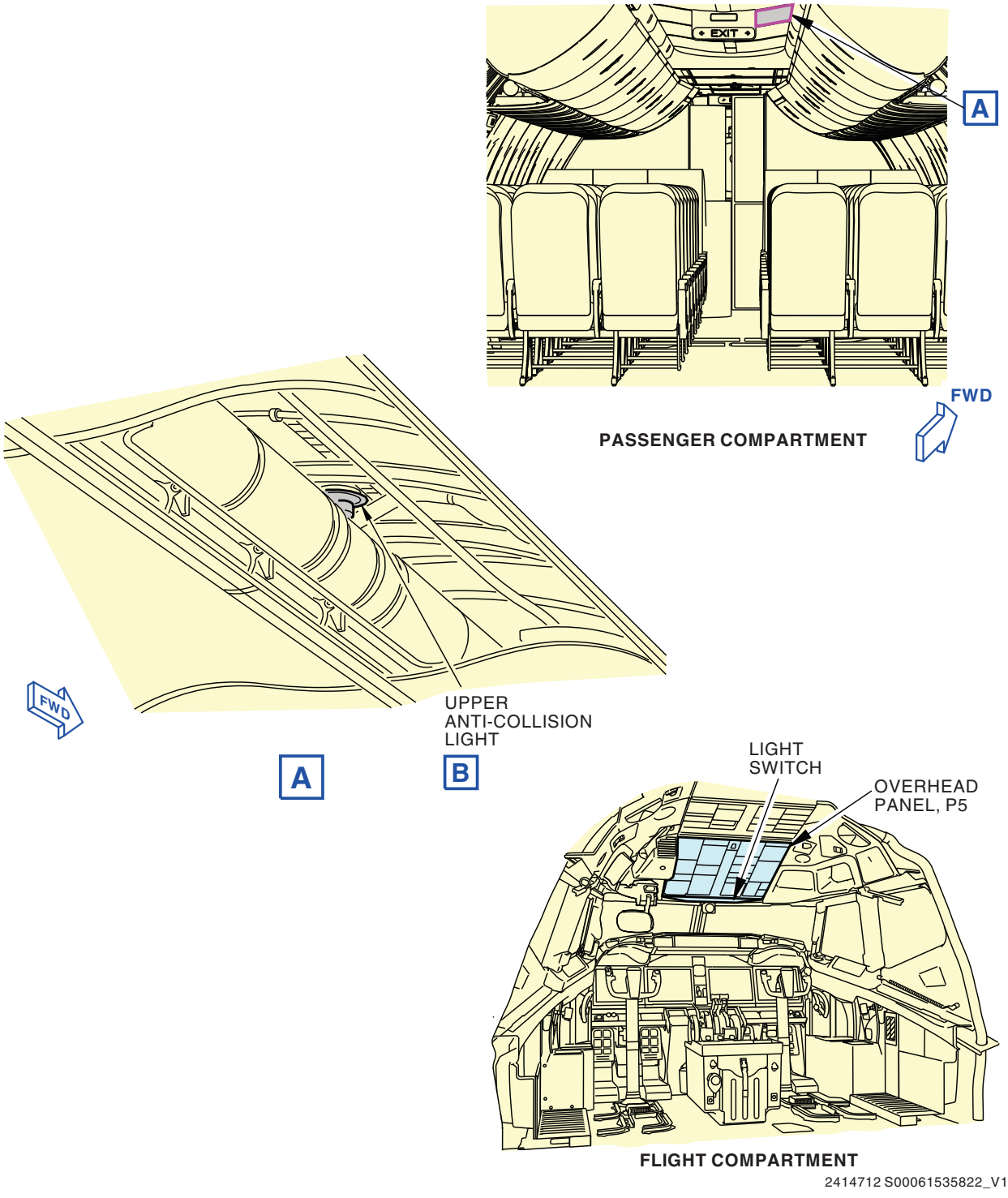
1) Make sure that the upper anti-collision light flashes (TASK 33-44-00-710-801).

(b) Set the ANTI COLLISION switch to OFF position.

1) Make sure that the upper anti-collision light goes off.

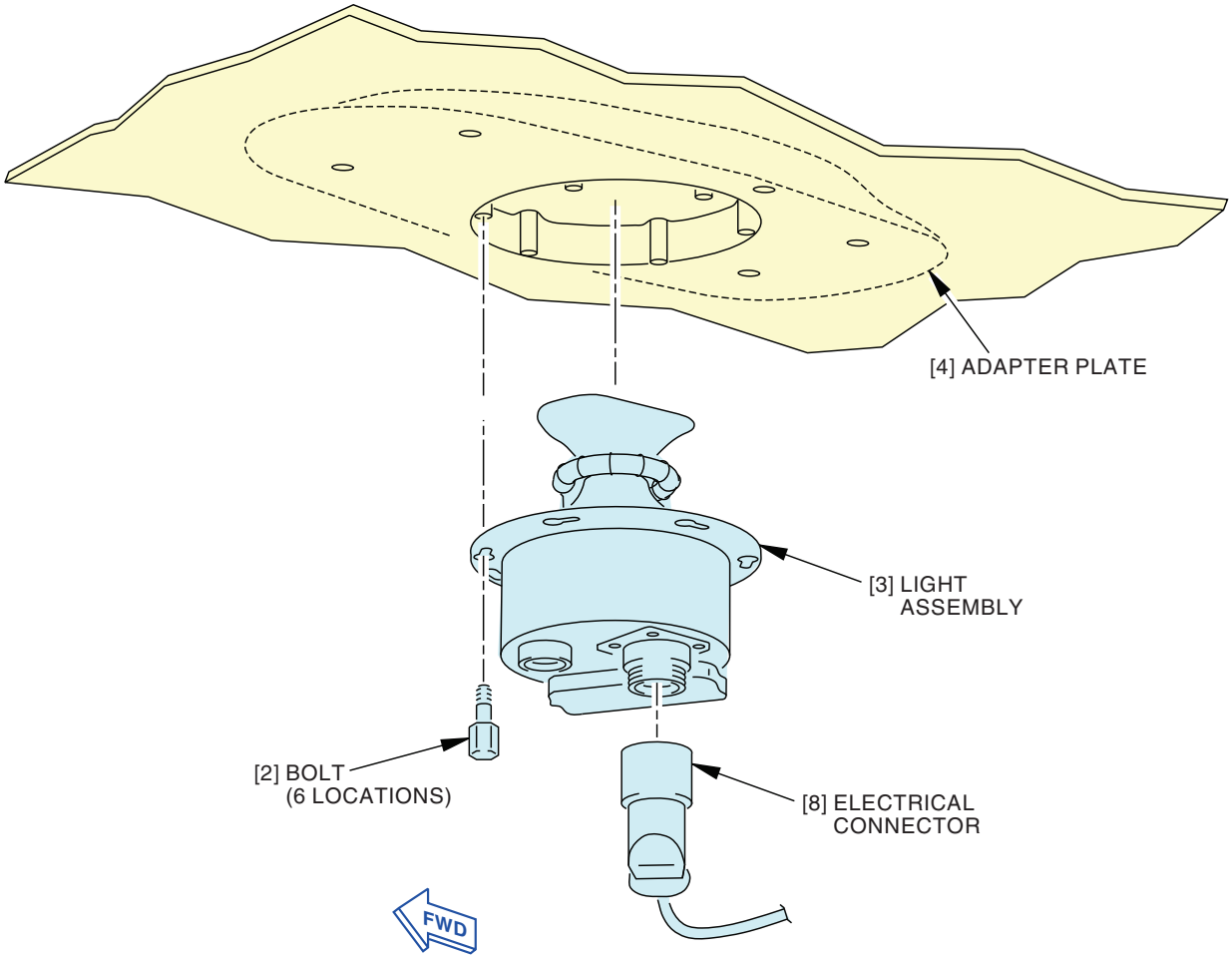
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Upper Anti-Collision Light - Light Assembly Replacement
Figure 202/33-44-01-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



UPPER ANTI-COLLISION LIGHT

B

2414714 S00061535824_V1

Upper Anti-Collision Light - Light Assembly Replacement
Figure 202/33-44-01-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-44-01-960-805

6. Upper Anti-Collision Light Lens Replacement

(Figure 203)

A. References

Reference	Title
51-21-21-100-801	Clean the Surface to be Painted (P/B 701)
51-21-41-370-801	Bonderite M-CR 1001 Application Process (P/B 701)
51-21-41-370-804	BMS10-128 Coating Application (P/B 701)
51-22-00-390-804	Injection Seal Application (P/B 201)

B. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50365	Agent - Peelable Parting (AC Products - AC962-73C) Production discontinued, use stock until depleted.	BAC5000
G50367	Agent - Peelable Parting (Aztec Chemical AZ 634-2)	MIL-PRF-6799, BAC5000
G50368	Agent - Peelable Parting (Rexco Chemical Company - Partall Coverall Film)	BAC5000

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Lens	33-44-01-01A-095	PTF ALL

D. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

E. Procedure

SUBTASK 33-44-01-010-004

(1) For access to the lens [7], go out of the airplane, on top of the fuselage.

SUBTASK 33-44-01-960-003

(2) Do these steps to replace the lens [7]:

(a) Loosen the screws [9] and lift off the lens [7].

(b) Clean the surface that is left exposed by the light assembly (TASK 51-21-21-100-801).

(c) If you change the aerodynamic light assembly to the non-aerodynamic light assembly, then do these steps:

NOTE: Make sure that to replace all components, including lens, adapter plate, lamp assembly, and power supply.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Apply coating, C00064, to the exposed surface (TASK 51-21-41-370-804).

2) Apply paint to the exposed surface (TASK 51-21-41-370-801).

(d) If you change the non-aerodynamic light assembly to the aerodynamic light assembly, do the following steps:

NOTE: Make sure that to replace all components, including lens, adapter plate, lamp assembly, and power supply.

1) Apply AC962-73C peelable parting agent, G50365, AZ 634-2 peelable parting agent, G50367, or Rexco Partall Coverall Film peelable parting agent, G50368 to the panel.

2) Apply an injection seal of sealant, A00247, to the gap between the panels above the light assembly (TASK 51-22-00-390-804).

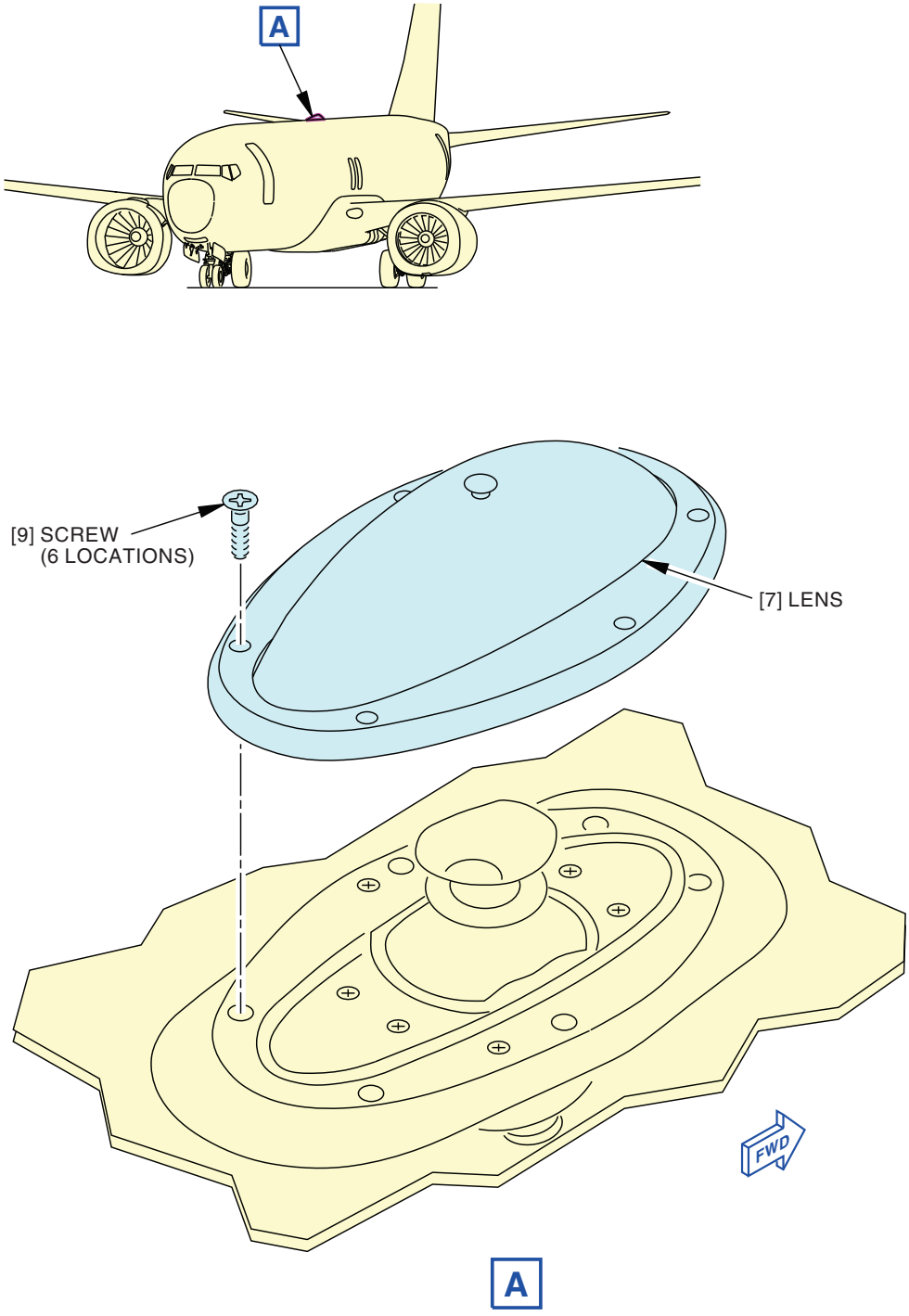
NOTE: Only apply sealant, A00247, to the part of the gap that the light assembly mounting base covers.

(e) If the new lens [7] is dirty, clean it with a clean, soft, dry cotton wiper, G00034.

(f) Put on the new lens [7] and tighten the screws [9].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414716 S00061535827_V2

Upper Anti-Collision Light Lens Replacement
Figure 203/33-44-01-990-809

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-44-01-960-802

7. Upper Anti-Collision Light - Power Supply Replacement

(Figure 204)

A. References

Reference	Title
25-21-45-000-801	Main Ceiling Panel - Removal (P/B 401)
25-21-45-400-801	Main Ceiling Panel - Installation (P/B 401)
25-24-31-000-801	Overhead Stowage Bin - Removal (P/B 401)
25-24-31-000-802	Overhead Stowage Bin Module - Removal (P/B 401)
25-24-31-400-801	Overhead Stowage Bin - Installation (P/B 401)
25-24-31-400-802	Overhead Stowage Bin Module - Installation (P/B 401)
33-44-00-710-801	Anti-Collision Lights - Operational Test (P/B 501)
WDM 33-44-11	Wiring Diagram Manual

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I
G50961	Coating - Corrosion Inhibiting Epoxy Primer - (BAC 377 Yellow Color)	BMS10-11 Type I

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
12	Power supply	33-44-01-01A-105	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

F. Procedure

SUBTASK 33-44-01-860-005

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

SUBTASK 33-44-01-010-002

(2) Do this task: Main Ceiling Panel - Removal, TASK 25-21-45-000-801.

NOTE: The power supply is installed to the right side of the anti-collision light and above overhead stowage bin.

SUBTASK 33-44-01-010-005

(3) Do this task: Overhead Stowage Bin Module - Removal, TASK 25-24-31-000-802 or Overhead Stowage Bin - Removal, TASK 25-24-31-000-801.

SUBTASK 33-44-01-960-002

WARNING DO NOT TOUCH THE POWER SUPPLY FOR 10 MINUTES AFTER YOU REMOVE THE POWER. AN ELECTRICAL SHOCK CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Do these steps to replace the power supply [12] (WDM 33-44-11):

NOTE: Do not discard the power supply. Usually, you can repair it with a new capacitor assembly. Do the repairs in a shop.

(a) Disconnect the electrical connectors [10].

(b) Loosen the fasteners [11] on the power supply.

(c) Remove the power supply [12].

(d) Clean the fay surfaces of the support bracket and power supply with abrasive (SWPM 20-20-00).

(e) Apply a thin layer of sealant, A50155, sealant, A50110, sealant, A50231, or sealant, A50296 to the fay surfaces on the bracket and the new power supply.

(f) Install the new power supply [12].

(g) Tighten the fasteners [11].

(h) Connect the electrical connectors [10].

(i) Measure the electrical bonding resistance, using an intrinsically safe approved bonding meter, COM-1550, between the power supply [12] and the fuselage (SWPM 20-20-00).

1) Make sure that the bonding resistance is 0.0025 ohm (2.5 milliohms) or less.

(j) Refinish the metal surfaces:

1) Apply a layer of coating, C00064 to the abraded surfaces of the support bracket and power supply.

2) Apply a layer of primer, C00259 or coating, G50961 to the abraded surfaces of the support bracket and power supply.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-44-01-410-003

(5) Do this task: Overhead Stowage Bin - Installation, TASK 25-24-31-400-801 or Overhead Stowage Bin Module - Installation, TASK 25-24-31-400-802.

SUBTASK 33-44-01-410-002

(6) Do this task: Main Ceiling Panel - Installation, TASK 25-21-45-400-801.

SUBTASK 33-44-01-860-006

(7) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

SUBTASK 33-44-01-710-002

(8) Do a test of the new power supply [12].

(a) At the overhead panel, P5, set the ANTI COLLISION switch for the upper anti-collision light to ON position.

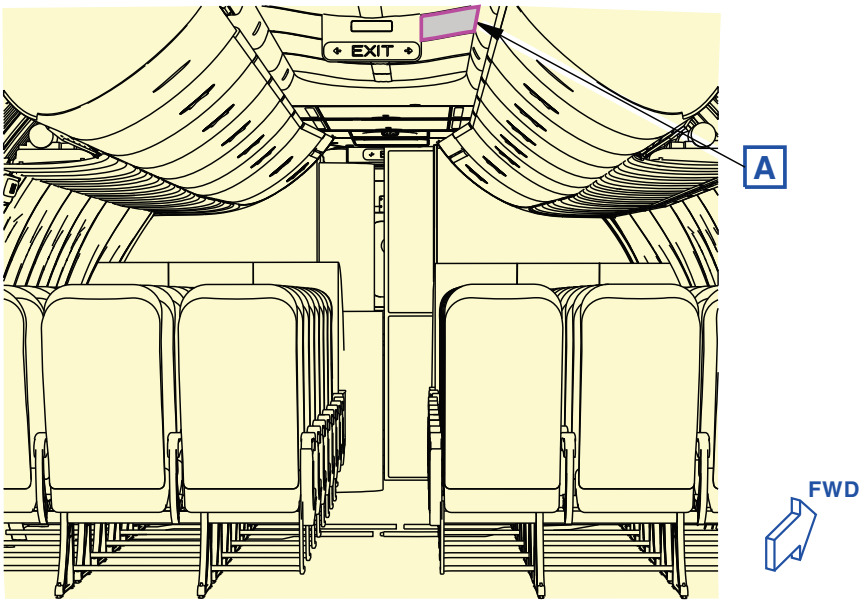
1) Make sure that the upper anti-collision light flashes (TASK 33-44-00-710-801).

(b) Set the ANTI COLLISION switch to OFF position.

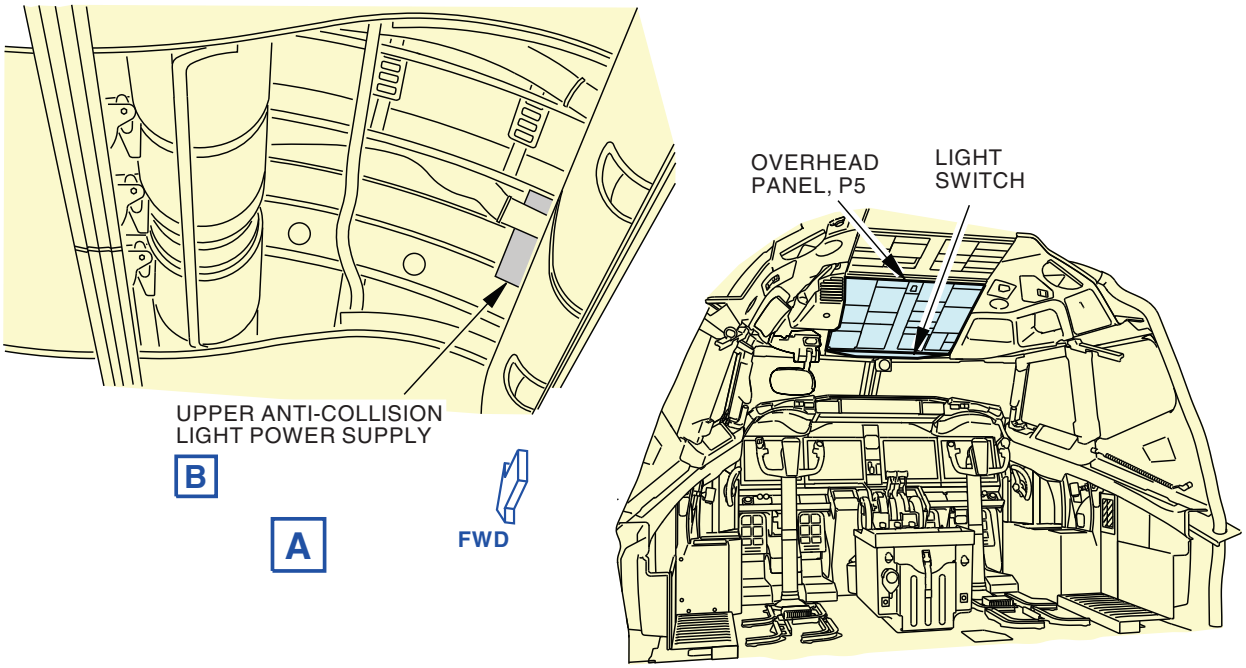
1) Make sure that the upper anti-collision light goes off.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PASSENGER COMPARTMENT

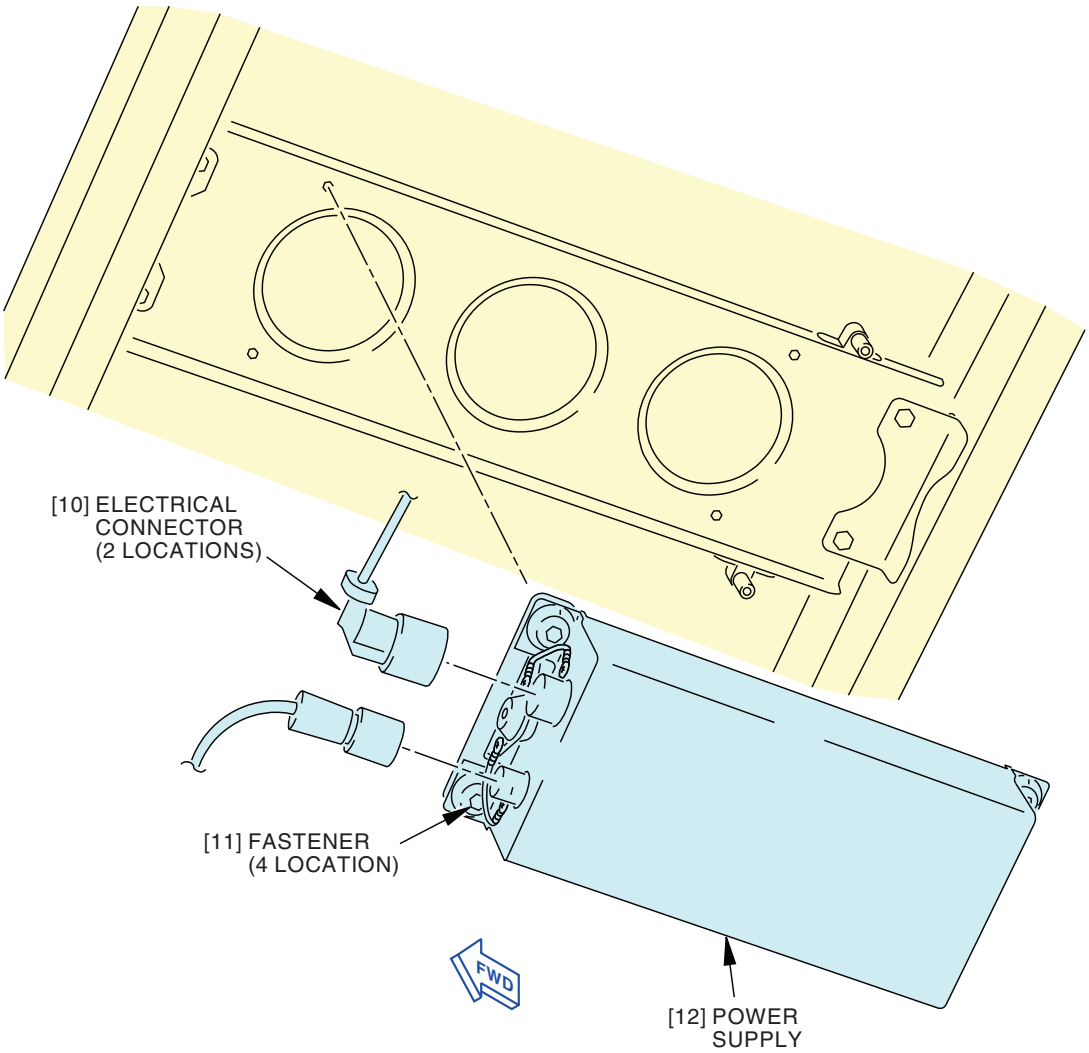


FLIGHT COMPARTMENT

2414717 S00061535829_V1

Upper Anti-Collision Light - Power Supply Replacement
Figure 204/33-44-01-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



UPPER ANTI-COLLISION LIGHT POWER SUPPLY

B

2414718 S00061535830_V1

Upper Anti-Collision Light - Power Supply Replacement
Figure 204/33-44-01-990-804 (Sheet 2 of 2)

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TASK 33-44-01-960-803

8. Upper Anti-Collision Light - Adapter Plate Replacement

(Figure 202, Figure 203, Figure 205)

A. General

(1) This task removes and installs the adapter plate for the upper anti-collision light.

(2) The upper anti-collision adapter plate will be referred to as the adapter plate.

B. References

Reference	Title
21-23-01-000-801	Overhead Distribution Duct Removal (P/B 201)
24-22-00-860-801	Supply Electrical Power (P/B 201)
25-21-45-000-801	Main Ceiling Panel - Removal (P/B 401)
25-21-45-400-801	Main Ceiling Panel - Installation (P/B 401)
33-44-00-710-801	Anti-Collision Lights - Operational Test (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
STD-297	Drill - Hand
STD-1328	Machine - Drilling, Electrical or Pneumatic

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50337	Sealant - Fuel Tank	BMS5-45 Class B
A50343	Sealant - PPG Aerospace PR-1772 Class B Low Weight Fuselage Sealant	BMS5-142 Type II Class B
B00130	Alcohol - Isopropyl	TT-I-735
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
C00862	Coating - Chemical Conversion - Bonderite M-CR 600 Aero (Formerly Alodine 600)	BAC5719 Class A, C or D, MIL-DTL-81706 Type I Class 1A or 3

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(Continued)

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Location Zones

Zone	Area
230	Subzone - Passenger Compartment - Body Station 360.00 to 663.75

F. Prepare for the Removal

SUBTASK 33-44-01-860-009

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

SUBTASK 33-44-01-010-003

(2) Do one of the steps below to get access to the adapter plate [4] (Figure 202):

NOTE: The overhead distribution duct is installed below and near the upper anti-collision light, which makes the adapter plate removal difficult.

(a) To remove the ceiling panel only, do this task: Main Ceiling Panel - Removal, TASK 25-21-45-000-801.

(b) To remove the ceiling panel and the section of the duct below the light, do this task: Overhead Distribution Duct Removal, TASK 21-23-01-000-801.

G. Procedure

SUBTASK 33-44-01-020-001

(1) Do these steps to remove the upper anti-collision light assembly [3] (Figure 202):

(a) Disconnect the electrical connector [8].

(b) Remove the bolts [2] on the bottom of the light assembly [3].

(c) Turn the light assembly [3] clockwise and pull it down from the adapter plate [4].

SUBTASK 33-44-01-020-002

(2) Do these steps to remove the upper anti-collision light lens [7] (Figure 203):

(a) For access to the lens [7], go out of the airplane, on top of the fuselage.

(b) Loosen the screws [9] and lift off the lens [7].

SUBTASK 33-44-01-020-003

(3) Remove the screws [13] from the gasket [14], on top of the fuselage (Figure 205).

SUBTASK 33-44-01-020-004

(4) Remove the bolts [15] from the adapter plate [4], on top of the fuselage (Figure 205).

(a) Remove the adapter plate [4].

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SUBTASK 33-44-01-120-001

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE. OBEY THE MATERIAL SAFETY DATA SHEETS (MSDS) FOR SOLVENTS. OBEY LOCAL REGULATIONS FOR THE CORRECT PROCEDURES TO USE OR DISCARD SOLVENTS. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Clean the following faying surfaces with solvent, B00184 or alcohol, B00130:

(a) aircraft skin

(b) adapter plate [4]

(c) gasket [14].

SUBTASK 33-44-01-910-001

(6) Manually apply Bonderite M-CR 600 Aero coating, C00862, to the abraded aircraft skin surface.

SUBTASK 33-44-01-320-001

(7) If you replace the adapter plate [4], then do these steps to back drill the new adapter plate [4] off-aircraft:

NOTE: You must drill the adapter plate off-aircraft.

(a) Use a hand drill, STD-297, or electrical or pneumatic drilling machine, STD-1328, and drill bit to back drill 12 holes to a diameter of 0.187 in. (0.475 cm) to 0.19 in. (0.48 cm).

NOTE: Use the 12 pilot holes in the adapter plate as a template.

(b) Increase the diameter of the pilot holes by hand countersinking the 12 holes to a diameter of 0.2966 in. (0.7534 cm) to 0.3116 in. (0.7915 cm) x 100 degrees.

SUBTASK 33-44-01-390-001

(8) Do these steps to prepare the adapter plate [4] and the aircraft skin for a sealed electrical faying surface bond (SWPM 20-20-00):

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH, YOUR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. SOLVENTS ARE DANGEROUS MATERIALS. SOLVENTS CAN BE FLAMMABLE. OBEY THE MATERIAL SAFETY DATA SHEETS (MSDS) FOR SOLVENTS. OBEY LOCAL REGULATIONS FOR THE CORRECT PROCEDURES TO USE OR DISCARD SOLVENTS. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Final clean the contact surface of the adapter plate [4] with a cotton wiper, G00034, soaked with solvent, B00184 or alcohol, B00130.

(b) Rub dry with a clean, dry cotton wiper, G00034.

(c) Continue to clean and dry the adapter plate [4] surface until the dry cotton wiper, G00034, stays clean.

(d) Apply sealant to the contact surface of the adapter plate [4].

1) Use sealant, A50337, sealant, A50231, or PR-1772 Class B Sealant, A50343, for the faying surface seal.

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SUBTASK 33-44-01-390-002

(9) Apply sealant, A00247, to the shank and threads of the twelve bolts [15].

SUBTASK 33-44-01-420-001

(10) Install the adapter plate [4] on the aircraft skin using the twelve bolts [15].

(a) Make sure that the sealant is continuously squeezed out along the edges of the adapter plate [4].

(b) If there are gaps, bubbles, or voids in the sealant squeeze out, then disassemble and apply more sealant.

(c) Shape the squeezed out sealant into a fillet seal.

(d) As an option, remove the extra squeezed out sealant.

NOTE: Make sure that the sealant that remains is flush with the mating part edges.

SUBTASK 33-44-01-760-001

(11) Measure the electrical bonding resistance, using an intrinsically safe approved bonding meter, COM-1550, between the adapter plate [4] and the aircraft skin (SWPM 20-20-00).

(a) Make sure that the bonding resistance is 0.0005 ohm (0.5 milliohm) or less.

SUBTASK 33-44-01-420-002

(12) Install the gasket [14] on the adapter plate [4] and tighten the screws [13] to a torque of 10 ±1 in-lb (1 ±0 N·m).

SUBTASK 33-44-01-420-003

(13) Install the lens [7] on the adapter plate [4] and tighten the screws [9].

(a) If the lens [7] is dirty, clean it with a clean, dry cotton wiper, G00034.

SUBTASK 33-44-01-420-004

(14) Do these steps to install the upper anti-collision light assembly [3]:

(a) Put the light assembly [3] in the adapter plate [4] and turn it counterclockwise.

(b) Install the bolts [2] and tighten to a torque of 10 ±1 in-lb (1 ±0 N·m).

(c) Connect the electrical connector [8].

(d) Measure the electrical bonding resistance, using an intrinsically safe approved bonding meter, COM-1550, between the light assembly [3] and the adapter plate [4] (SWPM 20-20-00).

1) Make sure that the bonding resistance is 0.0015 ohm (1.5 milliohms) or less.

SUBTASK 33-44-01-860-010

(15) If necessary, supply electrical power (TASK 24-22-00-860-801).

SUBTASK 33-44-01-710-003

(16) Do a test of the light assembly [3]:

(a) At the P5 overhead panel, set the ANTI COLLISION switch for the upper anti-collision light to ON position.

1) Make sure that the upper anti-collision light flashes (TASK 33-44-00-710-801).

(b) Set the ANTI COLLISION switch to OFF position.

1) Make sure that the upper anti-collision light goes off.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 33-44-01-420-005

(1) Do these steps to install the items that were removed for access to the adapter plate [4]:

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(a) To install the ceiling panel and the section of the duct below the light, do this task: Overhead Distribution Duct Removal, TASK 21-23-01-000-801.

(b) To install the ceiling panel only, do this task: Main Ceiling Panel - Installation, TASK 25-21-45-400-801.

SUBTASK 33-44-01-860-011

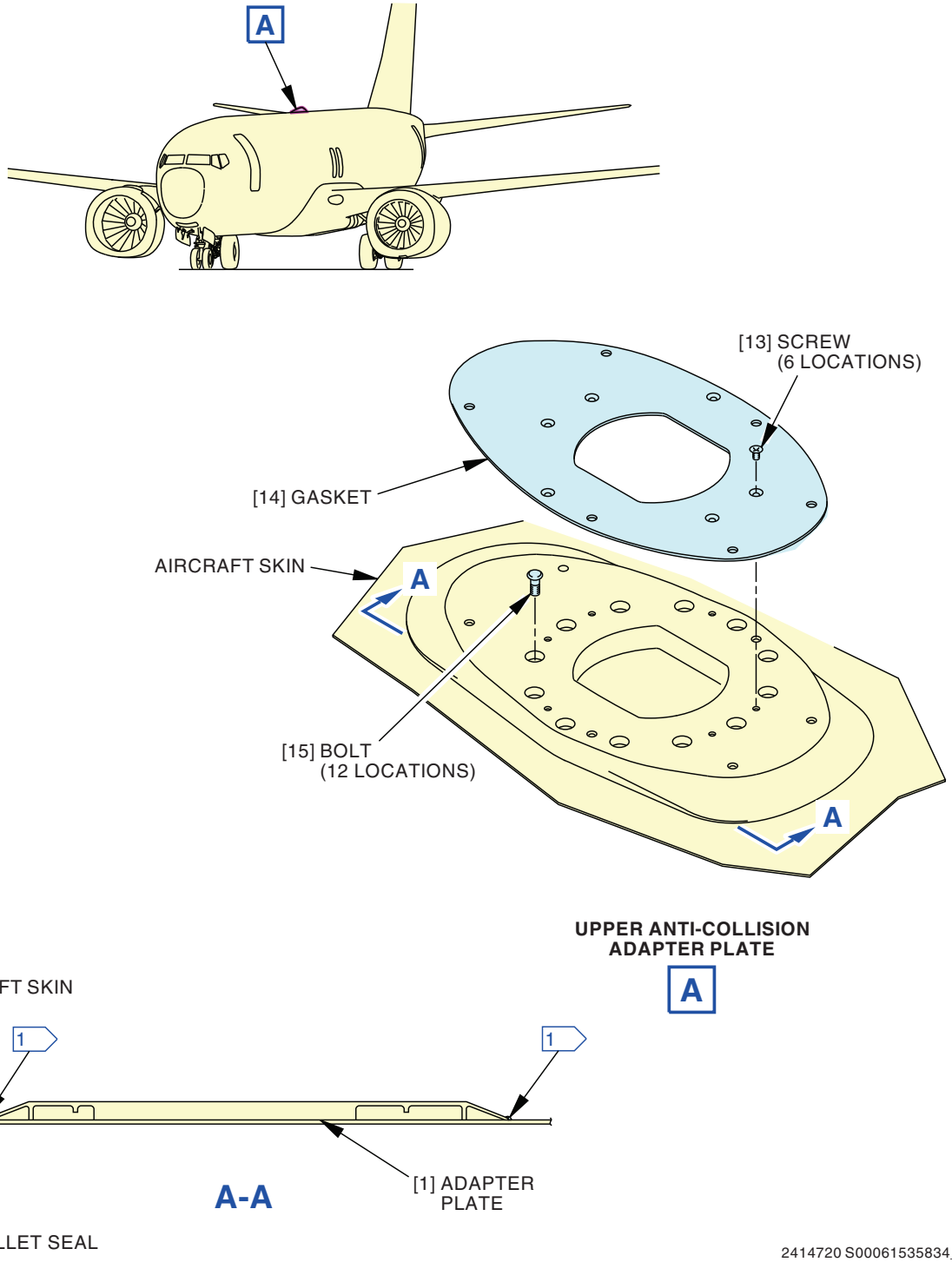
(2) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

——— END OF TASK ———

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Upper Anti-Collision Light - Adapter Plate Replacement
Figure 205/33-44-01-990-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LOWER ANTI-COLLISION LIGHT - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A deactivation of the lower anti-collision light

(2) An activation of the lower anti-collision light

(3) A replacement of the lower anti-collision light - light assembly

(4) A replacement of the lower anti-collision light lamp assembly

(5) A replacement of the lower anti-collision light lens

(6) A replacement of the lower anti-collision light power supply.

TASK 33-44-02-040-801

2. Lower Anti-Collision Light Deactivation

A. General

(1) This procedure removes electrical power from the lower anti-collision light.

B. Location Zones

Zone	Area
139	Keel Beam, (Part) Body Station 540.00 to Body Station 727.00

C. Procedure

SUBTASK 33-44-02-860-013

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

D. Lower Anti-Collision Light Deactivation Test

NOTE: This test is to make sure that the lower anti-collision light is in a zero energy condition.

SUBTASK 33-44-02-200-001

(1) Do these steps to make sure that the power is off:

(a) On the P5 overhead panel, set the ANTI COLLISION switch for the lower anti-collision light to ON.

1) Make sure that the lower anti-collision light stays off.

(b) Set the ANTI COLLISION switch for the lower anti-collision light to OFF.

1) Make sure that the lower anti-collision light stays off.

——— END OF TASK ———

TASK 33-44-02-440-801

3. Lower Anti-Collision Light Activation

A. General

(1) This procedure supplies electrical power to the lower anti-collision light.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Location Zones

Zone	Area
139	Keel Beam, (Part) Body Station 540.00 to Body Station 727.00

C. Procedure

SUBTASK 33-44-02-860-014

(1) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

D. Lower Anti-Collision Light Activation Test

NOTE: This test is to make sure that the lower anti-collision light operates.

SUBTASK 33-44-02-710-004

(1) Do these steps to make sure that the power is on:

(a) On the P5 overhead panel, set the ANTI COLLISION switch for the lower anti-collision light to ON.

1) Make sure that the lower anti-collision light comes on.

(b) Set the ANTI COLLISION switch for the lower anti-collision light to OFF.

1) Make sure that the lower anti-collision light goes off.

——— END OF TASK ———

TASK 33-44-02-960-802

4. Lower Anti-Collision Light - Light Assembly Replacement

(Figure 201)

A. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)
33-44-00-710-801	Anti-Collision Lights - Operational Test (P/B 501)
51-21-21-100-801	Clean the Surface to be Painted (P/B 701)
51-21-41-370-802	Apply Bonderite M-CR 600, Bonderite M-CR 1200, Bonderite M-CR 600RTU, or Bonderite M-CR 1200S Solution (P/B 701)
51-21-99-300-801	Decorative Exterior Paint System Application (P/B 701)
51-22-00-390-804	Injection Seal Application (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
WDM 33-44-11	Wiring Diagram Manual

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
B00130	Alcohol - Isopropyl	TT-I-735
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50365	Agent - Peelable Parting (AC Products - AC962-73C) Production discontinued, use stock until depleted.	BAC5000
G50367	Agent - Peelable Parting (Aztec Chemical AZ 634-2)	MIL-PRF-6799, BAC5000
G50368	Agent - Peelable Parting (Rexco Chemical Company - Partall Coverall Film)	BAC5000

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
6	Light assembly	33-44-02-01A-090	PTF ALL

E. Location Zones

Zone	Area
139	Keel Beam, (Part) Body Station 540.00 to Body Station 727.00

F. Prepare for the Replacement

SUBTASK 33-44-02-860-024

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

SUBTASK 33-44-02-860-030

(2) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Lower Anti-Collision Light Assembly Replacement

WARNING

WAIT 10 MINUTES AFTER YOU REMOVE ELECTRICAL POWER AND BEFORE YOU OPEN THE ANTI-COLLISION LIGHT UPPER AND LOWER FUSELAGE STROBE. THE ELECTRICAL POWER CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

SUBTASK 33-44-02-960-001

(1) Do these steps to replace the light assembly [6] (WDM 33-44-11):

NOTE: Do not discard the light assembly. Refer to the vendor manual for possible repairs. Do the repairs in a shop.

(a) Remove the captive screws [1] and spacers [15] to remove the lens [2].

(b) Remove the bolts [5].

(c) Pull down the light assembly [6].

(d) Disconnect the electrical connector [7].

(e) Remove the nut [8] and washer [9].

(f) Lift off the grounding wire [10].

(g) Clean the surface that is left exposed by the light assembly [6] (TASK 51-21-21-100-801).

(h) If the aerodynamic light assembly [6] is changed to the non-aerodynamic light assembly [6], do these steps:

1) Apply coating, C00064, to the exposed surface (TASK 51-21-41-370-802).

2) Apply paint to the exposed surface (TASK 51-21-99-300-801).

(i) If the non-aerodynamic light assembly [6] is changed to the aerodynamic light assembly [6], do these steps:

1) Apply AC962-73C peelable parting agent, G50365, AZ 634-2 peelable parting agent, G50367, or Rexco Partall Coverall Film peelable parting agent, G50368, to the panel.

2) Apply an injection seal of sealant, A00247, to the gap between the panels above the light assembly [6] (TASK 51-22-00-390-804).

NOTE: Only apply sealant, A00247, to the part of the gap that the light assembly mounting base covers.

(j) If it is necessary, remove the lens [2] from the new light assembly [6].

1) Loosen the captive screws [1].

2) Remove the spacers [15].

3) Remove the lens [2].

NOTE: Do not touch the light assembly lamp or reflector.

4) Keep the captive screws [1] for the installation.

(k) Connect the grounding wire [10] to the new light assembly [6]:

1) Install the washer [9] and nut [8] to connect the grounding wire [10].

(l) Connect the electrical connector [7] to the new light assembly [6].

(m) Measure the electrical resistance between the light assembly [6] and airplane structure (SWPM 20-20-00).

1) Use an intrinsically safe approved bonding meter, COM-1550.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Make sure that the resistance is 0.0015 ohm (1.5 milliohms) or less.

(n) Apply AC962-73C peelable parting agent, G50365, AZ 634-2 peelable parting agent, G50367, or Rexco Partall Coverall Film peelable parting agent, G50368, to the panel.

(o) Apply sealant, A00247, to the fay surface area on the aft side of the light assembly [6] mounting base.

NOTE: The sealant, A00247, must not extend 0.75 in. (1.90 cm) forward of and parallel to the gap between the panels.

1) Apply enough sealant, A00247, to fill the entire gap between the light assembly [6] mounting base and panels.

a) Make sure that the drain hole, adjacent to the light assembly [6], is clear of all sealant.

(p) Put the light assembly [6] in the opening.

NOTE: A light assembly with an arrow must be installed with the arrow pointed in the correct direction.

(q) Install the bolts [5] with corrosion-preventative compound, C00528.

(r) Apply the aerosmoother sealant, A00247, around the base of the light assembly [6].

(s) Apply a filet seal of sealant, A00247, around the base of the light assembly [6].

1) Make sure that the drain hole, adjacent to the light assembly [6], is clear of all sealant.

(t) Do the following steps to install the lens [2]:

1) Use a clean cotton wiper, G00034, soaked with alcohol, B00130, to clean the mating surfaces of the light assembly [6] base.

2) Install the captive screws [1] and spacers [15].

3) Tighten the captive screws [1] to 20 ±1 in-lb (2.3 ±0.1 N·m) to install the lens [2].

NOTE: It is acceptable to tighten the captive screws to 10 ±1 in-lb (1.1 ±0.1 N·m) if they fully engage at this value.

(u) If the plug [41] is installed on the bottom of the lens [2], remove and discard the plug [41].

(v) Make sure that the grommet [42] is not removed.

NOTE: The opening of the grommet becomes a drain hole.

(w) If the lens [2] is dirty, clean it with a clean, soft, dry cotton wiper, G00034.

H. Lower Anti-Collision Light - Light Assembly Test

SUBTASK 33-44-02-860-025

(1) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

SUBTASK 33-44-02-860-031

(2) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 33-44-02-710-001

(3) Do a test of the new light assembly [6]:

(a) On the P5 overhead panel, set the ANTI COLLISION switch for the lower anti-collision light to the ON position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

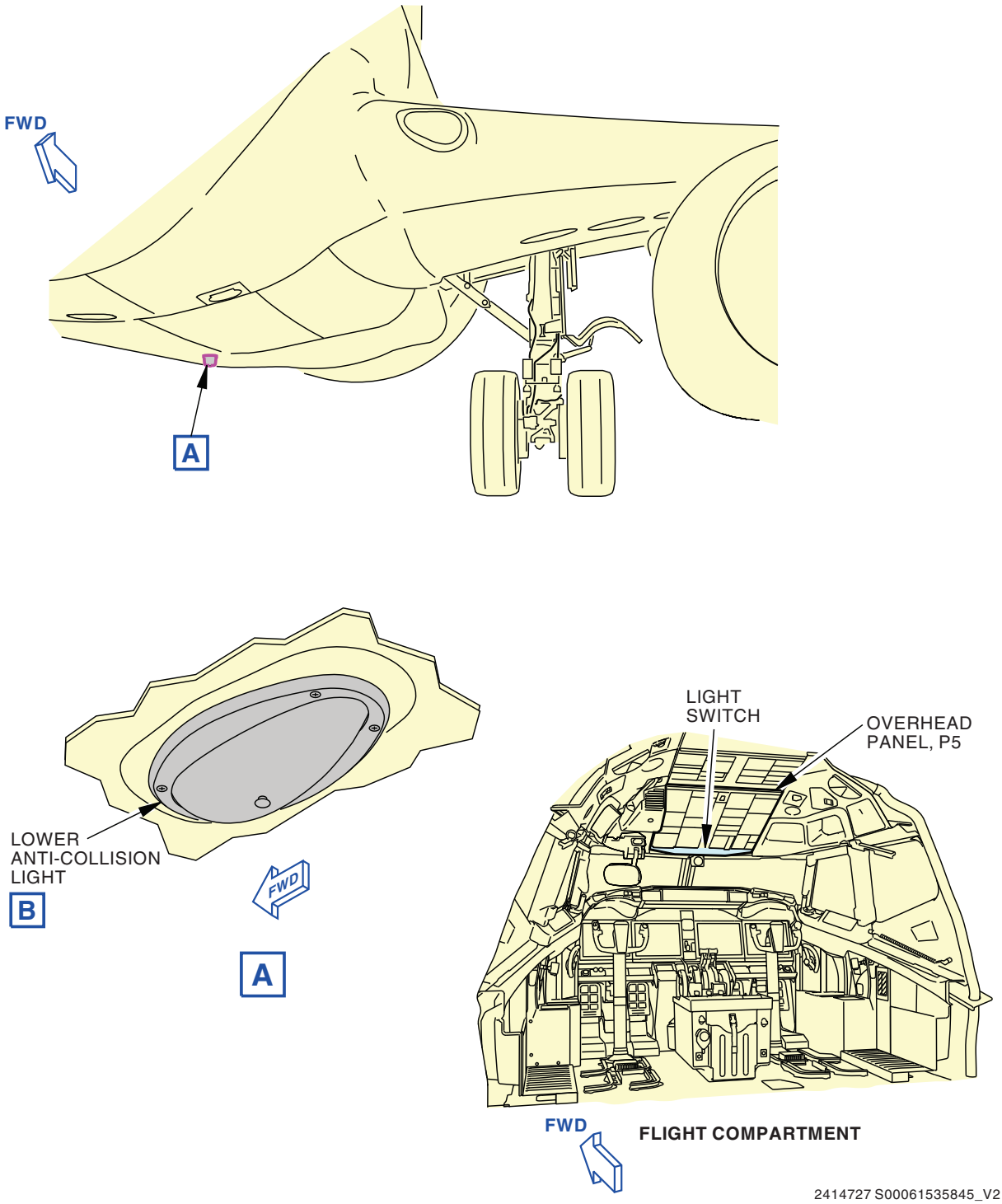
1) Make sure that the lower anti-collision light flashes (TASK 33-44-00-710-801).

(b) Set the ANTI COLLISION switch to the OFF position.

1) Make sure that the lower anti-collision light goes off.

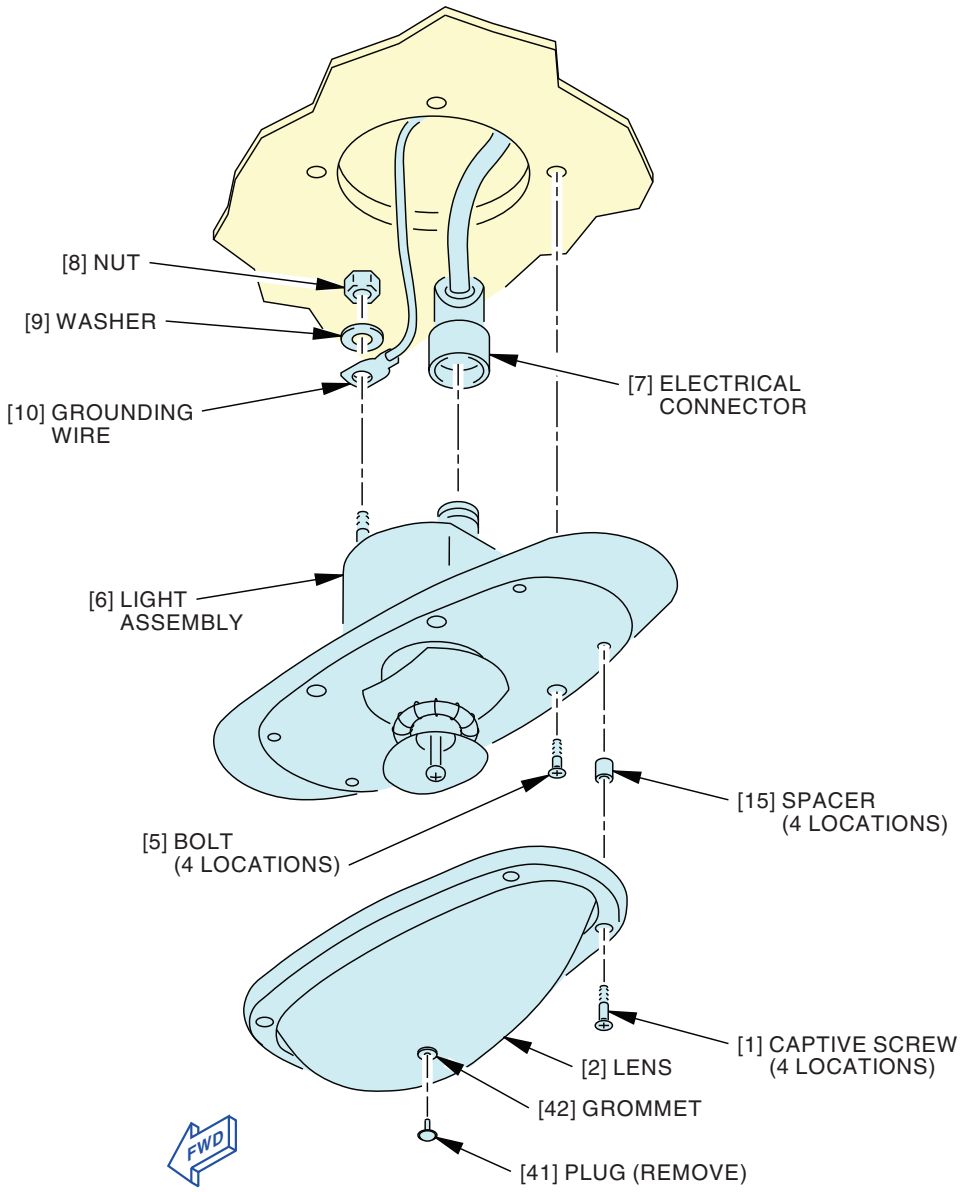
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Lower Anti-Collision Light - Light Assembly Replacement
Figure 201/33-44-02-990-802 (Sheet 1 of 2)

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LOWER ANTI-COLLISION LIGHT

B

2414728 S00061535846_V6

Lower Anti-Collision Light - Light Assembly Replacement
Figure 201/33-44-02-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-44-02-960-805

5. Lower Anti-Collision Light Lamp Assembly Replacement

(Figure 202)

A. General

(1) The lamp assembly in this task is also referred to as a reflector assembly by the vendor.

B. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)
33-44-00-710-801	Anti-Collision Lights - Operational Test (P/B 501)
WDM 33-44-11	Wiring Diagram Manual

C. Consumable Materials

Reference	Description	Specification
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Lamp assembly	33-44-02-01A-090	PTF ALL
		33-44-02-01A-137	PTF ALL

E. Location Zones

Zone	Area
139	Keel Beam, (Part) Body Station 540.00 to Body Station 727.00

F. Prepare for the Replacement

SUBTASK 33-44-02-860-026

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

SUBTASK 33-44-02-860-032

(2) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

G. Lower Anti-Collision Light Lamp Assembly Replacement

WARNING

WAIT 10 MINUTES AFTER YOU REMOVE ELECTRICAL POWER AND BEFORE YOU OPEN THE ANTI-COLLISION LIGHT UPPER AND LOWER FUSELAGE STROBE. THE ELECTRICAL POWER CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

SUBTASK 33-44-02-960-005

(1) Do these steps to replace the lamp assembly [4] (WDM 33-44-11):

NOTE: Do not discard the lamp assembly. Refer to the vendor manual for possible repairs. Do the repairs in a shop.

(a) Remove the captive screws [1] and spacers [15].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Lift away the lens [2].

(b) Remove the fastener [3] in the center of the lamp assembly [4].

(c) Replace the new lamp assembly [4].

(d) Tighten the fastener [3].

(e) Put on the lens [2].

1) Install the captive screws [1] and spacers [15].

2) Tighten the captive screws [1] to 20 ±1 in-lb (2.3 ±0.1 N·m).

NOTE: It is acceptable to tighten the captive screws to 10 ±1 in-lb (1.1 ±0.1 N·m) if they fully engage at this value.

3) If the lens [2] is dirty, clean it with a clean, soft, dry cotton wiper, G00034.

H. Lower Anti-Collision Light Lamp Assembly Test

SUBTASK 33-44-02-860-027

(1) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

SUBTASK 33-44-02-860-033

(2) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 33-44-02-710-005

(3) Do a test of the new lamp assembly [4]:

(a) On the P5 overhead panel, set the ANTI COLLISION switch for the lower anti-collision light to the ON position.

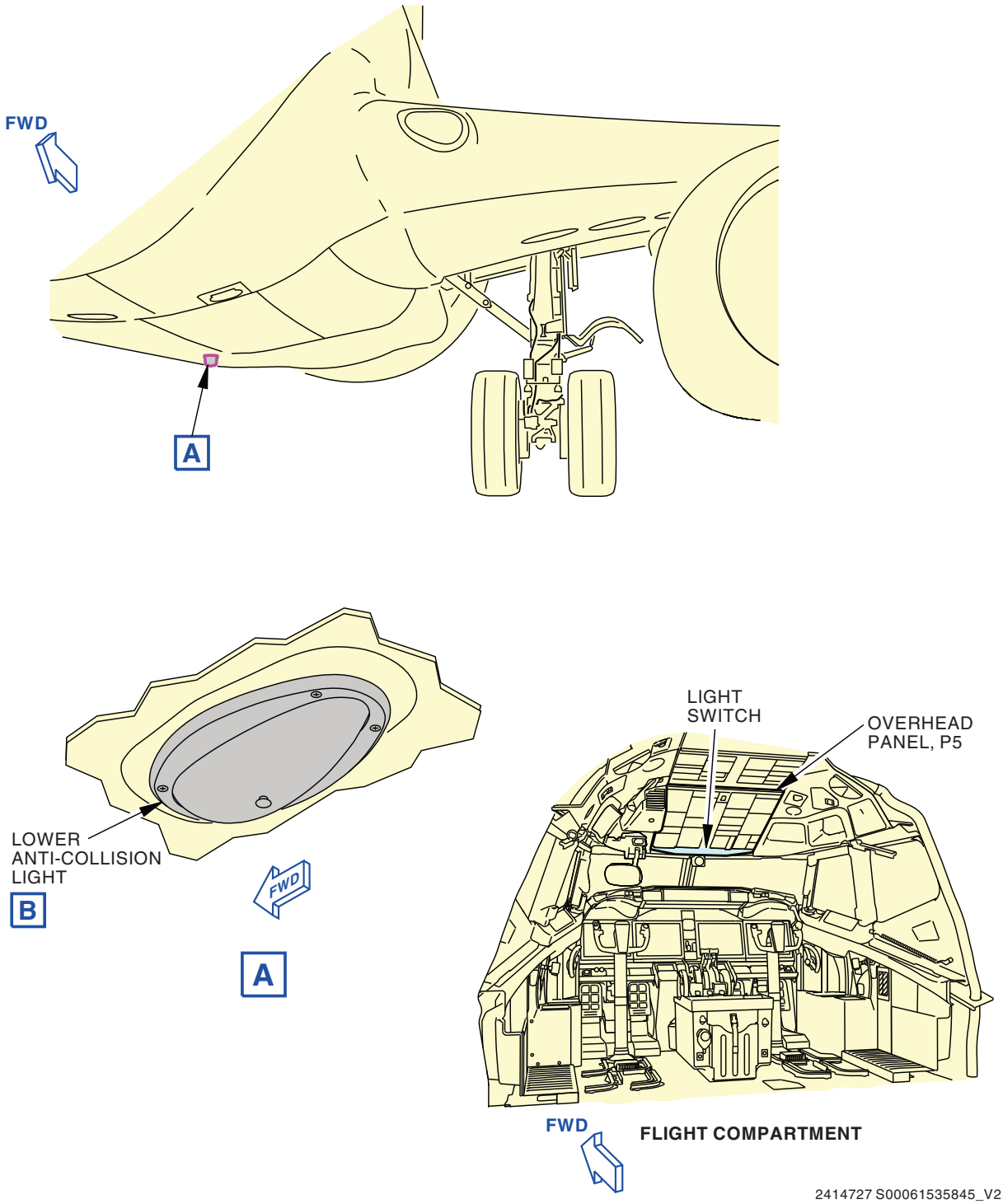
1) Make sure that the lower anti-collision light flashes (TASK 33-44-00-710-801).

(b) Set the ANTI COLLISION switch to the OFF position.

1) Make sure that the lower anti-collision light goes off.

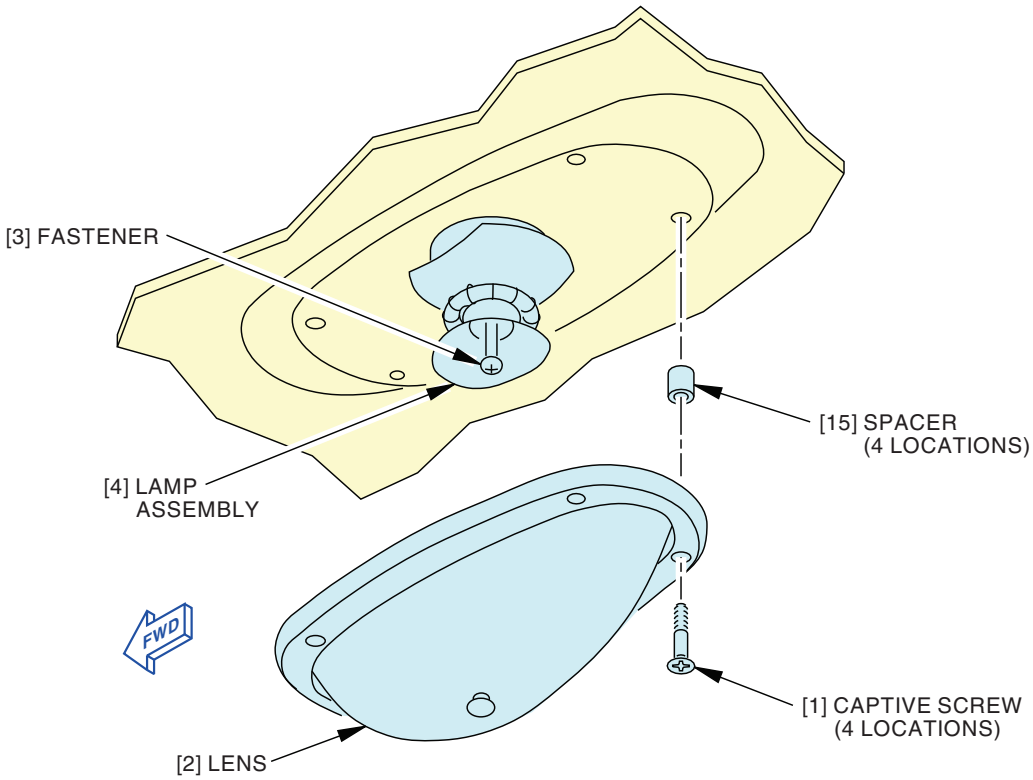
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Lower Anti-Collision Light Lamp Assembly Replacement
Figure 202/33-44-02-990-805 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LOWER ANTI-COLLISION LIGHT

B

2571698 S0000617051_V3

Lower Anti-Collision Light Lamp Assembly Replacement
Figure 202/33-44-02-990-805 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-44-02-960-806

6. Lower Anti-Collision Light Lens Replacement

(Figure 203)

A. Consumable Materials

Reference	Description	Specification
A00562	Adhesive - High Strength Silicone Rubber, One-Part - RTV157	
C00954	Primer - Adhesive Bonding - SS4004P RTV	BAC5010 Type 74
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Lens	33-44-02-01A-105	PTF ALL
42	Grommet	33-44-02-01A-107	PTF ALL

C. Location Zones

Zone	Area
139	Keel Beam, (Part) Body Station 540.00 to Body Station 727.00

D. Prepare for the Replacement

SUBTASK 33-44-02-860-028

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

E. Lower Anti-Collision Light Lens Replacement

SUBTASK 33-44-02-960-006

(1) Do these steps to replace the lens [2]:

(a) Loosen the captive screws [1].

(b) Remove the spacers [15].

(c) Lift off the lens [2].

(d) Put on the new lens [2].

1) Install the captive screws [1] and spacers [15].

2) Tighten the captive screws [1] to 20 ±1 in-lb (2 ±0 N·m).

NOTE: It is acceptable to tighten the captive screws to 10 ±1 in-lb (1 ±0 N·m) if they fully engage at this value.

3) If a plug [41] is installed on the bottom of the lens [2], remove and discard the plug [41].

4) Make sure that the grommet [42] is not removed.

NOTE: The opening of the grommet becomes a drain hole.

5) If it is necessary, replace the grommet [42], do these steps:

a) Remove the grommet [42].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO NOT GET PRIMER IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM PRIMER. PUT ON GOGGLES, AND GLOVES WHEN YOU USE PRIMER. PRIMER IS POISONOUS AND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

WARNING KEEP PRIMER AWAY FROM THE FLAME AND HEAT. PRIMER IS FLAMMABLE AND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

b) Apply SS4004P RTV primer, C00954, to the grommet [42] hole.

WARNING DO NOT GET ADHESIVE IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM ADHESIVE. PUT ON GOGGLES, AND GLOVES WHEN YOU USE ADHESIVE. ADHESIVE IS POISONOUS AND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

WARNING KEEP ADHESIVE AWAY FROM THE FLAME AND HEAT. ADHESIVE IS FLAMMABLE AND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

c) Apply RTV157 adhesive, A00562, to the new grommet [42].

d) Install the grommet [42] in lens [2].

6) If the lens [2] is dirty, clean it with a clean, soft, dry cotton wiper, G00034.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 33-44-02-860-029

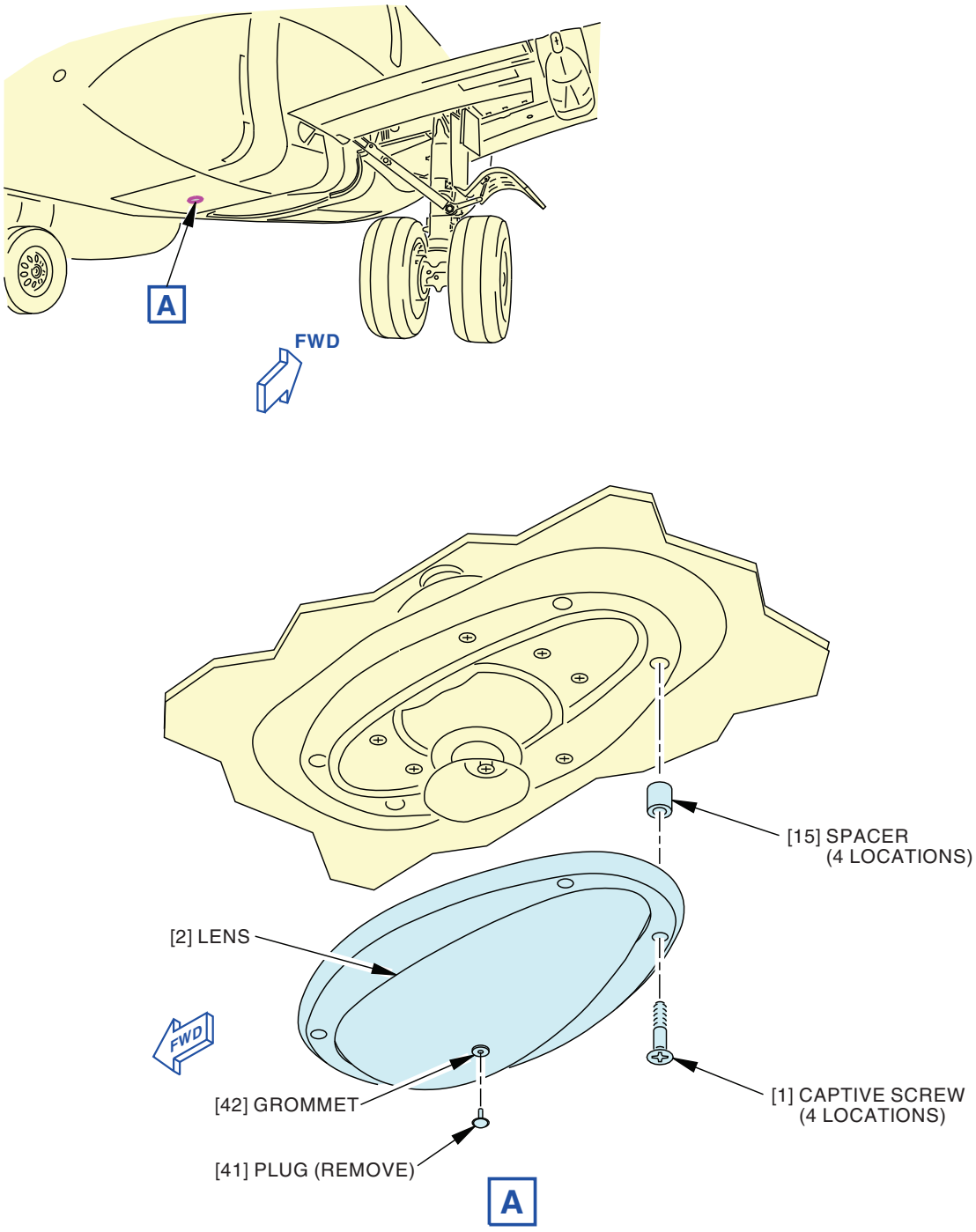
(1) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Lower Anti-Collision Light Lens Replacement
Figure 203/33-44-02-990-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-44-02-960-804

7. Lower Anti-Collision Light - Power Supply Replacement

(Figure 204)

A. General

(1) This task provides instruction to replace power supply for the lower anti-collision light assembly.

B. References

Reference	Title
33-44-00-710-801	Anti-Collision Lights - Operational Test (P/B 501)
51-22-00-160-801	Prepare For Sealing (P/B 201)
51-22-00-390-806	Fillet Seal Application (P/B 201)
53-51-21-000-802	Center Wing-To-Body Fairing Panel (Except 192AL, 192AR, 192HL, 192HR, 195BL, 195BR) Removal (P/B 401)
53-51-21-400-802	Center Wing-To-Body Fairing Panel (Except 192AL, 192AR, 192HL, 192HR, 195BL, 195BR) - Installation (P/B 401)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
WDM 33-44-11	Wiring Diagram Manual

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I
G50961	Coating - Corrosion Inhibiting Epoxy Primer - (BAC 377 Yellow Color)	BMS10-11 Type I

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
13	Power supply	33-44-02-02C-005	PTF ALL

F. Location Zones

Zone	Area
139	Keel Beam, (Part) Body Station 540.00 to Body Station 727.00
192	Lower Wing-To-Body Fairing - Under Wing Box

G. Access Panels

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

H. Lower Anti-Collision Light Power Supply Replacement

SUBTASK 33-44-02-860-005

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

SUBTASK 33-44-02-010-001

WARNING

OBEY THE SUBSEQUENT STEPS FOR ACCESS PANELS IDENTIFIED WITH A NITROGEN GENERATION SYSTEM PLACARD. IF THERE IS A LEAK IN THE NITROGEN GENERATION SYSTEM, IT WILL DECREASE THE OXYGEN IN THE AIR THAT YOU BREATHE. IF YOU BREATHE AIR THAT DOES NOT HAVE SUFFICIENT OXYGEN, DANGEROUS HEALTH CONDITIONS CAN QUICKLY OCCUR. THIS CAN CAUSE INJURY TO PERSONNEL.

(2) Remove this access panel:

(TASK 53-51-21-000-802)

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

SUBTASK 33-44-02-960-002

WARNING

DO NOT TOUCH THE POWER SUPPLY FOR 10 MINUTES AFTER YOU REMOVE THE POWER. AN ELECTRICAL SHOCK CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do these steps to replace the power supply [13] (WDM 33-44-11):

NOTE: Do not discard the power supply. Usually, you can repair it with a new capacitor assembly. Do the repairs in a shop.

(a) Disconnect the electrical connectors [11].

(b) Loosen the fasteners [12] on the power supply [13].

(c) Remove the sealant between the power supply [13] and structure (TASK 51-22-00-160-801).

(d) Remove the power supply [13].

(e) Clean the fay surface contact areas on the support bracket and power supply [13] with abrasive (SWPM 20-20-00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Refinish the metal surfaces.

1) Apply a layer of coating, C00064, to the surface contact areas on the support bracket and power supply [13].

2) Apply a layer of primer, C00259, or coating, G50961, to the surface contact areas on the support bracket and power supply [13].

(g) Apply a thin layer of sealant, A50155, sealant, A50110, sealant, A50231, or sealant, A50296, to the fay surface ground areas on the bracket and new power supply [13].

(h) Install the new power supply [13].

(i) Tighten the fasteners [12] to 20 ±2 in-lb (2.26 ±0.23 N·m).

(j) Connect the electrical connectors [11] to the new power supply [13].

(k) Measure the electrical bonding resistance between the power supply [13] and fuselage (SWPM 20-20-00).

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the electrical bonding resistance is 0.0005 ohm (0.5 milliohms) or less

(l) Apply a fillet seal around the power supply [13] (TASK 51-22-00-390-806).

SUBTASK 33-44-02-410-001

(4) Install this access panel:

(TASK 53-51-21-400-802)

Number	Name/Location
192BL	ECS Ram Air Inlet Mixing Duct Panel - Forward

SUBTASK 33-44-02-860-006

(5) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
B	12	C00111	EXTERIOR LIGHTING ANTI COLLISION RED

SUBTASK 33-44-02-710-002

(6) Do a test of the new power supply [13]:

(a) At the overhead panel, P5, set the ANTI COLLISION switch for the lower anti-collision light to the ON position.

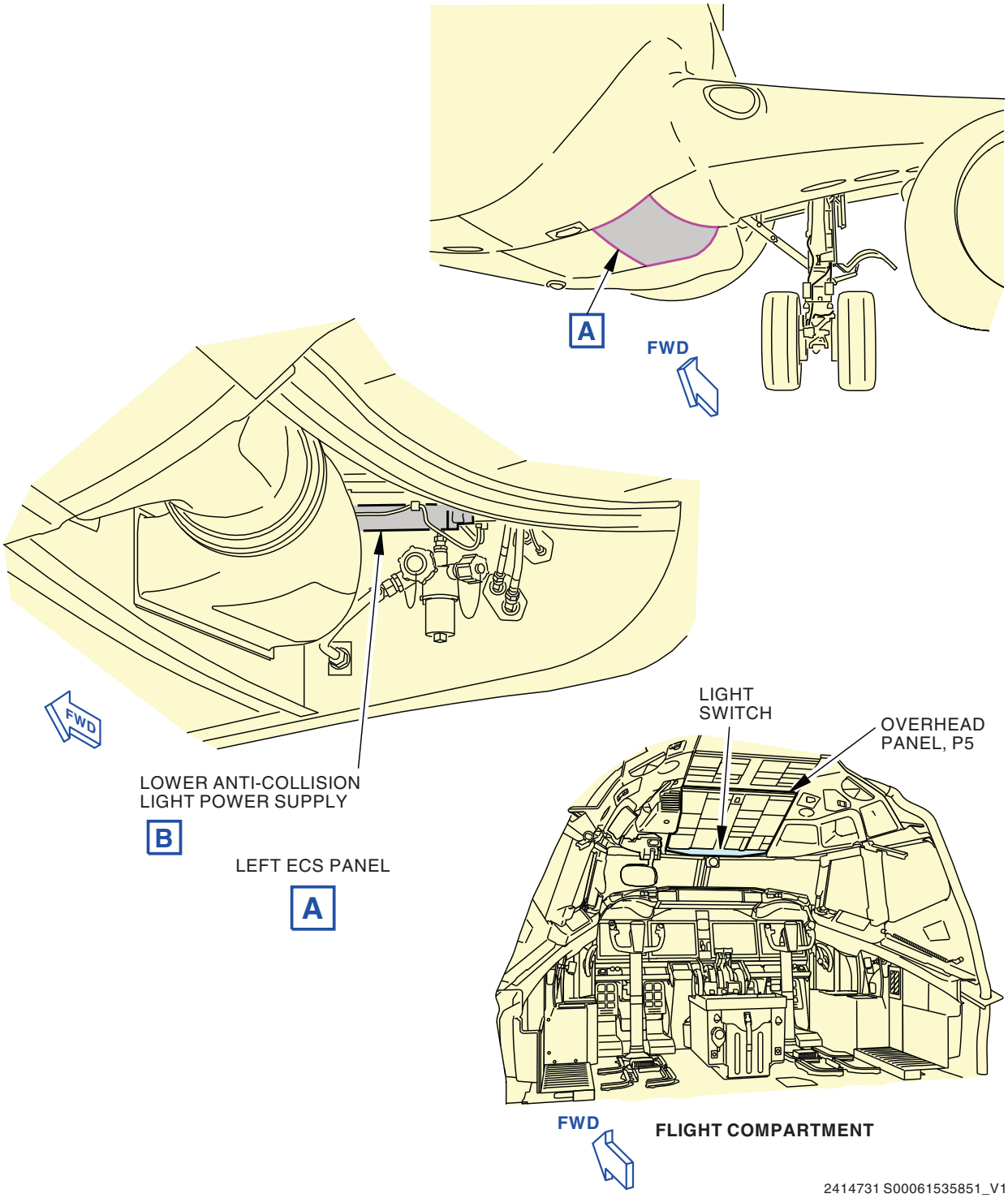
1) Make sure that the lower anti-collision light flashes (TASK 33-44-00-710-801).

(b) Set the ANTI COLLISION switch to the OFF position.

1) Make sure that the lower anti-collision light goes off.

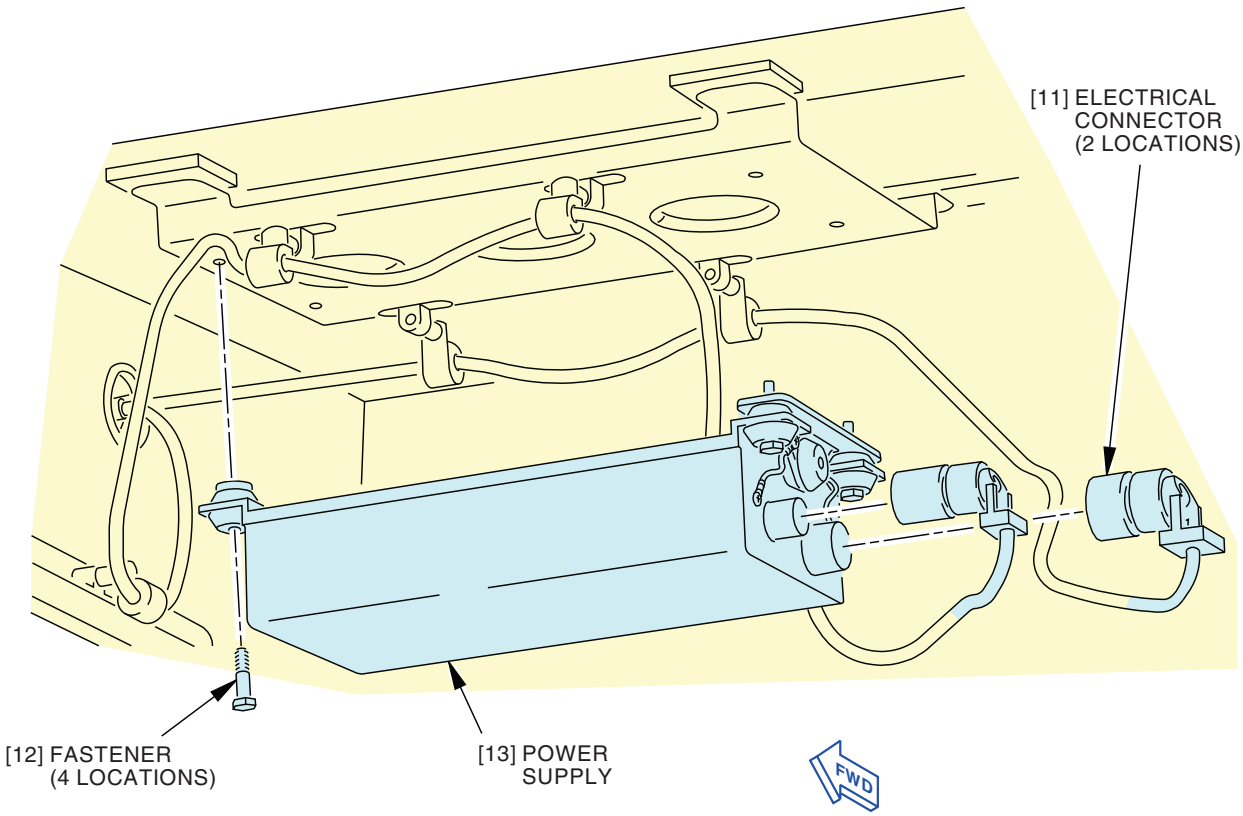
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Lower Anti-Collision Light - Power Supply Replacement
Figure 204/33-44-02-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LOWER ANTI-COLLISION LIGHT POWER SUPPLY

B

2414732 S00061535852_V1

Lower Anti-Collision Light - Power Supply Replacement
Figure 204/33-44-02-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TAILCONE ANTI-COLLISION LIGHT - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Tailcone Anti-Collision Light Deactivation

(2) Tailcone Anti-Collision Light Activation

(3) Tailcone Anti-Collision Light Replacement

(4) Tailcone Anti-Collision Light Transformer Replacement

(5) Tailcone Anti-Collision Light LED Module Replacement

(6) Tailcone Anti-Collision Light Housing Assembly Replacement.

B. The tailcone position light and tailcone anti-collision light are in the same light assembly.

TASK 33-44-04-040-801

2. Tailcone Anti-Collision Light Deactivation

A. General

(1) This procedure removes electrical power from the tailcone anti-collision lights.

B. Location Zones

Zone	Area
310	Fuselage - Body Station 1016.00 to Body Station 1260.00
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

C. Procedure

SUBTASK 33-44-04-860-013

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

D. Tailcone Anti-Collision Lights - Deactivation Test

NOTE: This test is to make sure that the tailcone anti-collision lights are in a zero energy condition.

SUBTASK 33-44-04-200-001

(1) Do these steps to make sure that the power is off:

(a) Set the POSITION switch for the tailcone anti-collision lights to STEADY on the P5 overhead panel.

1) Make sure that the tailcone anti-collision lights (left/right) stay off.

(b) Set the POSITION switch for the tailcone anti-collision lights to STROBE & STEADY on the P5 overhead panel.

1) Make sure that the tailcone anti-collision lights (left/right) stay off.

(c) Set the POSITION switch for the tailcone anti-collision lights to OFF.

1) Make sure that the tailcone anti-collision lights (left/right) stay off.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-44-04-440-801

3. Tailcone Anti-Collision Light Activation

A. General

(1) This procedure supplies electrical power to the tailcone anti-collision lights.

B. Location Zones

Zone	Area
310	Fuselage - Body Station 1016.00 to Body Station 1260.00
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

C. Procedure

SUBTASK 33-44-04-860-014

(1) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

D. Tailcone Anti-Collision Lights - Activation Test

NOTE: This test is to make sure that the tailcone anti-collision lights operate.

SUBTASK 33-44-04-710-006

(1) Do these steps to make sure that the power is on:

(a) Set the POSITION switch for the tailcone anti-collision lights to STROBE & STEADY on the P5 overhead panel.

1) Make sure that the tailcone anti-collision lights (left/right) flash continuously.

(b) Set the POSITION switch for the tailcone anti-collision lights to STEADY on the P5 overhead panel.

1) Make sure that the tailcone anti-collision lights (left/right) come on and stay on.

(c) Set the POSITION switch for the tailcone anti-collision lights to OFF.

1) Make sure that the tailcone anti-collision lights (left/right) go off.

——— END OF TASK ———

TASK 33-44-04-960-801

4. Tailcone Anti-Collision Light Replacement

A. References

Reference	Title
33-43-13-960-801	Tailcone Position Lights Replacement (P/B 201)

B. Location Zones

Zone	Area
310	Fuselage - Body Station 1016.00 to Body Station 1260.00
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tailcone Anti-Collision Light Replacement

SUBTASK 33-44-04-860-001

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

SUBTASK 33-44-04-960-001

(2) Do this to replace the light assembly: Tailcone Position Lights Replacement, TASK 33-43-13-960-801.

——— END OF TASK ———

TASK 33-44-04-960-802

5. Tailcone Anti-Collision Light Transformer Replacement

(Figure 201)

A. References

Reference	Title
WDM 33-44-12	Wiring Diagram Manual

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Consumable Materials

Reference	Description	Specification
A50473	Sealant - Brush (BMS5-95 Type I Class A-2 (See BAC5000 8.5.3 a. for preparation instructions))	BAC5000 Section 8.5.3 a.
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Tailcone anti-collision lights transfomer	33-44-04-01-010	PTF ALL

E. Location Zones

Zone	Area
310	Fuselage - Body Station 1016.00 to Body Station 1260.00
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

F. Tailcone Anti-Collision Light Transformer Replacement

SUBTASK 33-44-04-860-007

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

SUBTASK 33-44-04-960-003

(2) Do these steps to remove the tailcone anti-collision lights transfomer [1] (WDM 33-44-12):

(a) Disconnect the electrical connector [2].

(b) Remove the sealant between the tailcone anti-collision lights transfomer [1] and the mounting bracket.

(c) Remove the bolts [3] and washers [4].

(d) Remove the tailcone anti-collision lights transfomer [1].

(3) Do these steps to install a new tailcone anti-collision lights transfomer [1]:

(a) Clean the fay surface bonding area with solvent, B00083, and a clean cotton wiper, G00034.

(b) Put the new tailcone anti-collision lights transfomer [1] on the mounting bracket.

(c) Install the washers [4] and bolts [3].

(d) With the electrical bonding intrinsically safe approved bonding meter, COM-1550, make sure that the resistance between the tailcone anti-collision lights transfomer [1] and the fuselage measures no more than 0.0025 ohms (2.5 milliohms).

(e) Apply a polysulfide sealant, A50473 fillet seal between tailcone anti-collision lights transfomer [1] and the mounting bracket.

(f) Connect the electrical connector [2].

1) Clean the electrical connector [2] (SWPM 20-60-01).

2) Connect the electrical connector [2] (SWPM 20-60-06).

3) Use an intrinsically safe approved bonding meter, COM-1550, to measure the resistance between the electrical connector [2] braid and the housing (SWPM 20-20-00).

a) Make sure that the resistance is 0.003 ohm (3 milliohms) or less.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-44-04-860-008

(4) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

SUBTASK 33-44-04-710-003

(5) Do a test of the new tailcone anti-collision lights transfomer [1].

(a) At the P5 overhead panel, set the POSITION switch for the tailcone anti-collision light to the STROBE & STEADY mode.

1) Make sure that tailcone anti-collision lights (left/right) flash continuously.

(b) At the P5 overhead panel, set the POSITION switch for the tailcone anti-collision light to the STEADY mode.

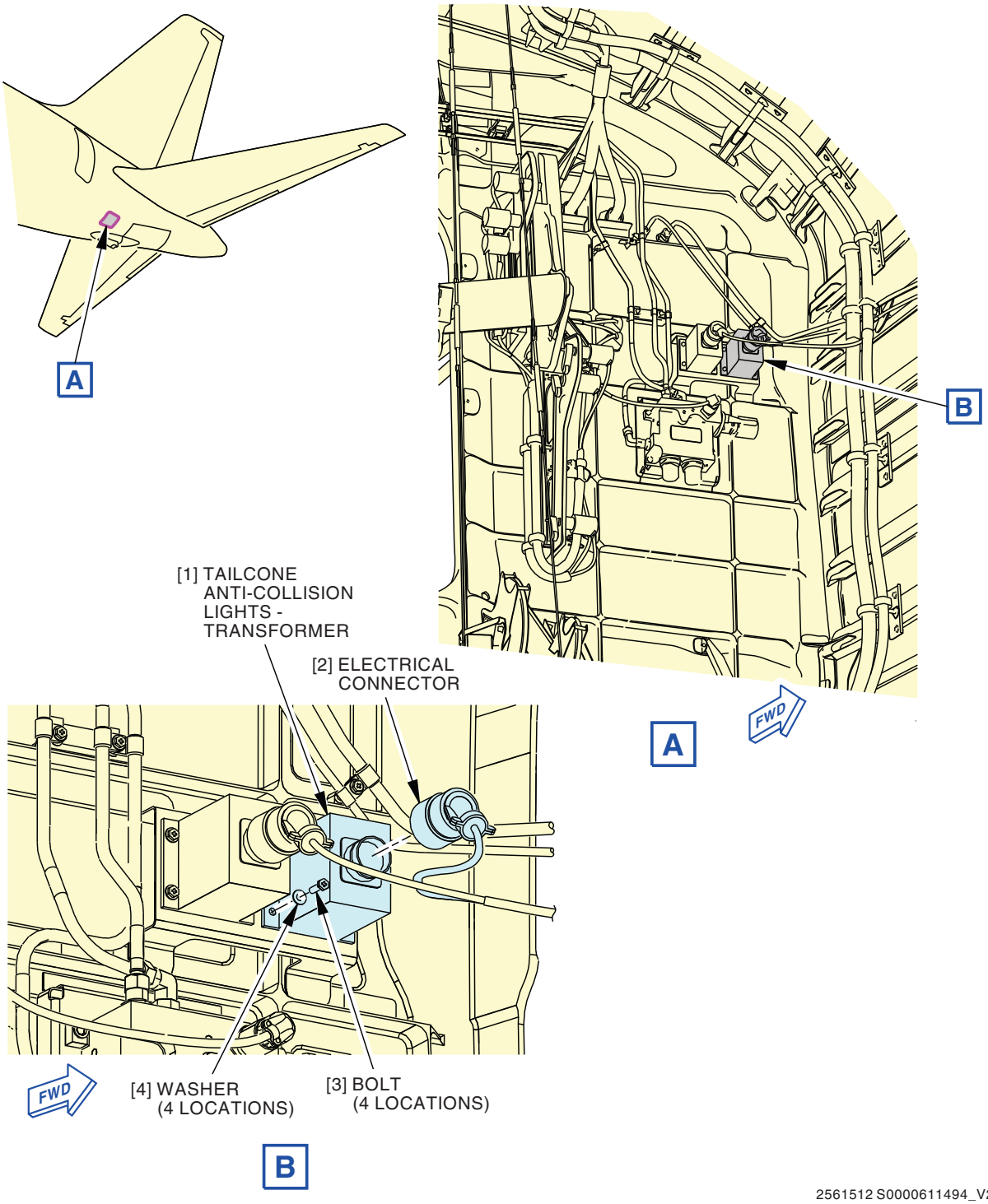
1) Make sure that the tailcone anti-collision lights (left/right) come on and stay on.

(c) Set the POSITION switch to the OFF mode.

1) Make sure that the tailcone anti-collision lights (left/right) go off.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Tailcone Anti-Collision Light Transformer Replacement
Figure 201/33-44-04-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-44-04-960-803

6. Tailcone Anti-Collision Light LED Module Replacement

(Figure 202)

A. General

(1) This procedure contains instructions for the replacement of the Light Emitting Diode (LED) modules installed on the tailcone anti-collision light.

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
20	LED module assembly	33-44-04-02-050	PTF ALL

C. Location Zones

Zone	Area
310	Fuselage - Body Station 1016.00 to Body Station 1260.00
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

D. Prepare for the Replacement

SUBTASK 33-44-04-860-023

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

SUBTASK 33-44-04-860-024

(2) Make sure that the Auxiliary Power Unit (APU) master switch, on the P5 forward overhead panel, is in the OFF position.

(a) Install the DO NOT OPERATE tag, STD-858, on the APU master switch.

SUBTASK 33-44-04-860-025

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

E. Tailcone Position Lights LED Module Replacement

SUBTASK 33-44-04-020-001

(1) Remove the housing assembly [30] (TASK 33-44-04-960-804).

SUBTASK 33-44-04-960-004

(2) Do these steps to replace the LED module assembly [20]:

(a) Use white cotton gloves to handle the housing assembly [30] and the LED module assembly [20].

(b) Remove the screws [21] that attach the LED module assembly [20] to the housing assembly [30].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Pull the LED module assembly [20] to disconnect it from the housing assembly [30].

(d) Replace the new LED module assembly [20].

(e) Install the screws [21] that hold the LED module assembly [20] to the housing assembly [30].

(f) Tighten the screws [21] until fully seated.

SUBTASK 33-44-04-420-001

(3) Install the housing assembly [30] (TASK 33-44-04-960-804).

SUBTASK 33-44-04-860-010

(4) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

SUBTASK 33-44-04-710-004

(5) Do a test of the new LED module assembly [20]:

(a) At the P5 overhead panel, set the POSITION switch for the tailcone anti-collision light to the STROBE & STEADY mode.

1) Make sure that the tailcone anti-collision lights (left/right) come on and flash continuously.

(b) At the P5 overhead panel, set the POSITION switch for the tailcone anti-collision light to the STEADY mode.

1) Make sure that the tailcone anti-collision lights (left/right) stop flashing and remain steady.

(c) Set the POSITION switch to the OFF mode.

1) Make sure that the tailcone anti-collision lights (left/right) go off.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 33-44-04-860-016

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

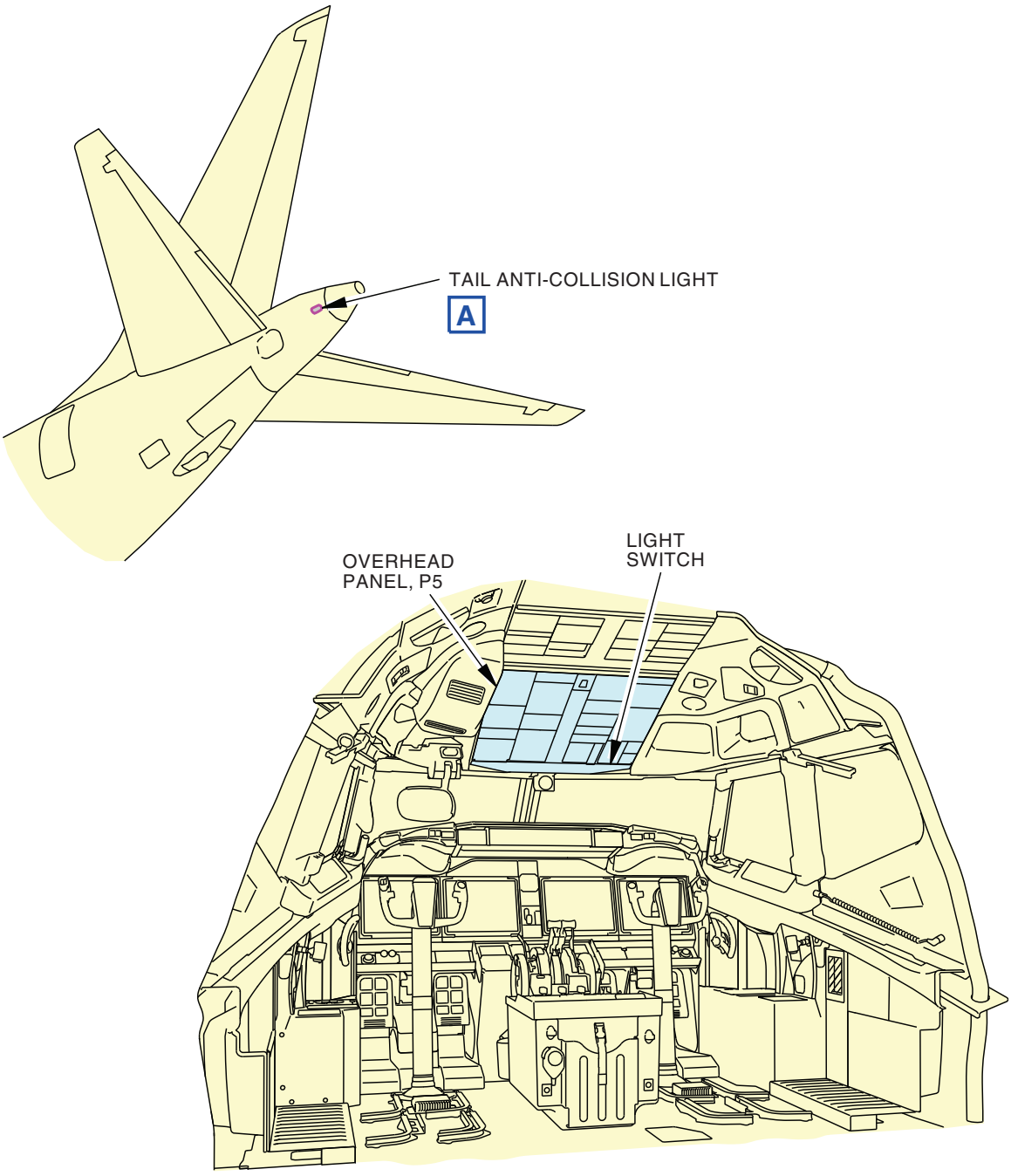
Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 33-44-04-020-002

(2) Remove the DO NOT OPERATE tag from the APU master switch, on the P5 forward overhead panel.

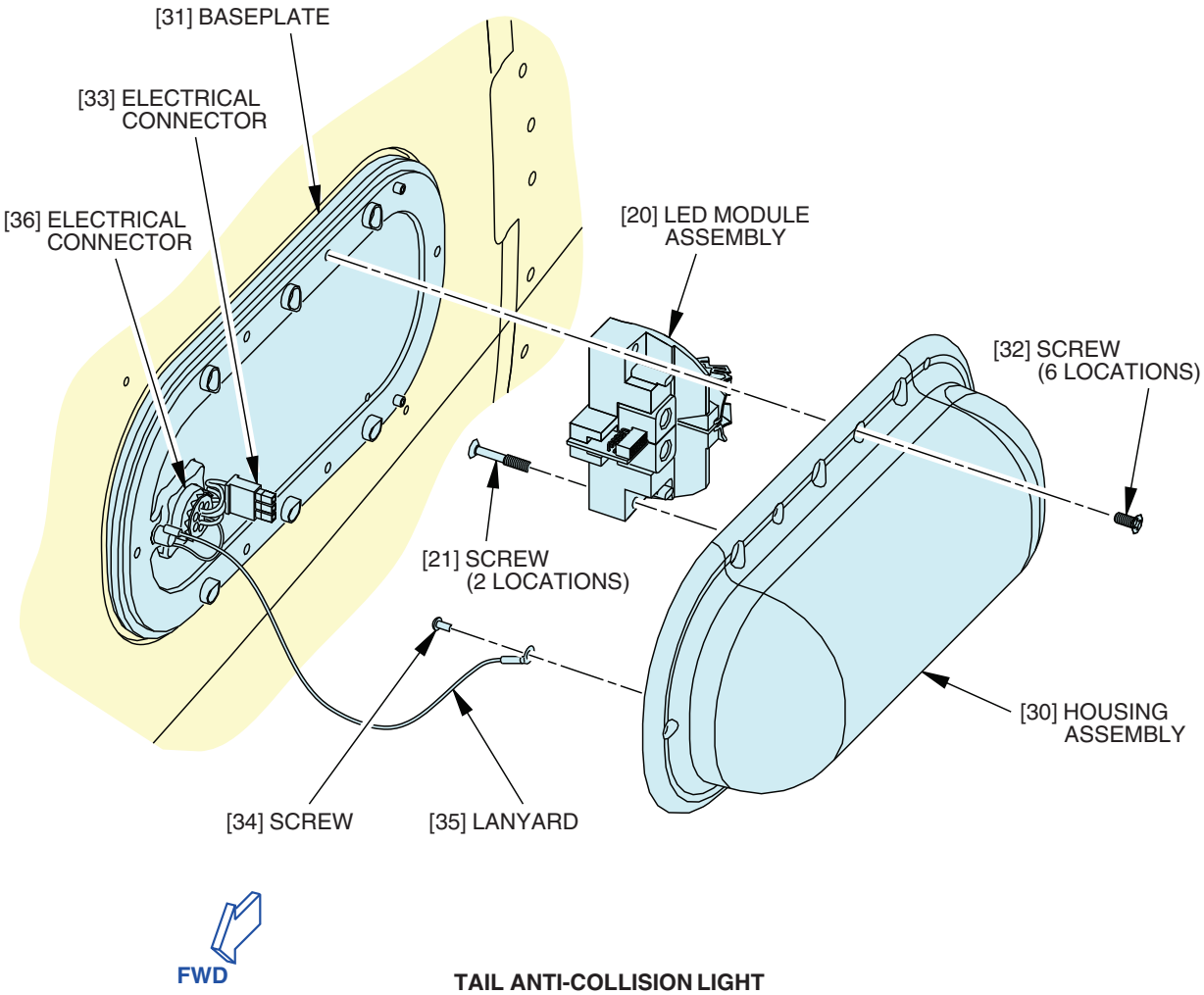
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Tailcone Anti-Collision Light LED Module Replacement
Figure 202/33-44-04-990-805 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TAIL ANTI-COLLISION LIGHT
(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)

A

2570017 S0000615941_V2

Tailcone Anti-Collision Light LED Module Replacement
Figure 202/33-44-04-990-805 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-44-04-960-804

7. Tailcone Anti-Collision Light Housing Assembly Replacement

(Figure 202)

A. General

(1) This procedure contains instructions for the replacement of the housing assembly installed on the tailcone anti-collision light.

B. References

Reference	Title
51-22-00 P/B 201	SEALS AND SEALING - MAINTENANCE PRACTICES
51-22-00-390-809	Aerodynamic Smoother Application (P/B 201)

C. Tools/Equipment

Reference	Description
STD-764	Scraper - Non-metallic
STD-810	Spatula - Fillet Smoothing, Hardwood or Plastic
STD-2483	Ohmmeter

D. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
30	Housing assembly	33-44-04-02-055	PTF ALL

F. Location Zones

Zone	Area
310	Fuselage - Body Station 1016.00 to Body Station 1260.00
317	Tail Cone Compartment - Left
318	Tail Cone Compartment - Right

G. Prepare for the Replacement

SUBTASK 33-44-04-860-020

(1) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

SUBTASK 33-44-04-860-026

(2) Make sure that the Auxiliary Power Unit (APU) master switch, on the P5 forward overhead panel, is in the OFF position.

(a) Install the DO NOT OPERATE tag, STD-858, on the APU master switch.

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SUBTASK 33-44-04-860-022

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

H. Tailcone Anti-Collision Light Housing Assembly Replacement

SUBTASK 33-44-04-960-005

(1) Do these steps to remove the housing assembly [30]:

(a) Use white cotton gloves to handle the housing assembly [30].

CAUTION ONLY USE APPROVED SCRAPERS ON THE AIRPLANE SKIN. SCRAPERS THAT ARE NOT APPROVED CAN MAKE SCRATCHES ON THE SKIN, AND CAUSE FATIGUE CRACKS.

(b) Carefully remove the aerodynamic fillet seal, with a non-metallic scraper, STD-764 (PAGEBLOCK 51-22-00/201).

(c) Remove the screws [32] that attach the housing assembly [30] to the baseplate [31].

(d) Pull the housing assembly [30] forward to disconnect it from the baseplate [31].

(e) Disconnect the electrical connector [33].

(f) Remove the screw [34] that attaches the lanyard [35] to the housing assembly [30].

(2) Do these steps to install the housing assembly [30]:

(a) Attach the lanyard [35] with screw [34] to the housing assembly [30].

(b) Use a digital ohmmeter, STD-2483 or equivalent, to measure the resistance between the shell of the electrical connector [36] and the primer-free area of the housing assembly [30].

1) Make sure that the resistance is 2.5 milliohms or less at 1 amp.

(c) Connect the electrical connector [33].

(d) Install the screws [32] that hold the housing assembly [30] to the baseplate [31].

(e) Apply an aerodynamic fillet seal around the housing assembly [30] with sealant, A00247 (TASK 51-22-00-390-809).

(f) Use the hardwood or plastic fillet smoothing spatula, STD-810, to make a smooth 45-degree fillet.

CAUTION DO NOT SEAL LOWER DRAIN HOLE OF THE TAILCONE ANTI-COLLISION LIGHT. LOWER DRAIN HOLE TO BE CLEAR OF ALL SEALANT.

(g) Apply sealant, A00247, to the upper drain hole (TASK 51-22-00-390-809).

(h) If the lens is dirty, clean it with a clean, soft, dry cotton wiper, G00034.

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SUBTASK 33-44-04-860-012

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	13	C04051	EXTERIOR LIGHTING POSITION LEFT
C	12	C04053	ANTI COLLISION LIGHT-TAIL CONE- WHITE

SUBTASK 33-44-04-710-005

(4) Do a test of the new housing assembly [30]:

(a) Set the POSITION switch for the tailcone anti-collision light, on the P5 overhead panel, to the STROBE & STEADY mode.

1) Make sure that tailcone anti-collision lights (left/right) flash continuously.

(b) Set the POSITION switch for the tailcone anti-collision light, on the P5 overhead panel, to the STEADY mode.

1) Make sure that the tailcone anti-collision lights (left/right) come on and stay on.

(c) Set the POSITION switch to the OFF mode.

1) Make sure that the tailcone anti-collision lights (left/right) go off.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 33-44-04-860-019

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	19	C01344	APU FIRE SW POWER

F/O Electrical System Panel, P6-4

Row	Col	Number	Name
A	14	C00033	AUX POWER UNIT CONT

SUBTASK 33-44-04-860-027

(2) Remove the DO NOT OPERATE tag from the APU master switch, on the P5 forward overhead panel.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WINGLET ANTI-COLLISION LIGHTS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Winglet Anti-Collision Light Deactivation

(2) Winglet Anti-Collision Light Activation

(3) Winglet Anti-Collision Light Removal

(4) Winglet Anti-Collision Light Installation

(5) Winglet Anti-Collision Light Transformer Replacement

(6) Winglet Anti-Collision Light LED Module and Power Supply Replacement.

TASK 33-44-13-040-801

2. Winglet Anti-Collision Light Deactivation

A. General

(1) This procedure removes electrical power from the winglet anti-collision lights.

B. Location Zones

Zone	Area
500	Left Wing
527	Left Winglet
600	Right Wing
627	Right Winglet

C. Procedure

SUBTASK 33-44-13-860-010

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	13	C04054	ANTI COLLISION LIGHTS-WINGLET-WHITE

D. Winglet Anti-Collision Lights Deactivation Test

NOTE: This test is to make sure that the winglet anti-collision lights are in a zero energy condition.

SUBTASK 33-44-13-200-001

(1) Do these steps to make sure that the power is off:

(a) Set the POSITION switch for the winglet anti-collision lights to STEADY on the P5 overhead panel.

1) Make sure that the winglet anti-collision lights (left/right) stay off.

(b) Set the POSITION switch for the winglet anti-collision lights to STROBE & STEADY on the P5 overhead panel.

1) Make sure that the winglet anti-collision lights (left/right) stay off.

(c) Set the POSITION switch for the winglet anti-collision lights to OFF.

1) Make sure that the winglet anti-collision lights (left/right) stay off.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-44-13-440-801

3. Winglet Anti-Collision Light Activation

A. General

(1) This procedure supplies electrical power to the winglet anti-collision lights.

B. Location Zones

Zone	Area
500	Left Wing
527	Left Winglet
600	Right Wing
627	Right Winglet

C. Procedure

SUBTASK 33-44-13-860-011

(1) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	13	C04054	ANTI COLLISION LIGHTS-WINGLET-WHITE

D. Winglet Anti-Collision Lights Activation Test

NOTE: This test is to make sure that the winglet anti-collision lights operate.

SUBTASK 33-44-13-710-003

(1) Do these steps to make sure that the power is on:

(a) Set the POSITION switch for the winglet anti-collision lights to STROBE & STEADY on the P5 overhead panel.

1) Make sure that the winglet anti-collision lights (left/right) come on and flash continuously.

(b) Set the POSITION switch for the winglet anti-collision lights to STEADY on the P5 overhead panel.

1) Make sure that the winglet anti-collision lights (left/right) go off.

(c) Set the POSITION switch for the winglet anti-collision lights to OFF on the P5 overhead panel.

1) Make sure that the winglet anti-collision lights (left/right) go off.

——— END OF TASK ———

TASK 33-44-13-000-801

4. Winglet Anti-Collision Light Removal

(Figure 201)

A. General

(1) This task contains instructions for the removal of the anti-collision light from the lower blade of the winglet.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-659	Platform - Maintenance Lift, 3 ft Minimum Height, 12.5 ft Maximum Height Part #: 1135-8000 Supplier: 00994 Opt Part #: 8662-010 Supplier: 00994
COM-2480	Platform - Mobile Elevating Work Platform SJ III Series Part #: SJIII 4726 Supplier: 3AF08 Part #: SJIII 4732 Supplier: 3AF08 Opt Part #: SJIII 4620 Supplier: 3AF08 Opt Part #: SJIII 4626 Supplier: 3AF08 Opt Part #: SJIII 4632 Supplier: 3AF08
SPL-768	Sealant Removal Tool, Non-Metallic Part #: ST982L-9 Supplier: 81205 Part #: ST982LF-9 Supplier: 81205 Part #: ST982LH-A-1 Supplier: 81205

D. Location Zones

Zone	Area
500	Left Wing
527	Left Winglet
600	Right Wing
627	Right Winglet

E. Prepare for the Removal

SUBTASK 33-44-13-860-001

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	13	C04054	ANTI COLLISION LIGHTS-WINGLET-WHITE

SUBTASK 33-44-13-860-003

(2) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 33-44-13-800-001

(3) Get a ladder, work platform, COM-2480, or maintenance platform, COM-659.

SUBTASK 33-44-13-800-002

(4) Do these steps to prepare metal support equipment (work platforms, work/maintenance stands, ladders):

NOTE: These steps apply to all metal support equipment in a 50.0 ft (15.2 m) radius of an open fuel tank.

(a) All support equipment must be in place before you begin the procedure.

(b) Bond the support equipment at an approved airplane bonding location.

(c) Ground the support equipment to the same ground as the airplane.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Winglet Anti-Collision Light Removal

SUBTASK 33-44-13-000-001

(1) Remove the anti-collision light [21]:

(a) Remove screws [22] from the anti-collision light [21].

(b) Remove the sealant from around the anti-collision light [21].

(c) Remove the anti-collision light [21] from the leading edge of the lower blade of the winglet.

SUBTASK 33-44-13-020-001

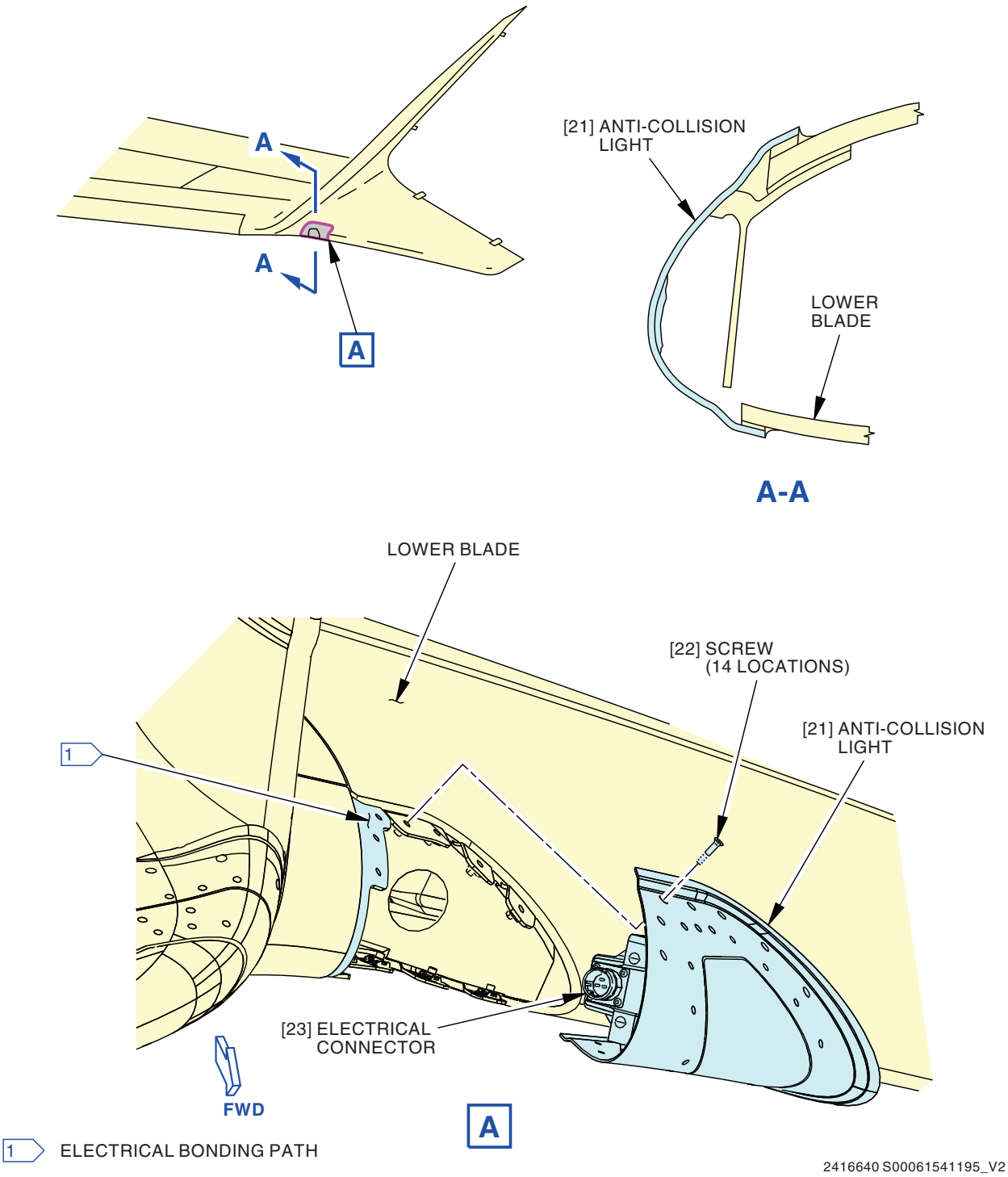
(2) Disconnect the electrical connector [23] from the anti-collision light [21].

SUBTASK 33-44-13-020-002

(3) Remove all sealant from leading edge of the winglet and the anti-collision light [21] with a sealant removal tool, SPL-768.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Anti-Collision Light Installation
Figure 201/33-44-13-990-809

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-44-13-400-801

5. Winglet Anti-Collision Light Installation

(Figure 201)

A. General

(1) This task contains instructions for the installation of the anti-collision light on the lower blade of the winglet.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
33-44-00-710-801	Anti-Collision Lights - Operational Test (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-659	Platform - Maintenance Lift, 3 ft Minimum Height, 12.5 ft Maximum Height Part #: 1135-8000 Supplier: 00994 Opt Part #: 8662-010 Supplier: 00994
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
COM-2480	Platform - Mobile Elevating Work Platform SJ III Series Part #: SJIII 4726 Supplier: 3AF08 Part #: SJIII 4732 Supplier: 3AF08 Opt Part #: SJIII 4620 Supplier: 3AF08 Opt Part #: SJIII 4626 Supplier: 3AF08 Opt Part #: SJIII 4632 Supplier: 3AF08

D. Consumable Materials

Reference	Description	Specification
A50343	Sealant - PPG Aerospace PR-1772 Class B Low Weight Fuselage Sealant	BMS5-142 Type II Class B
A50345	Sealant - PPG Aerospace PR-1776M Class B Low Weight Fuel Tank Sealant	BMS5-45 Class B

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
21	Anti-collision light	33-44-13-01-020	PTF ALL
		33-44-13-01-025	PTF ALL

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F. Location Zones

Zone	Area
500	Left Wing
527	Left Winglet
600	Right Wing
627	Right Winglet

G. Winglet Anti-Collision Light Installation

SUBTASK 33-44-13-100-001

(1) Clean the designated bond area of the anti-collision light [21] and the lower blade of the winglet to make an electrical bond.

SUBTASK 33-44-13-390-001

(2) Apply a layer of PR-1776M Class B Sealant, A50345, to the mounting surface between the anti-collision light [21] and the lower blade of the winglet to make a fay surface seal.

SUBTASK 33-44-13-420-001

(3) Connect the electrical connector [23] for the anti-collision light [21].

SUBTASK 33-44-13-400-001

(4) Install the anti-collision light [21]:

(a) Put the anti-collision light [21] in its position on the lower blade of the winglet.

(b) Apply a layer of PR-1776M Class B Sealant, A50345, to the screws [22].

(c) Install the screws [22] to attach the anti-collision light [21].

(d) Apply an aerodynamic fillet seal to the external gap around the anti-collision light [21] with PR-1772 Class B Sealant, A50343.

SUBTASK 33-44-13-760-001

(5) Measure the electrical resistance between the anti-collision light [21] and the lower blade of the winglet (SWPM 20-20-00).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the electrical resistance is 0.0005 ohm (0.5 milliohm) or less.

SUBTASK 33-44-13-860-004

(6) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	13	C04054	ANTI COLLISION LIGHTS-WINGLET-WHITE

SUBTASK 33-44-13-710-001

(7) Do this task: Anti-Collision Lights - Operational Test, TASK 33-44-00-710-801.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 33-44-13-080-001

(1) Remove the ladder, work platform, COM-2480, or maintenance platform, COM-659.

SUBTASK 33-44-13-860-006

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-44-13-960-801

6. Winglet Anti-Collision Light Transformer Replacement

(Figure 202)

A. References

Reference	Title
WDM 33-44-12	Wiring Diagram Manual

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

C. Consumable Materials

Reference	Description	Specification
A50473	Sealant - Brush (BMS5-95 Type I Class A-2 (See BAC5000 8.5.3 a. for preparation instructions))	BAC5000 Section 8.5.3 a.
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Location Zones

Zone	Area
527	Left Winglet
627	Right Winglet

E. Access Panels

Number	Name/Location
521AAB	Lower Leading Edge Access Panel
621ABB	Lower Leading Edge Access Panel

F. Procedure

SUBTASK 33-44-13-860-007

(1) Open the applicable circuit breaker and install the safety tag:

(WDM 33-44-12)

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CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	13	C04054	ANTI COLLISION LIGHTS-WINGLET-WHITE

SUBTASK 33-44-13-960-001

(2) Do these steps to replace the winglet anti-collision light transformer [1]:

(a) Open the applicable access panel:

Number	Name/Location
521AAB	Lower Leading Edge Access Panel
621ABB	Lower Leading Edge Access Panel

(b) Disconnect the electrical connector [4].

(c) Remove the bolts [3] and washers [2].

(d) Remove the winglet anti-collision light transformer [1].

(e) Clean the fay surface bonding area with solvent, B00083, and a clean cotton wiper, G00034.

(f) Install the new winglet anti-collision light transformer [1].

(g) Install the washers [2] and bolts [3].

(h) With the electrical bonding intrinsically safe approved bonding meter, COM-1550, make sure that the resistance between the winglet anti-collision light transformer [1] and the fuselage measures no more than 0.0025 ohms (2.5 milliohms).

(i) Apply a polysulfide sealant, A50473 fillet seal between the winglet anti-collision light transformer [1] and the mounting bracket.

(j) Connect the electrical connector [4].

SUBTASK 33-44-13-860-008

(3) Remove the safety tag and close the applicable circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	13	C04054	ANTI COLLISION LIGHTS-WINGLET-WHITE

SUBTASK 33-44-13-710-002

(4) Do a test of the winglet anti-collision light transformer [1].

(a) Set the POSITION switch for the winglet anti-collision lights to STROBE and STEADY on the P5 overhead panel.

1) Make sure that the winglet anti-collision lights (left/right) come on and strobe.

(b) Set the POSITION switch for the winglet anti-collision lights to STEADY.

1) Make sure that the winglet anti-collision lights (left/right) go off.

(c) Set the POSITION switch for the winglet anti-collision lights to OFF.

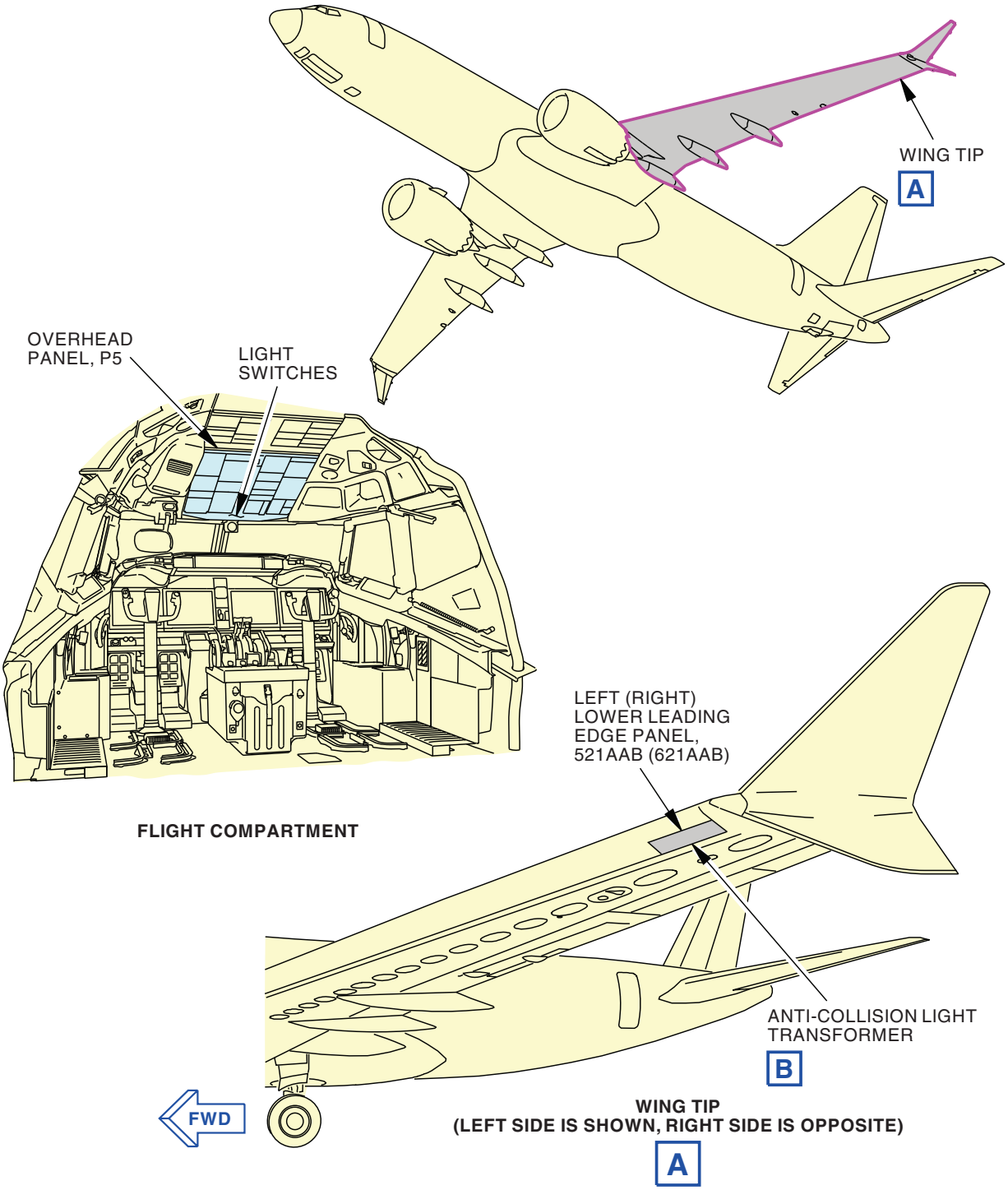
1) Make sure that the winglet anti-collision lights (left/right) go off.

(d) Close these access panels:

Number	Name/Location
521AAB	Lower Leading Edge Access Panel
621ABB	Lower Leading Edge Access Panel

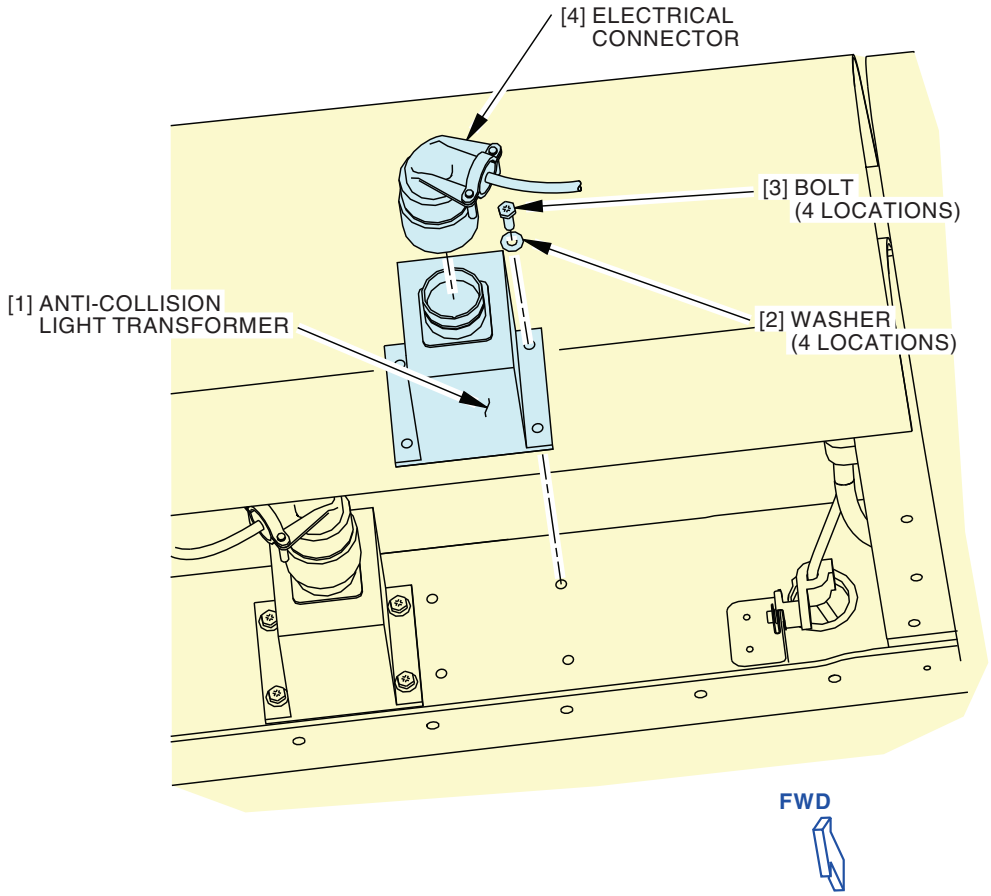
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Winglet Anti-Collision Light - Transformer Replacement
Figure 202/33-44-13-990-812 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ANTI-COLLISION LIGHT TRANSFORMER

B

2488865 S0000586428_V1

Winglet Anti-Collision Light - Transformer Replacement
Figure 202/33-44-13-990-812 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-44-13-960-802

7. Winglet Anti-Collision Light LED Module and Power Supply Unit Replacement

Figure 201, Figure 203

A. General

(1) This procedure contains instructions for the replacement of the following winglet anti-collision light parts:

(a) Winglet anti-collision light power supply unit

(b) Winglet anti-collision light LED module.

B. Consumable Materials

Reference	Description	Specification
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

C. Location Zones

Zone	Area
527	Left Winglet
627	Right Winglet

D. Procedure

SUBTASK 33-44-13-000-002

(1) Do this task: Winglet Anti-Collision Light Removal, TASK 33-44-13-000-801.

SUBTASK 33-44-13-020-003

(2) Remove the power supply unit [24] from winglet light assembly:

(a) Use white cotton gloves to handle the winglet light assembly.

(b) Remove the eight screws [27] that attaches the power supply unit [24] to the winglet light assembly.

(c) Remove the power supply unit [24] from the winglet light assembly.

(3) Remove LED module [25] from the power supply unit [24]:

(a) Use white cotton gloves to handle the LED module [25].

(b) Remove four screws [26] that attaches the LED module [25] to the power supply unit [24].

(c) Carefully pull the LED module [25] from the power supply unit.

(4) If power supply unit is defective, install LED module [25] to new power supply unit [24].

(a) Install the four screws [26] to secure the LED module [25] to the new power supply unit [24].

(5) If LED module [25] is defective, install new LED module [25] to the power supply unit [24].

(a) Install the four screws [26] to secure the LED module [25] to the new power supply unit [24].

SUBTASK 33-44-13-420-002

(6) Install thepower supply unit [24] to the winglet light assembly:

(a) Position the power supply unit [24] on to the winglet light assembly.

(b) Apply corrosion inhibiting material, G50136 to the eight screws [27] and install.

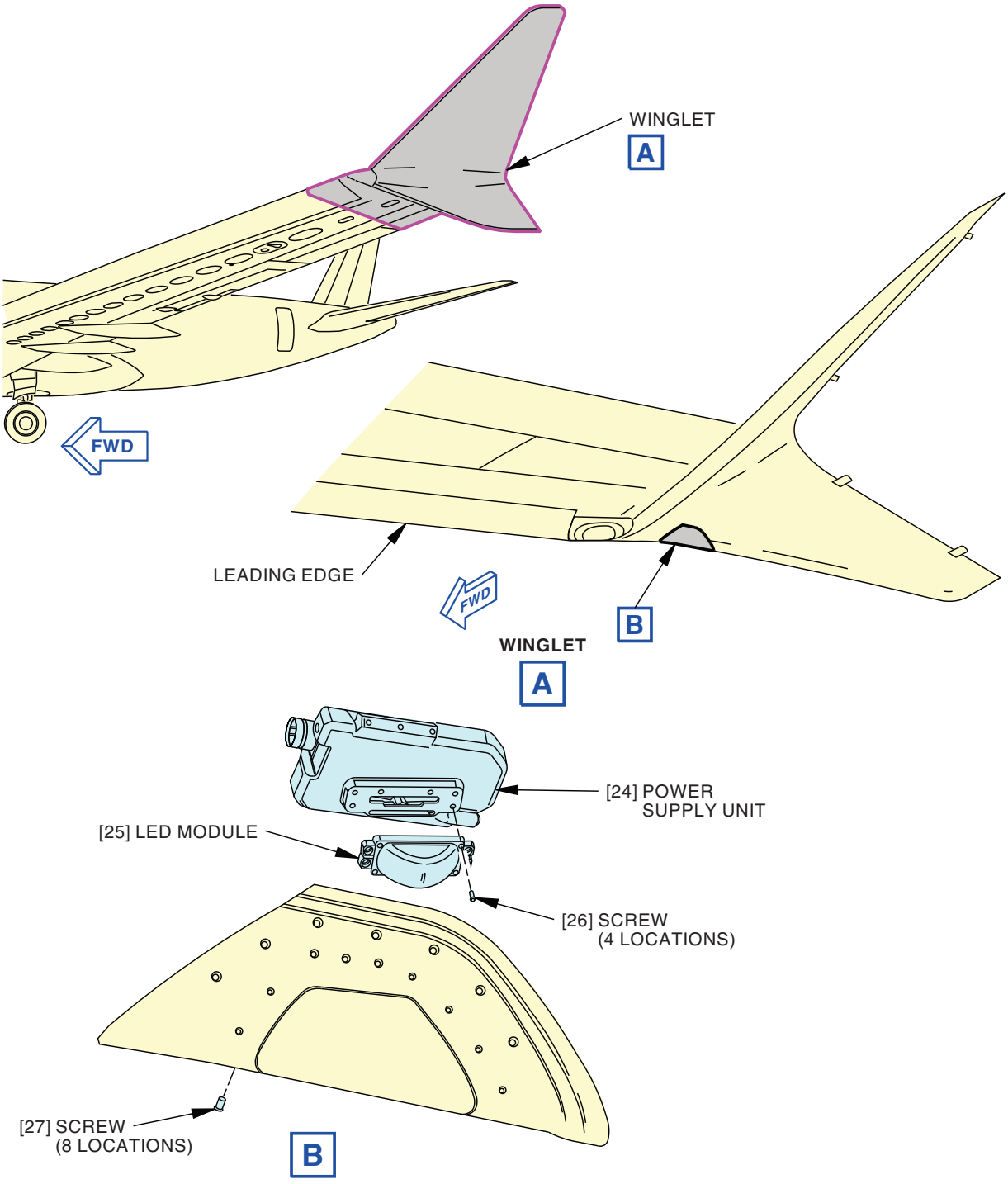
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-44-13-410-001

(7) Do this task: Winglet Anti-Collision Light Installation, TASK 33-44-13-400-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Winglet Anti-Collision LED Module and Power Supply Unit
Figure 203/33-44-13-990-813

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LOGO LIGHTS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Logo Light Deactivation

(2) Logo Light Activation

PTF ALL; AIRPLANES WITH LAMP LIGHTS

(3) Logo Light - Lamp Replacement

PTF ALL; AIRPLANES WITH LED MODULE ASSEMBLY LIGHTS

(4) Logo Light - LED Module Assembly Replacement

PTF ALL

(5) Logo Light - Light Assembly Replacement

TASK 33-49-00-040-801

2. Logo Light - Deactivation

(Figure 201)

A. General

(1) This procedure removes electrical power to the logo lights.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
332	Left Horizontal Stab - Front Spar to Rear Spar, Stab Sta 57.93 to Stab BL 249.81
342	Right Horizontal Stab - Front Spar to Rear Spar, Stab Sta 57.93 to Stab BL 249.81

C. Procedure

SUBTASK 33-49-00-860-001

(1) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	15	C00644	EXTERIOR LIGHTING LOGO ILLUM

D. Logo Light - Tryout

NOTE: This tryout is to make sure the Logo light is in a zero energy state.

SUBTASK 33-49-00-200-001

(1) Do the following steps to make sure the power has been removed:

(a) At the overhead panel, P5, set the logo light switch to the ON Position

(b) Set the GALLEY switch on the P5-13 panel to the ON position.

NOTE: On some airplanes the switch name is CAB/UTIL.

(c) Make sure that the logo lights are off.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-49-00-440-801

3. Logo Light - Activation

(Figure 201)

A. General

(1) This procedure adds electrical power to the logo lights.

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
332	Left Horizontal Stab - Front Spar to Rear Spar, Stab Sta 57.93 to Stab BL 249.81
342	Right Horizontal Stab - Front Spar to Rear Spar, Stab Sta 57.93 to Stab BL 249.81

C. Procedure

SUBTASK 33-49-00-860-002

(1) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	15	C00644	EXTERIOR LIGHTING LOGO ILLUM

——— END OF TASK ———

PTF ALL; AIRPLANES WITH LAMP LIGHTS

TASK 33-49-00-960-801

4. Logo Light - Lamp Replacement

(Figure 201)

A. References

Reference	Title
27-31-00-000-801	Elevator and Tab Control System - Deactivation (P/B 201)
27-31-00-400-801	Elevator and Tab Control System - Activation (P/B 201)
27-41-00-800-801	Horizontal Stabilizer Trim Control System - Deactivation (P/B 201)
27-41-00-800-802	Horizontal Stabilizer Trim Control System - Activation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
G01306	Gloves - Lint-free	

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Lamp	33-49-00-02-100	PTF ALL

D. Location Zones

Zone	Area
332	Left Horizontal Stab - Front Spar to Rear Spar, Stab Sta 57.93 to Stab BL 249.81
342	Right Horizontal Stab - Front Spar to Rear Spar, Stab Sta 57.93 to Stab BL 249.81

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH LAMP LIGHTS (Continued)

E. Procedure

SUBTASK 33-49-00-040-003

(1) Do this task: Horizontal Stabilizer Trim Control System - Deactivation, TASK 27-41-00-800-801.

SUBTASK 33-49-00-040-004

(2) Do this task: Elevator and Tab Control System - Deactivation, TASK 27-31-00-000-801.

SUBTASK 33-49-00-860-003

(3) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	15	C00644	EXTERIOR LIGHTING LOGO ILLUM

SUBTASK 33-49-00-010-001

(4) Open the hinged panel assembly to gain access to the logo light assembly.

NOTE: The hinged panel assembly is located between access panels 333EB (343EB) and 333FB (343FB).

SUBTASK 33-49-00-960-001

CAUTION

DO NOT TOUCH THE HALOGEN LAMP WITH YOUR BARE SKIN. USE A CLOTH TO HOLD THE HALOGEN LAMP. IF YOUR SKIN TOUCHES THE HALOGEN LAMP, YOU WILL DECREASE ITS LIFE.

(5) Do these steps to replace the lamp with lint-free gloves, G01306:

(a) Loosen the two fasteners [4] and remove the socket cover [3] on the bottom of the logo light housing.

NOTE: The lanyard [8] stays connected.

(b) Release the top spring [5] from the left hook [9] to remove the socket [2] (View C).

(c) Remove the socket [2].

(d) Release the lower spring [6] from the right hook [7] (View D).

(e) Remove the lamp [1] from the rear of the reflector assembly [11] (View E).

(f) Install the new lamp [1] to the reflector assembly [11].

1) Make sure the lamp [1] is flush into the reflector assembly [11].

(g) Latch the lower spring [6] to the right hook [7] to hold the lamp [1] in its position (View D).

NOTE: The “V” bend on the lower spring [6] is larger than the top spring [5].

1) Make sure the “V” bend of the lower spring [6] is against the edge of the lamp [1].

CAUTION

DO NOT USE TOO MUCH FORCE. TOO MUCH FORCE CAN CAUSE DAMAGE TO THE LAMP PINS.

(h) Carefully install the socket [2] to the lamp pins.

NOTE: You put the socket [2] to the lamp pins. You hold the socket [2] in its position by the top spring [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH LAMP LIGHTS (Continued)

1) Make sure the lower spring [6] is positioned under the socket [2] (View C).

CAUTION MAKE SURE THAT THE TOP SPRING IS CORRECTLY ALIGNED. IF YOU DO NOT ALIGN IT CORRECTLY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(i) Latch the top spring [5] to the left hook [9] to hold the socket [2] in its position (View C).

1) Make sure the socket [2] is seated firmly under the top spring [5].

(j) Install the socket cover [3] and tighten the fasteners [4].

SUBTASK 33-49-00-940-001

(6) Close the hinged panel assembly.

SUBTASK 33-49-00-860-004

(7) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	15	C00644	EXTERIOR LIGHTING LOGO ILLUM

SUBTASK 33-49-00-440-001

(8) If necessary, do this task: Horizontal Stabilizer Trim Control System - Activation, TASK 27-41-00-800-802.

SUBTASK 33-49-00-440-002

(9) If necessary, do this task: Elevator and Tab Control System - Activation, TASK 27-31-00-400-801.

SUBTASK 33-49-00-710-001

(10) Do a test of the new lamp:

(a) Set the GALLEY switch on the P5-13 panel to the ON position.

NOTE: On some airplanes the switch name is CAB/UTIL.

(b) At the overhead panel, P5, set the switch for the logo lights to the on mode.

1) Make sure that the lamp comes on.

2) It can be necessary to adjust the direction of the light beam. Turn the adjustment screws below the housing until the light beam is on the logo.

(c) Set the GALLEY switch on the P5-13 panel to the OFF position.

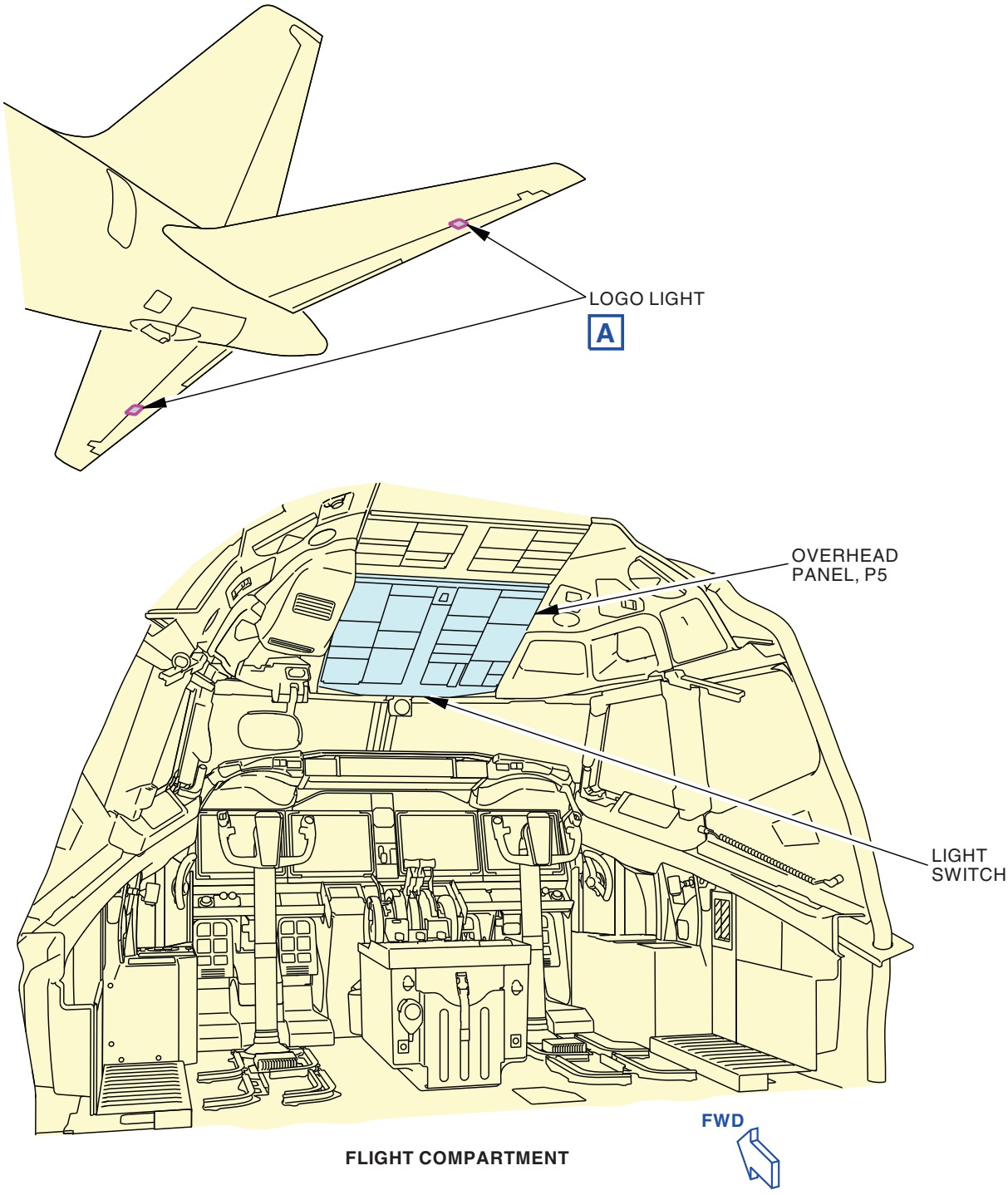
NOTE: On some airplanes the switch name is CAB/UTIL.

(d) Set the switch for the logo lights to the off mode.

1) Make sure that the lamp goes off.

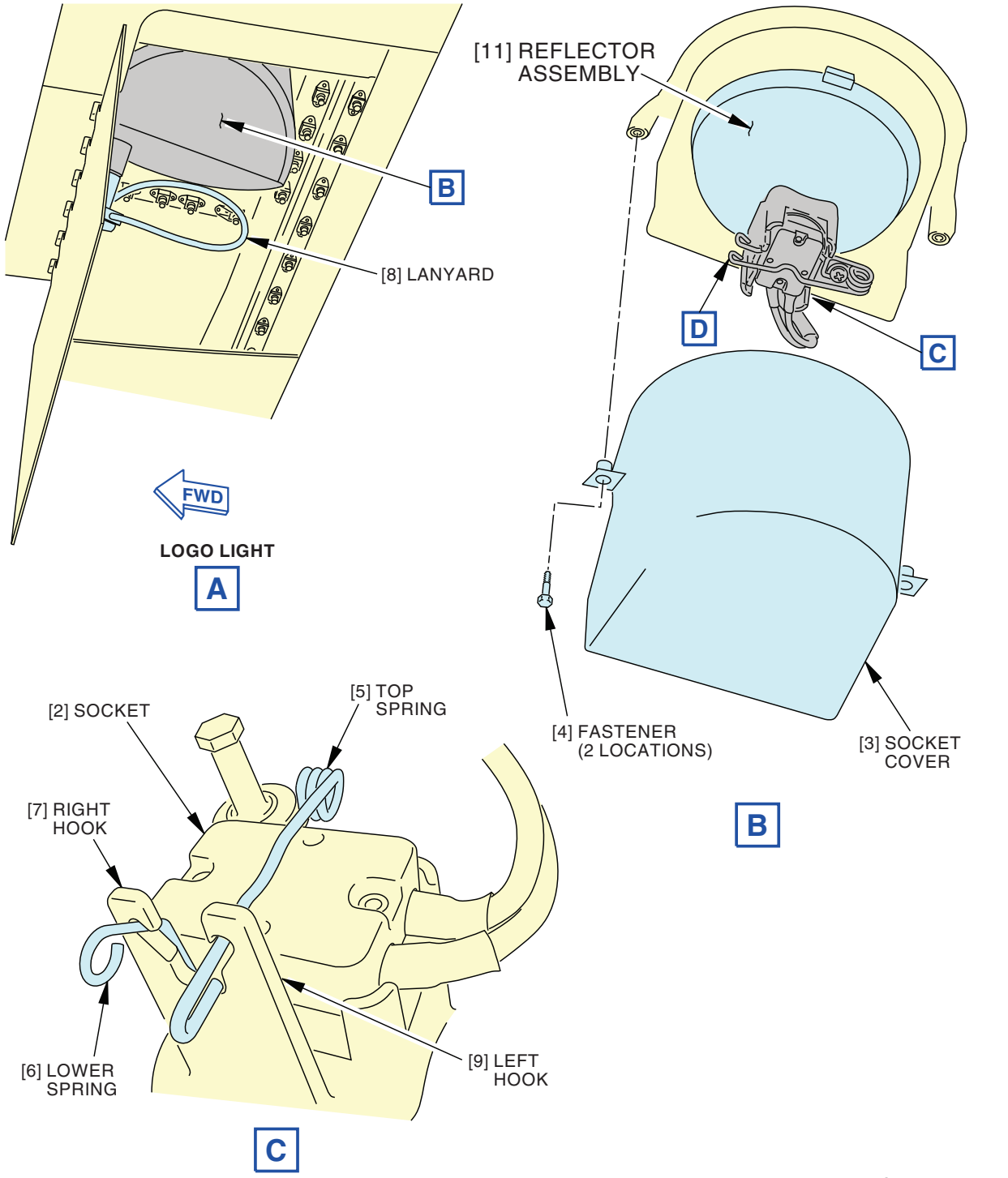
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



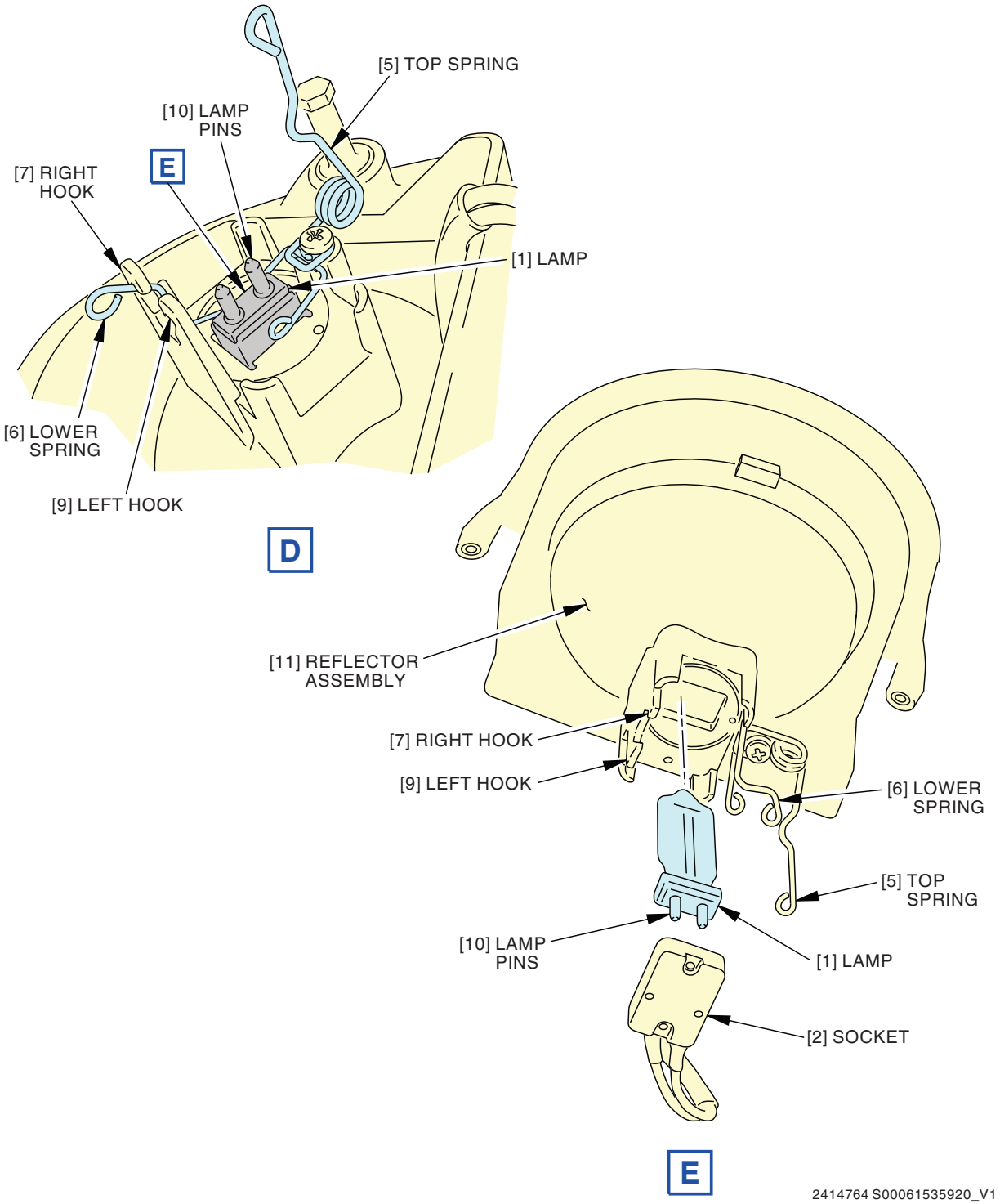
Logo Light - Lamp Replacement
Figure 201/33-49-00-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Logo Light - Lamp Replacement
Figure 201/33-49-00-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Logo Light - Lamp Replacement
Figure 201/33-49-00-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH LED MODULE ASSEMBLY LIGHTS

TASK 33-49-00-960-803

5. Logo Light - LED Module Assembly Replacement

(Figure 202)

A. General

(1) This procedure gives the steps necessary for the logo light LED module assembly replacement.

B. References

Reference	Title
27-31-00-000-801	Elevator and Tab Control System - Deactivation (P/B 201)
27-31-00-400-801	Elevator and Tab Control System - Activation (P/B 201)
27-41-00-800-801	Horizontal Stabilizer Trim Control System - Deactivation (P/B 201)
27-41-00-800-802	Horizontal Stabilizer Trim Control System - Activation (P/B 201)

C. Location Zones

Zone	Area
332	Left Horizontal Stab - Front Spar to Rear Spar, Stab Sta 57.93 to Stab BL 249.81
342	Right Horizontal Stab - Front Spar to Rear Spar, Stab Sta 57.93 to Stab BL 249.81

D. Procedure

SUBTASK 33-49-00-040-005

(1) Do this task: Horizontal Stabilizer Trim Control System - Deactivation, TASK 27-41-00-800-801.

SUBTASK 33-49-00-040-006

(2) Do this task: Elevator and Tab Control System - Deactivation, TASK 27-31-00-000-801.

SUBTASK 33-49-00-860-009

(3) Make sure that this circuit breaker is open and has safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	15	C00644	EXTERIOR LIGHTING LOGO ILLUM

SUBTASK 33-49-00-010-002

(4) Open the hinged panel assembly to gain access to the logo light assembly.

NOTE: The hinged panel assembly is located between access panels 333EB and 333FB (343EB and 343FB on the right wing).

SUBTASK 33-49-00-960-003

(5) Do these steps to replace the LED module assembly [30]:

(a) Remove the captive screw [33] from the LED module assembly [30].

(b) Remove the captive screws [31].

(c) Pull down the LED module assembly [30].

(d) Disconnect the connector [32] at J2 from the LED module assembly [30].

(e) Connect the connector [32] at J2 to a new LED module assembly [30].

(f) Put the new LED module assembly [30] in position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH LED MODULE ASSEMBLY LIGHTS (Continued)

(g) Install the captive screws [31].

(h) Install the captive screw [33] to the LED module assembly [30].

SUBTASK 33-49-00-940-002

(6) Close the hinged panel assembly.

SUBTASK 33-49-00-860-010

(7) Close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	15	C00644	EXTERIOR LIGHTING LOGO ILLUM

SUBTASK 33-49-00-440-003

(8) Do this task: Horizontal Stabilizer Trim Control System - Activation, TASK 27-41-00-800-802.

SUBTASK 33-49-00-440-004

(9) Do this task: Elevator and Tab Control System - Activation, TASK 27-31-00-400-801.

SUBTASK 33-49-00-710-003

(10) Do a test of the new LED module assembly:

(a) Set the GALLEY switch on the P5-13 panel to the ON position.

NOTE: On some airplanes the switch name is CAB/UTIL.

(b) At the overhead panel, P5, set the switch for the logo lights to the ON mode.

1) Make sure that the LED module assembly comes on.

2) It can be necessary to adjust the direction of the light beam. Turn the adjustment screws below the housing until the light beam is on the logo.

(c) Set the GALLEY switch on the P5-13 panel to the OFF position.

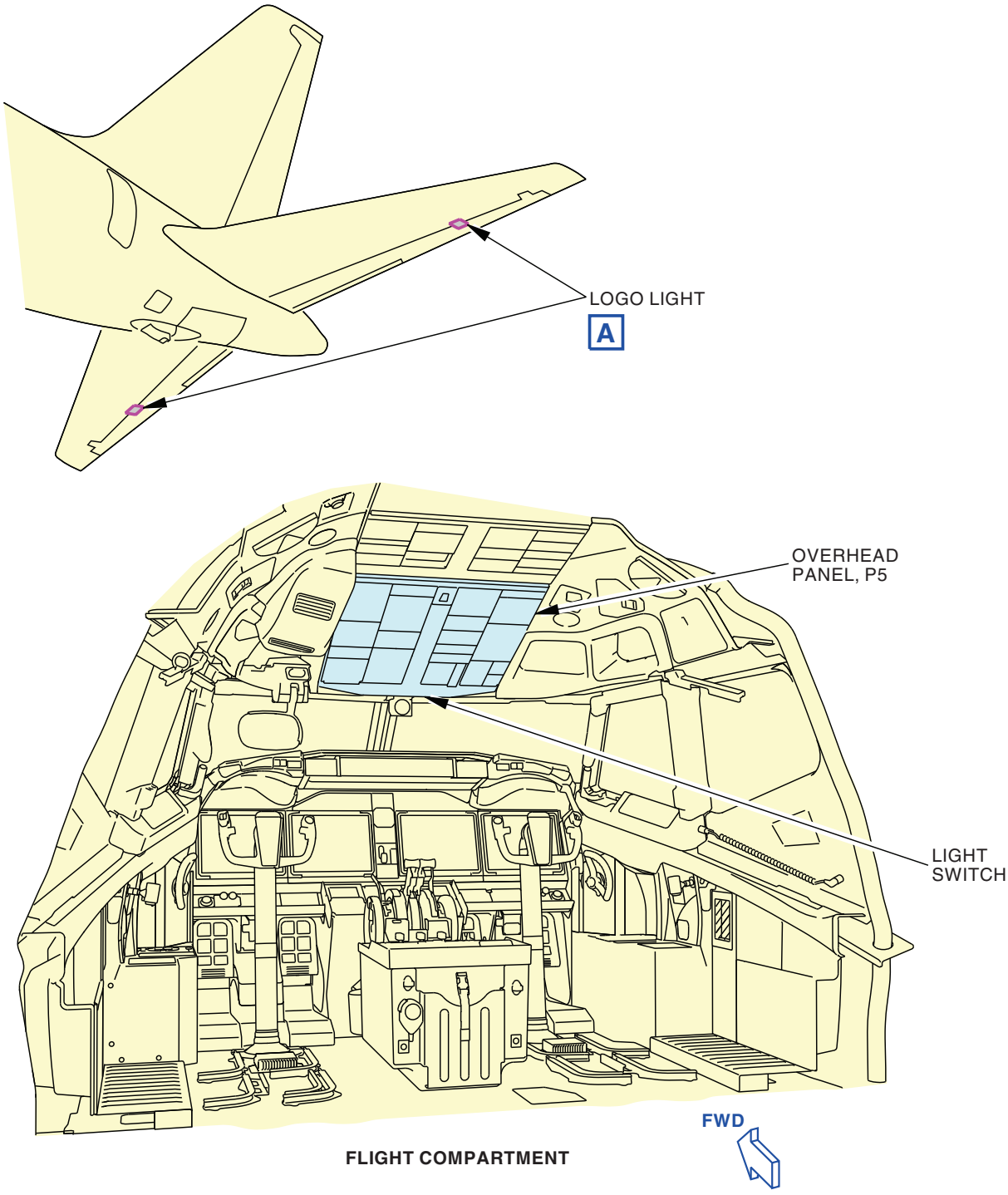
NOTE: On some airplanes the switch name is CAB/UTIL.

(d) Set the switch for the logo lights to the OFF mode.

1) Make sure the LED module assembly goes off.

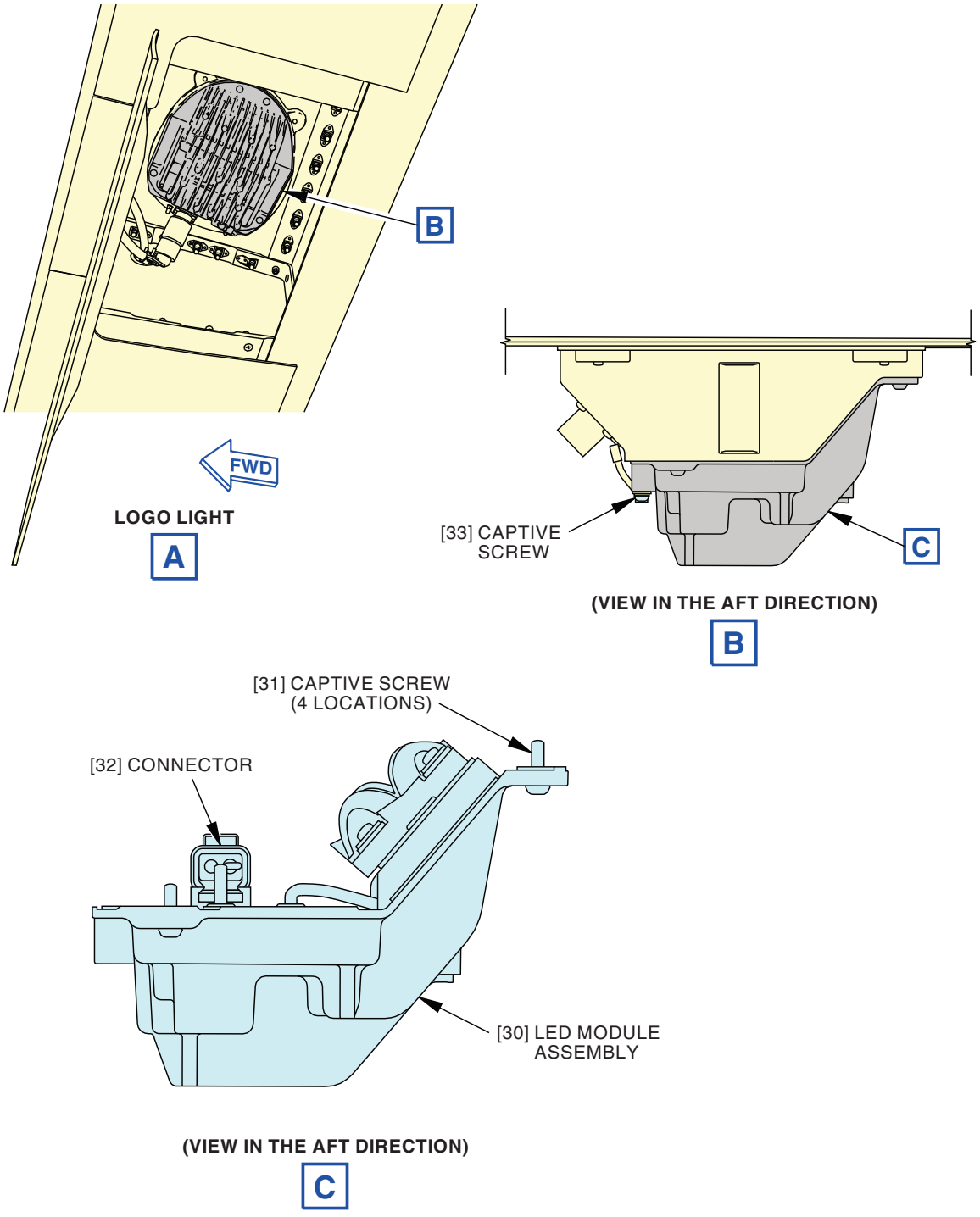
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Logo Light - LED Module Assembly Replacement
Figure 202/33-49-00-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Logo Light - LED Module Assembly Replacement
Figure 202/33-49-00-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL

TASK 33-49-00-960-802

6. Logo Light - Light Assembly Replacement

(Figure 203)

A. General

PTF ALL; AIRPLANES WITH LAMP LIGHTS

(1) When you replace the light assembly, do not discard it. Usually, you can repair the assembly with a new transformer. Do the repairs in a shop.

PTF ALL

B. References

Reference	Title
27-31-00-000-801	Elevator and Tab Control System - Deactivation (P/B 201)
27-31-00-400-801	Elevator and Tab Control System - Activation (P/B 201)
27-41-00-800-801	Horizontal Stabilizer Trim Control System - Deactivation (P/B 201)
27-41-00-800-802	Horizontal Stabilizer Trim Control System - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659

D. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
B00065	Alcohol - Denatured, Ethyl (Ethanol)	AMS 3002 (Supersedes O-A-396)
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G01306	Gloves - Lint-free	
G50952	Abrasive - Silicon Carbide, 180 Grit	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description	Specification
G51464	Abrasive - Aluminum Oxide paper, 60 grit or finer	
G51465	Abrasive - Garnet sandpaper, 240 (7/0) grit	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
22	Light assembly	33-49-00-01-010	PTF ALL
		33-49-00-01-015	PTF ALL

F. Location Zones

Zone	Area
332	Left Horizontal Stab - Front Spar to Rear Spar, Stab Sta 57.93 to Stab BL 249.81
342	Right Horizontal Stab - Front Spar to Rear Spar, Stab Sta 57.93 to Stab BL 249.81

G. Access Panels

Number	Name/Location
333FB	Horizontal Stabilizer, Access Panel, Trailing Edge
343FB	Horizontal Stabilizer, Access Panel - T.E. Area

H. Logo Light - Light Assembly Replacement

SUBTASK 33-49-00-040-001

(1) Do this task: Horizontal Stabilizer Trim Control System - Deactivation, TASK 27-41-00-800-801.

SUBTASK 33-49-00-040-002

(2) Do this task: Elevator and Tab Control System - Deactivation, TASK 27-31-00-000-801.

SUBTASK 33-49-00-860-007

(3) Open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	15	C00644	EXTERIOR LIGHTING LOGO ILLUM

SUBTASK 33-49-00-960-002

(4) Do these steps to replace the light assembly [22]:

(a) Remove the bolts [21] on the top of the light assembly [22].

(b) Lift up the light assembly [22].

(c) Disconnect the electrical connector [23].

(d) Replace the light assembly [22].

1) Make sure a lamp is installed in the new light assembly [22].

a) Do not touch the new lamp with your hands. Use lint-free gloves, G01306

(e) Connect the electrical connector [23].

(f) Put the light assembly [22] in its position.

(g) Open these access panels to access to the back of the light assembly:

Number	Name/Location
333FB	Horizontal Stabilizer, Access Panel, Trailing Edge
343FB	Horizontal Stabilizer, Access Panel - T.E. Area

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO NOT BREATHE THE PARTICLES, OR GET THEM IN YOUR MOUTH, EYES, OR ON YOUR SKIN. PUT ON A RESPIRATOR, GOGGLES, AND GLOVES AND OTHER EQUIPMENT FOR PROTECTION. THESE PARTICLES CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(h) Prepare the surface around the edges of the joint between the lower surface of the light assembly [22] and airplane structure.

1) Use 60 grit or finer wet/dry aluminum oxide sandpaper, G51464, 180 grit silicon carbide abrasive, G50952, garnet 240 grit (7/0) or finer sandpaper, G51465, to clean the surface.

a) Sand until a bright aluminum bonding surface is visible.

WARNING DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. PUT ON A RESPIRATOR, PROTECTIVE SPLASH GOGGLES, AND GLOVES WHEN YOU USE SOLVENTS. KEEP THE SOLVENTS AWAY FROM SPARKS, FLAME, AND HEAT. SOLVENTS ARE POISONOUS AND FLAMMABLE. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

2) Clean the surface with cotton wiper, G00034, moist with alcohol, B00065.

(i) Apply a fillet seal with sealant, A02315, around the edges of the joint between the lower surface of the light assembly [22] and airplane structure.

(j) Close these access panels:

Number	Name/Location
333FB	Horizontal Stabilizer, Access Panel, Trailing Edge
343FB	Horizontal Stabilizer, Access Panel - T.E. Area

(k) Install the bolts [21] with the corrosion-preventative compound, C00528.

(l) With the intrinsically safe approved bonding meter, COM-1550, make sure the resistance between the light assembly [22] and the fuselage measures no more than 0.0025 ohms.

SUBTASK 33-49-00-860-008

(5) Remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	15	C00644	EXTERIOR LIGHTING LOGO ILLUM

SUBTASK 33-49-00-440-005

(6) Do this task: Horizontal Stabilizer Trim Control System - Activation, TASK 27-41-00-800-802.

SUBTASK 33-49-00-440-006

(7) Do this task: Elevator and Tab Control System - Activation, TASK 27-31-00-400-801.

SUBTASK 33-49-00-710-002

(8) Do a test of the light assembly:

(a) Set the GALLEY switch on the P5-13 panel to the ON position.

NOTE: On some airplanes the switch name is CAB/UTIL.

(b) At the overhead panel, P5, set the switch for the logo lights to the on mode.

1) Make sure that the light comes on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) If it is necessary to adjust the direction of the light beam, do these steps:
   a) Open the access panel below the logo light.
   b) Turn the adjustment screws below the housing until the light beam is on the logo.
   c) Close the access panel.

(c) Set the GALLEY switch on the P5-13 panel to the OFF position.

NOTE: On some airplanes the switch name is CAB/UTIL.

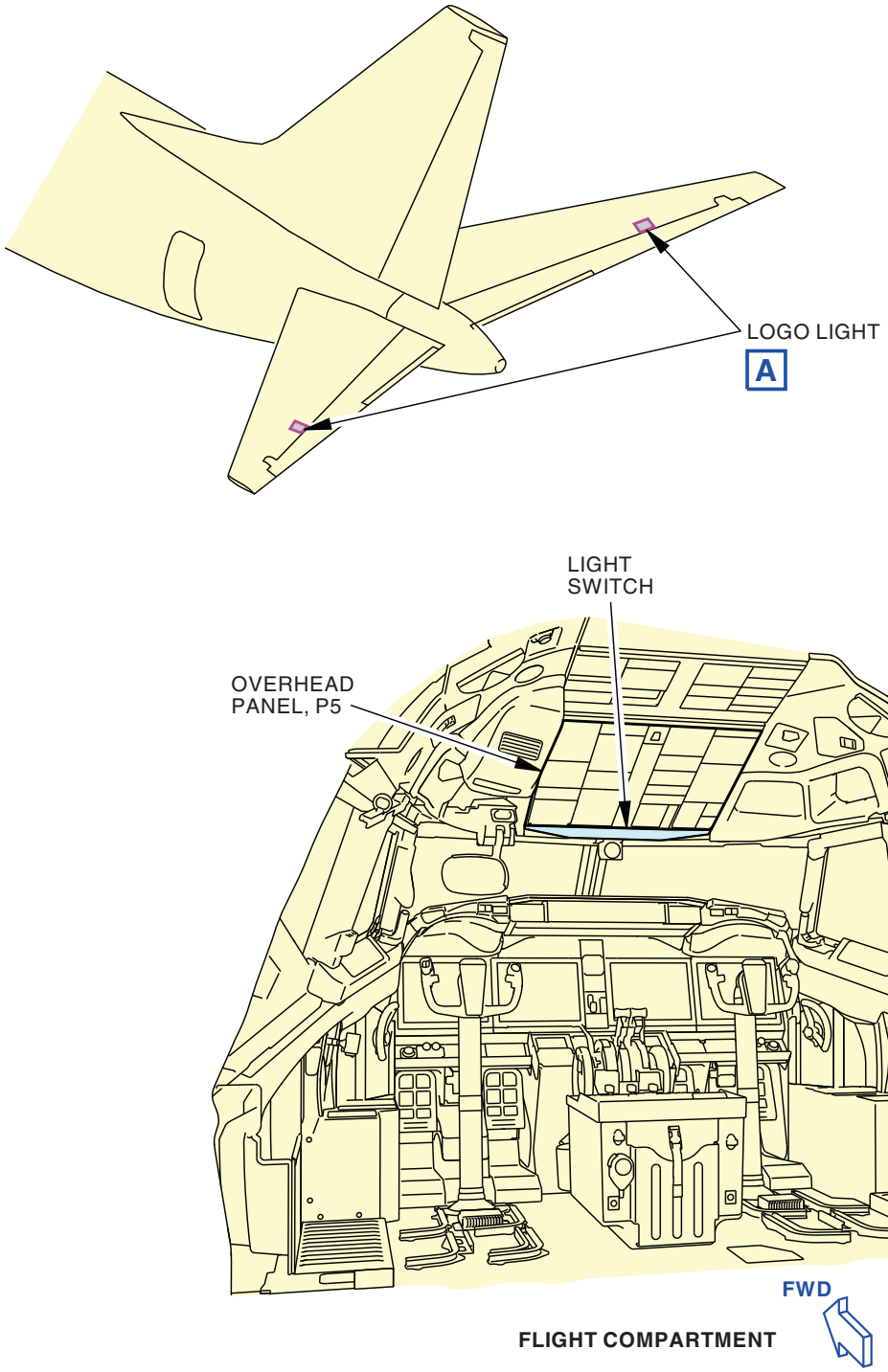
(d) Set the logo lights switch to the off mode.
   1) Make sure that the light goes off.

SUBTASK 33-49-00-390-001

(9) With sealant, A02315, apply an aerodynamic seal around the perimeter of the light assembly.

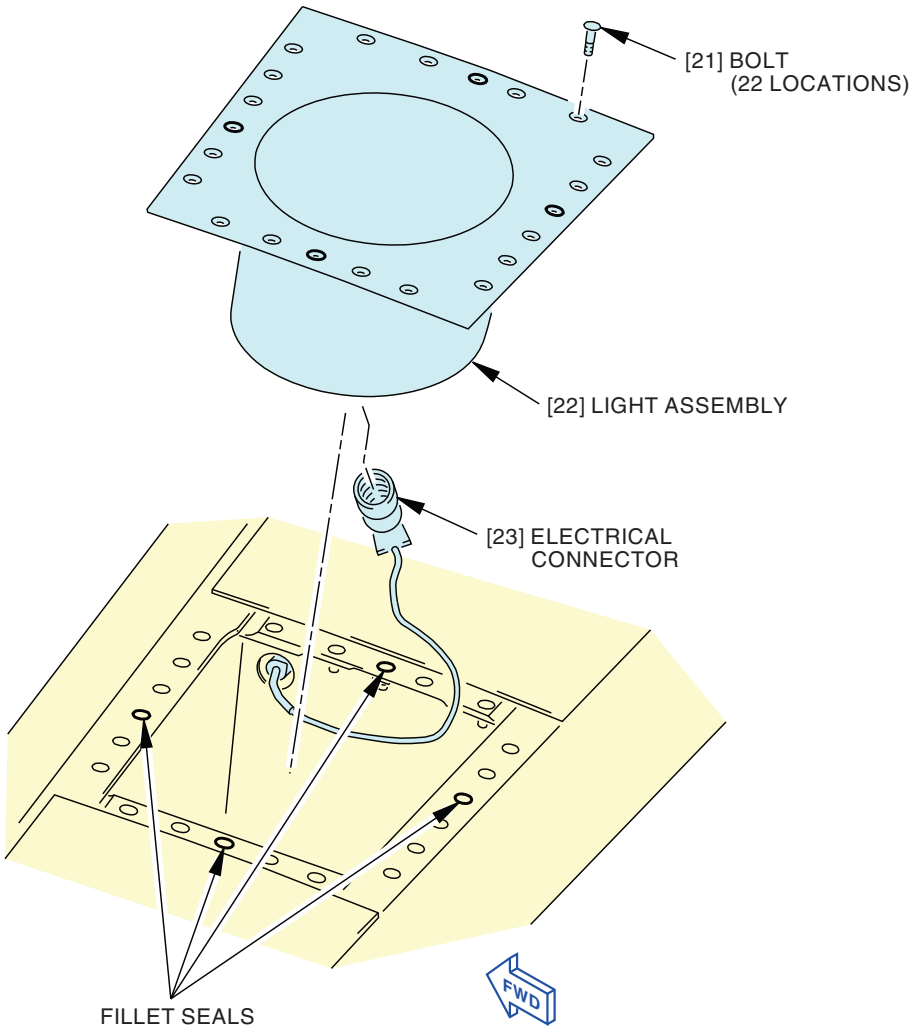
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Logo Light - Light Assembly Replacement
Figure 203/33-49-00-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LOGO LIGHT
(EXAMPLE)

A

2414766 S00061535923_V3

Logo Light - Light Assembly Replacement
Figure 203/33-49-00-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EMERGENCY LIGHTS - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Emergency Lights - Operational Check

(2) Emergency Lights - Operational Test

TASK 33-51-00-710-801

2. Emergency Lights Operational Check

(Figure 501, Figure 502, Figure 503, Figure 504 and Figure 505)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This check makes sure that the emergency lights come on.

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Emergency Lights Operational Check

SUBTASK 33-51-00-860-004

(1) Make sure that the BSI LED lighting system, on the aft attendant control panel, is turned off.

SUBTASK 33-51-00-860-007

(2) At the P5 overhead panel, set the DOME switch to the OFF mode.

SUBTASK 33-51-00-710-009

(3) Do a check of the emergency lights as follows:

(a) Set the EMER EXIT LIGHTS switch, on the P5 forward overhead panel, to the ON position, or on the aft attendant control panel, press the EMERGENCY EXIT LIGHTS switch.

NOTE: The power supply operates the emergency lights. Use the lights for a minimum time to do the test.

1) Make sure that the emergency dome light comes on (Figure 501).

2) Make sure that these emergency lights in the passenger compartment come on:

- Aisle light (Figure 502)
- Overdoor exit sign (Figure 503)
- Overwing hatch handle lights (Figure 503)
- Exit locator sign (Figure 503)
- Exit indicator light (Figure 503)
- If applicable, floor proximity light (Figure 504)
- If applicable, flood light.

3) Make sure that the exterior lights come on (Figure 505).

(b) Set the EMER EXIT LIGHTS switch, on the P5 overhead panel, to the OFF position, or on the aft attendant control panel, to the NORMAL position.

1) Make sure that the emergency dome light goes off (Figure 501).

2) Make sure that these emergency lights in the passenger compartment go off:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- Aisle light (Figure 502)
- Overdoor exit sign (Figure 503)
- Overwing hatch handle lights (Figure 503)
- Exit locator sign (Figure 503)
- Exit indicator light (Figure 503)
- If applicable, floor proximity light (Figure 504)
- If applicable, flood light.

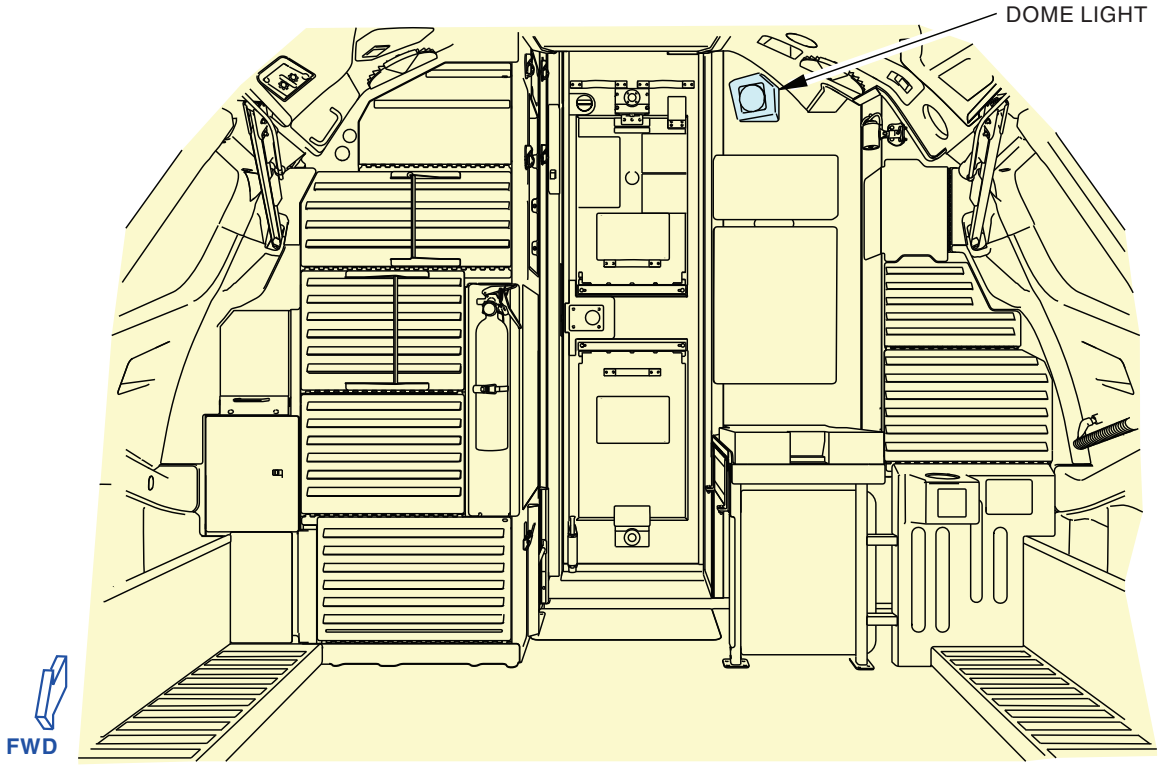
3) Make sure that the exterior lights go off (Figure 505).

SUBTASK 33-51-00-860-008

(4) At the P5 overhead panel, set the DOME switch to the DIM or BRIGHT mode as necessary.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

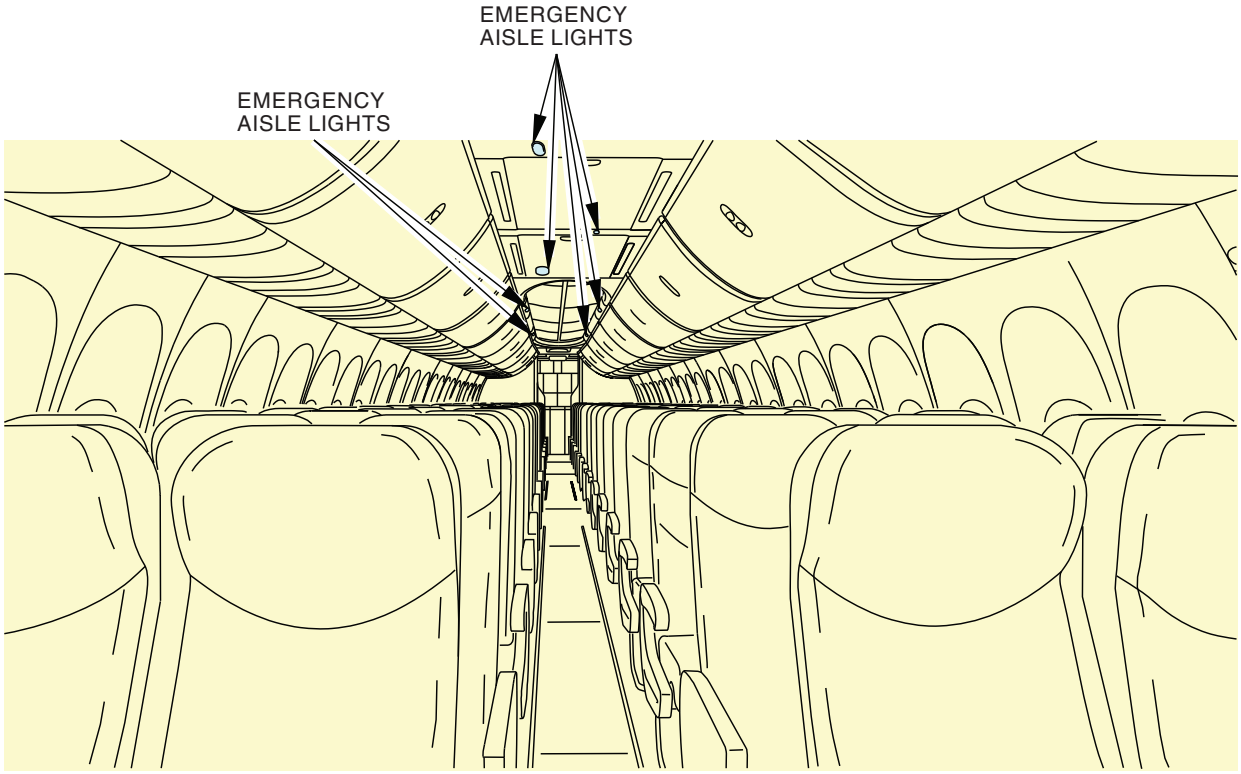


**FLIGHT COMPARTMENT
(EXAMPLE)**

3043043 S0000809935_V2

**Emergency Lights - Dome Lights
Figure 501/33-51-00-990-813**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

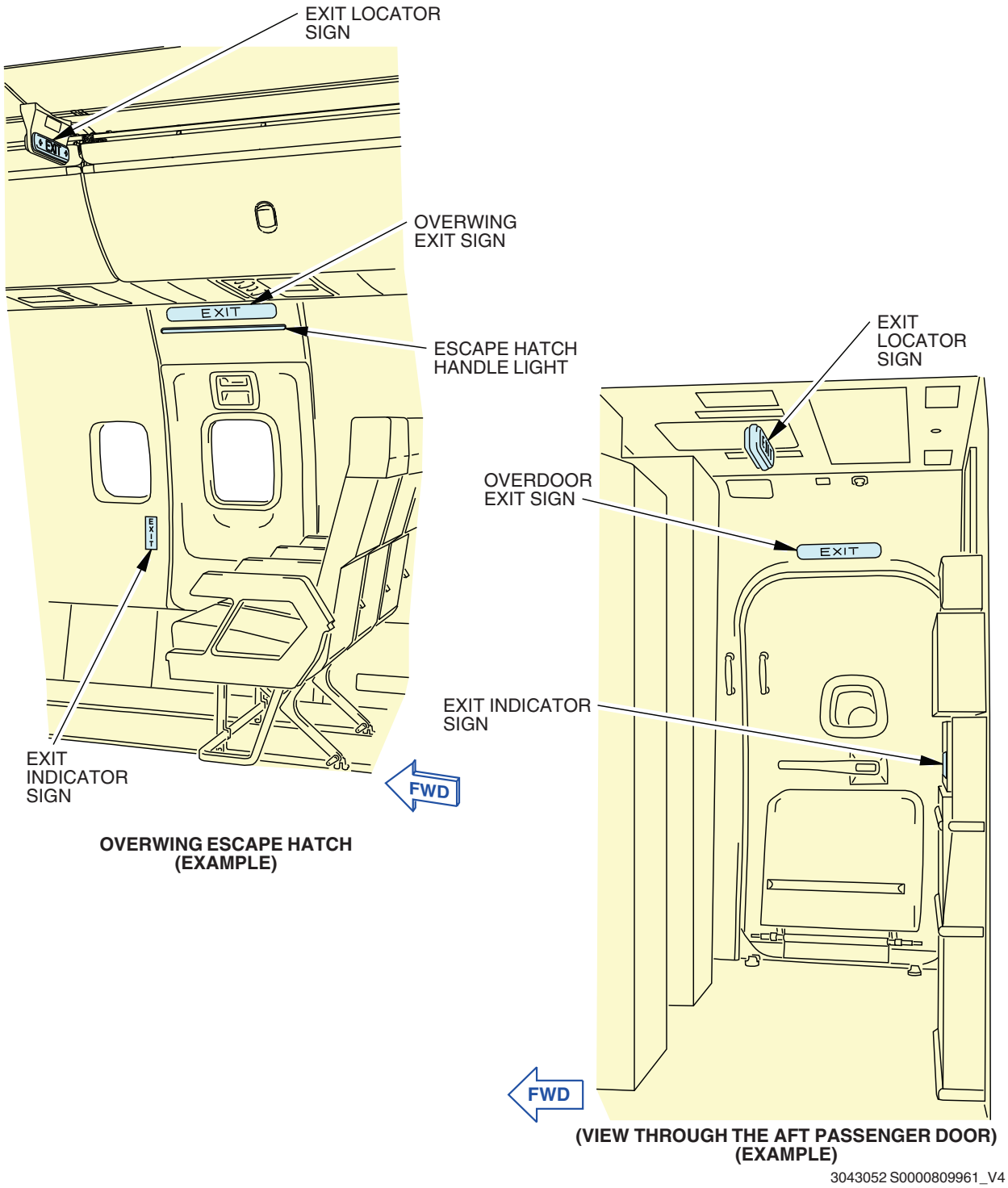


**PASSENGER COMPARTMENT
(EXAMPLE)**

3043063 S0000809963_V2

**Emergency Lights - Emergency Aisle Lights (EAL)
Figure 502/33-51-00-990-814**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



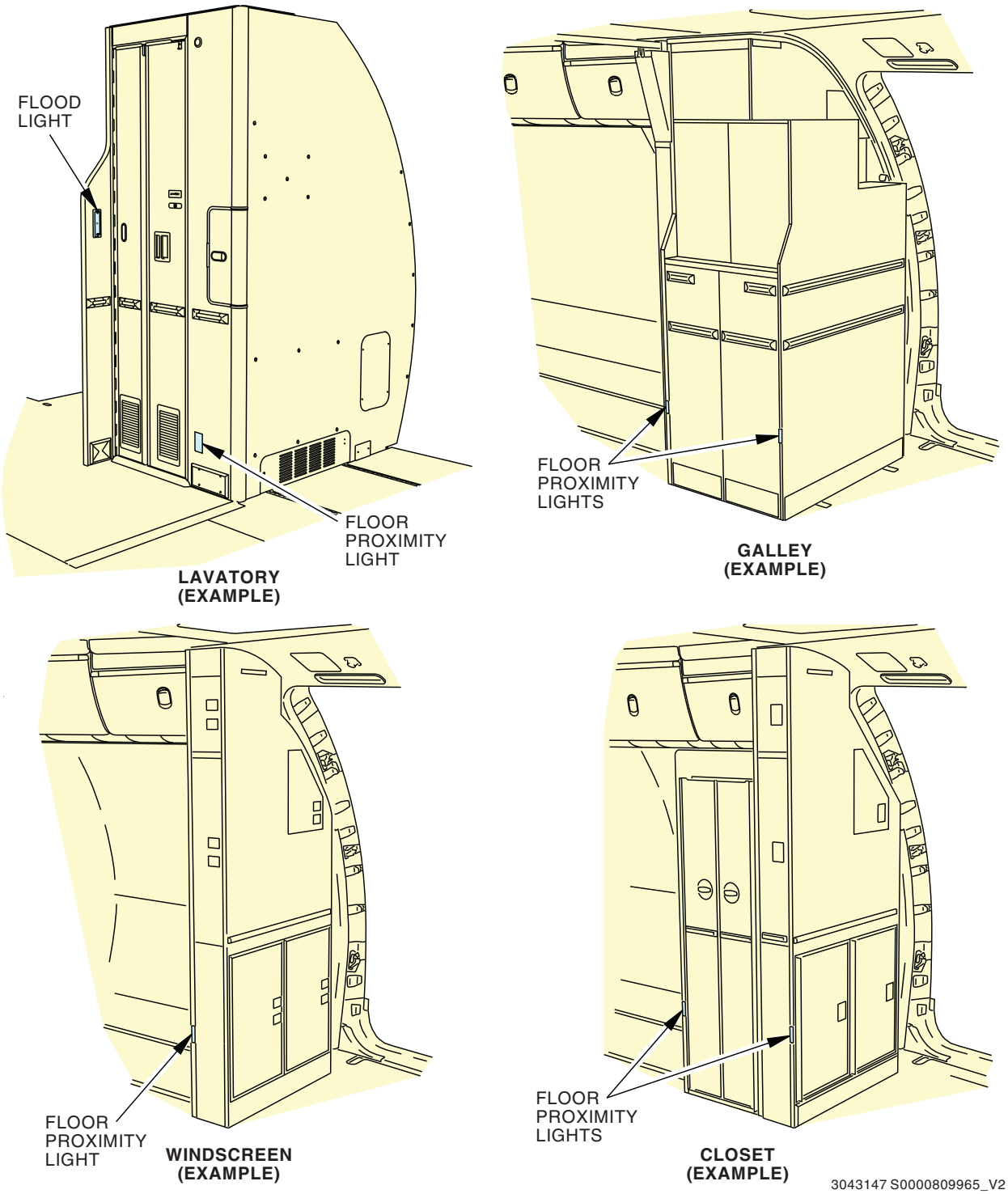
OVERWING ESCAPE HATCH (EXAMPLE)

(VIEW THROUGH THE AFT PASSENGER DOOR) (EXAMPLE)

3043052 S0000809961_V4

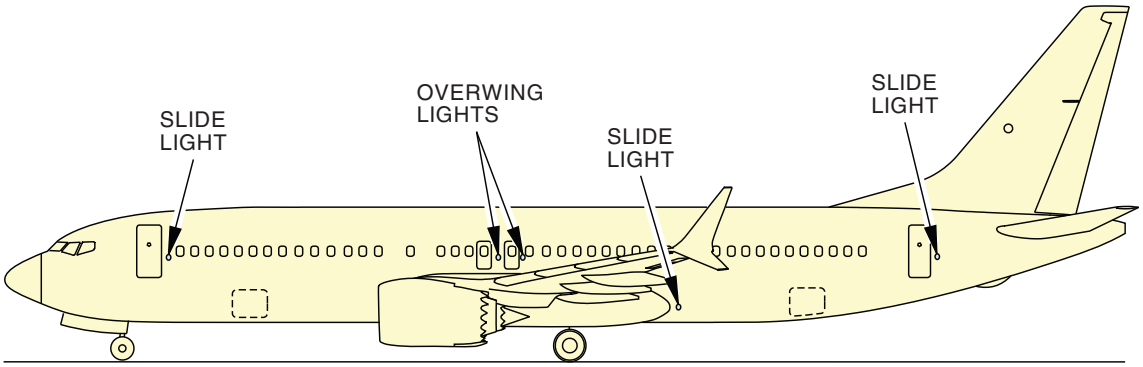
Emergency Lights - Exit Signs
Figure 503/33-51-00-990-815

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Emergency Lights - Floor Proximity Lights
Figure 504/33-51-00-990-816

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

3043145 S0000809967_V2

Emergency Lights - Overwing/Slide Lights
Figure 505/33-51-00-990-817

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-51-00-720-801

3. Emergency Lights Operational Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This test makes sure that all components in the emergency lighting system operate.

B. References

Reference	Title
33-51-06-610-801	Power Supply - Charge the Battery Packs (P/B 201)

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Emergency Lights Operational Test

SUBTASK 33-51-00-610-001

(1) Make sure that the battery packs are fully charged (TASK 33-51-06-610-801).

SUBTASK 33-51-00-710-002

(2) At the aft attendant panel, make sure that the BSI LED lighting system is turned off.

(3) At the P5 overhead panel, set the DOME switch to the OFF mode.

(4) Do a check of the emergency lighting system with the EMER EXIT LIGHTS switch on the forward overhead panel, P5, as follows:

(a) At the forward overhead panel, P5, set the pilots EMER EXIT LIGHTS switch to the ON position.

1) Make sure that the emergency dome light comes on (Figure 501).

2) Make sure that the exterior lights come on (Figure 505).

3) Make sure that these emergency lights in the passenger compartment come on:

- Aisle light (Figure 502)
- Overdoor exit sign (Figure 503)
- Overwing hatch handle lights (Figure 503)
- Exit locator sign (Figure 503)
- Exit indicator light (Figure 503)
- If applicable, floor proximity light (Figure 504)
- If applicable, flood light.

(b) Set the EMER EXIT LIGHTS switch to the OFF position.

1) Make sure that the emergency dome light goes off (Figure 501).

2) Make sure that the NOT ARMED indication shows on the forward overhead panel, P5.

3) Make sure that these emergency lights in the passenger compartment go off:

- Aisle light (Figure 502)
- Overdoor exit sign (Figure 503)
- Overwing hatch handle lights (Figure 503)
- Exit locator sign (Figure 503)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- Exit indicator light (Figure 503)
- If applicable, floor proximity light (Figure 504)
- If applicable, flood light.

4) Make sure that the exterior lights go off (Figure 505).

(c) Set the switch to the ARMED mode.

1) Make sure that the lights stay off.

2) Make sure that the NOT ARMED indication on the forward overhead panel, P5, goes off.

(d) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	10	C01357	PASS CABIN LIGHTING EMER CHGR L AFT
A	11	C00250	PASS CABIN LIGHTING EMER CHGR FWD
B	10	C01358	PASS CABIN LIGHTING EMER CHGR R AFT

1) Make sure that the emergency dome light comes on (Figure 501).

2) Make sure that the exterior lights come on (Figure 505).

3) Make sure that these emergency lights in the passenger compartment come on:

- Aisle light (Figure 502)
- Overdoor exit sign (Figure 503)
- Overwing hatch handle lights (Figure 503)
- Exit locator sign (Figure 503)
- Exit indicator light (Figure 503)
- If applicable, floor proximity light (Figure 504)
- If applicable, flood light.

(e) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	10	C01357	PASS CABIN LIGHTING EMER CHGR L AFT
A	11	C00250	PASS CABIN LIGHTING EMER CHGR FWD
B	10	C01358	PASS CABIN LIGHTING EMER CHGR R AFT

1) Make sure that the emergency dome light goes off (Figure 501).

2) Make sure that the exterior lights go off (Figure 505).

3) Make sure that these emergency lights in the passenger compartment go off:

- Aisle light (Figure 502)
- Overdoor exit sign (Figure 503)
- Overwing hatch handle lights (Figure 503)
- Exit locator sign (Figure 503)
- Exit indicator light (Figure 503)
- If applicable, floor proximity light (Figure 504)
- If applicable, flood light.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Set the switch to the OFF position.

SUBTASK 33-51-00-710-003

(5) Do a check of the emergency lighting system with the aft attendant EMERGENCY EXIT LIGHT switch.

(a) At the aft attendant panel, make sure that the BSI LED lighting system is turned off.

(b) At the aft attendant panel, press the EMERGENCY EXIT LIGHTS switch.

1) Make sure that these emergency lights in the passenger compartment come on and stay on for a minimum of 5 minutes:

- Aisle light (Figure 502)
- Overdoor exit sign (Figure 503)
- Overwing hatch handle lights (Figure 503)
- Exit locator sign (Figure 503)
- Exit indicator light (Figure 503)
- If applicable, floor proximity light (Figure 504)
- If applicable, flood light.

2) Make sure that the exterior lights come on (Figure 505).

3) Make sure that the emergency dome light comes on (Figure 501).

(c) Press the EMERGENCY EXIT LIGHTS switch again to turn the lights off.

1) Make sure that these emergency lights in the passenger compartment go off:

- Aisle light (Figure 502)
- Overdoor exit sign (Figure 503)
- Overwing hatch handle lights (Figure 503)
- Exit locator sign (Figure 503)
- Exit indicator light (Figure 503)
- If applicable, floor proximity light (Figure 504)
- If applicable, flood light.

2) Make sure that the exterior lights go off (Figure 505).

3) Make sure that the emergency dome light goes off (Figure 501).

SUBTASK 33-51-00-860-006

(6) At the P5 overhead panel, set the DOME switch to the DIM or BRIGHT mode as necessary.

SUBTASK 33-51-00-610-002

(7) Do this task: Power Supply - Charge the Battery Packs, TASK 33-51-06-610-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EXIT SIGNS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Exit Sign - Lamp Replacement

(2) Exit Sign - Light Assembly Replacement

TASK 33-51-01-960-801

2. Exit Sign - Lamp Replacement

(Figure 201)

A. Location Zones

Zone	Area
200	Upper Half of Fuselage

B. Prepare for the Replacement

SUBTASK 33-51-01-040-003

(1) Make sure that the EMER EXIT LIGHTS switch, on the P5 overhead panel, is set to the OFF position.

SUBTASK 33-51-01-040-004

(2) Make sure that the EMERGENCY EXIT LIGHTS switch, on the applicable attendant panel, is set to the NORMAL position.

C. Replace the Exit Sign Lamp

SUBTASK 33-51-01-960-001

WARNING DO NOT TOUCH THE SOCKET OR THE METAL END OF THE LAMP. WHEN ELECTRICAL POWER IS SUPPLIED TO THE LAMP, ELECTRICAL SHOCK CAN OCCUR. THIS CAN CAUSE INJURIES TO PERSONNEL.

(1) Do these steps to replace the applicable lamp:

(a) Remove the applicable lens [1] or lens [3].

NOTE: If an exit sign has a lens on two sides, remove only one lens.

1) For the light assembly [14], remove the screws [6] and o-rings [7] to remove the lens [3].

2) For the light assembly [23], loosen the screw [8] to remove the lens [1].

(b) Replace the applicable lamp [2], lamp [4], or lamp [5].

(c) Install the applicable lens [1] or lens [3].

1) For the light assembly [14], install the screws [6] and o-rings [7].

2) For the light assembly [23], tighten the screw [8].

SUBTASK 33-51-01-710-001

(2) Do a test of the new lamp.

NOTE: The power supply operates the emergency lights. Use the lights for a minimum time to do the test.

(a) Set the EMERGENCY EXIT LIGHT switch, on the applicable attendant panel, to the ON position.

1) Make sure that the new lamp comes on.

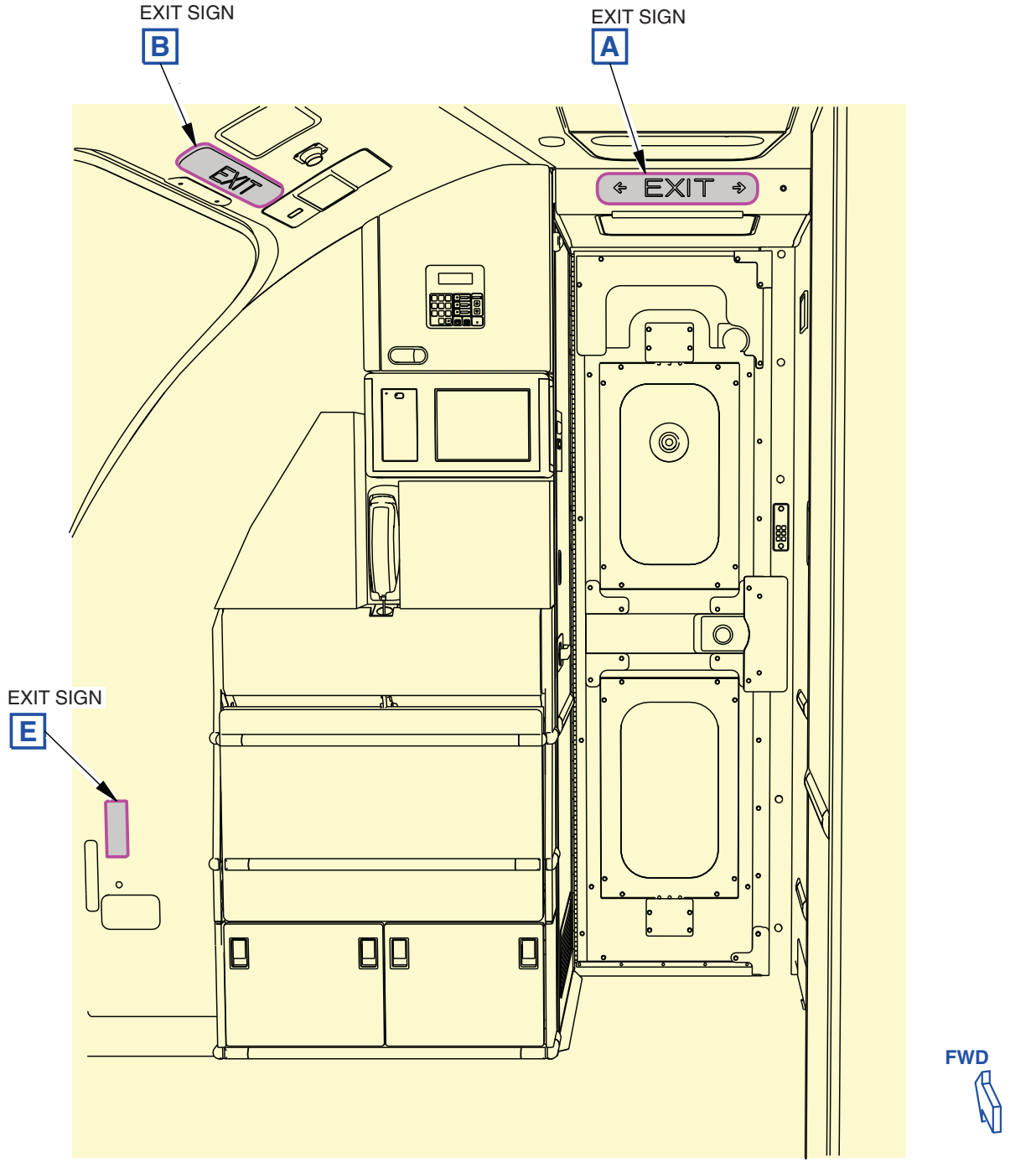
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Set the EMERGENCY EXIT LIGHT switch, on the applicable attendant panel, to the NORMAL position.

1) Make sure that all emergency lights go off.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

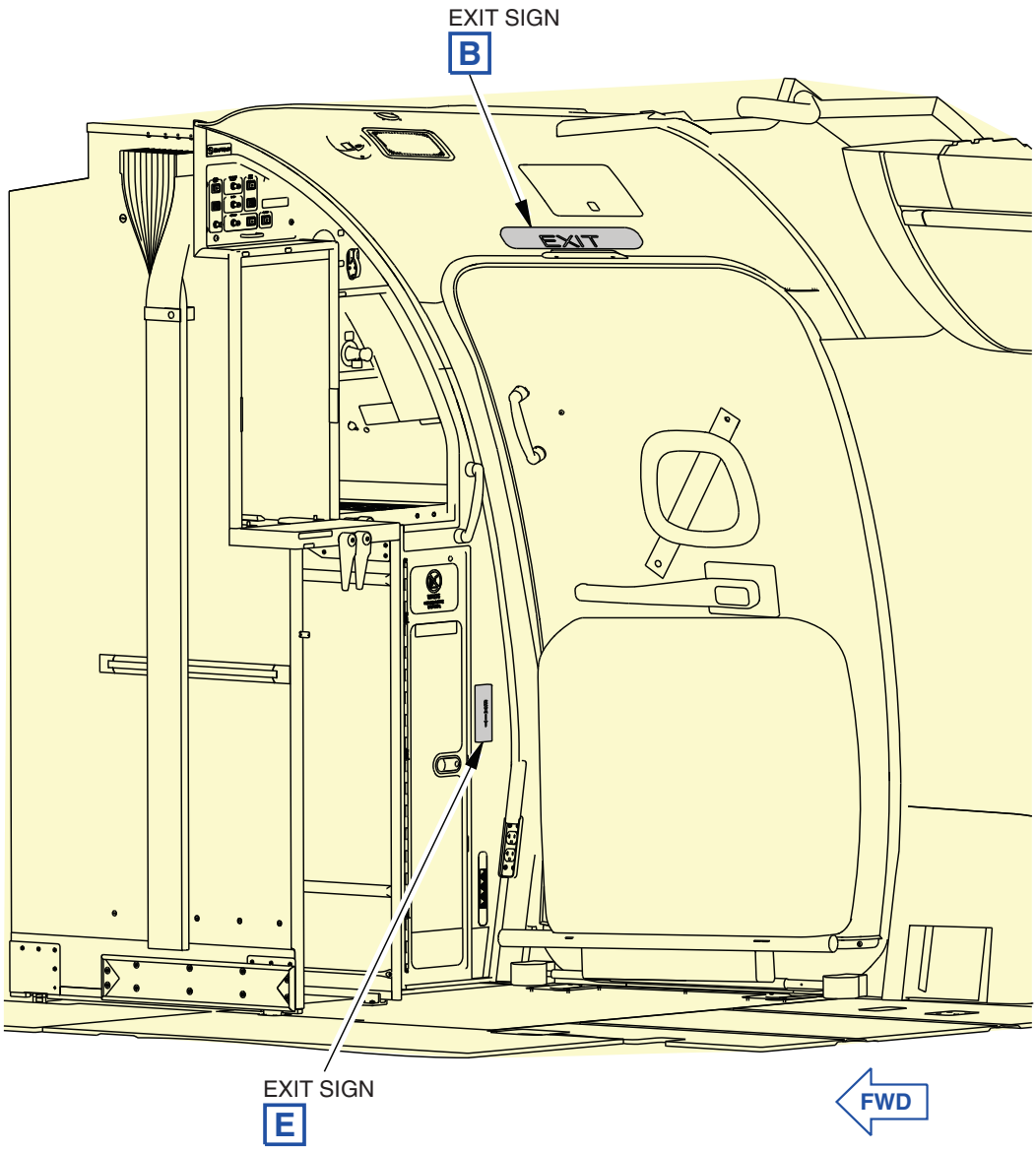


FORWARD ENTRY AREA (EXAMPLE)

2414767 S00061535932_V3

Exit Sign - Lamp Replacement
Figure 201/33-51-01-990-801 (Sheet 1 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

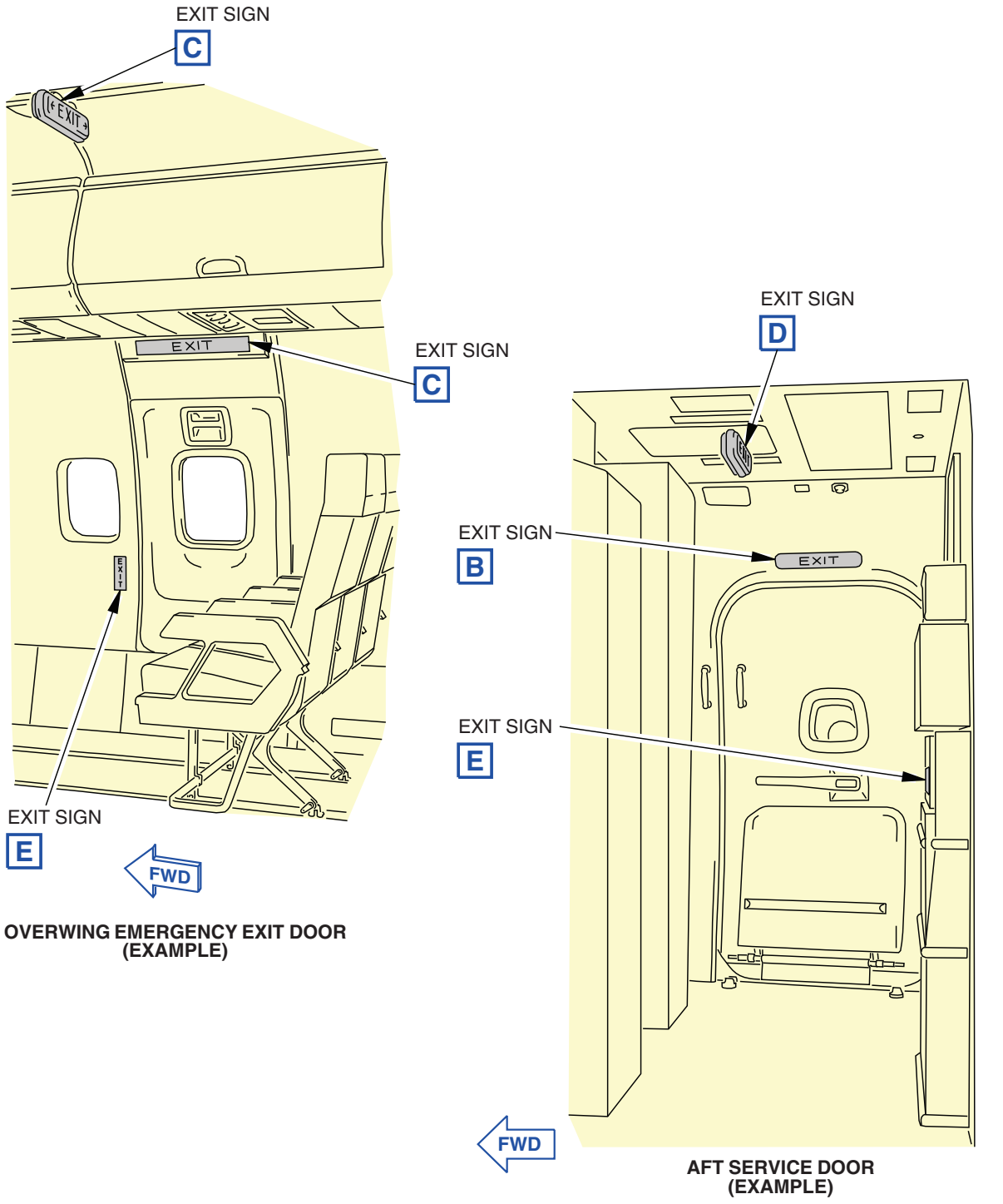


**FORWARD SERVICE DOOR
(EXAMPLE)**

3122869 S0000856930_V1

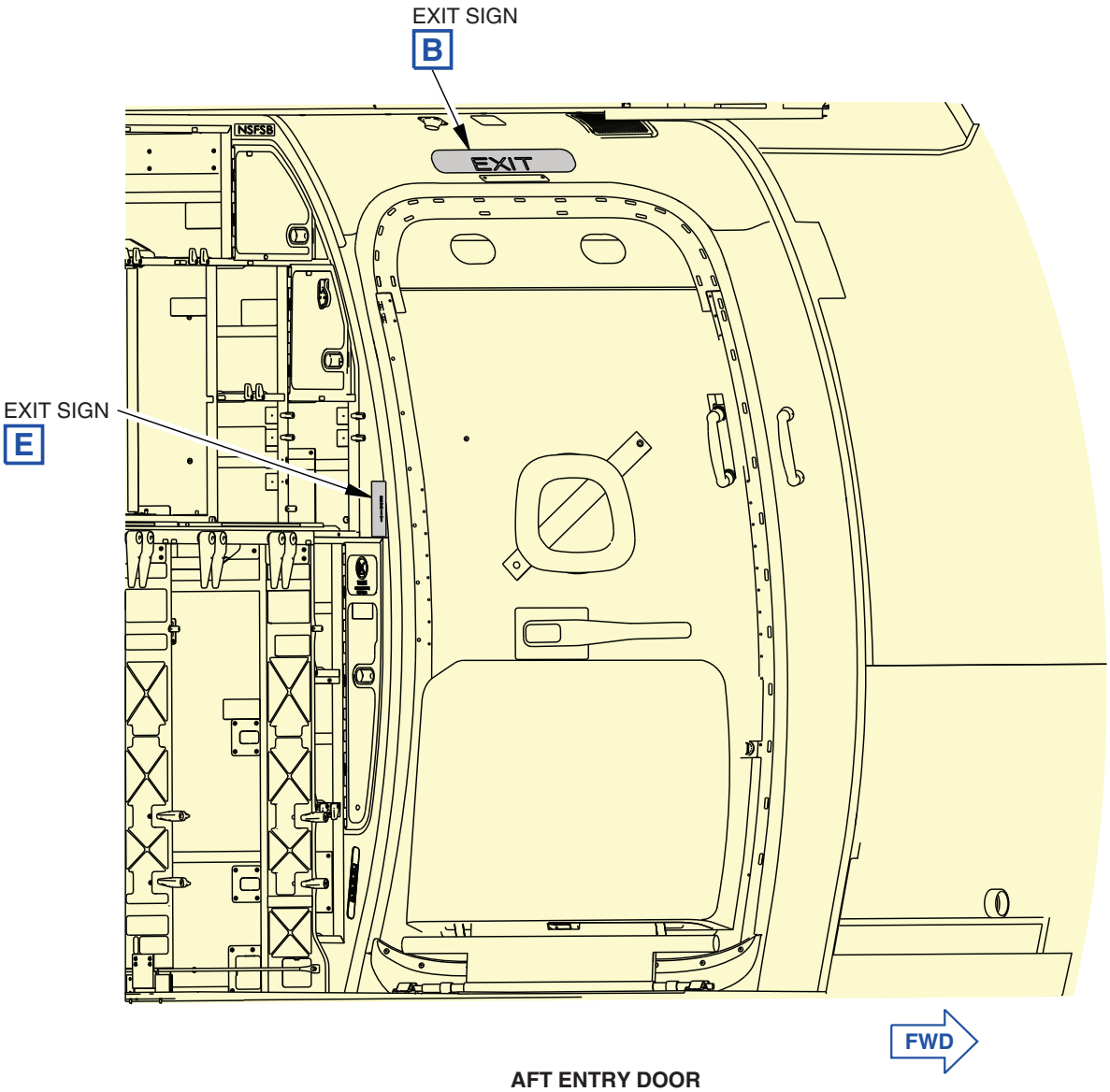
**Exit Sign - Lamp Replacement
Figure 201/33-51-01-990-801 (Sheet 2 of 8)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Exit Sign - Lamp Replacement
Figure 201/33-51-01-990-801 (Sheet 3 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

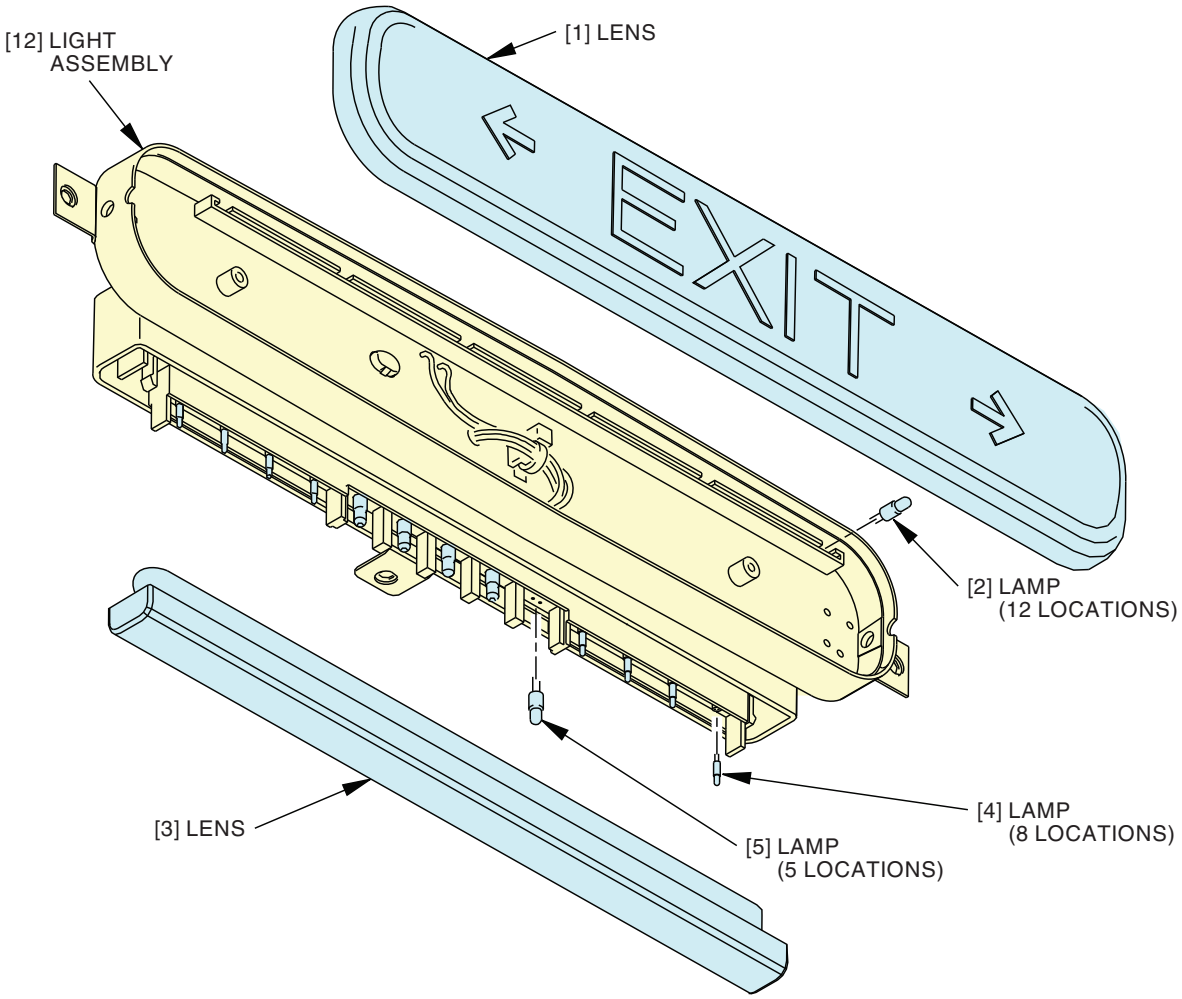


**AFT ENTRY DOOR
(EXAMPLE)**

3122867 S0000856931_V1

**Exit Sign - Lamp Replacement
Figure 201/33-51-01-990-801 (Sheet 4 of 8)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



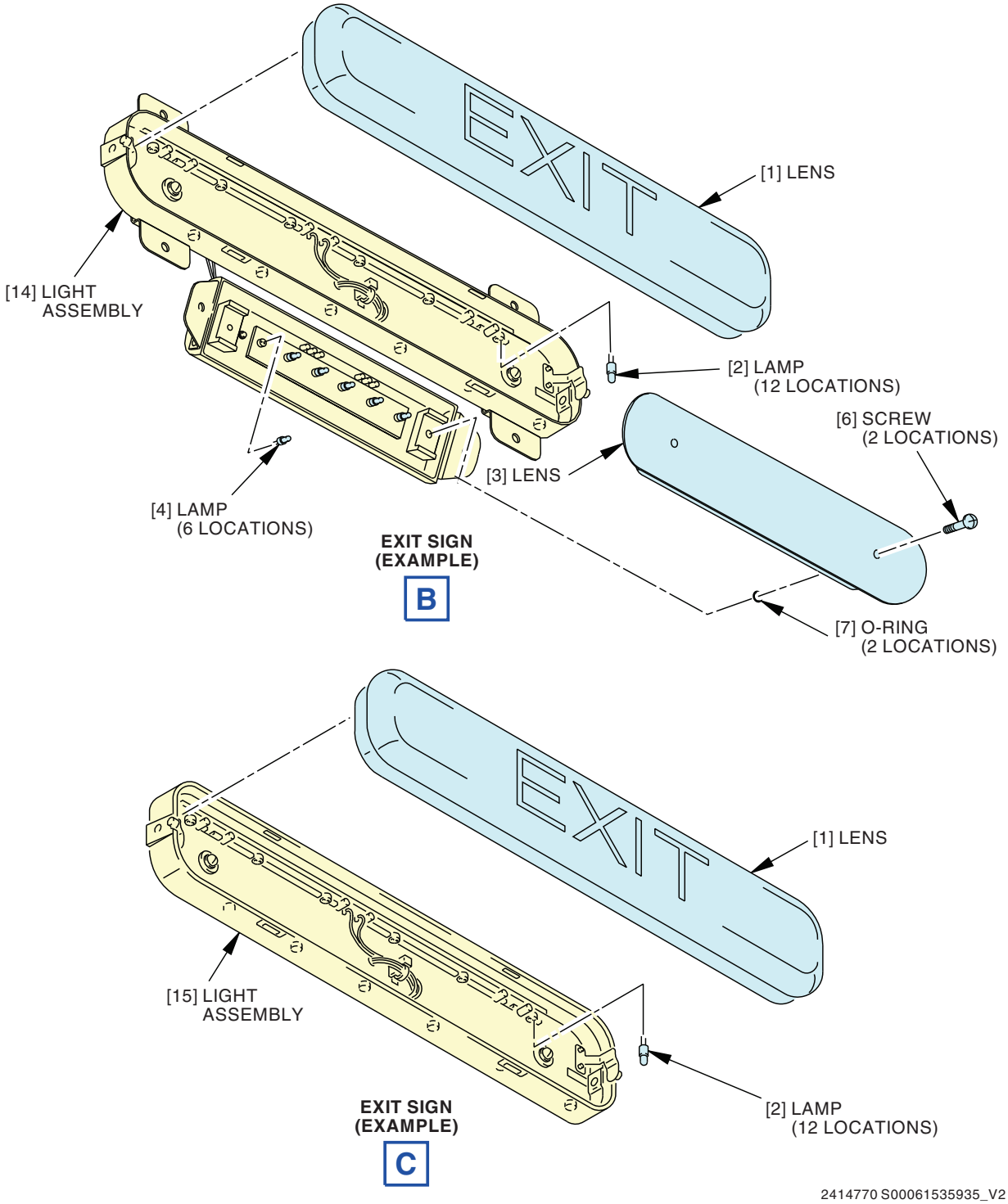
EXIT SIGN (EXAMPLE)

A

3062377 S0000825522_V1

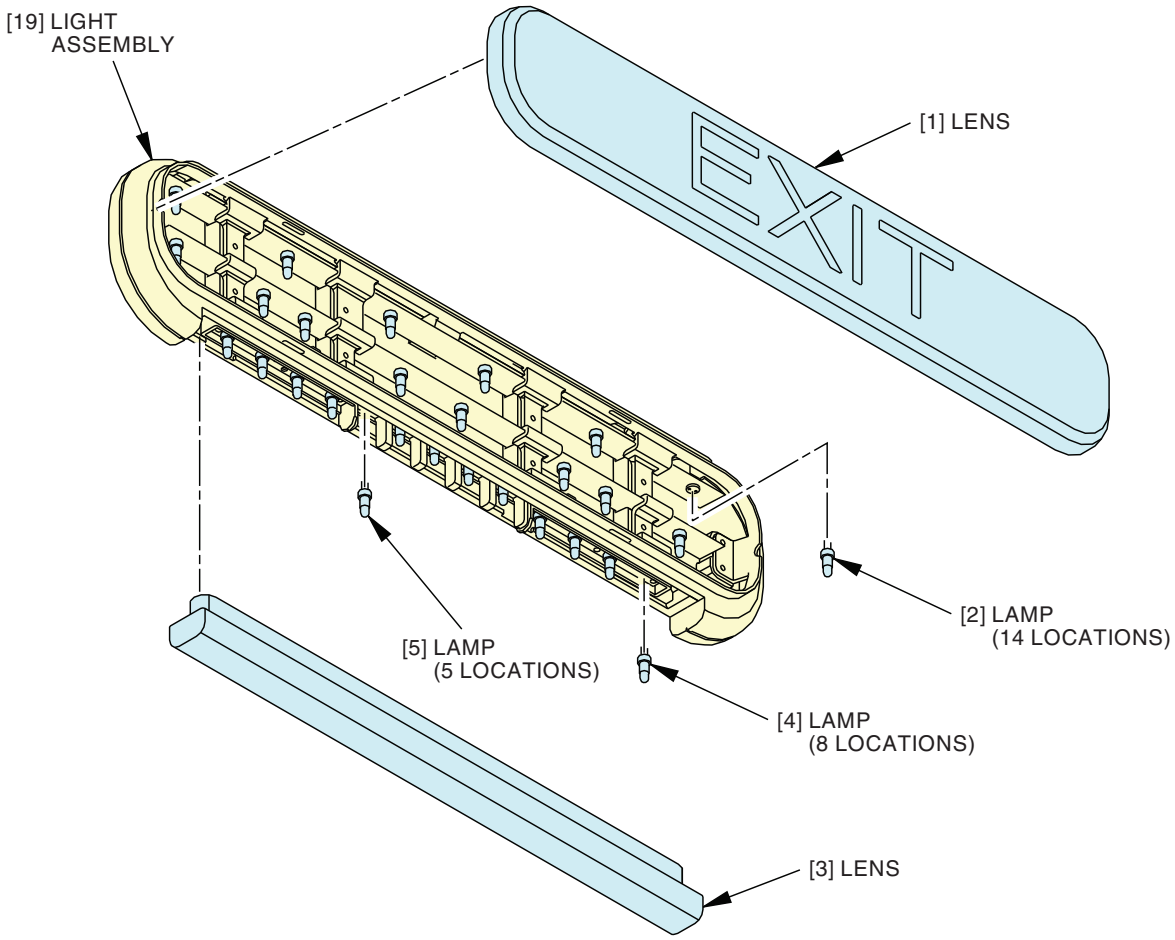
Exit Sign - Lamp Replacement
Figure 201/33-51-01-990-801 (Sheet 5 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Exit Sign - Lamp Replacement
Figure 201/33-51-01-990-801 (Sheet 6 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



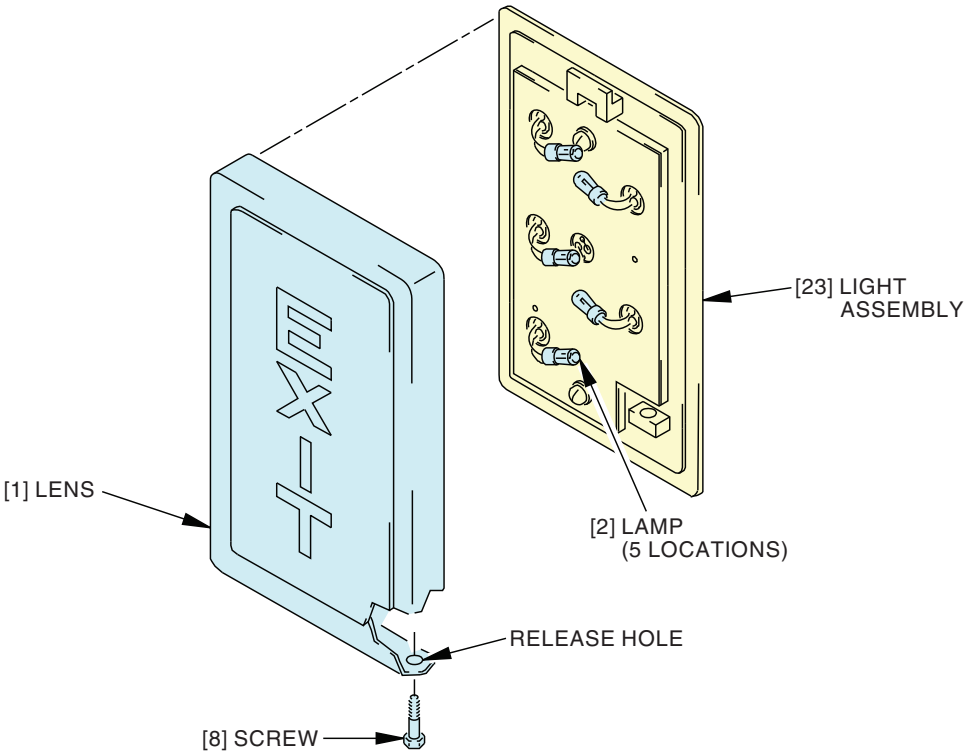
EXIT SIGN (EXAMPLE)

D

2908931 S0000695057_V2

Exit Sign - Lamp Replacement
Figure 201/33-51-01-990-801 (Sheet 7 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414772 S00061535937_V2

Exit Sign - Lamp Replacement
Figure 201/33-51-01-990-801 (Sheet 8 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-51-01-960-802

3. Exit Sign - Light Assembly Replacement

(Figure 202)

A. References

Reference	Title
25-21-24-000-801	Aft Door Doorway Header Panel Removal (P/B 401)
25-21-24-400-801	Aft Door Doorway Header Panel Installation (P/B 401)
25-21-30-020-801	Forward Entry Door Doorway Header Panel Removal (P/B 401)
25-21-30-420-801	Forward Entry Door Doorway Header Panel Installation (P/B 401)
25-21-34-000-801	Forward Galley Door Doorway Header Panel Removal (P/B 401)
25-21-34-400-801	Forward Galley Door Doorway Header Panel Installation (P/B 401)
25-21-71-000-805	Close Out Ceiling Panel Removal (P/B 401)
25-21-71-000-807	Aft Entry Ceiling Panel Removal (P/B 401)
25-21-71-400-805	Close Out Ceiling Panel Installation (P/B 401)
25-21-71-400-807	Aft Entry Ceiling Panel Installation (P/B 401)
25-23-61-000-801	Passenger Service Unit - Removal (P/B 401)
25-23-61-400-801	Passenger Service Unit - Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
G00291	Tape - Aluminum Foil, 3M 425	AMS-T-23397 / L-T-80
G50532	Tape - Silicone, Foam	BMS1-68

C. Location Zones

Zone	Area
118	Electrical and Electronics Compartment - Right
200	Upper Half of Fuselage

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Replacement

SUBTASK 33-51-01-410-003

(1) Open the applicable circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	8	C01403	PASSENGER CABIN PASS CALL LEFT
A	9	C00073	PASSENGER CABIN CREW CALL
A	10	C01357	PASS CABIN LIGHTING EMER CHGR L AFT
A	11	C00250	PASS CABIN LIGHTING EMER CHGR FWD
B	8	C01404	PASSENGER CABIN PASS CALL RIGHT
B	10	C01358	PASS CABIN LIGHTING EMER CHGR R AFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-51-01-040-001

(2) Make sure that the EMER EXIT LIGHTS switch, on the P5 overhead panel, is set to the OFF position.

SUBTASK 33-51-01-040-002

(3) Make sure that the EMERGENCY EXIT LIGHT switch, on the applicable attendant panel, is set to the NORMAL position.

SUBTASK 33-51-01-010-002

(4) To get access to the P92 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 33-51-01-410-004

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(5) Open the applicable circuit breaker and install the safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
E	11	C00594	WINDOW LIGHT RIGHT
E	12	C00776	WINDOW LIGHT LEFT

SUBTASK 33-51-01-010-003

(6) If it is necessary, do one of the steps that follow to get access to the applicable light assembly:

(a) Remove the applicable door header, do the applicable task:

NOTE: To access the light assemblies on the emergency exit door headers, the door header does not need to be removed.

- Aft Door Doorway Header Panel Removal, TASK 25-21-24-000-801
- Forward Entry Door Doorway Header Panel Removal, TASK 25-21-30-020-801
- Forward Galley Door Doorway Header Panel Removal, TASK 25-21-34-000-801.

(b) Remove the adjacent passenger service unit, do this task: Passenger Service Unit - Removal, TASK 25-23-61-000-801.

(c) Open the lowered ceiling close out door (TASK 25-21-71-000-805).

(d) Open the aft entry ceiling access panel (TASK 25-21-71-000-807).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Replace the Exit Sign Light Assembly

SUBTASK 33-51-01-020-005

WARNING DO NOT TOUCH THE SOCKET OR THE METAL END OF THE LAMP. WHEN ELECTRICAL POWER IS SUPPLIED TO THE LAMP, ELECTRICAL SHOCK CAN OCCUR. THIS CAN CAUSE INJURIES TO PERSONNEL.

(1) Do these steps to remove the applicable light assembly:

(a) Remove the applicable lens [1] and lens [3].

1) For the light assembly [14], remove the screws [6] and o-rings [7].

2) For the light assembly [23], loosen the screw [8].

(b) For the light assembly [14], remove the screws [13].

(c) For the light assembly [19], remove the screws [17] and washers [18].

(d) For the light assembly [22], remove the access panel [21].

(e) Remove the screws [9] and washers [10].

(f) Disconnect the electrical connectors [11].

(g) Remove the light assembly.

SUBTASK 33-51-01-420-001

(2) Do these steps to install the applicable light assembly:

(a) Make sure that lamps are installed in the new light assembly.

(b) If it is necessary, replace the tape as follows:

1) For light assembly [12], use 3M 425 Aluminum Foil Tape, G00291.

2) For light assembly [14], use 3M 425 Aluminum Foil Tape, G00291.

3) For light assembly [15], use 3M 425 Aluminum Foil Tape, G00291.

4) For light assembly [19], use tape, G50532.

5) For light assembly [20], use tape, G50532.

6) For light assembly [22], use tape, G50532.

(c) Connect the electrical connectors [11].

(d) Install the screws [9] and washers [10].

1) If applicable, install the brackets [16] with the screws [9] and washers [10].

(e) For the light assembly [14], install the screws [13].

(f) For the light assembly [19], install the screws [17] and washers [18].

(g) For the light assembly [22], install the access panel [21].

(h) Install the applicable lens [1] and lens [3].

1) For the light assembly [14], install the screws [6] and o-rings [7].

2) For the light assembly [23], tighten the screw [8].

SUBTASK 33-51-01-410-011

(3) If it is necessary, do one of the steps that follow to close access to the applicable light assembly:

(a) Install the applicable door header, do the applicable task:

- Aft Door Doorway Header Panel Installation, TASK 25-21-24-400-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- Forward Entry Door Doorway Header Panel Installation, TASK 25-21-30-420-801
- Forward Galley Door Doorway Header Panel Installation, TASK 25-21-34-400-801.

(b) Install the adjacent passenger service unit, do this task: Passenger Service Unit - Installation, TASK 25-23-61-400-801.

(c) Close the lowered ceiling close out door (TASK 25-21-71-400-805).

(d) Close the aft entry ceiling access panel (TASK 25-21-71-400-807).

SUBTASK 33-51-01-710-002

(4) Do a test of the new light assembly:

NOTE: The power supply operates the emergency lights. Use the lights for a minimum time to do the test.

(a) Set the EMERGENCY EXIT LIGHT switch, on the applicable attendant panel, to the ON position.

1) Make sure that the new exit sign comes on.

(b) Set the EMERGENCY EXIT LIGHT switch, on the applicable attendant panel, to the NORMAL position.

1) Make sure that all emergency lights go off.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 33-51-01-410-008

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(1) Remove the safety tag and close the applicable circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
E	11	C00594	WINDOW LIGHT RIGHT
E	12	C00776	WINDOW LIGHT LEFT

SUBTASK 33-51-01-410-010

(2) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-51-01-410-009

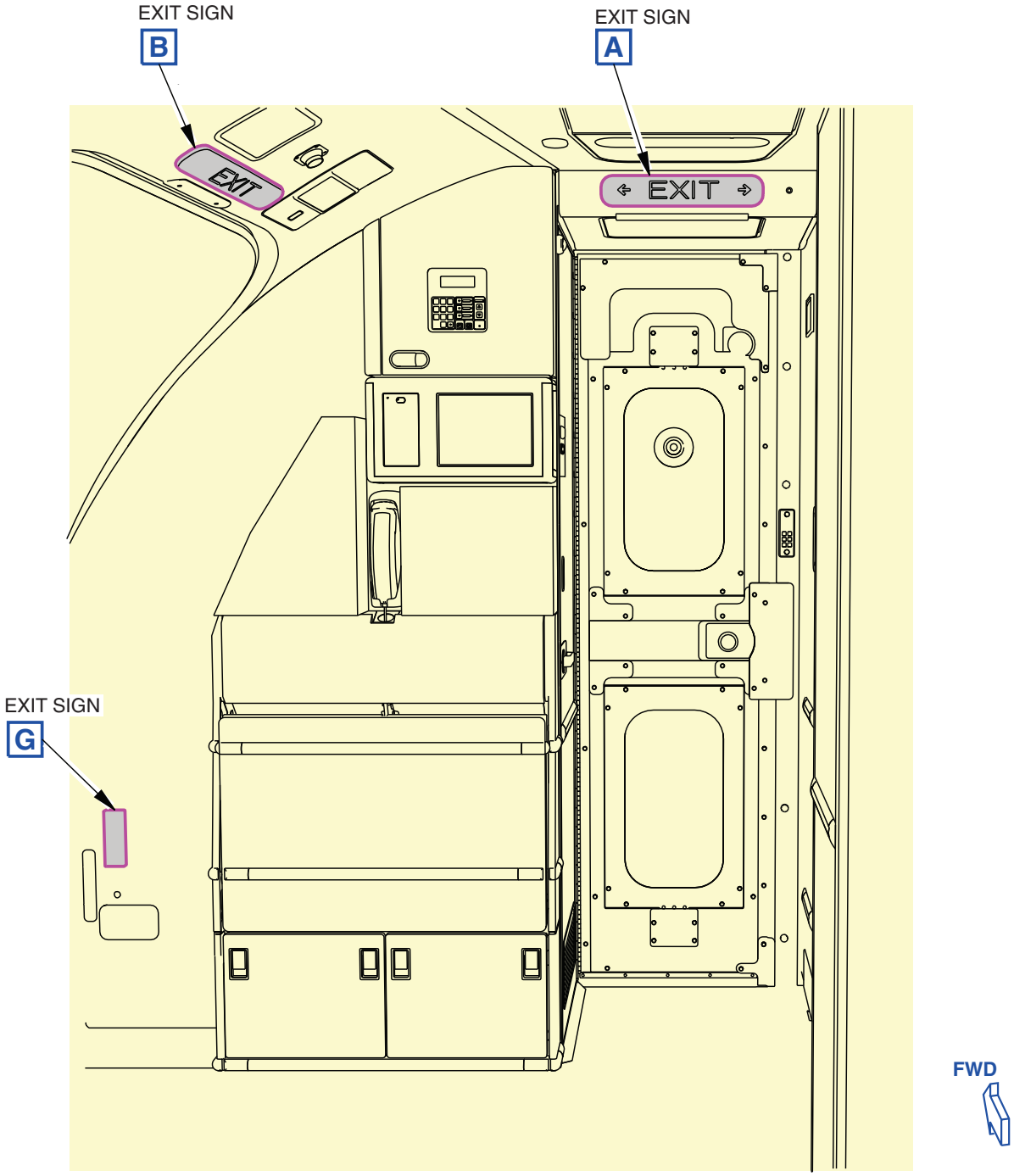
(3) Remove the safety tag and close the applicable circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	8	C01403	PASSENGER CABIN PASS CALL LEFT
A	9	C00073	PASSENGER CABIN CREW CALL
A	10	C01357	PASS CABIN LIGHTING EMER CHGR L AFT
A	11	C00250	PASS CABIN LIGHTING EMER CHGR FWD
B	8	C01404	PASSENGER CABIN PASS CALL RIGHT
B	10	C01358	PASS CABIN LIGHTING EMER CHGR R AFT

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

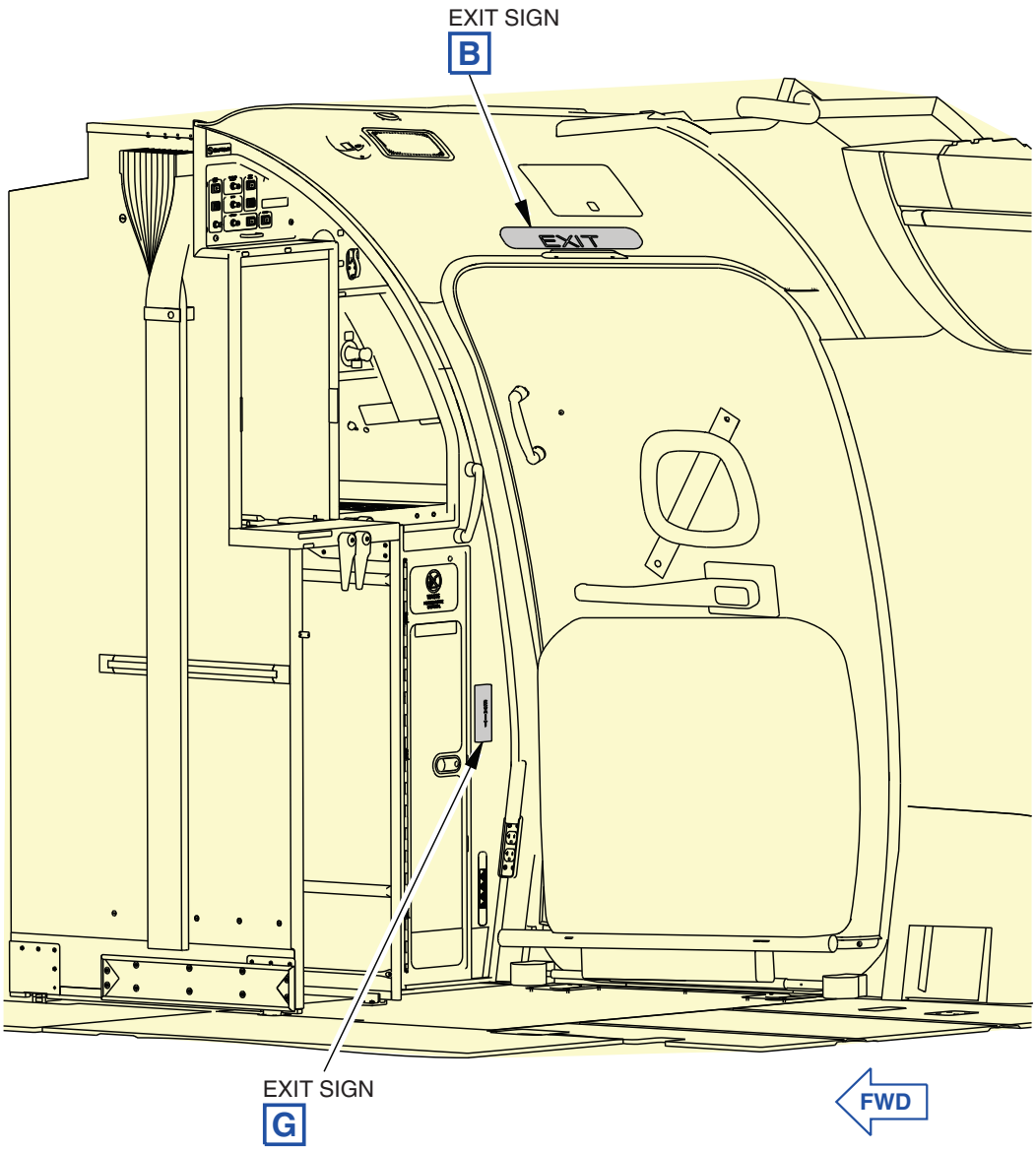


**FORWARD ENTRY AREA
(EXAMPLE)**

3123176 S0000857412_V1

**Exit Sign - Light Assembly Replacement
Figure 202/33-51-01-990-802 (Sheet 1 of 9)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

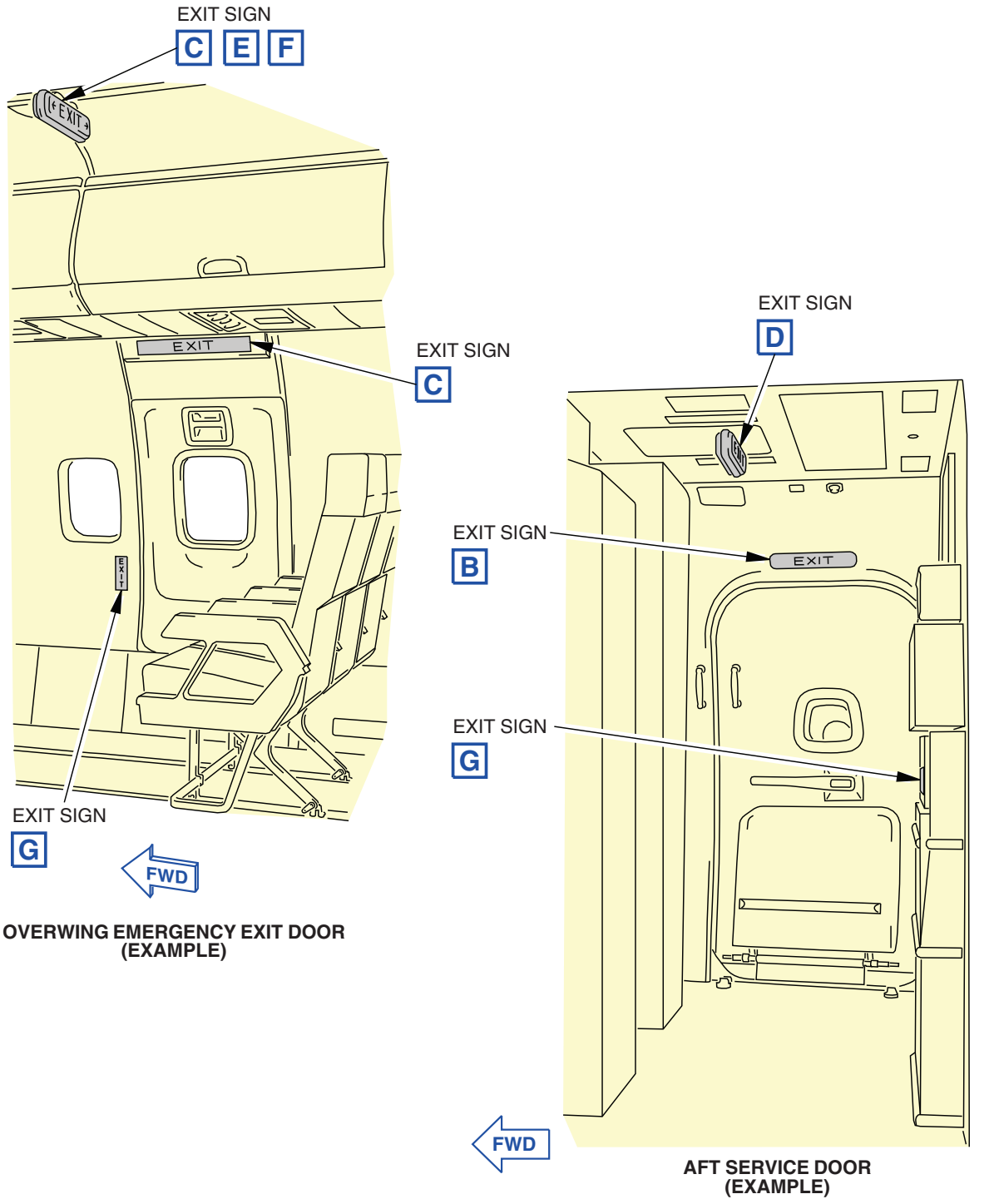


**FORWARD SERVICE DOOR
(EXAMPLE)**

3122850 S0000856932_V1

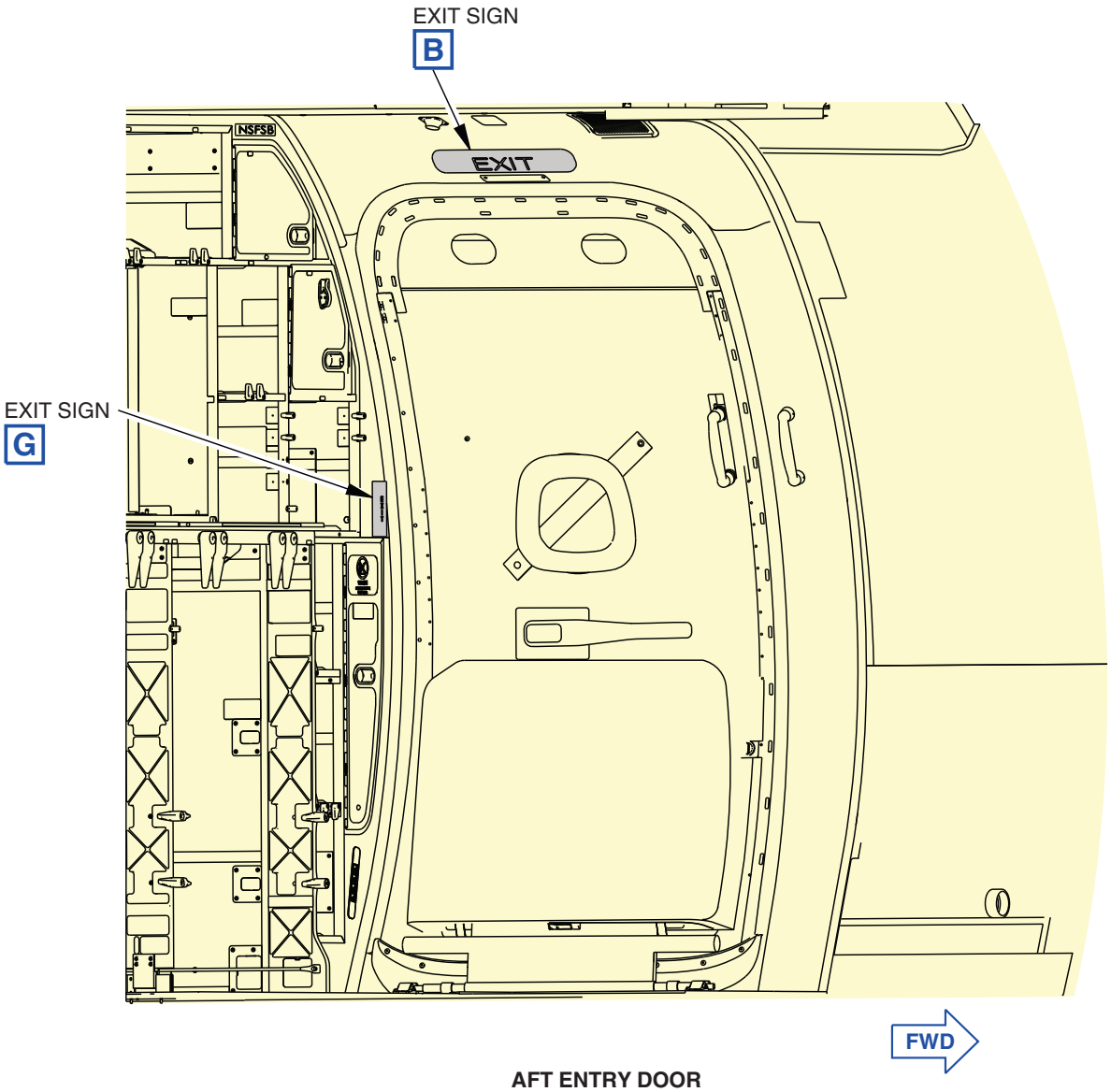
**Exit Sign - Light Assembly Replacement
Figure 202/33-51-01-990-802 (Sheet 2 of 9)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Exit Sign - Light Assembly Replacement
Figure 202/33-51-01-990-802 (Sheet 3 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

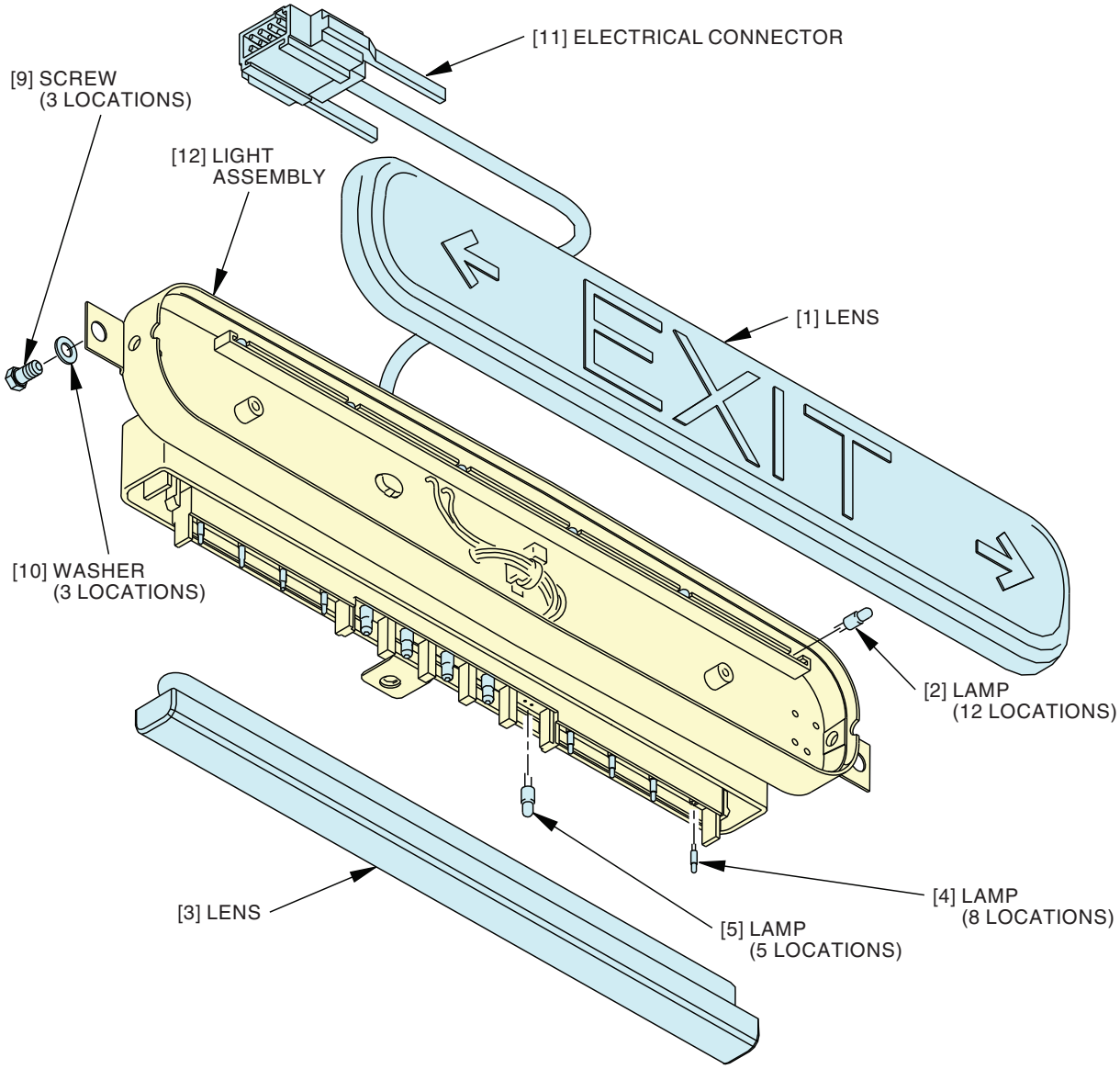


**AFT ENTRY DOOR
(EXAMPLE)**

3122868 S0000856933_V1

**Exit Sign - Light Assembly Replacement
Figure 202/33-51-01-990-802 (Sheet 4 of 9)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



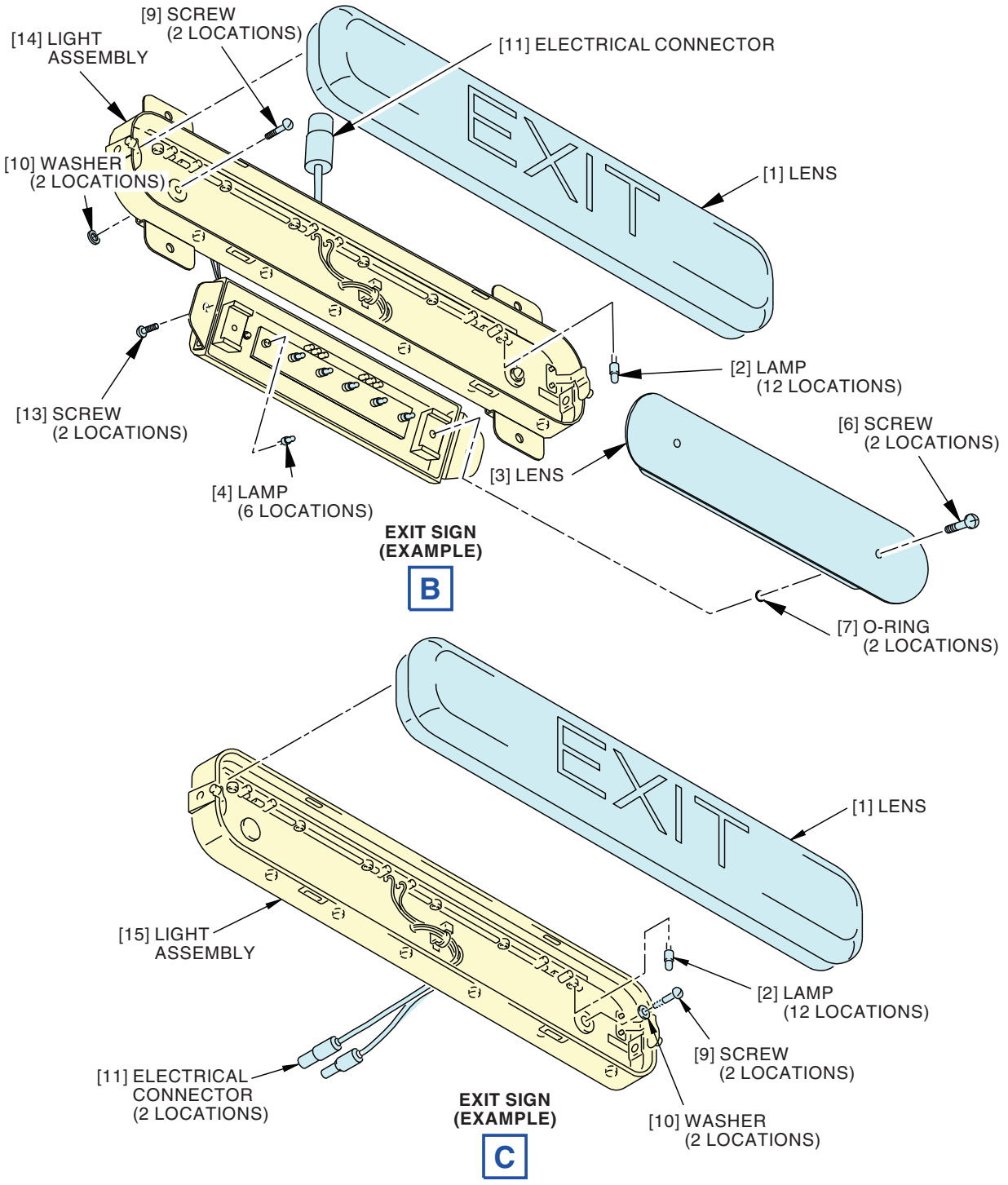
EXIT SIGN
(EXAMPLE)

A

3062517 S0000825525_V1

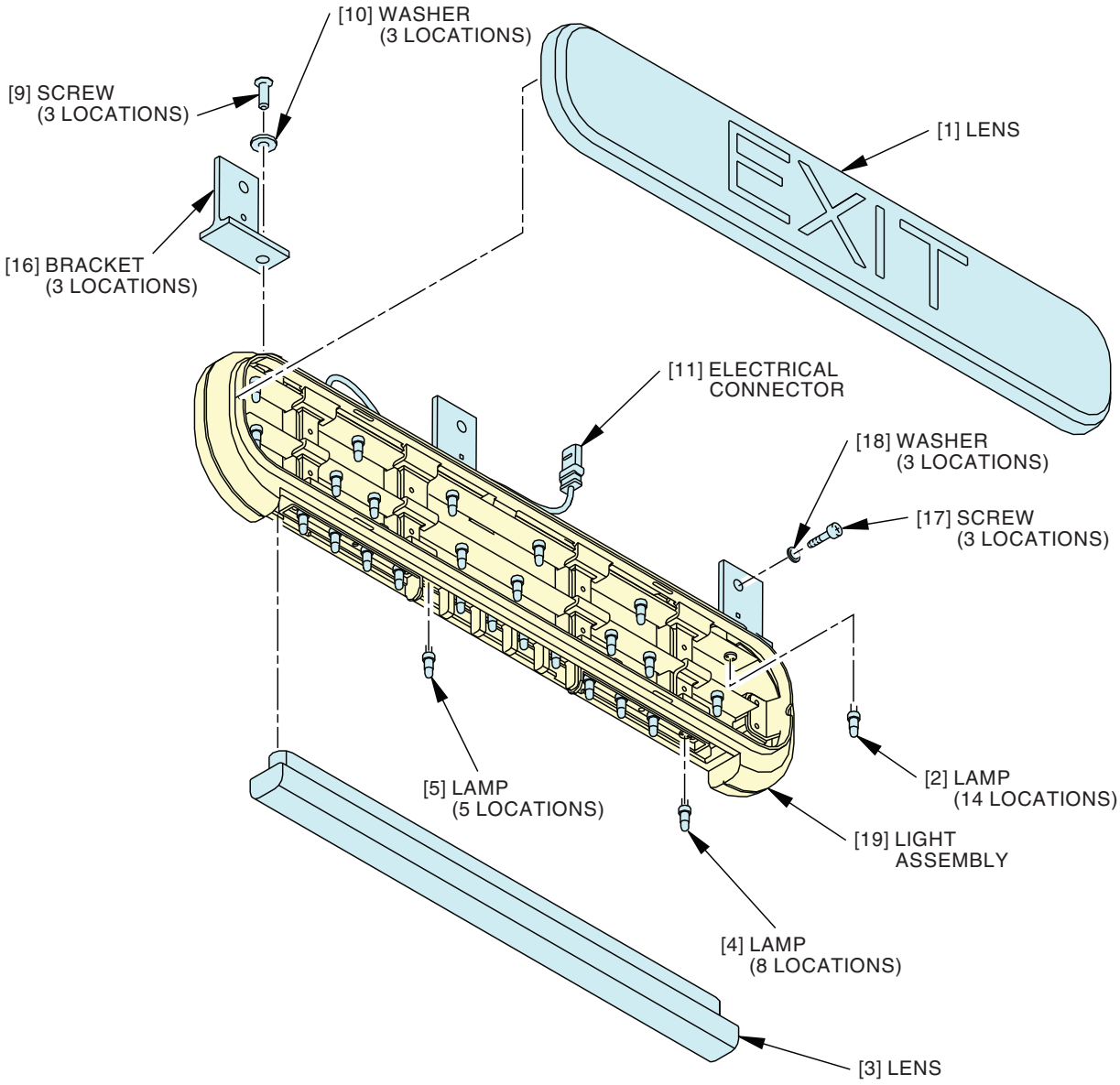
Exit Sign - Light Assembly Replacement
Figure 202/33-51-01-990-802 (Sheet 5 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Exit Sign - Light Assembly Replacement
Figure 202/33-51-01-990-802 (Sheet 6 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



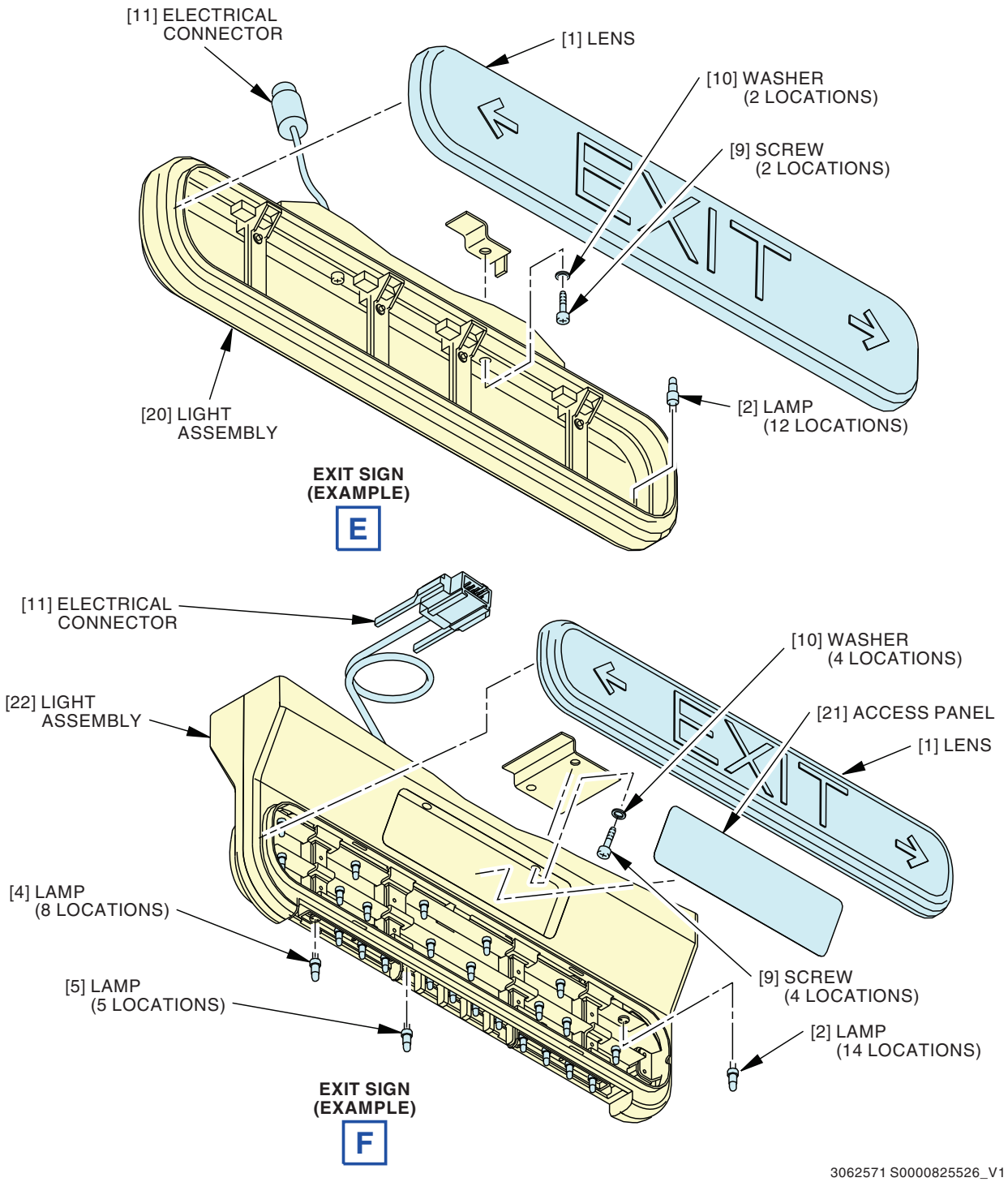
EXIT SIGN
(EXAMPLE)

D

2908937 S0000695058_V2

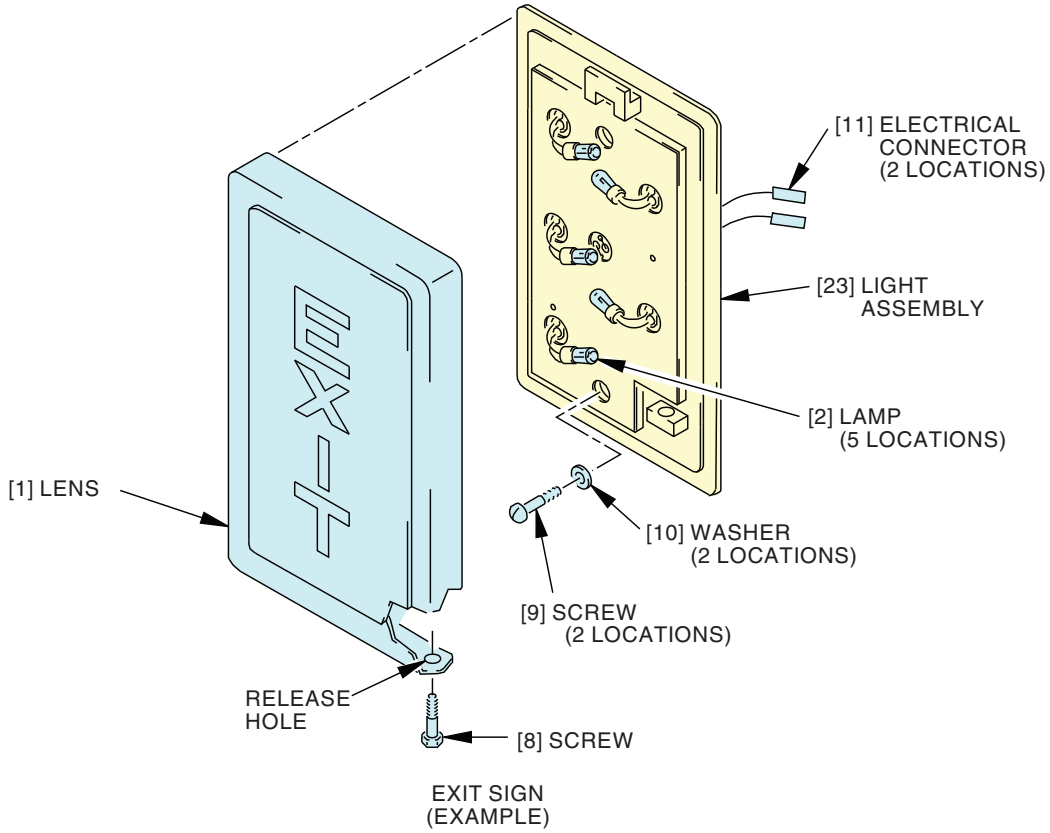
Exit Sign - Light Assembly Replacement
Figure 202/33-51-01-990-802 (Sheet 7 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Exit Sign - Light Assembly Replacement
Figure 202/33-51-01-990-802 (Sheet 8 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414776 S00061535942_V2

Exit Sign - Light Assembly Replacement
Figure 202/33-51-01-990-802 (Sheet 9 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AREA LIGHTS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Emergency Aisle Light (EAL) - Lens Replacement

(2) Emergency Aisle Light (EAL) - Assembly Replacement.

TASK 33-51-03-960-801

2. Emergency Aisle Light (EAL) - Lens Replacement

(Figure 201)

A. General

(1) This task includes the steps to replace the Emergency Aisle Light (EAL) lens.

B. Tools/Equipment

Reference	Description
STD-549	Knife - Putty, Broad Blade

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Lens	33-51-03-94-060	PTF ALL

D. Location Zones

Zone	Area
200	Upper Half of Fuselage

E. Procedure

SUBTASK 33-51-03-010-001

(1) Do these steps to remove the lens [4]:

(a) Use a laminated card for this procedure.

NOTE: The laminated card must be thin and sturdy, to slide between the valance and the lens. This will prevent damage to the valance.

(b) Insert and slide the edge of the card between the rim of the lens [4] and the valance.

(c) Move the card around and stop at the top of the lens [4].

1) Make sure that the card is in a horizontal position between the lens [4] and the valance.

(d) Insert the broad blade putty knife, STD-549 (or equivalent), between the card and the lens [4].

(e) Pull the handle of the broad blade putty knife, STD-549 (or equivalent) in your direction, away from the valance.

NOTE: You can twist the handle of the putty knife slightly, if necessary.

1) Remove the lens [4].

SUBTASK 33-51-03-410-001

(2) Do these steps to install the lens [4]:

(a) Align the lens against the EAL housing assembly [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the two prongs of the lens [4] align with the top of the EAL housing assembly [5].

NOTE: The top part of the lens has two prongs. The lower part has one prong.

(b) Engage the single prong of the lens [4] behind the lower rim of the EAL housing assembly [5].

(c) Push the top of the lens [4] against the EAL housing assembly [5].

1) Make sure that the two prongs of the lens [4] snap into their positions.

SUBTASK 33-51-03-710-001

(3) Do a test of the light.

NOTE: The power supply operates the emergency lights. Use the lights for a minimum time to do the test.

(a) At the pilots overhead panel, P5, or the attendant panel, set the emergency light switch to the ON mode.

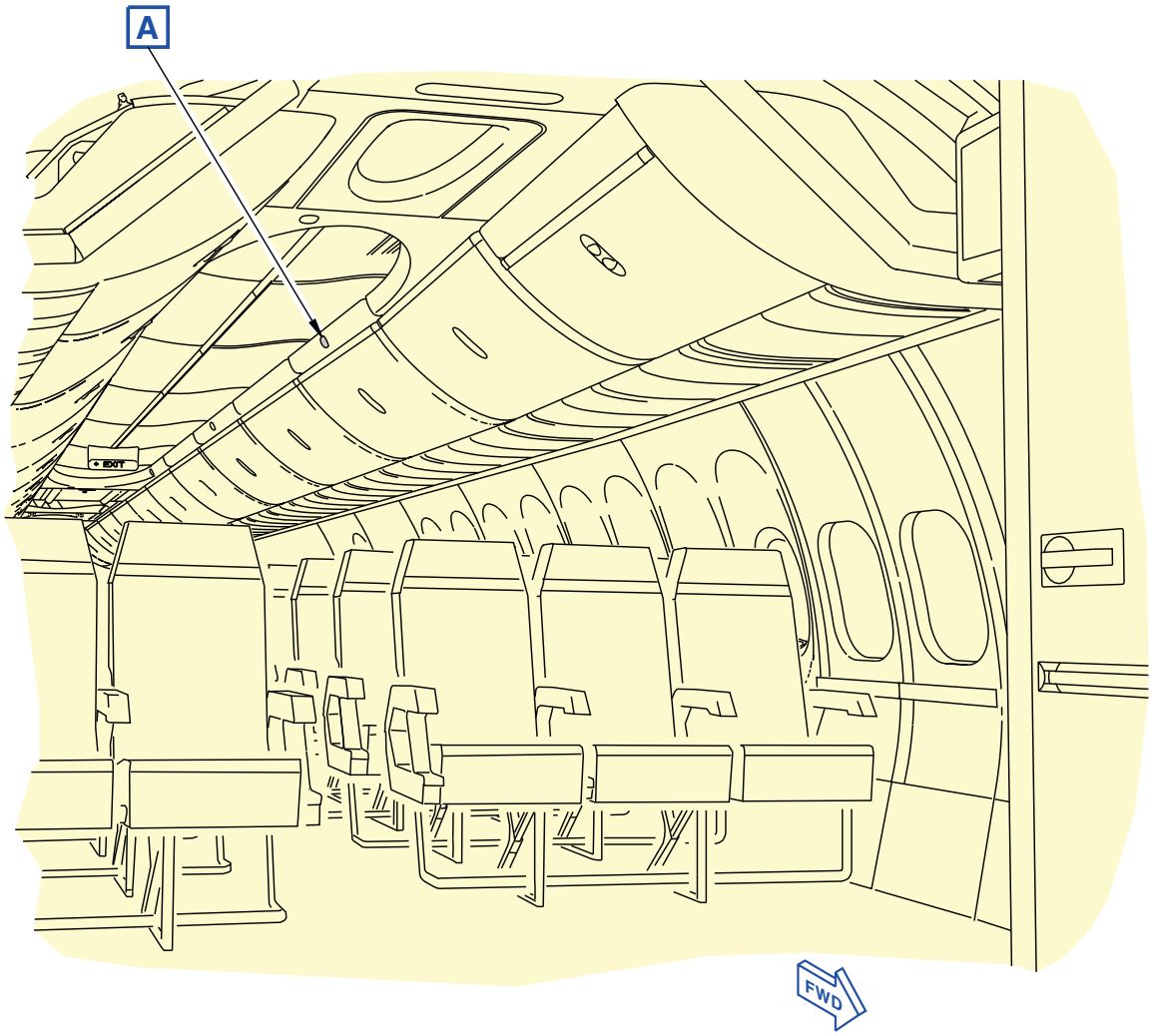
1) Make sure that the light comes on.

(b) Set the emergency light switch on the overhead panel, P5, to the OFF mode or the emergency light switch on the attendant panel to the NORMAL mode.

1) Make sure that all emergency lights go off.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

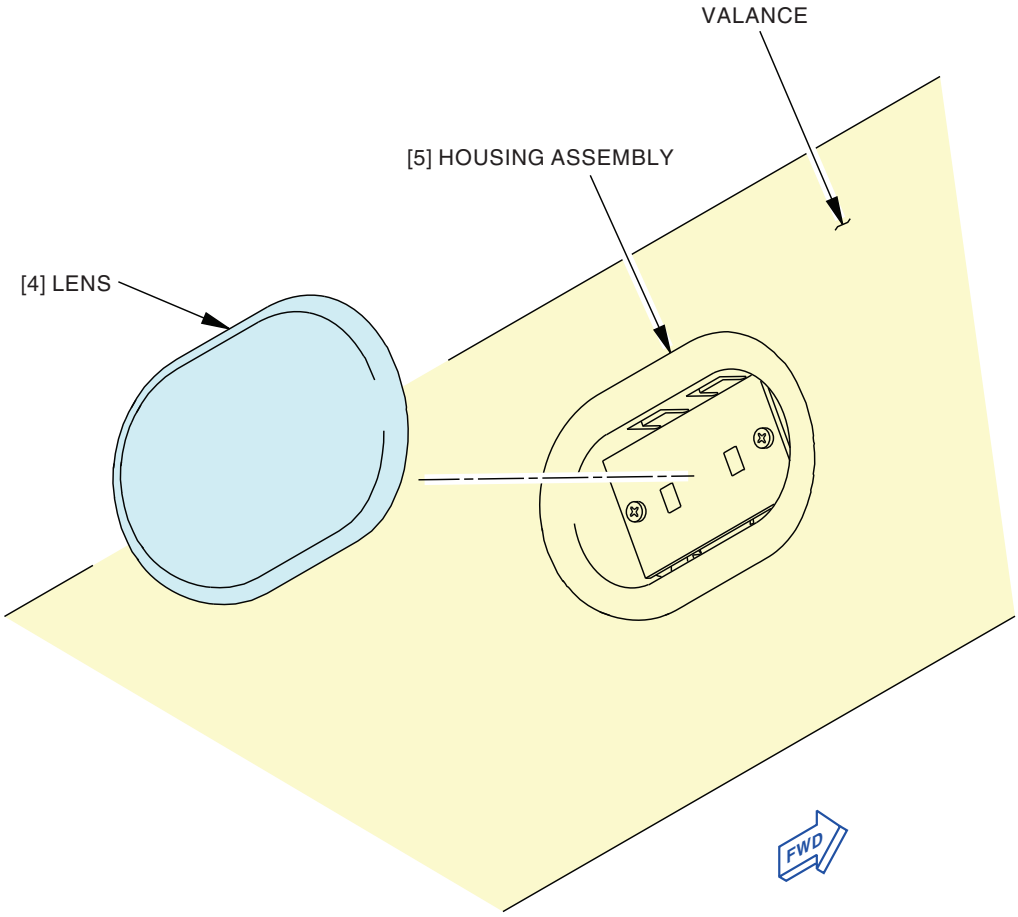


PASSENGER COMPARTMENT

2414778 S00061535952_V1

Emergency Aisle Light (EAL) - Lens Replacement
Figure 201/33-51-03-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EMERGENCY AISLE LIGHT (EAL)
(EXAMPLE)

A

2414779 S00061535953_V1

Emergency Aisle Light (EAL) - Lens Replacement
Figure 201/33-51-03-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-51-03-960-802

3. Emergency Aisle Light (EAL) – Assembly Replacement

(Figure 202)

A. General

(1) This task includes the steps to replace the emergency aisle light (EAL) assembly [8].

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
8	Emergency aisle light (EAL) assembly	33-51-03-04-050	PTF ALL
		33-51-03-12-050	PTF ALL
		33-51-03-90A-085	PTF ALL
		33-51-03-90A-380	PTF ALL
		33-51-03-90D-040	PTF ALL
		33-51-03-90N-040	PTF ALL
		33-51-03-92-055	PTF ALL
		33-51-03-92-160	PTF ALL
11	Gasket	33-51-03-94-080	PTF ALL

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Emergency Aisle Light (EAL) Assembly Replacement

SUBTASK 33-51-03-860-001

(1) Make sure that the pilot emergency light switch, on the P5 overhead panel, is set to the OFF position.

SUBTASK 33-51-03-860-002

(2) Make sure that the attendant emergency light switch in the passenger compartment is set to the NORMAL position.

SUBTASK 33-51-03-010-002

(3) Do these steps to remove the emergency aisle light (EAL) assembly [8]:

(a) Remove the lens (TASK 33-51-03-960-801).

NOTE: You can remove the EAL assembly with the lens installed.

(b) Get access behind the valance assembly.

(c) Disconnect the electrical connector [9].

(d) Remove the retaining ring [10].

(e) Remove the gasket [11].

1) Replace the gasket [11] if damaged.

(f) Remove the emergency aisle light (EAL) assembly [8].

SUBTASK 33-51-03-410-002

(4) Do these steps to install the emergency aisle light (EAL) assembly [8]:

(a) Install the emergency aisle light (EAL) assembly [8] from the outer side (decorative side) of the valance assembly.

1) Put the electrical connector [9] through the hole of the valance assembly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Align the top of the emergency aisle light (EAL) assembly [8] with the top of the hole of the valance assembly.

NOTE: You can identify the top of the EAL housing assembly by its two prong slots.

(c) Get access behind the valance assembly.

(d) Install the new gasket [11].

(e) Install the retaining ring [10] that attaches the emergency aisle light (EAL) assembly [8] to the valance assembly.

1) Make sure that the emergency aisle light (EAL) assembly [8] attaches into the correct position.

(f) Connect the electrical connector [9].

(g) Install the lens (TASK 33-51-03-960-801).

NOTE: You can install the EAL assembly with the lens installed.

SUBTASK 33-51-03-710-002

(5) Do a test of the new light.

NOTE: The power supply operates the emergency lights. Use the lights for a minimum time to do the test.

(a) On the P5 overhead panel, or the attendant panel, set the emergency light switch to the ON position.

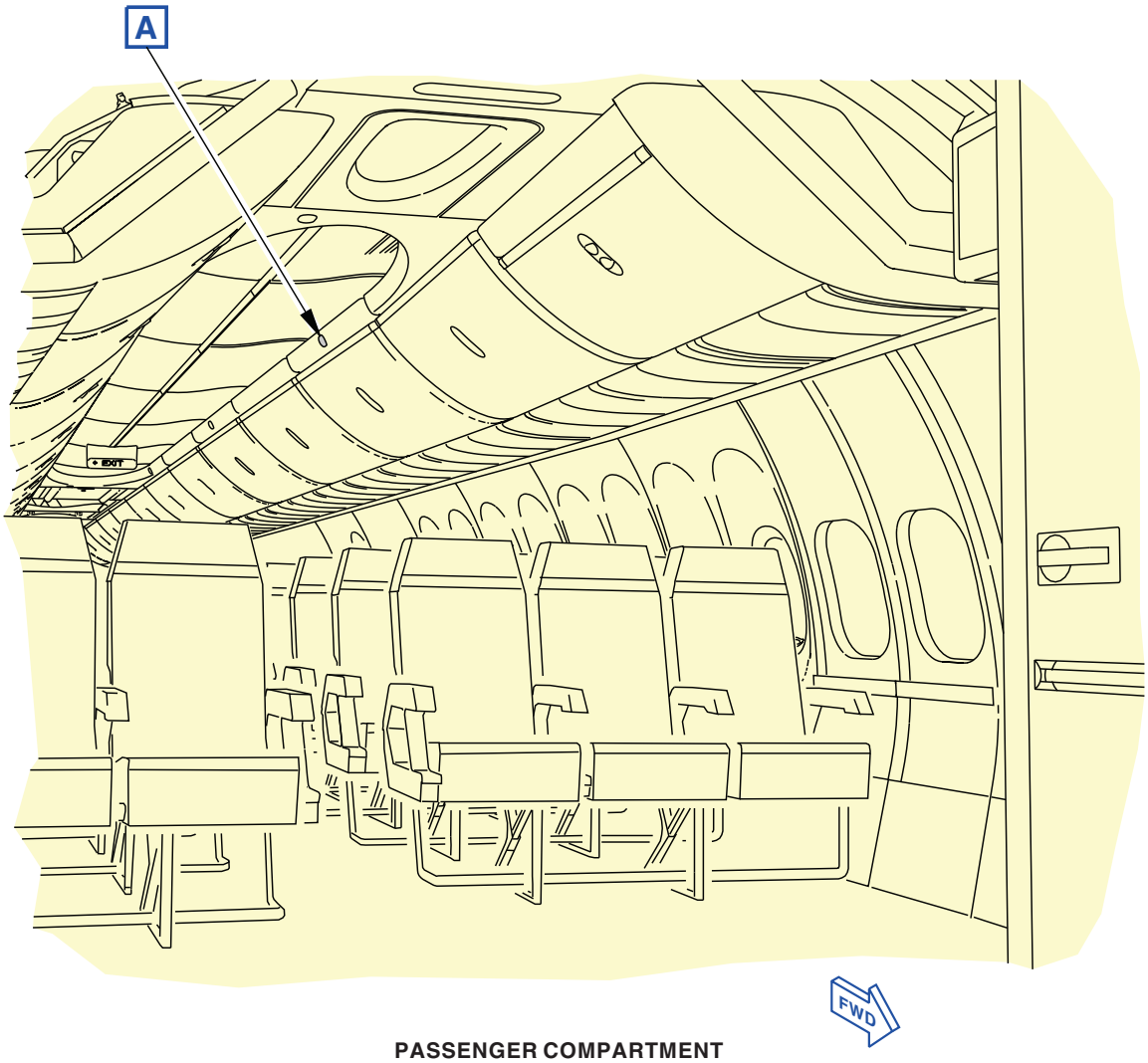
1) Make sure that the light comes on.

(b) Set the emergency light switch, on the P5 overhead panel, to the OFF position or the emergency light switch on the attendant panel to the NORMAL position.

1) Make sure that all emergency lights go off.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

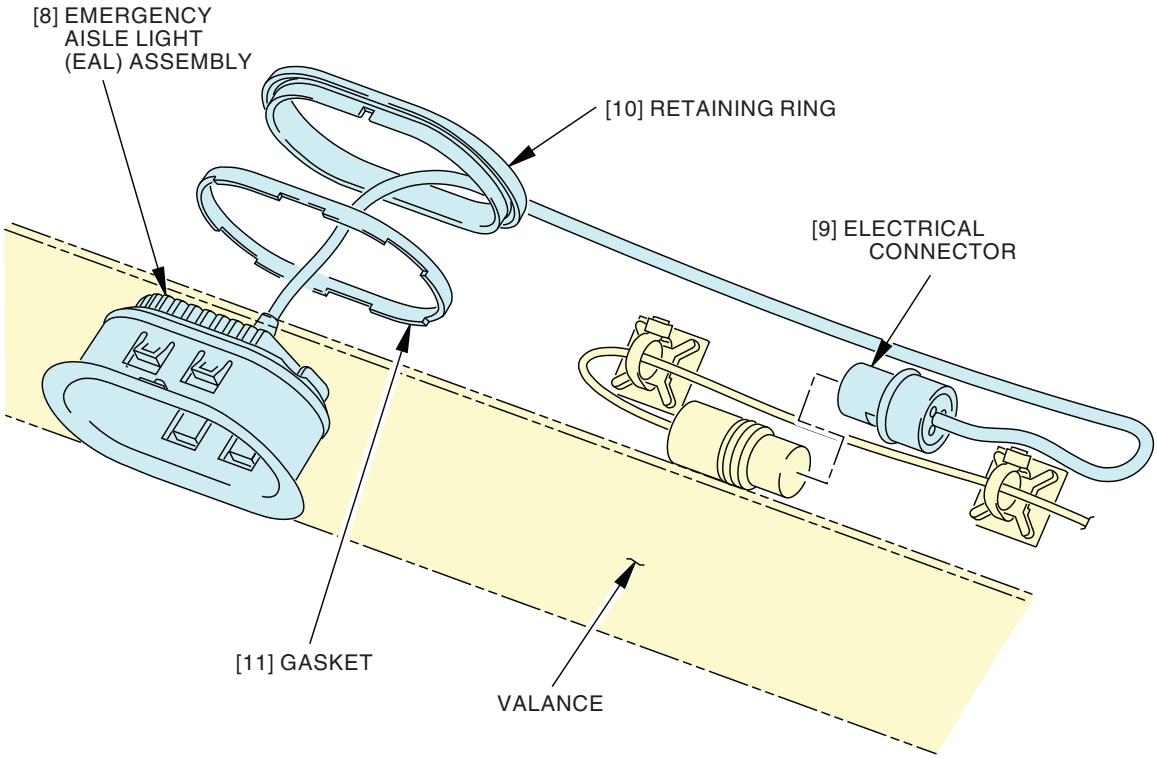


PASSENGER COMPARTMENT

2414780 S00061535955_V1

Emergency Aisle Light (EAL) - Assembly Replacement
Figure 202/33-51-03-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EMERGENCY AISLE LIGHT (EAL) (EXAMPLE)

A

2414781 S00061535956_V1

Emergency Aisle Light (EAL) - Assembly Replacement
Figure 202/33-51-03-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SLIDE/OVERWING LIGHTS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Slide/Overwing Light - Lamp Replacement

(2) Slide/Overwing Light - Light Assembly Replacement.

TASK 33-51-04-960-801

2. Slide/Overwing Light - Lamp Replacement

(Figure 201)

A. General

(1) This task includes all exterior emergency lights. The lights are installed on the fuselage adjacent to the doors and over the wings.

B. References

Reference	Title
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
WDM 33-51-12	Wiring Diagram Manual
WDM 33-51-14	Wiring Diagram Manual

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-659	Platform - Maintenance Lift, 3 ft Minimum Height, 12.5 ft Maximum Height Part #: 1135-8000 Supplier: 00994 Opt Part #: 8662-010 Supplier: 00994
COM-2480	Platform - Mobile Elevating Work Platform SJ III Series Part #: SJIII 4726 Supplier: 3AF08 Part #: SJIII 4732 Supplier: 3AF08 Opt Part #: SJIII 4620 Supplier: 3AF08 Opt Part #: SJIII 4626 Supplier: 3AF08 Opt Part #: SJIII 4632 Supplier: 3AF08

D. Consumable Materials

Reference	Description	Specification
G02309	Material - Moldable Plastic Shim	BMS8-338

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Lamp	33-51-04-06A-075	PTF ALL
8	Lamp	33-51-04-08A-085	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Location Zones

Zone	Area
200	Upper Half of Fuselage

G. Slide/Overwing Light Lamp Replacement

SUBTASK 33-51-04-860-001

(1) Make sure that the pilots emergency light switch, on the P5 overhead panel, is set to the OFF mode.

SUBTASK 33-51-04-860-002

(2) Make sure that the attendants emergency light switch in the passenger compartment is set to the NORMAL mode.

SUBTASK 33-51-04-040-001

(3) If it is necessary, deactivate the applicable flap system:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE TRAILING EDGE FLAP BEFORE YOU DO WORK ON THE FLAP SYSTEM. THE FLAPS MOVE QUICKLY. THEY CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(a) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

WARNING DO THE DEACTIVATION PROCEDURE FOR THE LEADING EDGE FLAPS. THE ACCIDENTAL MOVEMENT OF THE FLAPS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(b) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 33-51-04-480-001

(4) If it is necessary, get access to the applicable slide/overwing light with a ladder, work platform, COM-2480, or maintenance platform, COM-659.

SUBTASK 33-51-04-960-001

WARNING DO NOT TOUCH THE SOCKETS OR THE METAL ENDS OF THE LAMP. SHOCK CAN OCCUR WHEN YOU SUPPLY ELECTRICAL POWER TO THE LIGHT.

(5) Do these steps to replace the applicable lamp (WDM 33-51-12 through WDM 33-51-14):

(a) Loosen the fasteners [2] or fasteners [6] and remove the lens assembly [3] or lens assembly [7].

NOTE: The lanyard stays connected.

(b) Remove the lamp [4] or lamp [8] from the socket.

1) Remove the screw [15] and washer [16] that attach the lamp [4] or lamp [8] to the socket with the retainer [17].

(c) Replace the lamp [4] or lamp [8].

(d) Install the lamp [4] or lamp [8] in the socket.

1) Install the screw [15] and washer [16] that attach the lamp [4] or lamp [8] to the socket with the retainer [17].

(e) If the shims on the lens assembly [3] or lens assembly [7] gasket are damaged, replace them.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Remove the unwanted shim material.
2) Apply new shim material, G02309, in the applicable areas of the rubber gasket.

NOTE: Use only the necessary quantity. Adjacent compartments in the rubber gasket will help contain the shim material. Keep shim material away from the lens fasteners.

3) Install the lens assembly [3] or lens assembly [7].
   a) Tighten the fasteners [2] or fasteners [6] until the lens assembly [3] or lens assembly [7] is aligned with the fuselage.
4) Let the shim material dry for six hours.
5) Remove the lens assembly [3] or lens assembly [7].
6) Clean and examine the shims.
   a) Make sure that the shim thickness is no more than 0.125 in. (0.318 cm).

(f) Install the lens assembly [3] or lens assembly [7].
   1) Tighten the fasteners [2] or fasteners [6].

SUBTASK 33-51-04-080-001

(6) Remove the ladders, work platform, COM-2480, or maintenance platform, COM-659, from the work area.

SUBTASK 33-51-04-440-001

(7) If deactivated, activate the flap system:
   (a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.
   (b) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 33-51-04-710-001

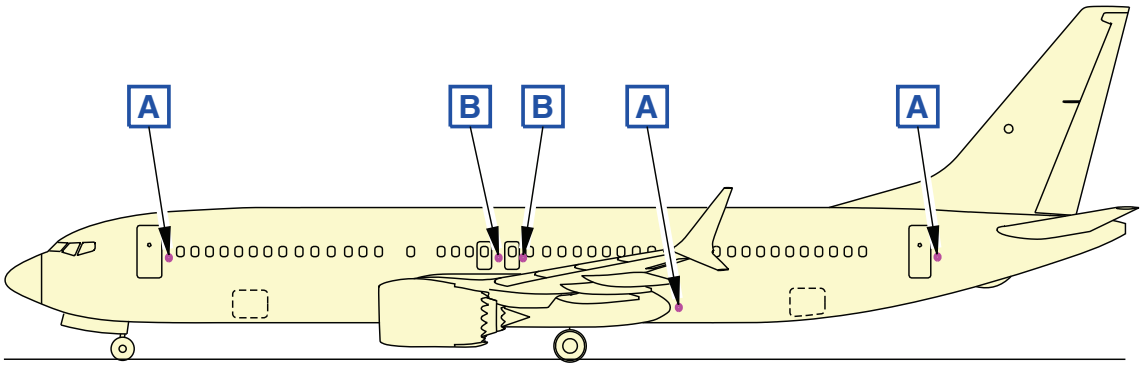
(8) Do a test of the new lamp [4] or lamp [8]:

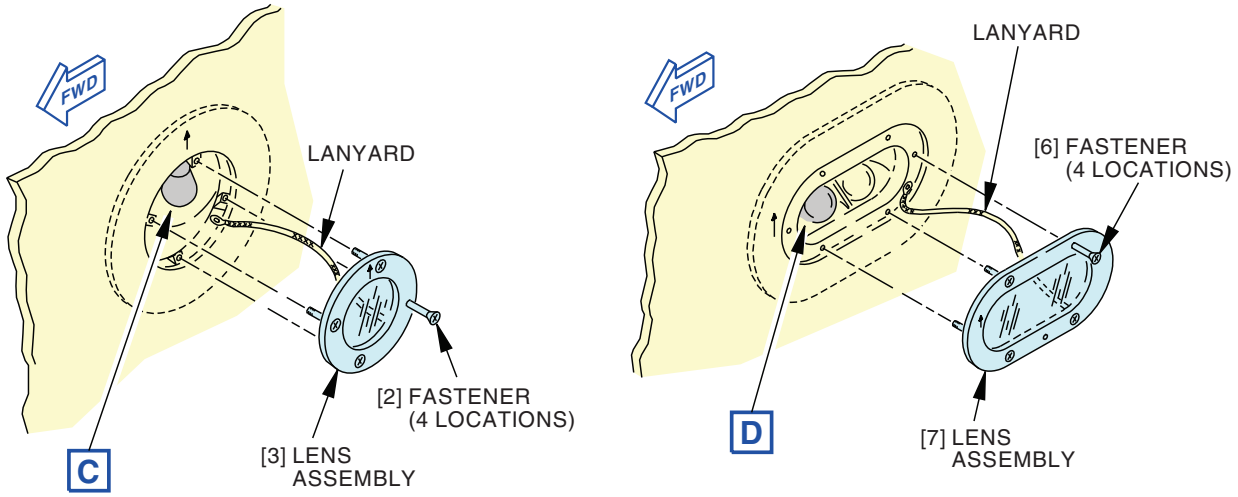
NOTE: The battery packs supply electrical power to the emergency lights. Use the lights for a minimum time to do the test.

   (a) On the P5 pilots overhead panel, or the attendant panel, set the emergency light switch to the ON mode.
      1) Make sure that the new lamp [4] or lamp [8] comes on.
   (b) Set the emergency light switch, on the P5 overhead panel, to the OFF mode or emergency light switch on the attendant panel to the NORMAL mode.
      1) Make sure that all emergency lights go off.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details





SLIDE LIGHT (EXAMPLE)

A

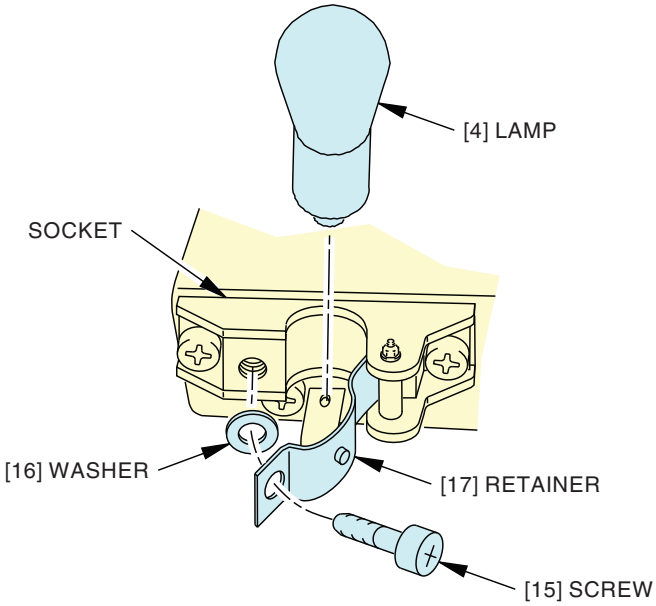
OVERWING LIGHT (EXAMPLE)

B

2414782 S00061535960_V2

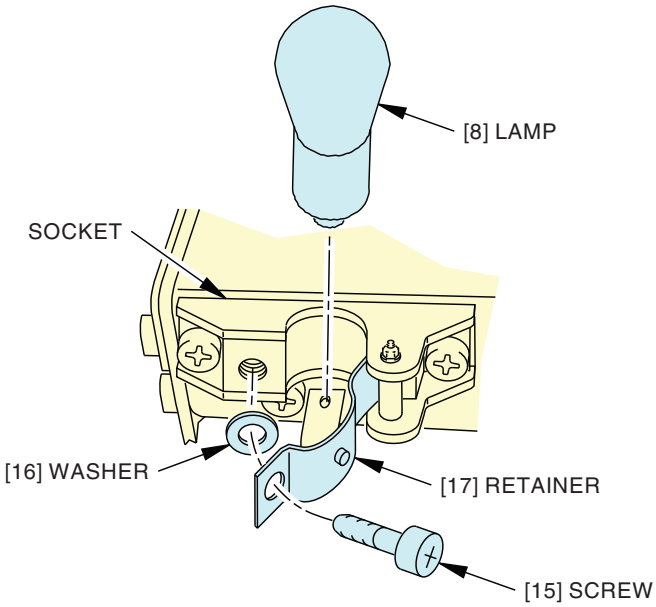
Slide Light - Lamp Replacement
Figure 201/33-51-04-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

C



(ONE LAMP IS SHOWN, ANOTHER LAMP IS EQUIVALENT)
(EXAMPLE)

D

3038014 S0000806749_V1

Slide Light - Lamp Replacement
Figure 201/33-51-04-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-51-04-960-802

3. Slide/Overwing Light - Light Assembly Replacement

(Figure 202)

A. General

(1) This task includes all exterior emergency lights. The lights are installed on the fuselage adjacent to the doors and over the wings.

B. References

Reference	Title
25-21-46-000-801	Sidewall Panel - Removal (P/B 401)
27-51-00-040-801	Trailing Edge Flap System Deactivation (P/B 201)
27-51-00-440-801	Trailing Edge Flap System Reactivation (P/B 201)
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-659	Platform - Maintenance Lift, 3 ft Minimum Height, 12.5 ft Maximum Height Part #: 1135-8000 Supplier: 00994 Opt Part #: 8662-010 Supplier: 00994
COM-2480	Platform - Mobile Elevating Work Platform SJ III Series Part #: SJIII 4726 Supplier: 3AF08 Part #: SJIII 4732 Supplier: 3AF08 Opt Part #: SJIII 4620 Supplier: 3AF08 Opt Part #: SJIII 4626 Supplier: 3AF08 Opt Part #: SJIII 4632 Supplier: 3AF08

D. Consumable Materials

Reference	Description	Specification
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
11	Light assembly	33-51-04-01A-005	PTF ALL
		33-51-04-02A-010	PTF ALL
		33-51-04-03A-005	PTF ALL

F. Location Zones

Zone	Area
200	Upper Half of Fuselage

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Slide/Overwing Light Assembly Replacement

SUBTASK 33-51-04-010-001

(1) Remove the applicable sidewall panel to get access to the light assembly [11] from the inside of the cabin (TASK 25-21-46-000-801).

SUBTASK 33-51-04-020-001

(2) From the inside of the cabin, disconnect the electrical connector [14] from the light assembly [11].

SUBTASK 33-51-04-040-003

(3) If it is necessary, deactivate the applicable flap system:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE TRAILING EDGE FLAP BEFORE YOU DO WORK ON THE FLAP SYSTEM. THE FLAPS MOVE QUICKLY. THEY CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(a) Do this task: Trailing Edge Flap System Deactivation, TASK 27-51-00-040-801.

WARNING DO THE DEACTIVATION PROCEDURE FOR THE LEADING EDGE FLAPS. THE ACCIDENTAL MOVEMENT OF THE FLAPS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(b) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 33-51-04-480-002

(4) If it is necessary, get access to the applicable slide/overwing light with a ladder, work platform, COM-2480, or maintenance platform, COM-659.

SUBTASK 33-51-04-020-002

(5) Loosen the bolts [12] and collars [13] to remove the light assembly [11].

SUBTASK 33-51-04-110-001

(6) Clean the fay surface bonding area of the contact surface with solvent, B00148, and a clean cotton wiper, G00034.

SUBTASK 33-51-04-160-001

(7) Rub dry with a clean, dry cotton wiper, G00034.

SUBTASK 33-51-04-160-002

(8) Continue to clean and dry the fay surface bonding area until the dry cotton wiper, G00034, stays clean.

SUBTASK 33-51-04-390-001

(9) Apply sealant, A50231, to the contact surface of the light assembly [11] housing.

SUBTASK 33-51-04-390-002

(10) Apply sealant, A50231, to the shank and threads of the bolts [12].

SUBTASK 33-51-04-420-001

(11) Install the light assembly [11] on the aircraft skin with the bolts [12] and collars [13].

(a) Make sure that sealant, A50231, is continuously squeezed out along the edges of the light assembly [11] housing.

(b) If there are clearances, bubbles, or voids in sealant, A50231, do these steps:

1) Squeeze out sealant, A50231.

2) Remove the light assembly [11] from the aircraft skin.

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3) Apply more sealant, A50231.

(c) Shape squeezed out sealant, A50231, into a fillet seal.

(d) As an option, remove the extra squeezed out sealant, A50231.

1) Make sure that sealant, A50231, that remains is flush with the mating part edges.

SUBTASK 33-51-04-420-002

(12) From the inside of the cabin, connect the electrical connector [14] to the light assembly [11].

SUBTASK 33-51-04-710-002

(13) Do a test of the light assembly [11]:

NOTE: The battery packs supply electrical power to the emergency lights. Use the lights for a minimum amount of time to do the test.

(a) On the P5 overhead panel, or the attendant control panel, set the emergency light switch to the ON mode.

1) Make sure that the applicable emergency light comes on.

(b) Set the emergency light switch, on the P5 overhead panel, to the OFF mode or the emergency light switch on the attendant panel to the NORMAL mode.

1) Make sure that the applicable emergency lights go off.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 33-51-04-410-001

(1) Install the applicable sidewall panel (TASK 25-21-46-000-801).

SUBTASK 33-51-04-080-002

(2) Remove the ladders, work platform, COM-2480, or maintenance platform, COM-659, from the work area.

SUBTASK 33-51-04-440-003

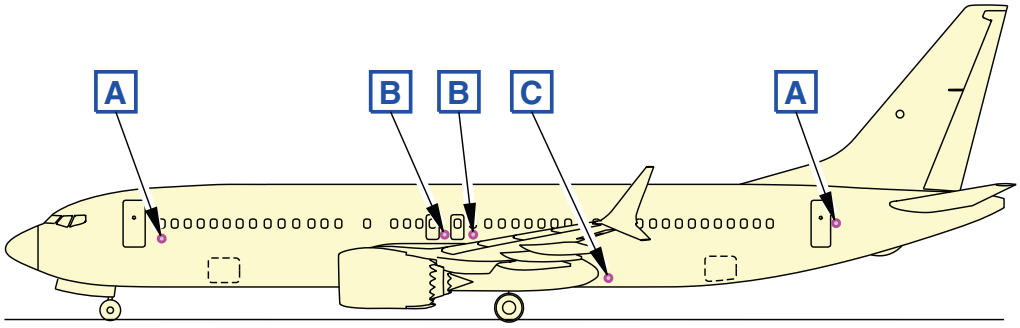
(3) If deactivated, activate the flap system:

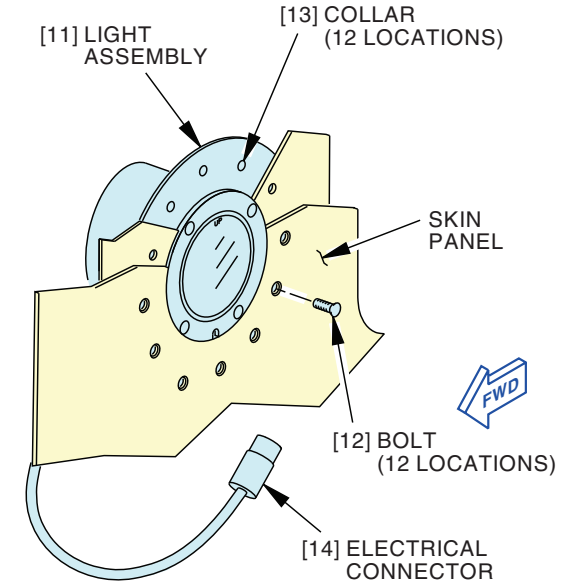
(a) Do this task: Trailing Edge Flap System Reactivation, TASK 27-51-00-440-801.

(b) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

——— END OF TASK ———

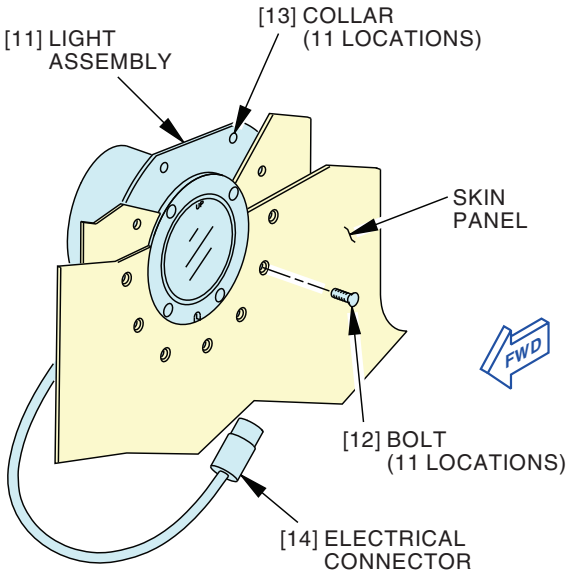
ECCN 9E991 BOEING PROPRIETARY - See title page for details





SLIDE LIGHT ASSEMBLY
(FOWARD RIGHT, AFT LEFT, AFT RIGHT)

A



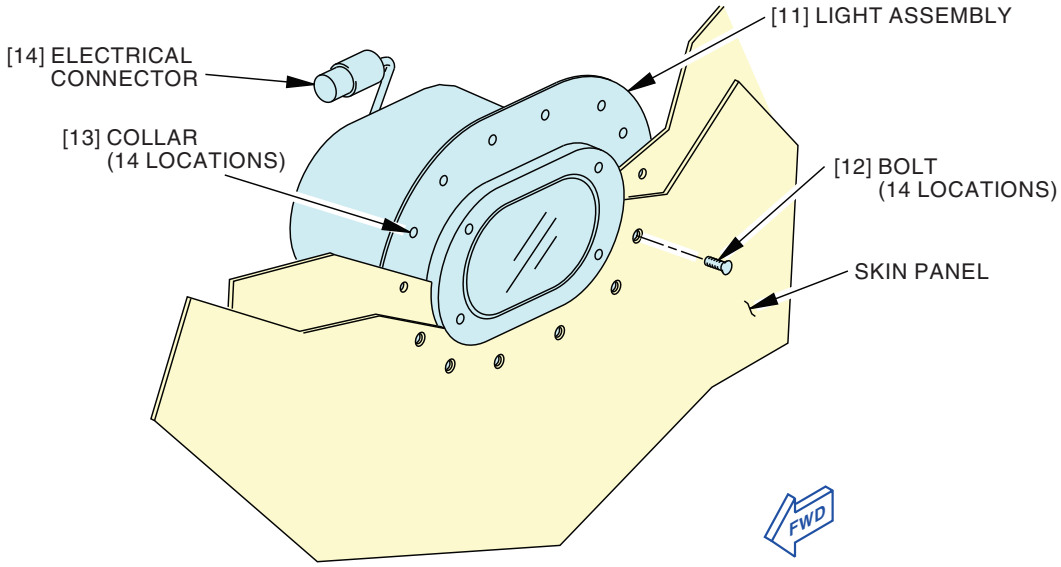
SLIDE LIGHT ASSEMBLY
(FOWARD LEFT ONLY)

A

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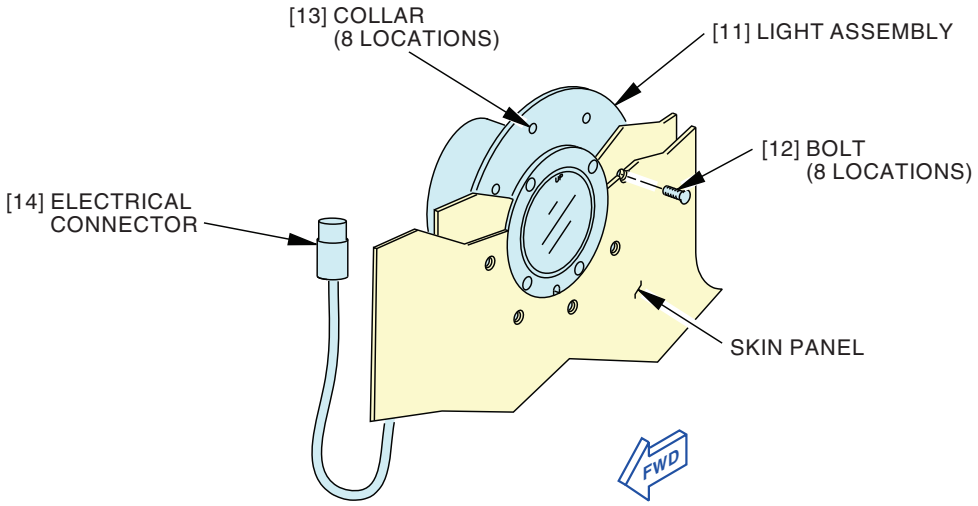
Slide Light - Light Assembly Replacement
Figure 202/33-51-04-990-802 (Sheet 1 of 2)

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OVERWING LIGHT ASSEMBLY

B



SLIDE LIGHT ASSEMBLY

C

2414784 S00061535963_V1

Slide Light - Light Assembly Replacement
Figure 202/33-51-04-990-802 (Sheet 2 of 2)

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POWER SUPPLIES - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

NOTE: See WDM 33-51-11 through WDM 33-51-22 for Power Supplies/Battery Packs quantity and locations.

(1) Power Supply - Charge the Battery Packs

(2) Power Supply - Fuse Replacement

(3) Power Supply - Battery Pack Replacement

(4) Power Supply - Power Supply Replacement

(5) Power Supply - Battery Pack Capacity Check

(6) Power Supply - Battery Pack Deep-Cycle Procedure

TASK 33-51-06-610-801

2. Power Supply - Charge the Battery Packs

A. General

(1) This procedure has steps to charge battery packs for the emergency lights.

B. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Power Supply Battery Packs Charging

SUBTASK 33-51-06-860-001

(1) Make sure that the pilots emergency light switch, on the P5 overhead panel, is set to the OFF mode.

SUBTASK 33-51-06-860-002

(2) Make sure that the attendants emergency light switch, in the passenger compartment, is set to the NORMAL mode.

SUBTASK 33-51-06-860-003

(3) To charge the battery packs, do this task: Supply Electrical Power, TASK 24-22-00-860-801.

NOTE: The battery packs in the emergency light power supplies are continuously charged when electrical power is supplied to the airplane, unless the emergency light switches are set to the on mode. If the battery packs are fully drained, maximum time necessary to charge them is 90 minutes.

(a) If the airplane power was removed for 6 days or more, do not activate the emergency light system for a minimum of 90 minutes after electrical power is applied.

1) Do not do a functional test until the batteries are charged for a minimum of 90 minutes.

NOTE: The system must be fully charged to operate correctly.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-51-06-960-801

3. Power Supply - Fuse Replacement

(Figure 201)

A. General

(1) This procedure has steps to replace the fuse for the power supply for the emergency exit lights.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
WDM 33-51-11 thru 33-51-22	Wiring Diagram Manual
WDM 33-51-12 thru 33-51-22	Wiring Diagram Manual

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Power Supply Fuse Replacement

SUBTASK 33-51-06-860-016

(1) Make sure that the pilots' emergency light switch, on the P5 overhead panel, is set to the OFF mode.

SUBTASK 33-51-06-860-005

(2) Open the applicable circuit breaker and install the safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	10	C01357	PASS CABIN LIGHTING EMER CHGR L AFT
A	11	C00250	PASS CABIN LIGHTING EMER CHGR FWD
B	10	C01358	PASS CABIN LIGHTING EMER CHGR R AFT

SUBTASK 33-51-06-040-001

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(3) Before the power supply fuse is touched, do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 33-51-06-010-001

(4) Get access to the power supply:

NOTE: Use the Wiring Diagram Manual (WDM) to identify the power supply locations (WDM 33-51-11 thru 33-51-22).

(a) For the power supplies near the forward entry door, do the following steps:

1) Open the applicable access panel.
2) Support the battery access door.
3) Remove the screws [9].
4) Open the battery access door.

(b) Open the access panel in the lowered ceiling panel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Remove the sidewall carpet riser.

SUBTASK 33-51-06-960-001

(5) Do these steps to replace the fuse:

(a) Loosen the fasteners [3] on the battery pack.

CAUTION MAKE SURE TO KEEP THE BATTERY PACK PARALLEL TO THE BATTERY CHARGER SURFACE. DAMAGE TO THE BATTERY PACK CAN OCCUR.

(b) Lift the battery pack [4] to disconnect its pins from the bottom of the power supply.

(c) Replace each damaged fuse [8].

(d) Connect the pins on the battery pack [4] with the bottom of the power supply.

(e) Tighten the fasteners [3].

SUBTASK 33-51-06-710-001

(6) Do a test of the new fuse:

(a) Remove the safety tag and close the applicable circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	10	C01357	PASS CABIN LIGHTING EMER CHGR L AFT
A	11	C00250	PASS CABIN LIGHTING EMER CHGR FWD
B	10	C01358	PASS CABIN LIGHTING EMER CHGR R AFT

(b) Set the test switch [7] to the ON mode.

1) Make sure that the lights and signs connected to the power supply come on.

NOTE: Use the WDM to identify each of the lights and signs (WDM 33-51-12 thru 33-51-22).

2) If the lights and sign do not come on, supply electrical power to the airplane to charge the battery packs and do the test again (TASK 33-51-06-610-801).

NOTE: The battery packs in the emergency light power supplies are continuously charged when electrical power is supplied to the airplane, unless the emergency light switches are set to the ON mode. If the battery packs are fully drained, maximum time necessary to charge them is 90 minutes.

NOTE: The system must be fully charged to operate correctly.

(c) After approximately one minute, make sure that all the emergency lights and signs go off automatically.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 33-51-06-410-004

(1) For the power supplies near the forward entry door, do the following steps:

(a) Close the battery access door as follows:

1) Close and hold the battery access door.

2) Install the screws [9] through the clip and to the battery access door.

3) Tighten the screws [9].

(b) Close the access panel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-51-06-410-001

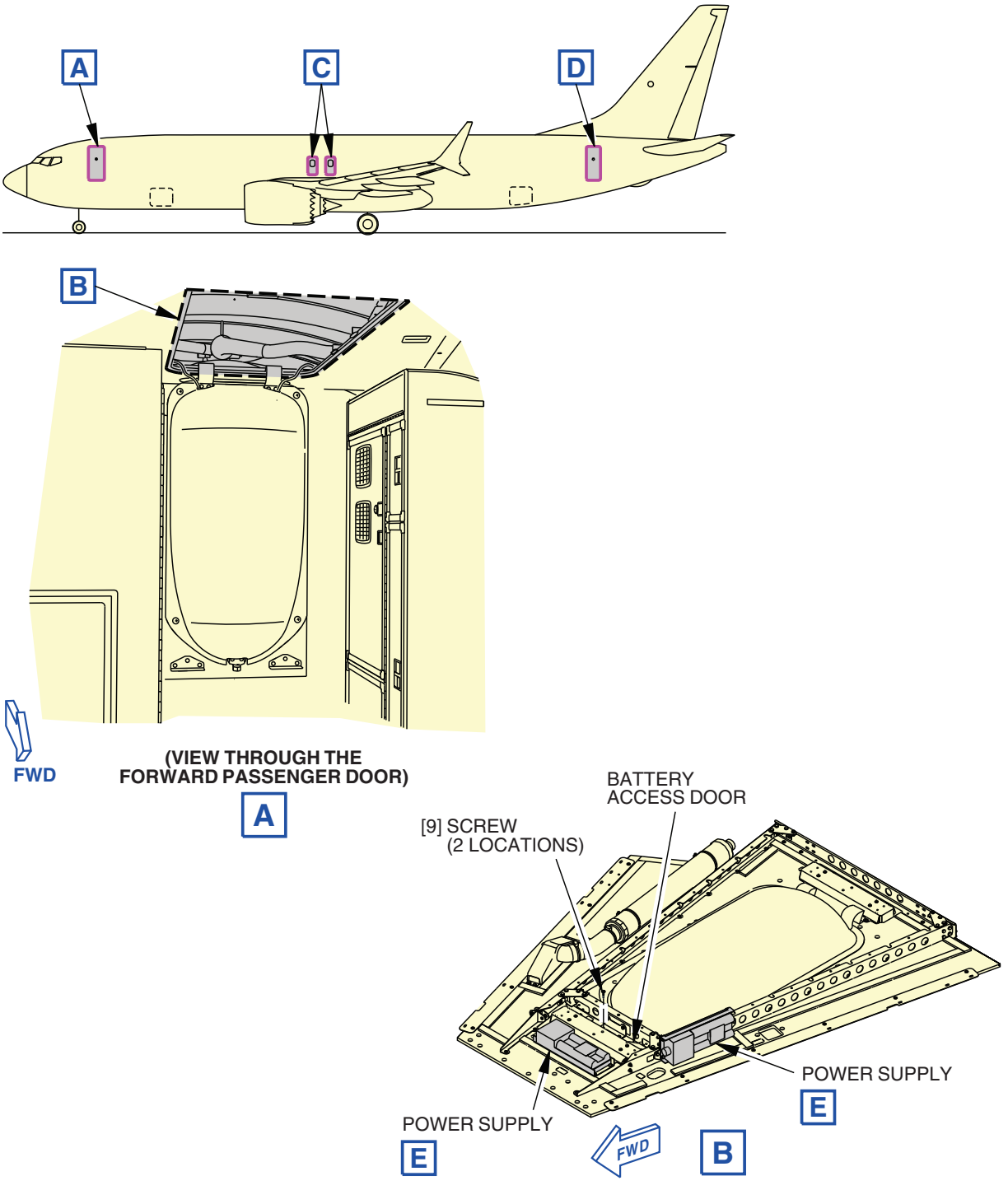
(2) Close the access panel in the lowered ceiling panel.

SUBTASK 33-51-06-420-002

(3) Install the sidewall carpet riser.

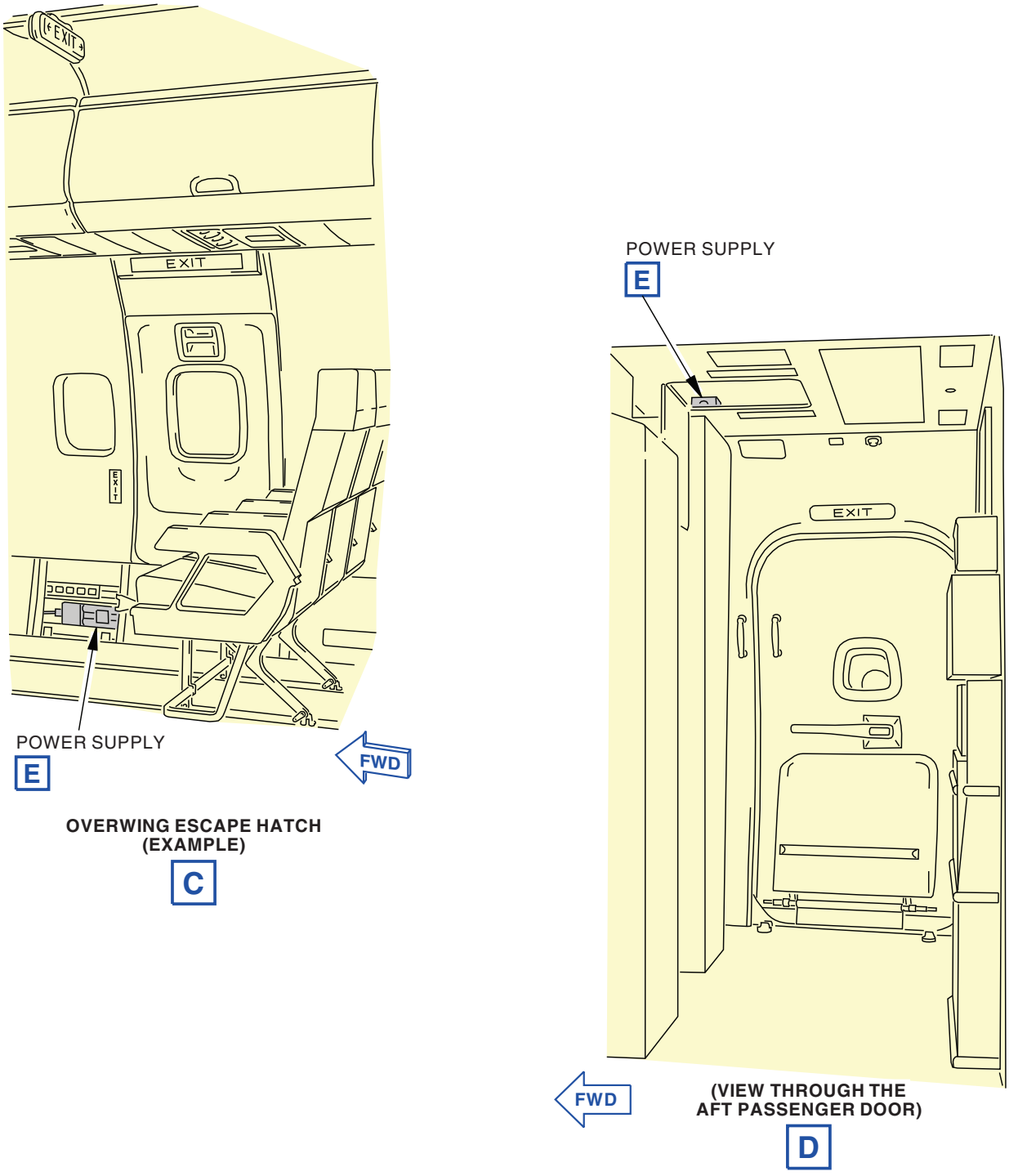
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



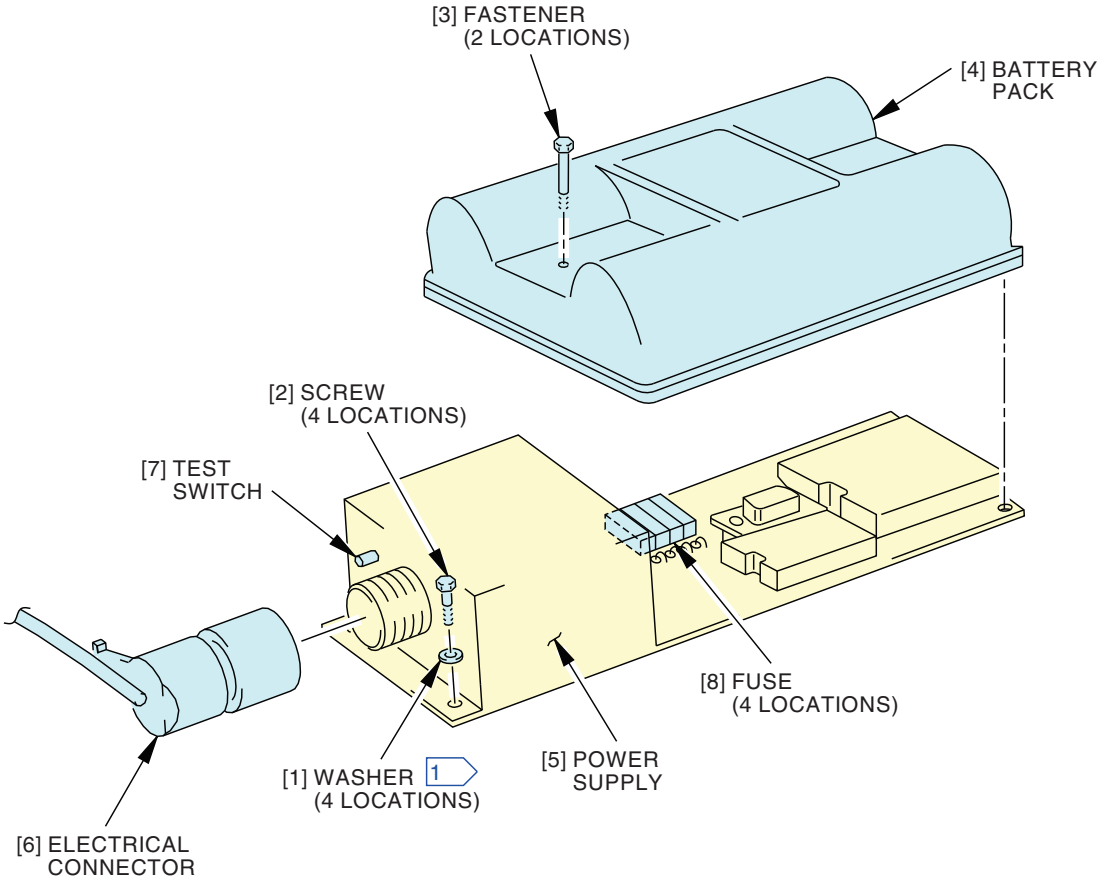
Power Supply - Power Supply/Battery Pack/Fuse Replacement
Figure 201/33-51-06-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Power Supply - Power Supply/Battery Pack/Fuse Replacement
Figure 201/33-51-06-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



POWER SUPPLY
(EXAMPLE)

E

1 IF APPLICABLE

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Power Supply - Power Supply/Battery Pack/Fuel Replacement
Figure 201/33-51-06-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 33-51-06-960-802

4. Power Supply - Battery Pack Replacement

(Figure 201)

A. General

(1) This procedure has steps to replace the battery pack for emergency lights.

(2) When you replace the battery pack, do not discard it. Refer to the vendor instructions for possible repairs.

NOTE: Ignore the expiration dates (if installed) on these battery packs.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
WDM 33-51-12 thru 33-51-22	Wiring Diagram Manual

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Power Supply Battery Pack Replacement

SUBTASK 33-51-06-860-017

(1) Make sure that the pilots emergency light switch, on the P5 overhead panel, is set to the OFF mode.

SUBTASK 33-51-06-860-006

(2) Open the applicable circuit breaker and install the safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	10	C01357	PASS CABIN LIGHTING EMER CHGR L AFT
A	11	C00250	PASS CABIN LIGHTING EMER CHGR FWD
B	10	C01358	PASS CABIN LIGHTING EMER CHGR R AFT

SUBTASK 33-51-06-040-002

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(3) Before the power supply battery pack is touched, do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 33-51-06-010-002

(4) Get access to the power supply that contains the battery pack:

NOTE: Use the Wiring Diagram Manual (WDM) to identify power supply locations (WDM 33-51-12 thru 33-51-22).

(a) For the power supplies near the forward entry door, do the following steps:

1) Open the applicable access panel.

2) Support the battery access door.

3) Remove the screws [9].

4) Open the battery access door.

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(b) Open the access panel in the lowered ceiling panel.

(c) Remove the sidewall carpet riser.

SUBTASK 33-51-06-960-002

(5) Do these steps to remove the battery pack:

(a) Loosen the fasteners [3].

CAUTION MAKE SURE TO KEEP THE BATTERY PACK PARALLEL TO THE BATTERY CHARGER SURFACE. DAMAGE TO THE BATTERY PACK CAN OCCUR.

(b) Lift away the battery pack [4] to disconnect its pins from the base of the power supply.

SUBTASK 33-51-06-960-003

(6) Do these steps to install the battery pack:

NOTE: The battery pack must be fully charged before installation.

(a) Align the connector pins with the base of the power supply, then install the battery pack [4].

(b) Tighten the fasteners [3].

SUBTASK 33-51-06-710-002

(7) Do a test of the new battery pack:

(a) Remove the safety tag and close the applicable circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	10	C01357	PASS CABIN LIGHTING EMER CHGR L AFT
A	11	C00250	PASS CABIN LIGHTING EMER CHGR FWD
B	10	C01358	PASS CABIN LIGHTING EMER CHGR R AFT

(b) Set the test switch [7] to the ON mode.

1) Make sure that the lights and signs connected to the battery pack come on.

NOTE: Use the WDM to identify the individual lights and signs (WDM 33-51-12 thru 33-51-22).

2) If the lights and sign do not come on, supply electrical power to the airplane to charge the battery packs, then do the test again (TASK 33-51-06-610-801).

NOTE: The battery packs in the emergency light power supplies are continuously charged when electrical power is supplied to the airplane, unless the emergency light switches are set to the ON mode. If the battery packs are fully drained, maximum time necessary to charge them is 90 minutes.

NOTE: The system must be fully charged to operate correctly.

(c) After approximately one minute, make sure that all emergency lights and signs go off automatically.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 33-51-06-410-005

(1) For the power supplies near the forward entry door, do the following steps:

(a) Close the battery access door as follows:

1) Close and hold the battery access door.

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2) Install the screws [9] through the clip and to the battery access door.

3) Tighten the screws [9].

(b) Close the access panel.

SUBTASK 33-51-06-410-002

(2) Close the access panel in the lowered ceiling panel.

SUBTASK 33-51-06-420-004

(3) Install the sidewall carpet riser.

——— END OF TASK ———

TASK 33-51-06-960-803

5. Power Supply - Power Supply Replacement

(Figure 201)

A. General

(1) This procedure has steps to replace a power supply for the emergency lights.

(2) When you replace the power supply, do not discard it. Refer to the vendor instructions for possible repairs.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
WDM 33-51-12 thru 33-51-22	Wiring Diagram Manual

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Power supply	33-51-06-12-045	PTF ALL

D. Location Zones

Zone	Area
200	Upper Half of Fuselage

E. Power Supply Replacement

SUBTASK 33-51-06-860-018

(1) Make sure that the pilots emergency light switch, on the P5 overhead panel, is set to the OFF mode.

SUBTASK 33-51-06-860-007

(2) Open the applicable circuit breaker and install the safety tag:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	10	C01357	PASS CABIN LIGHTING EMER CHGR L AFT
A	11	C00250	PASS CABIN LIGHTING EMER CHGR FWD
B	10	C01358	PASS CABIN LIGHTING EMER CHGR R AFT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 33-51-06-040-003

CAUTION

DO NOT TOUCH THE CONNECTOR PINS, OR OTHER CONDUCTORS. IF YOU TOUCH THESE CONDUCTORS, ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE COMPONENTS.

(3) Before the power supply is touched, do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 33-51-06-010-003

(4) Get access to the power supply:

NOTE: Use the Wiring Diagram Manual (WDM) to identify power supply locations (WDM 33-51-12 thru 33-51-22).

(a) For the power supplies near the forward entry door, do the following steps:

1) Open the applicable access panel.
2) Support the battery access door.
3) Remove the screws [9].
4) Open the battery access door.

(b) Open the access panel in the lowered ceiling panel.

(c) Remove the sidewall carpet riser.

SUBTASK 33-51-06-960-004

(5) Do these steps to replace the power supply:

(a) Disconnect the electrical connector [6].

(b) Remove the screws [2] and washers [1] (if applicable).

(c) Remove the power supply [5].

(d) Install the new power supply [5].

(e) Install the washers [1] (if applicable) and screws [2] .

(f) Connect the electrical connector [6].

SUBTASK 33-51-06-710-003

(6) Do a test of the new power supply:

(a) Remove the safety tag and close the applicable circuit breaker:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
A	10	C01357	PASS CABIN LIGHTING EMER CHGR L AFT
A	11	C00250	PASS CABIN LIGHTING EMER CHGR FWD
B	10	C01358	PASS CABIN LIGHTING EMER CHGR R AFT

(b) Set the test switch [7] to the ON mode.

1) Make sure that the lights and signs connected to the power supply come on.

NOTE: Use the WDM to identify the individual lights and signs (WDM 33-51-12 thru 33-51-22).

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2) If the lights and signs do not come on, supply electrical power to the airplane to charge the battery packs and do the test again (TASK 33-51-06-610-801).

NOTE: The battery packs in the emergency light power supplies are continuously charged when electrical power is supplied to the airplane, unless the emergency light switches are set to the ON mode. If the battery packs are fully drained, maximum time necessary to charge them is 90 minutes.

NOTE: The system must be fully charged to operate correctly.

(c) After approximately one minute, make sure that all emergency lights and signs go off automatically.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 33-51-06-410-006

(1) For the power supplies near the forward entry door, do the following steps:

(a) Close the battery access door as follows:

1) Close and hold the battery access door.

2) Install the screws [9] through the clip and to the battery access door.

3) Tighten the screws [9].

(b) Close the access panel.

SUBTASK 33-51-06-410-003

(2) Close the access panel in the lowered ceiling panel.

SUBTASK 33-51-06-420-006

(3) Install the sidewall carpet riser.

——— END OF TASK ———

TASK 33-51-06-200-801

6. Power Supply - Battery Pack Capacity Check

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure has steps to check the capacity of the battery pack for the emergency lights.

(2) The capacity check makes sure that each battery pack can operate its emergency lights and signs for a minimum of 15 minutes.

B. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
WDM 33-51-12 thru 33-51-22	Wiring Diagram Manual

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Power Supply Battery Pack Capacity Check

SUBTASK 33-51-06-860-008

(1) Make sure that electrical power is applied (TASK 24-22-00-860-801).

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SUBTASK 33-51-06-860-009

(2) Make sure that the battery packs are fully charged (TASK 33-51-06-610-801).

NOTE: The battery packs in the emergency light power supplies are continuously charged when electrical power is supplied to the airplane, unless the emergency light switches are set to the ON mode. If power has been on for 90 minutes and no emergency lights have been tested, the battery packs can be considered fully charged. If the battery packs are fully drained, maximum time necessary to charge them is 90 minutes.

NOTE: The system must be fully charged to operate correctly.

SUBTASK 33-51-06-210-001

(3) Do the battery pack capacity check:

(a) Set the pilots emergency light switch, on the P5 overhead panel, to the ON mode.

NOTE: You will note the amount of time the emergency lights stay on for each battery pack.

(b) Leave the switch in the ON mode until all emergency lights go off.

(c) Make sure that the emergency lights and signs stay on for a minimum of 15 minutes.

1) Identify each battery pack that does not operate its emergency lights and signs for a minimum of 15 minutes.

NOTE: Use the WDM (WDM 33-51-12 thru 33-51-22). It is not necessary to identify individual lights and signs. It is only necessary to identify emergency light power supplies where unsatisfactory battery packs may be installed.

(d) Set the pilots emergency light switch, on the P5 overhead panel, to the OFF mode.

(e) Charge the battery packs (TASK 33-51-06-610-801).

NOTE: The battery packs in the emergency light power supplies are continuously charged when electrical power is supplied to the airplane, unless the emergency light switches are set to the ON mode. If the battery packs are fully drained, maximum time necessary to charge them is 90 minutes.

NOTE: The system must be fully charged to operate correctly.

SUBTASK 33-51-06-860-010

(4) If there are battery packs that do not operate correctly during the capacity check, then do these steps:

NOTE: The battery packs have been re-charged following the capacity check.

(a) To deep cycle the battery packs, do this task: Power Supply - Battery Pack Deep-Cycle Procedure, TASK 33-51-06-600-801

SUBTASK 33-51-06-810-001

(5) Retest any battery packs that did not stay on for 15 minutes above.

(a) Set the pilots emergency light switch, on the P5 overhead panel, to the ON mode.

NOTE: You will note the amount of time the emergency lights stay on that had previously failed.

(b) Leave the switch in the ON mode for a minimum of 15 minutes.

(c) Make sure that the emergency lights stayed on for a minimum of 15 minutes.

SUBTASK 33-51-06-860-011

(6) If the battery pack operated correctly during the retest, put the airplane back to its usual condition.

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(a) Set the pilots emergency light switch, on the P5 panel, to the OFF mode.

(b) Charge the battery pack (TASK 33-51-06-610-801).

NOTE: The battery pack in the emergency lights power supply is continuously charged when electrical power is supplied to the airplane, unless the emergency light switches are set to the ON mode. If the battery pack is fully drained, maximum time necessary to charge it is 90 minutes.

NOTE: The system must be fully charged to operate correctly.

SUBTASK 33-51-06-860-012

(7) If the battery pack did not operate correctly during the retest, replace it.

(a) Do this task: Power Supply - Battery Pack Replacement, TASK 33-51-06-960-802.

(b) Set the pilots emergency light switch, on the P5 overhead panel, to the OFF mode.

(c) Charge the battery pack (TASK 33-51-06-610-801).

NOTE: The battery pack in the emergency lights power supply is continuously charged when electrical power is supplied to the airplane, unless the emergency light switches are set to the ON mode. If the battery pack is fully drained, maximum time necessary to charge it is 90 minutes.

NOTE: The system must be fully charged to operate correctly.

——— END OF TASK ———

TASK 33-51-06-600-801

7. Power Supply - Battery Pack Deep-Cycle Procedure

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure has steps to check the deep cycle of the battery pack for the emergency lights.

(2) The deep-cycle procedure removes the electrical power in the battery packs down to the power supply lower level cutoff and then charges them to their capacity.

B. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)

C. Location Zones

Zone	Area
200	Upper Half of Fuselage

D. Power Supply Battery Pack Deep-Cycle Procedure

SUBTASK 33-51-06-860-014

(1) Make sure that electrical power is applied (TASK 24-22-00-860-801).

SUBTASK 33-51-06-610-001

(2) Do the battery pack deep-cycle procedure:

(a) Set the pilots emergency light switch, on the P5 overhead panel, to the ON mode.

(b) Keep the switch on until all emergency lights and signs go off.

NOTE: The removal of electrical power from the battery packs can take up to 90 minutes.

(c) After all emergency lights and signs go off, set the switch to the OFF mode.

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(d) Charge the battery packs (TASK 33-51-06-610-801).

NOTE: The battery packs in the emergency light power supplies are continuously charged when electrical power is supplied to the airplane, unless the emergency light switches are set to the on mode. If the battery packs are fully drained, maximum time necessary to charge them is 90 minutes.

NOTE: The system must be fully charged to operate correctly.

——— END OF TASK ———

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FLOOR PROXIMITY LIGHTS ON GALLEYS, LAVATORIES, CLOSETS, AND WINDSCREENS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Floor Proximity Light - Lamp Replacement

(2) Floor Proximity Light - Light Assembly Replacement.

TASK 33-51-14-960-801

2. Floor Proximity Light - Lamp Replacement

(Figure 201)

A. References

Reference	Title
WDM 33-51-21	Wiring Diagram Manual

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Procedure

SUBTASK 33-51-14-960-001

WARNING DO NOT TOUCH THE SOCKETS OR THE METAL ENDS OF THE LAMP. SHOCK CAN OCCUR WHEN YOU SUPPLY ELECTRICAL POWER TO THE LIGHT.

(1) Do these steps to replace the applicable lamp (WDM 33-51-21):

(a) Remove the lens [3] or lens [6] from the light assembly [1] or light assembly [4].

(b) Replace the lamp [2] or lamp [5].

(c) Install the lens [3] or lens [6] to the light assembly [1] or light assembly [4].

SUBTASK 33-51-14-710-001

(2) Do a test of the new lamp [2] or lamp [5]:

NOTE: The emergency lights are energized by the power supplies. Use the lights for a minimum time to do the test.

(a) At the attendant panel in the passenger compartment, set the emergency light switch to the ON mode.

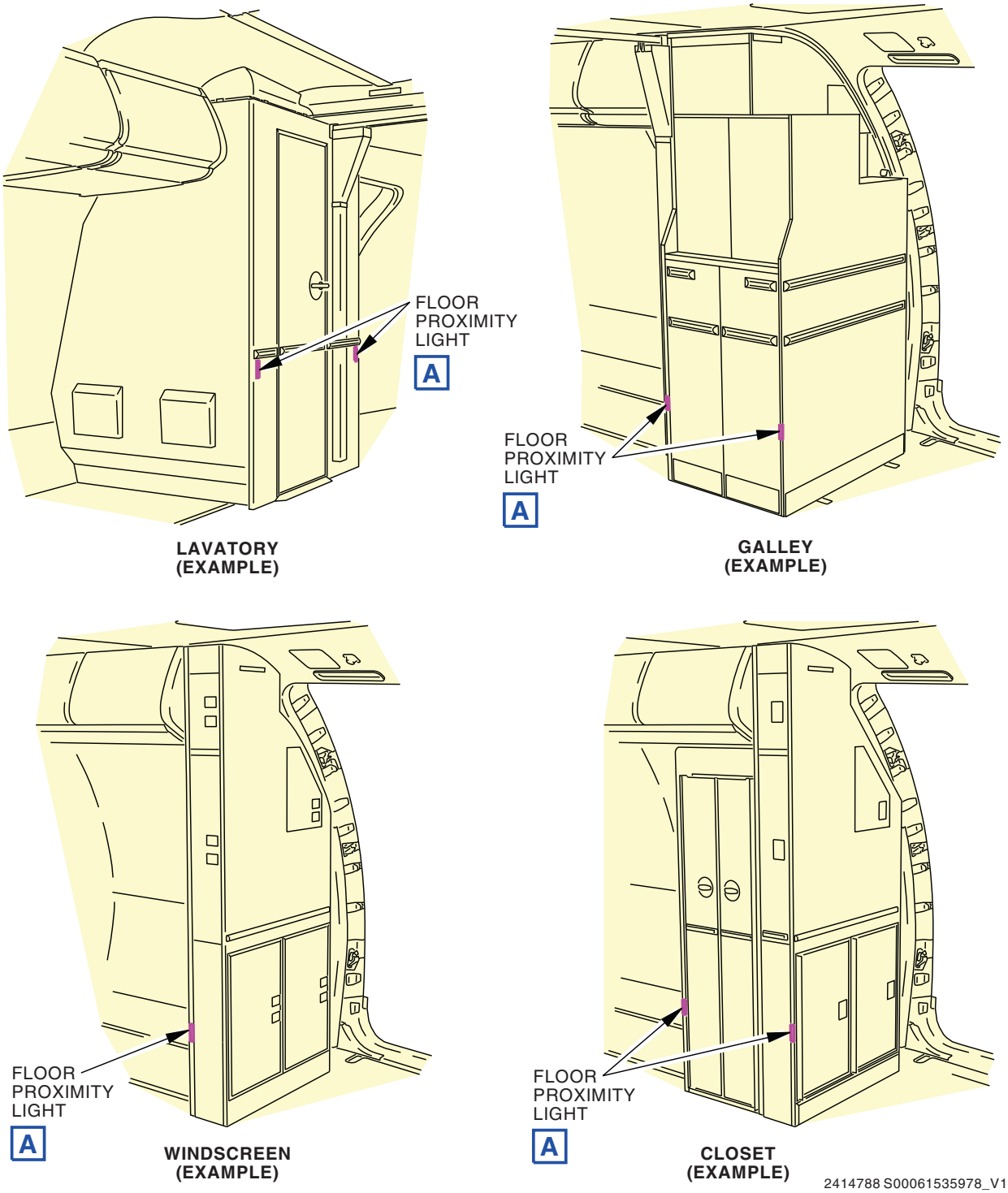
1) Make sure that the lamp comes on.

(b) Set the switch to the NORMAL mode.

1) Make sure that all emergency lights and signs go off.

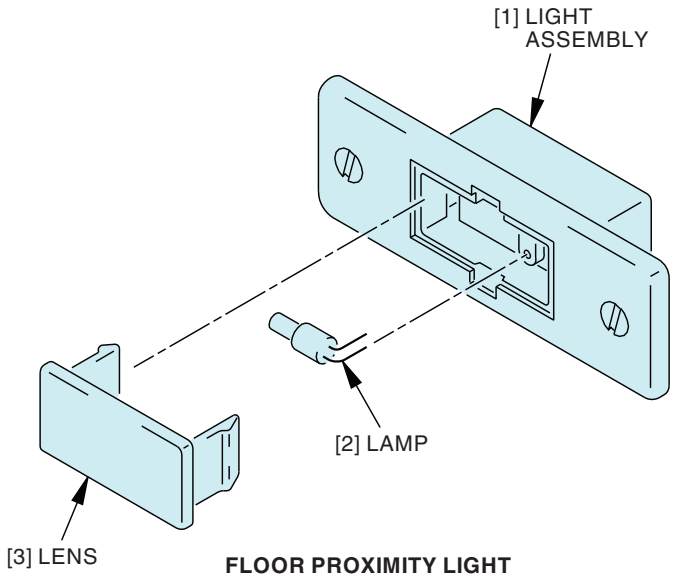
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



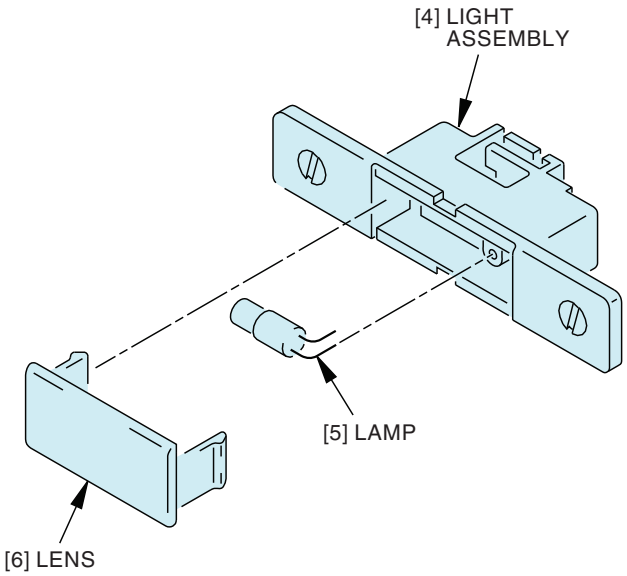
Floor Proximity Light - Lamp Replacement
Figure 201/33-51-14-990-801 (Sheet 1 of 2)

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FLOOR PROXIMITY LIGHT (EXAMPLE)

A



FLOOR PROXIMITY LIGHT (EXAMPLE)

A

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Floor Proximity Light - Lamp Replacement
Figure 201/33-51-14-990-801 (Sheet 2 of 2)

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TASK 33-51-14-960-802

3. Floor Proximity Light - Light Assembly Replacement

(Figure 202)

A. References

Reference	Title
WDM 33-51-21	Wiring Diagram Manual

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Procedure

SUBTASK 33-51-14-960-002

(1) Do these steps to replace the applicable light assembly (WDM 33-51-21):

(a) Remove the fasteners [7] or fasteners [9].

(b) Pull out the light assembly [1] or light assembly [4].

(c) Disconnect the electrical wires [8] or electrical wires [10].

(d) Connect the electrical wires [8] or electrical wires [10] to the new light assembly [1] or light assembly [4].

1) Make sure that the new light assembly [1] or light assembly [4] includes a lamp (TASK 33-51-14-960-801).

(e) Put in the new light assembly [1] or light assembly [4].

1) Make sure that the electrical wires [8] or electrical wires [10] are installed correctly in their usual position.

(f) Install the fasteners [7] or fasteners [9].

SUBTASK 33-51-14-710-002

(2) Do a test of the new light assembly [1] or light assembly [4]:

NOTE: The emergency lights are energized by the power supplies. Use the lights for a minimum time to do the test.

(a) At the attendant panel in the passenger compartment, set the emergency light switch to the ON mode.

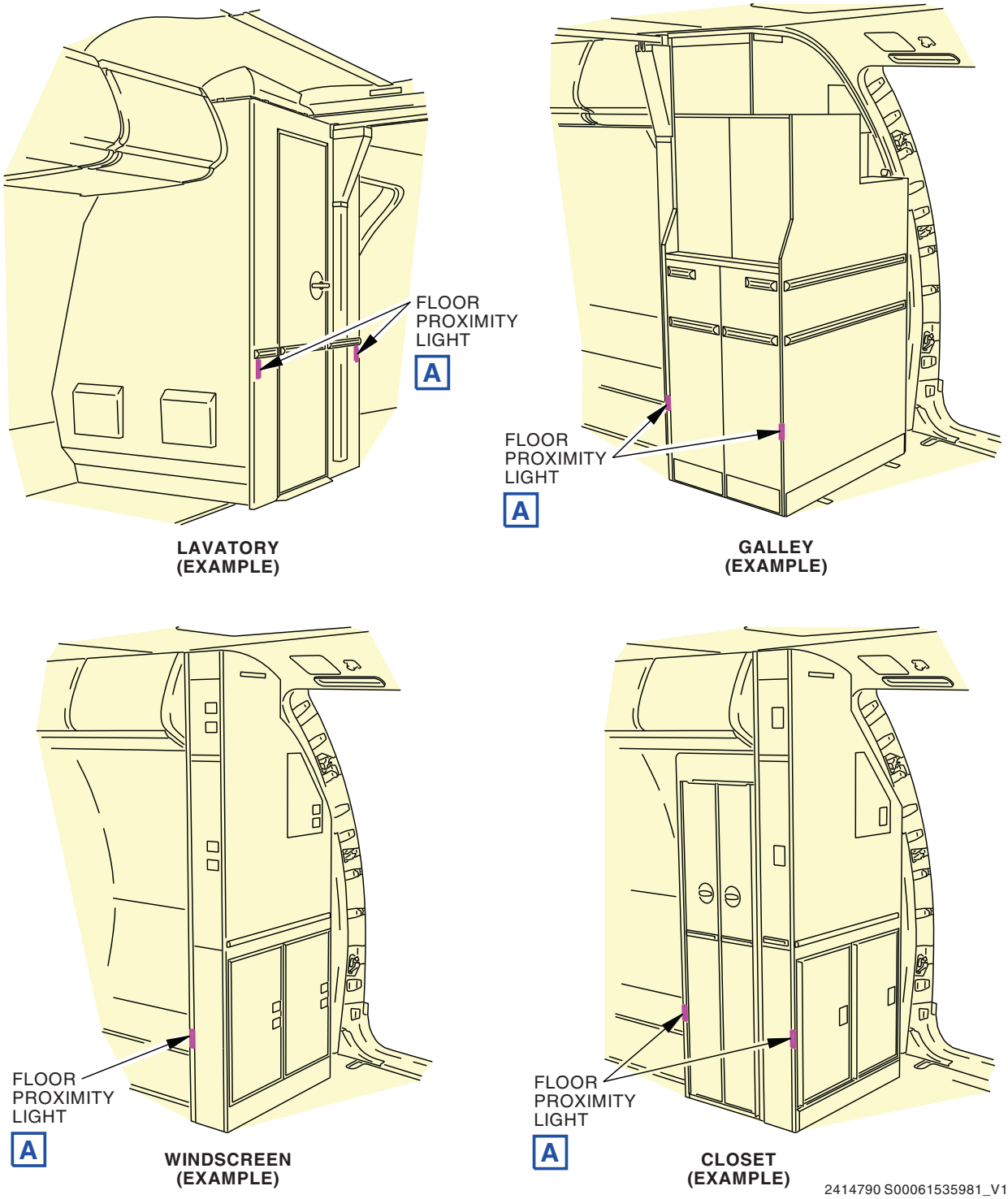
1) Make sure that the light comes on.

(b) Set the switch to the NORMAL mode.

1) Make sure that all emergency lights and signs go off.

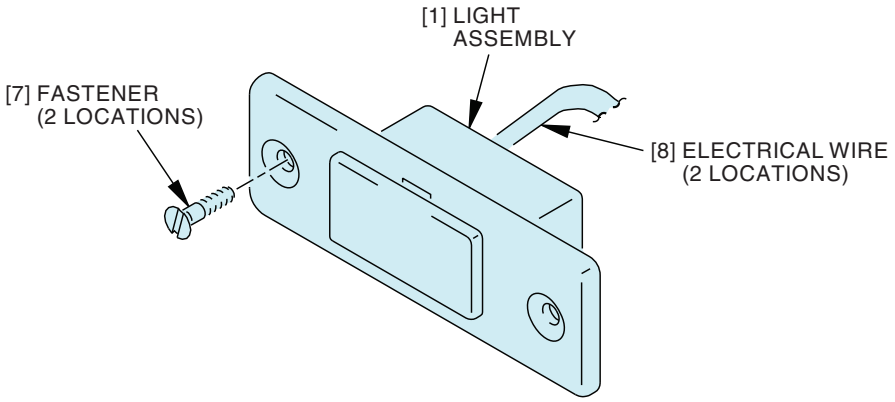
——— END OF TASK ———

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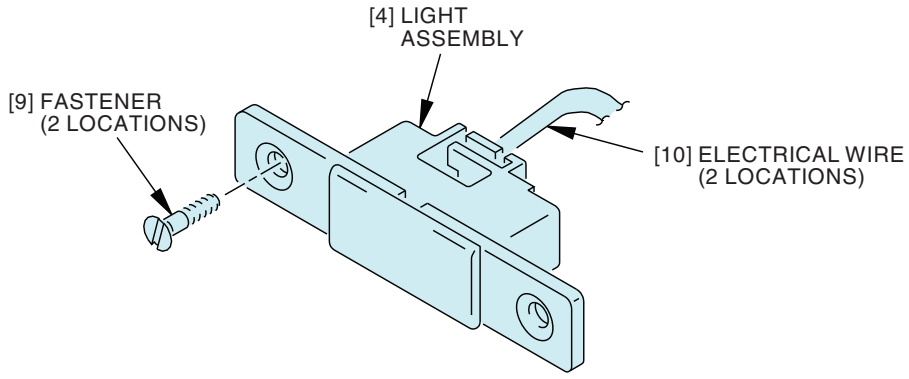
Floor Proximity Light - Light Assembly Replacement
Figure 202/33-51-14-990-802 (Sheet 1 of 2)

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FLOOR PROXIMITY LIGHT (EXAMPLE)

A



FLOOR PROXIMITY LIGHT (EXAMPLE)

A

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Floor Proximity Light - Light Assembly Replacement
Figure 202/33-51-14-990-802 (Sheet 2 of 2)

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FLOOR PROXIMITY LIGHTS, PHOTOLUMINESCENT - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Floor Proximity Light - Charge the Photoluminescent Strips

(2) Floor Proximity Light - Photoluminescent Strip Replacement

(3) Floor Proximity Light - Photoluminescent Strip Cleaning

(4) Floor Proximity Light - Photoluminescent Strip Inspection

B. Do not use this procedure for the exit signs that identify the doors and escape hatches.

TASK 33-51-15-860-801

2. Floor Proximity Light - Charge the Photoluminescent Strips

A. General

(1) The photoluminescent strips are energized when light is present. This procedure uses the passenger cabin lights to supply a continuous charge to all sections of the photoluminescent strips.

(2) The procedure uses three types of Boeing Sky Interior (BSI) light scenes to charge photoluminescent strips.

(a) Standard BSI light scene.

(b) Airline custom BSI light scene.

1) The photoluminescent strips charging criteria is to use the White - White - Amber (WWA) light types, such as the Center Overhead Stowage Lights, Crossbin Lights and Direct Lights. This charging criterion is limited to WWA light types because of their ability to charge photoluminescent strips as compared to the full color Light Emitting Diode (LED) lights.

2) The photoluminescent charge criteria that are limited to the W1 (cool white light source) and W2 (warm white light source) colors with High or Medium intensities. This is because these colors and intensities have the ability to charge the photoluminescent strips.

(c) Attendant Control Panel (ACP) Inoperative (30% Brightness).

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Charge the Photoluminescent Strips with the Standard BSI Light Scene

SUBTASK 33-51-15-860-001

(1) Do these steps to charge the photoluminescent strips with the standard BSI light scene:

(a) At the applicable ACP, set the Passenger Seating Area scene to White BRIGHT.

NOTE: Passenger compartment lighting is very important to the performance of the photoluminescent floor proximity system. Use the applicable maintenance procedures to correctly maintain these lights.

(b) To do the initial charging of the photoluminescent strips, select the scenarios from the applicable table (Table 201 or Table 202).

NOTE: The tables below contain charging information and can be used to determine how long the strips remain illuminated (Maximum Duration Allowed (Hours)).

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Table 201/33-51-15-993-801 BSI First Flight Charging Requirement for Photoluminescent Strips (Blue or Indigo Color Strips)

Charge Scenario (Light Scene)	Maximum Photoluminescent Duration Allowed (Hours)	Additional Steps to Follow for charging
First flight of the day Bins closed (White BRIGHT)		Close overhead bins during charging. Cabin activity is limited to minor aisle traffic of crew and personnel.
5 minute charge	3.5	
10 minute charge	5.0	
15 minute charge	7.5	
First flight of the day Bins open No Passengers (White BRIGHT)		Bin doors can be open during charging. Cabin activity is limited to minor aisle traffic or crew and personnel.
10 minute charge	5.0	
15 minute charge	7.5	

Table 202/33-51-15-993-802 BSI First Flight Charging Requirement for Photoluminescent Strips (Not Blue and Not Indigo Color Strips)

Charge Scenario (Light Scene)	Maximum Photoluminescent Duration Allowed (Hours)	Additional Steps to Follow for charging
First flight of the day Bins closed (White BRIGHT)		Close overhead bins during charging. Cabin activity is limited to minor aisle traffic of crew and personnel.
5 minute charge	8.0	
10 minute charge	9.5	
15 minute charge	12.5	
First flight of the day Bins open No Passengers (White BRIGHT)		Bin doors can be open during charging. Cabin activity is limited to minor aisle traffic or crew and personnel.
10 minute charge	9.5	
15 minute charge	12.5	

(c) Photoluminescent duration can be extended more than the initial charge, by using the extended charge scenarios from the applicable table (Table 203 or Table 204).

NOTE: The photoluminescent duration times listed in Table 203 and Table 204 differ from those listed in Table 201 and Table 202 because of differences in the charge time starting conditions.

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Table 203/33-51-15-993-803 BSI Quick Turnaround and In-Flight Charging Requirement (Blue or Indigo Color Strips)

Charge Scenario (Light Scene)	Maximum Photoluminescent Duration Allowed (Hours)	Additional Steps to Follow for charging
Continuous Flight	No limit if ceiling lights and sidewall lights stay on.	If the Attendant Control Panel loses communication with the ceiling lights and sidewall lights, these lights will go to default medium white. Flight duration can be extended continuously.
In-Flight Charging (Taxi/Take-off) 10 minute charge (Meal/Beverage) 15 minute charge (Day Cruise) 15 minute charge	 2.5 3.5 3.5	 Charging must begin prior to previous discharge duration ending. Strips are assumed to be shadowed during meal service by galley cart and flight attendants which are stationary for 5 minutes.
Quick turn with bin doors open and passengers in seats (White BRIGHT) 10 minute charge 15 minute charge	 5.0 7.5	Charging must begin prior to previous discharge duration ending. Bin doors can be open during charging. Passenger loading and unloading periods cannot be included in the charge time. Passengers can be seated on the airplane.

Table 204/33-51-15-993-804 BSI Quick Turnaround and In-Flight Charging Requirement (Not Blue and Not Indigo Color Strips)

Charge Scenario (Light Scene)	Maximum Photoluminescent Duration Allowed (Hours)	Additional Steps to Follow for charging
Continuous Flight	No limit if ceiling lights and sidewall lights stay on.	If the Attendant Control Panel loses communication with the ceiling lights and sidewall lights, these lights will go to default medium white. Flight duration can be extended continuously.
In-Flight Charging (Taxi/Take-off) 10 minute charge (Meal/Beverage) 15 minute charge (Day Cruise) 15 minute charge	 3.5 5.0 5.0	 Charging must begin prior to previous discharge duration ending. Strips are assumed to be shadowed during meal service by galley cart and flight attendants which are stationary for 5 minutes.

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Table 204/33-51-15-993-804 BSI Quick Turnaround and In-Flight Charging Requirement (Not Blue and Not Indigo Color Strips) (Continued)

Charge Scenario (Light Scene)	Maximum Photoluminescent Duration Allowed (Hours)	Additional Steps to Follow for charging
In-Flight Charging - Double Charge at 3 Hours and 6.5 Hours (Meal/Beverage) 15 minute charge (Day Cruise) 30 minute charge	7.5 7.5	Charging must begin prior to previous discharge duration ending. Strips are assumed to be shadowed during meal service by galley cart and flight attendants which are stationary for 5 minutes. To get double charging credit, strips must be charged within 3 hours ± 15 minutes after prior charge. The second charge must take place before 6.5 hours have elapsed from initial charge.
Quick turn with bin doors open and passengers in seats (White BRIGHT) 10 minute charge 15 minute charge	 9.5 12.5	Charging must begin prior to previous discharge duration ending. Bin doors can be open during charging. Passenger loading and unloading periods cannot be included in the charge time. Passengers can be seated on the airplane.

D. Charge the Photoluminescent Strips with the Airline Custom BSI Light Scene

SUBTASK 33-51-15-860-002

(1) Do these steps to charge the photoluminescent strips with the Airline custom BSI light scene:

(a) At the applicable ACP, set the Passenger Seating Area scene to White BRIGHT.

NOTE: Passenger compartment lighting is very important to the performance of the photoluminescent floor proximity system. Use the applicable maintenance procedures to correctly maintain these lights.

(b) To do the initial charging of the photoluminescent strips, select the scenario from the applicable table (Table 205 or Table 206).

NOTE: The tables below contain charging information and can be used to determine how long the strips remain illuminated (Maximum Duration Allowed (Hours)).

NOTE: These tables give guidance on photoluminescent strip charging. The airline must coordinate approval of these charging guidance and placard visibility when using custom scenes with their regulatory agency.

NOTE: W1 and W2 are two different colors:

- W1 - Cool white light source with Medium/High intensity
- W2 - Warm white light source with Medium/High intensity.

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Table 205/33-51-15-993-805 BSI First Flight Charging Requirement for Photoluminescent Strips (Blue or Indigo Color Strips)

Charge Scenario (Light Scene)	Maximum Photoluminescent Duration Allowed (Hours)	Additional Steps to Follow for charging
First flight of the day Bins closed No passenger (Crossbin/COS/Direct W1/W2 High) 15 minute charge	3.5	Close overhead bins during charging. Cabin activity is limited to minor aisle traffic of crew and personnel. Passengers will shadow the system and are not allowed on board during charging.

Table 206/33-51-15-993-806 BSI First Flight Charging Requirement for Photoluminescent Strips (Not Blue and Not Indigo Color Strips)

Charge Scenario (Light Scene)	Maximum Photoluminescent Duration Allowed (Hours)	Additional Steps to Follow for charging
First flight of the day Bins closed No passenger (Crossbin/COS/Direct W1/W2 High) 15 minute charge	7.0	Close overhead bins during charging. Cabin activity is limited to minor aisle traffic of crew and personnel. Passengers will shadow the system and are not allowed on board during charging.

(c) Photoluminescent duration can be extended more than the initial charge, by using the extended charge scenarios from the applicable table (Table 207 or Table 208).

NOTE: The photoluminescent duration times listed in Table 207 and Table 208 differ from those listed in Table 205 and Table 206 because of differences in the charge time starting conditions.

NOTE: These tables give guidance on photoluminescent strip charging. The airline must coordinate approval of these charging guidance and placard visibility when using custom scenes with their regulatory agency.

NOTE: W1 and W2 are two different colors:

- W1 - Cool white light source with Medium/High intensity
- W2 - Warm white light source with Medium/High intensity.

Table 207/33-51-15-993-807 BSI Quick Turnaround and In-Flight Charging Requirement (Blue or Indigo Color Strips)

Charge Scenario (Light Scene)	Maximum Photoluminescent Duration Allowed (Hours)	Additional Steps to Follow for charging
In-Flight Charging (Crossbin/COS/Direct W1/W2 Medium) 15 minute charge	2.5	Charging must begin prior to previous discharge duration ending. Strips are assumed to be shadowed during meal service by galley cart and flight attendants which are stationary for 5 minutes.
(Crossbin/COS/Direct W1/W2 High) 15 minute charge	3.5	

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Table 207/33-51-15-993-807 BSI Quick Turnaround and In-Flight Charging Requirement (Blue or Indigo Color Strips) (Continued)

Charge Scenario (Light Scene)	Maximum Photoluminescent Duration Allowed (Hours)	Additional Steps to Follow for charging
Quick turnaround Bin closed No passengers. (Crossbin/COS/Direct W1/W2 High) 15 minute charge	3.5	Close overhead bins during charging. Cabin activity is limited to minor aisle traffic of crew and personnel. Passengers will shadow the system and are not allowed on board during charging.

Table 208/33-51-15-993-808 BSI Quick Turnaround and In-Flight Charging Requirement (Not Blue and Not Indigo Color Strips)

Charge Scenario (Light Scene)	Maximum Photoluminescent Duration Allowed (Hours)	Additional Steps to Follow for charging
In-Flight Charging (Crossbin/COS/Direct W1/W2 Medium) 15 minute charge	3.5	Charging must begin prior to previous discharge duration ending. Strips are assumed to be shadowed during meal service by galley cart and flight attendants which are stationary for 5 minutes.
(Crossbin/COS/Direct W1/W2 High) 15 minute charge	5.0	
In-Flight Charging - Double Charge at 3 Hours and 6.5 Hours (Crossbin/COS/Direct W1/W2 Medium) 15 minute charge	6.0	Charging must begin prior to previous discharge duration ending. Strips are assumed to be shadowed during meal service by galley cart and flight attendants which are stationary for 5 minutes. To get double charging credit, strips must be charged within 3 hours ± 15 minutes after prior charge. The second charge must take place before 6.5 hours have elapsed from initial charge.
(Crossbin/COS/Direct W1/W2 High) 15 minute charge	7.5	
Quick turnaround Bin closed No passengers. (Crossbin/COS/Direct W1/W2 High) 15 minute charge	7.0	Close overhead bins during charging. Cabin activity is limited to minor aisle traffic of crew and personnel. Passengers will shadow the system and are not allowed on board during charging.

E. Charge the Photoluminescent Strips with the Attendant Control Panel Inoperative

SUBTASK 33-51-15-860-003

(1) Do these steps to charge the photoluminescent strips with an ACP inoperative (30% Brightness - Medium White Setting):

NOTE: Passenger compartment lighting is very important to the performance of the photoluminescent floor proximity system. Use the applicable maintenance procedures to correctly maintain these lights.

(a) To do the charging of the photoluminescent strips, use the table below:

NOTE: The table below contains charging information and can be used to determine how long the strips remain illuminated (Maximum Duration Allowed (Hours)).

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Table 209/33-51-15-993-809 BSI Charging Requirement for the Attendant Control Panel Inoperative (30% Brightness)

Charge Scenario (Light Scene)	Maximum Photoluminescent Duration Allowed (Hours)	Additional Steps to Follow for charging
Continuous Flight	No limit with ceiling lights and sidewall lights on	If the Attendant Control Panel loses communication with the ceiling lights and sidewall lights, these lights will go to default medium white. Flight duration can be extended continuously.
First flight of the day Bins closed No passengers (White Medium) 15 minute charge	7.0	Close overhead bins during charging. Cabin activity is limited to minor aisle traffic of crew and personnel. Passengers will shadow the system and are not allowed on board during charging.

END OF TASK

TASK 33-51-15-960-801

3. Floor Proximity Light - Photoluminescent Strip Replacement

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task has the steps to replace sections or complete lengths of the photoluminescent strips.

(a) Replacement strip sections must be a minimum of 18 in. (457.2 mm) long.

(b) Overwing exit aisle strips must be replaced as one piece.

B. References

Reference	Title
51-22-00-160-801	Prepare For Sealing (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A00635	Adhesive - RTV 108 Translucent Silicone Rubber RTV Paste, One-part	BAC5010 Type 60 Class A Grade 2 / MIL-A-46106 Type 1
A50038	Adhesive - Fast-setting, 2 Part Epoxy, Medium Viscosity	BMS5-123 Type I Class 2 Clear
A50464	Tape - Vinyl, Adhesive Coated	BMS5-179
A50555	Adhesive - 2 Part Epoxy, Permabond ET500	
A50556	Adhesive - 2 Part Epoxy, Araldite Fusion	
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
B00148	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B00184	Solvent - Presealing, Cleaning Solvent	BMS11-7
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G02129	Tape - 223S (use until stock depleted)	

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(Continued)

Reference	Description	Specification
G50156	Tape - Carpet, Double-sided, 16 oz/sq. yd., Black. Storage Life 18 Months.	BMS5-133 Type II Class 2

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Photoluminescent insert	33-51-15-18-005	PTF ALL

E. Location Zones

Zone	Area
200	Upper Half of Fuselage

F. Photoluminescent Strip Replacement

SUBTASK 33-51-15-960-001

(1) Do these steps to replace the photoluminescent strip:

(a) Use a smooth tool to separate the tape holding the light strip onto the floor panels.

(b) Remove the old tape that held the strip onto the floor panels.

(c) Clean the floor panels:

1) Apply solvent, B00184, or solvent, B00148, or solvent, B00083, with a clean oil-free cotton wiper, G00034.

2) Rub the surface to remove the tape.

3) Use a clean cotton wiper, G00034, to remove the solvent before it evaporates.

(d) Do these steps to prepare the new photoluminescent strips that do not have end caps:

1) If it is necessary, trim the new strip to fit.

NOTE: If this is a strip or a portion of strip to be mounted between existing strips, you will want to leave room to seal the exposed ends of the new strip and still maintain the appropriate gap between each end of the existing strips.

2) Do this task: Prepare For Sealing, TASK 51-22-00-160-801.

CAUTION

SEALANT MUST NOT EXTEND OVER PERIPHERY OF PHOTOLUMINESCENT STRIP.

3) Carefully seal the exposed end of all strips with small beads of adhesive, A50038.

(e) Do these steps to prepare the new photoluminescent strips that have end caps [3] (Figure 201):

NOTE: If this is a strip or a portion of strip to be mounted between existing strips, you will want to leave room to seal the exposed ends of the new strip and still maintain the appropriate gap between each end of the existing strips.

1) If the end caps [3] are installed, remove the end caps [3] from each end of the cover sleeve [2].

2) If it is necessary, do these steps to trim the new photoluminescent insert [1] to fit.

a) Remove the color filter [4] and photoluminescent insert [1] in the cover sleeve [2].

b) Cut the cover sleeve [2] to the applicable length.

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c) Cut the color filter [4] and photoluminescent insert [1] to 0.039 in. (1.0 mm) - 0.079 in. (2.0 mm) shorter than the length of the cover sleeve [2].

d) Clean the cover sleeve [2], color filter [4], and photoluminescent insert [1] from dust and unwanted material.

e) Put the color filter [4] and photoluminescent insert [1] into the cover sleeve [2].

3) Do this task: Prepare For Sealing, TASK 51-22-00-160-801.

4) Apply small beads of Permabond ET500 epoxy adhesive, A50555, Araldite Fusion epoxy adhesive, A50556, or RTV 108 adhesive, A00635, to each end of the cover sleeve [2].

5) If the end caps [3] were removed, install the end caps [3] to each end of the cover sleeve [2].

6) Remove the excess adhesive, from the photoluminescent strip.

(f) If it is necessary, clean the floor panels:

1) Apply solvent, B00184, or solvent, B00148, or solvent, B00083, with a clean oil-free cotton wiper, G00034.

2) Use a clean cotton wiper, G00034, to remove the solvent before it evaporates.

(g) Clean the photoluminescent strips (TASK 33-51-15-100-801).

(h) Do these steps if the new photoluminescent strips do not have a filler (Figure 201):

1) Apply new tape, G50156, to the floor panel.

2) Using the new piece(s) of photoluminescent strip as a guide, mark tape, G50156, where the ends of the strip will be located.

(i) Place small lateral strips of ¾ in. (19 mm) wide Nitto 223S Tape, G02129, or tape, A50464, to hide all visible tape, G50156, and floor panels between the strips.

(j) Place the strip in its location, maintaining the gap indicated below between the new strip and existing strips:

1) At the expansion joint, STA 520, maintain a 0.30 in. (7.62 mm) gap between the new strip and adjoining strip.

2) At all other locations, maintain a 0.10 in. (2.54 mm) gap between the new strip and existing strips.

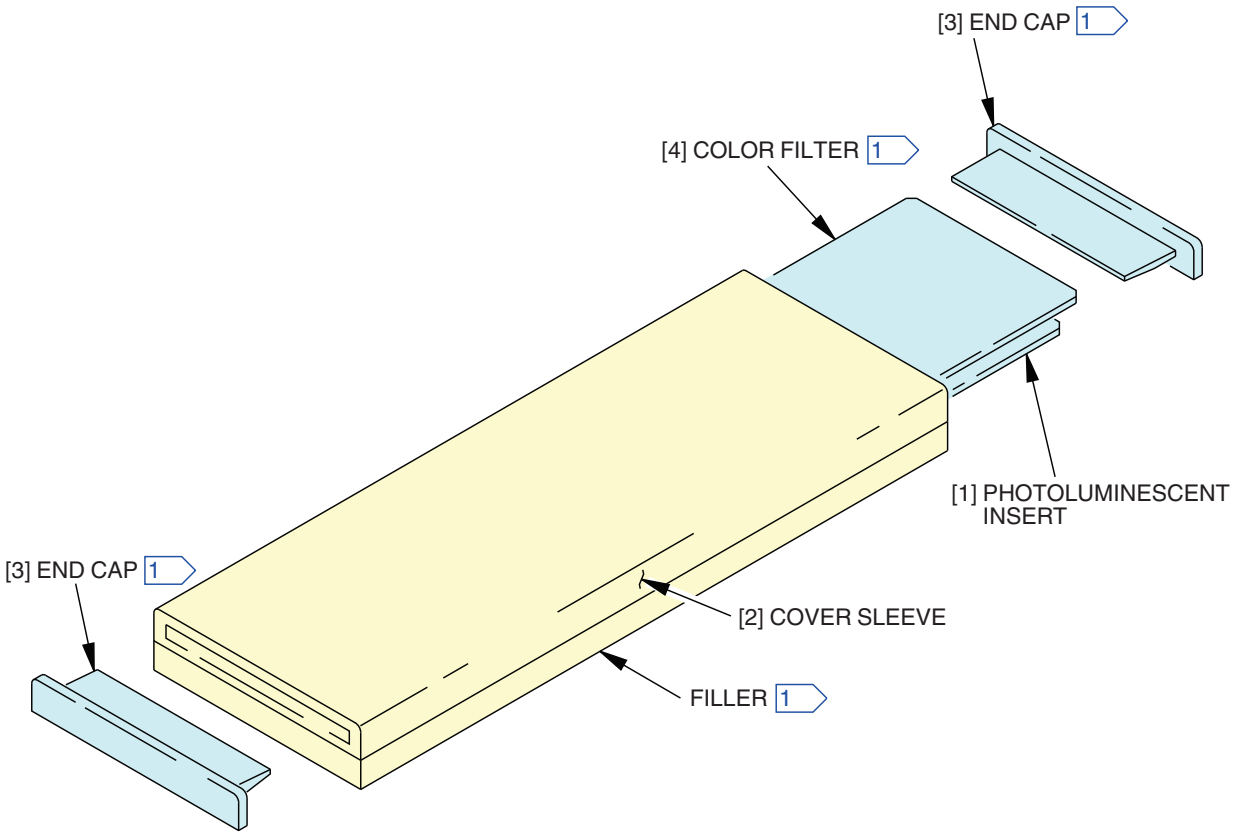
(k) Press the strip down firmly along it's length to attach it in place.

SUBTASK 33-51-15-210-002

(2) Do this task: Floor Proximity Light - Photoluminescent Strip Inspection, TASK 33-51-15-210-801.

——— END OF TASK ———

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COLOR PHOTOLUMINESCENT STRIP (TYPICAL)

1 IF APPLICABLE

2414792 S00061535987_V3

Floor Proximity Light - Photoluminescent Strip Replacement
Figure 201/33-51-15-990-801

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TASK 33-51-15-100-801

4. Floor Proximity Light - Photoluminescent Strip Cleaning

A. Consumable Materials

Reference	Description	Specification
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
B00541	Cleaner - General Purpose Household Detergent	

B. Location Zones

Zone	Area
200	Upper Half of Fuselage

C. Photoluminescent Strip Cleaning

SUBTASK 33-51-15-100-001

CAUTION

MAKE SURE THAT YOU APPLY THE SOLVENT TO THE LIGHT STRIP. IF THE SOLVENT GETS ON THE TRACK OR THE ENDS OF THE TRACK, DAMAGE TO THE AIRPLANE CAN OCCUR.

(1) Do this task to clean installed photoluminescent strips.

(a) Use a cloth dampened with general purpose household detergent cleaner, B00541, and water and clean the strips.

(b) If a stronger cleaner is necessary, use a cloth dampened with solvent, B00083. Then clean the strips.

SUBTASK 33-51-15-410-001

(2) If it is necessary to remove photoluminescent strips to clean unwanted material, do these tasks:

(a) Do an inspection of the removed strips (TASK 33-51-15-210-801).

(b) Install the strips (TASK 33-51-15-960-801).

——— END OF TASK ———

TASK 33-51-15-210-801

5. Floor Proximity Light - Photoluminescent Strip Inspection

A. Location Zones

Zone	Area
200	Upper Half of Fuselage

B. Photoluminescent Strip Inspection

SUBTASK 33-51-15-210-001

(1) Do this procedure to examine the light strips.

(a) FOR STANDARD PHOTOLUMINESCENT STRIP;

There can be no more than 5.0 in. (12.7 cm) of length of photoluminescent strip between 15.0 in. (38.1 cm) and 72.0 in. (182.9 cm) that will not give out light.

(b) FOR COLORED PHOTOLUMINESCENT STRIP;

There can be no more than 5.0 in. (12.7 cm) of length of colored photoluminescent strip between 15.0 in. (38.1 cm) and 79.0 in. (200.7 cm) that will not give out light.

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(c) If the strips are dirty, do this task: TASK 33-51-15-100-801.

(d) For sections of light strip that cannot be cleaned to give out light, do this task: TASK 33-51-15-960-801.

——— END OF TASK ———

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FLOOR PROXIMITY LIGHTS, CABIN LIGHT INTENSITY - ADJUSTMENT/TEST

1. General

A. This procedure contains this task:

(1) The measuring of the available light intensity of the cabin lighting.

TASK 33-51-15-860-802

2. Cabin Lighting - Measure the Available Light Intensity

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure measures the amount of light available to provide the charge for the photoluminescent floor proximity strips.

(2) The photoluminescent floor proximity strips are energized when light is present.

(3) Depending on the type of light-meter selected, these conversion factors will allow readings in Foot-Candles (Fc) or Lux to be used:

(a) Fc x 10.764 = Lux

(b) Lux x 0.0929 = Fc.

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-11030	Sensor - Illuminance (Photometric) Part #: U23-01-101 Supplier: 23673
COM-11031	Meter - Light (Photometer) Part #: U68188 Supplier: 23673
COM-23586	Meter - Light (with Light Sensor) Part #: EA33 Supplier: 093V7 Part #: XRP-3000 Supplier: 8W1E4

C. Consumable Materials

Reference	Description	Specification
G00270	Tape - Scotch Flatback Masking 250	ASTM D6123 (Supersedes A-A-883)

D. Location Zones

Zone	Area
200	Upper Half of Fuselage

E. Prepare for the Test

SUBTASK 33-51-15-980-001

(1) Move the airplane out of direct sunlight or daylight conditions.

NOTE: You can move the airplane inside a hanger or take the readings at night.

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SUBTASK 33-51-15-860-006

(2) Make sure that the airplane interior is in fully operational passenger provisions configuration.

NOTE: Provisions to include all seats, bins, galleys, lavs, etc. installed for normal use.

SUBTASK 33-51-15-280-001

(3) Do these steps to mark light intensity measurement points on the aisleway carpet (Figure 501):

(a) Place a piece of Scotch Flatback Masking Tape 250, G00270, centered between the photoluminescent strips, on the aisleway carpet at the centerline of the forward cabin door.

(b) Use a ruler or measuring tape to measure 40 in. (1016 mm) aft of that piece and put another piece of tape on the carpet.

(c) In like fashion, place pieces of tape every 40 in. (1016 mm) throughout the rest of the length of the cabin, all the way back to, and including, the centerline of the most aft passenger door.

SUBTASK 33-51-15-970-002

(4) Make several copies of (Table 501) to use for recording data at each measurement point.

SUBTASK 33-51-15-860-007

(5) Do these steps to set the lights for the cabin light level test:

(a) At the applicable attendant control panel, set the Passenger Seating Area to the White Bright scene.

(b) Set all reading lights to the off mode.

(c) Close all window shades, lavatory doors, cabin egress doors, and all overhead bin doors.

(d) Maintain the above conditions for a minimum of 30 minutes to stabilize the light source.

F. Available Light Intensity Measuring

SUBTASK 33-51-15-860-005

(1) Do these steps to measure the light intensity throughout the passenger cabin (Figure 501):

NOTE: One person should perform the tests while another person records the data.

(a) Use the light meter, COM-11031, in combination with the illuminance photometric sensor, COM-11030, or the standalone light meter, COM-23586.

1) If the light meter, COM-11031, and the illuminance photometric sensor, COM-11030, are used, do these steps:

a) Connect the cable from the light meter, COM-11031, to the illuminance photometric sensor, COM-11030.

b) Turn the light meter, COM-11031, on.

c) Zero or initialize the light meter, COM-11031, if it is necessary, then make sure to turn on the following settings:

<1> 0 to 200 Lux Range (or approximately 0-25 Fc)

<2> Daylight Mode (if applicable)

<3> Zero or initialize (if it is necessary).

(b) Do these steps to measure the light intensity:

1) Place the light meter, COM-23586, or the illuminance photometric sensor, COM-11030, that was previously connected to the light meter, COM-11031, on the light strip on one side of the aisle at the point to be measured.

a) Make sure that the sensor points up towards the cabin ceiling.

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2) Make sure that no object is in the path of the light to the light meter, COM-23586, or illuminance photometric sensor, COM-11030.

3) Record the measurement point number and the light level value in the Table 501.

Table 501/33-51-15-993-810 Cabin Light Intensity Test Light Meter Readings

Measurement Point	Measured Light (LUX)	
	Left Side	Right Side
Subtotal This Page:		
Subtotal Previous Page:		
Overall Total:		
Average (Last Page Only):		

4) Move the light meter, COM-23586, or illuminance photometric sensor, COM-11030, onto the strip on the other side of the aisle and measure and record the light level value on the data sheet for that side of the aisle.

5) Repeat these steps for other points along the cabin floor.

SUBTASK 33-51-15-970-001

(2) Do these steps to calculate the average light level in the passenger cabin:

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(a) Add all the recorded readings from all data sheets for values measured on each side of the aisle and record those totals on the last sheet.

NOTE: You will have two totals, one for the right side and one for the left side.

(b) Divide each total by the number of readings taken for that side.

(c) Make sure that the average value for each side is 30 Lux (2.787 Fc) minimum.

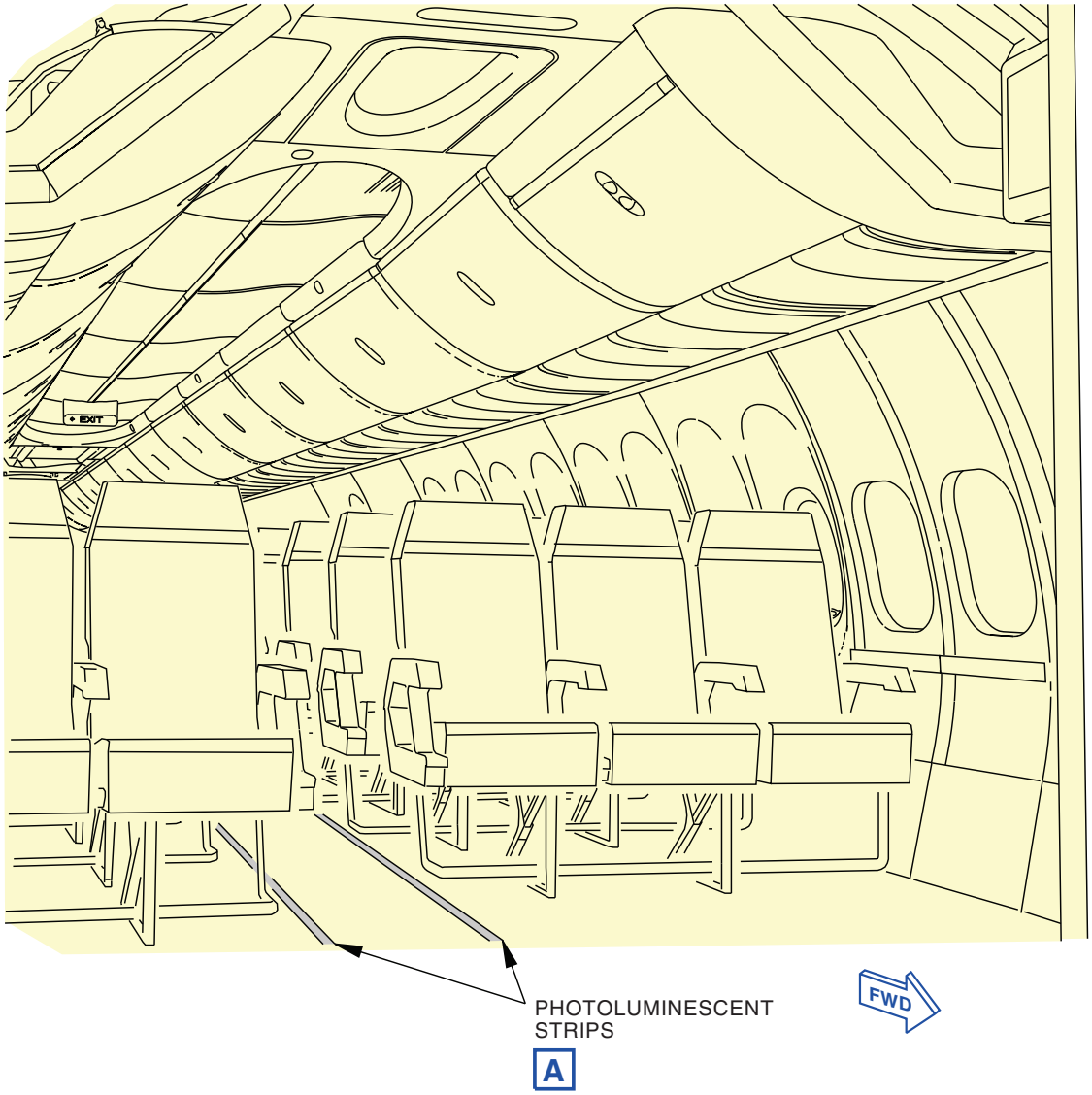
1) If you do not get 30 Lux (2.787 Fc) minimum, clean the light lens or replace the applicable light assembly.

SUBTASK 33-51-15-940-001

(3) Return the airplane to it's usual condition.

——— END OF TASK ———

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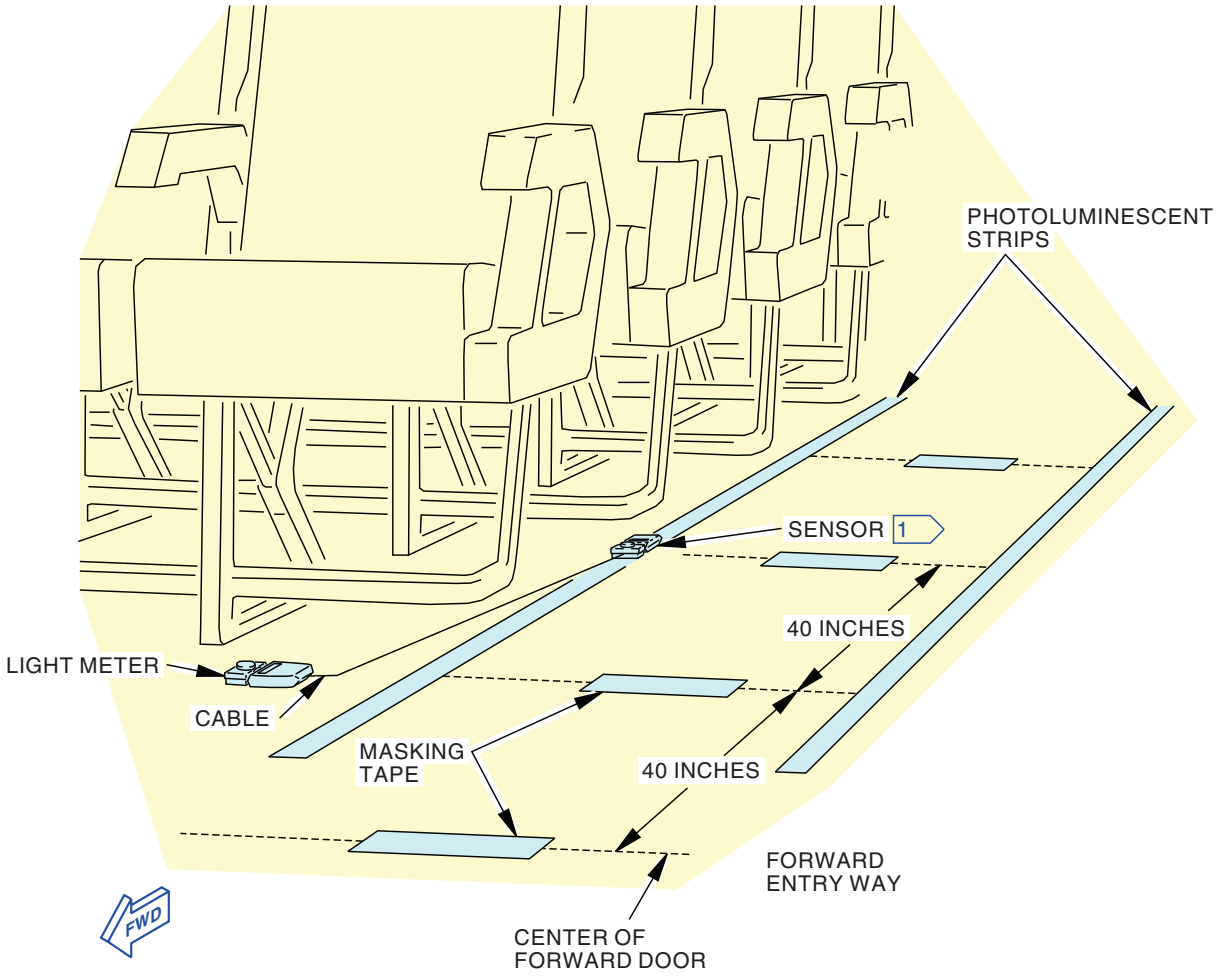


**PASSENGER COMPARTMENT
(EXAMPLE)**

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**Floor Proximity Lights - Cabin Light Intensity Test
Figure 501/33-51-15-990-802 (Sheet 1 of 2)**

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1 SENSOR POINTS UP TO CABIN CEILING.

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Floor Proximity Lights - Cabin Light Intensity Test
Figure 501/33-51-15-990-802 (Sheet 2 of 2)

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